Server Migration Service

API Reference

Issue	37
Date	2026-07-06

HUAWEI CLOUD COMPUTING TECHNOLOGIES CO., LTD.

Huawei Cloud Computing Technologies Co., Ltd.

Address: Huawei Cloud Data Center Jiaoxinggong Road
Qianzhong Avenue
Gui'an New District
Gui Zhou 550029
People's Republic of China

Website: https://www.huaweicloud.com/intl/en-us/

Contents

1 Before You Start

Welcome to *Server Migration Service API Reference*. Server Migration Service (SMS) helps you migrate applications and data from on-premises x86 physical servers or VMs on other private or public clouds to Elastic Cloud Servers (ECSs) on Huawei Cloud.

This document describes how to use SMS application programming interfaces (APIs) to perform operations such as creating, deleting, and querying migration tasks. For details about all supported operations, see **API Overview**.

If you plan to access SMS through an API, ensure that you are familiar with SMS concepts. For details, see **Service Overview**.

Endpoints

An endpoint is the **request address** for calling an API. Endpoints vary depending on services and regions. For the SMS endpoints, see **Regions and Endpoints**.

The following table lists SMS endpoints. Select a desired one based on the service requirements.

Table 1-1 SMS endpoint

Region Name	Region Code	Endpoint
AP-Singapore	ap-southeast-3	sms.ap-southeast-3.myhuaweicloud.com

Concepts

- Account

  An account is created upon successful registration. The account has full access permissions for all of its cloud services and resources. It can be used to reset user passwords and grant user permissions. The account is a payment entity, which should not be used directly to perform routine management. For security purposes, create Identity and Access Management (IAM) users and grant them permissions for routine management.

- User

  An IAM user is created by an account in IAM to use cloud services. Each IAM user has its own identity credentials (password and access keys).

  The account name, username, and password are required for API authentication.

- Region

  Regions are divided based on geographical location and network latency. Public services, such as Elastic Cloud Server (ECS), Elastic Volume Service (EVS), Object Storage Service (OBS), Virtual Private Cloud (VPC), Elastic IP (EIP), and Image Management Service (IMS), are shared within the same region. Regions are classified into universal regions and dedicated regions. A universal region provides universal cloud services for common tenants. A dedicated region provides specific services for specific tenants.

  For details, see **Region and AZ**.

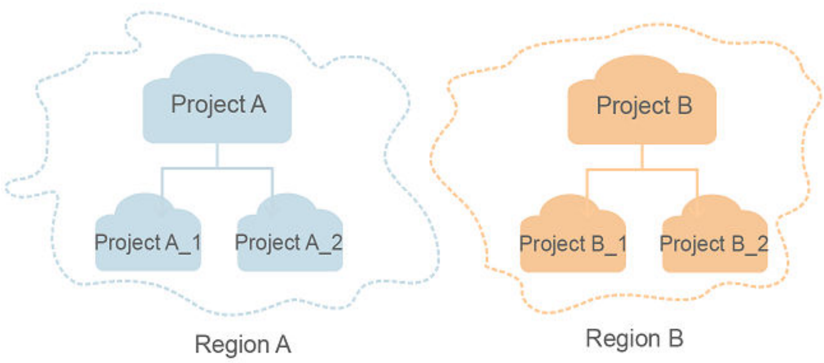
- AZ

  An AZ contains one or more physical data centers. Each AZ has independent cooling, fire extinguishing, moisture-proof, and electricity facilities. Within an AZ, compute, network, storage, and other resources are logically divided into multiple clusters. AZs within a region are interconnected using high-speed optical fibers to allow you to build cross-AZ high-availability systems.

- Project

  Projects group and isolate resources (including compute, storage, and network resources) across physical regions. A default project is provided for each region, and subprojects can be created under each default project. Users can be granted permissions to access all resources in a specific project. If you need more refined access control, create subprojects under a default project and create resources in subprojects. Then you can assign users the permissions required to access only the resources in the specific subprojects.

  Figure 1-1 Project isolation model



- Enterprise project

  Enterprise projects group and manage resources across regions. Resources in different enterprise projects are logically isolated. An enterprise project can contain resources of multiple regions, and resources can be added to or removed from enterprise projects.

  For details about enterprise projects and about how to obtain enterprise project IDs, see **Enterprise Management User Guide**.

2 API Overview

The APIs provided by the SMS service are the SMS APIs. By using the SMS APIs, you can use all the functions provided by the SMS service, including querying the source server list, creating migration tasks, and viewing the migration progress.

Table 2-1 SMS API list

Type	Description
Source server management	APIs used for reporting information about source servers, querying the source server list, and deleting the source servers
Task management	APIs used for managing migration tasks, including creating, starting, stopping, querying, and deleting a migration task.
Command management	APIs used for obtaining the commands from the service end and reporting the command execution results to the service end
Template management	APIs used for querying the template list and the information about a template with a specified ID, and adding, modifying, and deleting template information

3 Calling APIs

3.1 Making an API Request

This section describes the structure of a REST API request, and uses the IAM API for **creating an IAM user** as an example to demonstrate how to call an API.

Request URI

A request URI is in the following format:

{URI-scheme}://{Endpoint}/{resource-path}?{query-string}

Although a request URI is included in the request header, most programming languages or frameworks require the request URI to be transmitted separately.

Table 3-1 URI parameter description

Parameter	Description
URI-scheme	Protocol used to transmit requests. All APIs use HTTPS.
Endpoint	Domain name or IP address of the server bearing the REST service. The endpoint varies between services in different regions. It can be obtained from **Regions and Endpoints**. For example, the endpoint of IAM in region **CN-Hong Kong** is **iam.ap-southeast-1.myhuaweicloud.com**.
resource-path	Access path of an API for performing a specified operation. Obtain the path from the URI of an API. For example, the **resource-path** of the API used to obtain a user token is **/v3/auth/tokens**.
query-string	Query parameter, which is optional. Ensure that a question mark (?) is included before each query parameter that is in the format of *Parameter name=Parameter value*. For example, **?limit=10** indicates that a maximum of 10 data records will be displayed.

IAM is a global service. You can create an IAM user using the endpoint of IAM in any region. For example, to create an IAM user in the **CN-Hong Kong** region, obtain the endpoint of IAM (**iam.ap-southeast-1.myhuaweicloud.com**) for this region and the **resource-path** (**/v3.0/OS-USER/users**) in the URI of the API for **creating an IAM user**. Then construct the URI as follows:

```
https://iam.ap-southeast-1.myhuaweicloud.com/v3.0/OS-USER/users
```

Figure 3-1 Example URI

NOTE

To simplify the URI display in this document, each API is provided only with a **resource-path** and a request method. The **URI-scheme** of all APIs is **HTTPS**, and the endpoints of all APIs in the same region are identical.

Request Methods

The HTTP protocol defines the following request methods that can be used to send a request to the server.

Table 3-2 HTTP methods

Method	Description
GET	Requests the server to return specified resources.
PUT	Requests the server to update specified resources.
POST	Requests the server to add resources or perform special operations.
DELETE	Requests the server to delete specified resources, for example, an object.
HEAD	Same as GET except that the server must return only the response header.
PATCH	Requests the server to update partial content of a specified resource. If the resource does not exist, a new resource will be created.

For example, in the case of the API for **creating an IAM user**, the request method is **POST**. An example request is as follows:

```
POST https://iam.ap-southeast-1.myhuaweicloud.com/v3.0/OS-USER/users
```


Request Header

You can also add additional header fields to a request, such as the fields required by a specified URI or HTTP method. For example, to request for the authentication information, add **Content-Type**, which specifies the request body type.

Common request header fields are as follows.

Table 3-3 Common request header fields

Parameter	Description	Mandatory	Example Value
Host	Specifies the server domain name and port number of the resources being requested. The value can be obtained from the URL of the service API. The value is in the format of *Hostname:Port number*. If the port number is not specified, the default port is used. The default port number for **https** is **443**.	No This field is mandatory for AK/SK authentication.	code.test.com or code.test.com:443
Content-Type	Specifies the type (or format) of the message body. The default value **application/json** is recommended. Other values of this field will be provided for specific APIs if any.	Yes	application/json
Content-Length	Specifies the length of the request body. The unit is byte.	No	3495
X-Project-Id	Specifies the project ID. Obtain the project ID by following the instructions in **Obtaining a Project ID**.	No This field is mandatory for requests that use AK/SK authentication in the Dedicated Cloud (DeC) scenario or multi-project scenario.	e9993fc787d94b6c886cbaa340f9c0f4

Parameter	Description	Mandatory	Example Value
X-Auth-Token	Specifies the user token. It is a response to the API for **obtaining a user token** (This is the only API that does not require authentication). After the request is processed, the value of **X-Subject-Token** in the response header is the token value.	No This field is mandatory for token authentication.	The following is part of an example token: MIIPAgYJKoZIhvcNAQcCo...ggg1BBIINPXsidG9rZ

NOTE

In addition to supporting authentication using tokens, APIs support authentication using AK/SK, which uses SDKs to sign a request. During the signature, the **Authorization** (signature authentication) and **X-Sdk-Date** (time when a request is sent) headers are automatically added in the request.

For more details, see "Authentication Using AK/SK" in **Authentication**.

The following shows an example request of the API for **creating an IAM user** when AK/SK authentication is used:

```
POST https://iam.ap-southeast-1.myhuaweicloud.com/v3.0/OS-USER/users
Content-Type: application/json
X-Sdk-Date: 20240416T095341Z
Authorization: SDK-HMAC-SHA256 Access=****************, SignedHeaders=content-type;host;x-sdk-date,
Signature=****************
```

(Optional) Request Body

This part is optional. A request body is generally sent in a structured format (for example, JSON or XML), which is specified by **Content-Type** in the request header. It is used to transfer content other than the request header. If the request body contains full-width characters, these characters must be coded in UTF-8.

The request body varies depending on APIs. Certain APIs do not require the request body, such as the APIs requested using the GET and DELETE methods.

The following shows an example request (a request body included) of the API for **creating an IAM user**. You can learn about request parameters and related description from this example. The bold parameters need to be replaced for a real request.

- **accountid**: account ID of an IAM user
- **username**: name of an IAM user
- **email**: email of an IAM user
- **************: login password of an IAM user

```
POST https://iam.ap-southeast-1.myhuaweicloud.com/v3.0/OS-USER/users
Content-Type: application/json
X-Sdk-Date: 20240416T095341Z
```



```
Authorization: SDK-HMAC-SHA256 Access=***************, SignedHeaders=content-type;host;x-sdk-date, 
Signature=***************

{
    "user": {
        "domain_id": "accountid",
        "name": "username",
        "password": "**********",
        "email": "email",
        "description": "IAM User Description"
    }
}
```

If all data required for the API request is available, you can send the request to call the API through **curl**, **Postman**, or coding.

3.2 Authentication

Requests for calling an API can be authenticated using either of the following methods:

- AK/SK authentication: Requests are encrypted using AK/SK pairs. AK/SK authentication is recommended because it is more secure than token authentication.
- Token authentication: Requests are authenticated using tokens.

AK/SK Authentication

NOTE

- AK/SK authentication supports API requests with a body not larger than 12 MB. For API requests with a larger body, token authentication is recommended.
- An AK/SK pair can either be permanent or temporary. If it is temporary, the **X-Security-Token** field must be included in the request header. The value is the security token of the temporary AK/SK pair.
- API Gateway checks the time format and compares the request time with the time when API Gateway received the request. If the time difference exceeds 15 minutes, API Gateway will reject the request. So, the local time on the client must be synchronized with the clock server to avoid a large offset in the value of **X-Sdk-Date** in the request header.

In AK/SK authentication, an AK/SK pair is used to sign requests and the signature is then added to the requests for authentication.

- AK: access key ID, which is a unique identifier used in conjunction with a secret access key to sign requests cryptographically.
- SK: secret access key, which is used in conjunction with an AK to sign requests cryptographically. It identifies a request sender and prevents the request from being modified.

In AK/SK authentication, you can use an AK/SK pair to sign requests based on the signature algorithm or using the signing SDK. For details about how to sign requests and use the signing SDK, see **API Request Signing Guide**.

NOTE

The signing SDK is only used for signing requests and is different from the SDKs provided by services.

Token Authentication

NOTE

- A token remains valid for 24 hours after it is generated. You can cache a token and reuse it for authentication instead of generating a new one each time.
- Before using a token, ensure that it has sufficient time remaining before expiration. Using a near-expiry token may cause API call failures.

A token specifies temporary permissions in a computer system. During API authentication using a token, the token is added to requests to get permissions for calling the API. You can obtain a token by calling the **Obtaining a User Token** API.

IMS is a project-level service. When you call the API, set **auth.scope** in the request body to **project**.

```
{
    "auth": {
        "identity": {
            "methods": [
                "password"
            ],
            "password": {
                "user": {
                    "name": "username",    // IAM user name
                    "password": $ADMIN_PASS,  // IAM user password. You are advised to store it in ciphertext in 
the configuration file or an environment variable and decrypt it when needed to ensure security.
                    "domain": {
                        "name": "domainname"   // Name of the account that the IAM user belongs to
                    }
                }
            }
        },
        "scope": {
            "project": {
                "name": "xxxxxxxx"    // Project name
            }
        }
    }
}
```

After a token is obtained, the **X-Auth-Token** header field must be added to requests to specify the token when calling other APIs. For example, if the token is **ABCDEFG....**, **X-Auth-Token: ABCDEFG....** can be added to a request as follows:

```
POST https://iam.ap-southeast-1.myhuaweicloud.com/v3.0/OS-USER/users
Content-Type: application/json
X-Auth-Token: ABCDEFG....
```

3.3 Response

Status Code

After sending a request, you will receive a response, including a status code, response header, and response body.

A status code is a group of digits, ranging from 1xx to 5xx. It indicates the status of a request. For more information, see **Status Codes**.

For example, if status code **201** is returned for calling the API used to **create an IAM user**, the request is successful.

Response Header

Similar to a request, a response also has a header, for example, **Content-Type**.

Figure 3-2 shows the response header for the API used to **create an IAM user**.

NOTE

For security purposes, you are advised to set the token in ciphertext in configuration files or environment variables and decrypt it when using it.

Figure 3-2 Header fields of the response to the request for creating an IAM user

```
"X-Frame-Options": "SAMEORIGIN",
"X-IAM-ETag-id": "2562365939-d8f6f12921974cb097338ac11fceac8a",
"Transfer-Encoding": "chunked",
"Strict-Transport-Security": "max-age=31536000; includeSubdomains;",
"Server": "api-gateway",
"X-Request-Id": "af2953f2bcc67a42325a69a19e6c32a2",
"X-Content-Type-Options": "nosniff",
"Connection": "keep-alive",
"X-Download-Options": "noopen",
"X-XSS-Protection": "1; mode=block;",
"X-IAM-Trace-Id": "token_[redacted]_null_af2953f2bcc67a42325a69a19e6c32a2",
"Date": "Tue, 21 May 2024 09:03:40 GMT",
"Content-Type": "application/json; charset=utf8"
```

(Optional) Response Body

The body of a response is often returned in a structured format (for example, JSON or XML) as specified in the **Content-Type** header field. The response body transfers content except the response header.

The following shows part of the response body for the API used to **create an IAM user**.

```
{
    "user": {
        "id": "c131886aec...",
        "name": "IAMUser",
        "description": "IAM User Description",
        "areacode": "",
        "phone": "",
        "email": "***@***.com",
        "status": null,
        "enabled": true,
        "pwd_status": false,
        "access_mode": "default",
        "is_domain_owner": false,
        "xuser_id": "",
        "xuser_type": "",
        "password_expires_at": null,
        "create_time": "2024-05-21T09:03:41.000000",
        "domain_id": "d78cbac1.........",
        "xdomain_id": "30086000........",
        "xdomain_type": "",
```

```
        "default_project_id": null
    }
}
```

If an error occurs during API calling, an error code and a message will be displayed. The following shows an error response body.

```
{
    "error_msg": "The request message format is invalid.",
    "error_code": "IMG.0001"
}
```

In the response body, **error_code** is an error code, and **error_msg** provides information about the error.

4 Application Examples

4.1 Creating a Migration Task

Scenarios

This section describes how to create a migration task by calling APIs. For details, see **Calling APIs**.

Involved APIs

- **Obtaining a User Token Through Password Authentication**: This API is used to obtain a user token for authentication.
- **Registering a Source Server with SMS**: This API is used to register a source server with SMS.
- **Creating a Migration Task**: This API is used to create a migration task using a target AK/SK pair.
- **Querying a Migration Task**: This API is used to query the details of a migration task with a specified ID to check whether the migration task is successfully created.

Prerequisites

You have obtained an AK/SK pair for accessing the target platform. For details about how to obtain an AK/SK pair, see **Where Can I Obtain Access Keys (AK and SK)?**

Procedure

Step 1 Obtain an IAM user token.

- API

  URI format: POST /v3/auth/tokens

  For details, see **Obtaining a User Token Through Password Authentication**.
- Example request

```
POST: https://{iam_endpoint}/v3/auth/tokens
```


Obtain *{endpoint}* from **Regions and Endpoints**.

Body:

```
{
    "auth": {
        "identity": {
            "methods": [
                "password"
            ],
            "password": {
                "user": {
                    "name": "username",
                    "domain": {
                        "name": "domainname"
                    },
                    "password": "******"
                }
            }
        },
        "scope": {
            "project": {
                "id": "0215ef11e49d4743be23dd97a1561xxx"
            }
        }
    }
}
```

In the response header, the value of **X-Subject-Token** is the token.

```
X-Subject-Token:MIIDkgYJKoZIhvcNAQcCoIIDgzCCA38CAQExDTALBglghkgBZQMEAgEwgXXXXX...
```

Step 2 Report the source server information and register the source server.

- API

  URI format:

  ```
  POST /v3/sources
  ```

  For details, see **Registering a Source Server with SMS**.

- Example request

  ```
  POST  https://sms.ap-southeast-1.myhuaweicloud.com/v3/sources
  ```

  Header:

  ```
  Content-Type: application/json
  X-Auth-Token: "Token"
  ```

  Body:

  ```
  {
      "os_type": "LINUX",
      "name": "bike-centos",
      "os_version": "CENTOS_7_4_64BIT",
      "linux_block_check": "{\"release_type\": \"CENTOS\", \"release_version\": \"7.4.1708\", 
  \"kernel_simplification\": \"3.10.0\", \"architecture\": \"x86_64\", \"kernel_version\": 
  \"3.10.0-1062.1.1.el7.x86_64\"}",
      "kernel_version": "3.10.0-1062.1.1.el7.x86_64",
      "virtualization_type": "HVM",
      "paravirtualization": true,
      "firmware": "BIOS",
      "has_rsync": true,
      "boot_loader": "GRUB",
      "disks": [{
          "name": "/dev/vda",
          "size": 42949672960,
          "device_use": "BOOT",
          "partition_style": "MBR",
          "physical_volumes": [{
              "name": "/dev/vda1",
              "size": 42948624384,
              "device_use": "OS",
  ```



```
            "used_size": 2854862848,
            "inode_size": "256",
            "file_system": "ext4",
            "mount_point": "/"
        }]
    }],
    "volume_groups": [],
    "cpu_quantity": 1,
    "memory": 1038716928,
    "networks": [{
        "name": "eth0",
        "ip": "192.168.77.77",
        "mac": "ef05f3911eeccb12a0a8931dc198af84e848b0e9e3edd0812805429fc649xxxx"
    }],
    "ip": "192.168.77.77",
    "agent_version": "1.2.3-beta"
}
```

- Example response

```
{
         "id": "33b798a8-4f80-49ce-8b8a-18a85adfe13e"
}
```

Step 3 Create a migration task for the source server.

- API

  URI format:

```
POST /v3/tasks
```

  For details, see **Creating a Migration Task**.

- Example request

  Header:

```
Content-Type: application/json
X-Auth-Token: "Token"
```

  Body:

```
{
    "auto_install_pvdriver" : true,
    "auto_start" : true,
        "syncing": false,
    "migration_ip" : "172.16.0.xxx",
    "name" : "MigrationTask",
    "os_type" : "WINDOWS",
    "project_id" : "05825205120026802xxxx01721bc1xxx",
    "project_name" : "project_name",
    "region_id" : "region_id",
    "region_name" : "region_name",
    "source_server" : {
        "id" : "9a01cb97-3eec-440e-xxxx-04016a8d7502"
    },
    "start_target_server" : true,
    "target_server" : {
        "name" : "name",
        "vm_id" : "6dac09d8-5835-4888-xxxx-787453c4e1d4",
        "disks" : [{
                "device_use" : "OS",
                "disk_id" : "354419d9-bd41-4b50-xxxx-e4f6f57c6xxx",
                "name" : "Disk 0",
                "physical_volumes" : [{
                        "device_use" : "BOOT",
                        "file_system" : "NTFS",
                        "index" : 1,
                        "name" : "(Reserved)",
                        "size" : 367001600,
                        "used_size" : 31244288,
                        "uuid" : "\\\\?\\Volume{b99b1c6f-75ef-xxxx-80b3-806e6f6e6963}\\"
                    }, {
```



```
                    "device_use" : "OS",
                    "file_system" : "NTFS",
                    "index" : 2,
                    "name" : "C:\\",
                    "size" : 42580574208,
                    "used_size" : 16148279296,
                    "uuid" : "\\\\?\\Volume{b99b1c70-75ef-xxxx-80b3-806e6f6e6963}\\"
                }
            ],
            "size" : 42949672960,
            "used_size" : 42949672960
        }, {
            "device_use" : "NORMAL",
            "disk_id" : "ef409e5f-2321-49d1-xxxx-027fc93985ef",
            "name" : "Disk 1",
            "physical_volumes" : [{
                    "file_system" : "NTFS",
                    "index" : 1,
                    "name" : "D:\\",
                    "size" : 53684994048,
                    "used_size" : 3462647808,
                    "uuid" : "\\\\?\\Volume{9f817be3-2cea-xxxx-816f-e995816380e2}\\"
                }
            ],
            "size" : 53687091200,
            "used_size" : 53687091200
        }
    ],
    "btrfs_list" : null
  },
  "type" : "MIGRATE_BLOCK",
  "use_public_ip" : false
}
```

- Example response

```
{
        "id": "8abda8635e09d185015e09d188dd0001xx"
}
```

Step 4 View the migration task status.

- API

  URI format:

```
GET /v3/tasks/{task_id}
```

  For details, see "Querying a Migration Task by Task ID."

- Example request

```
GET  https://sms.ap-southeast-1.myhuaweicloud.com/v3/tasks/
8abda8635e09d185015e09d188dd0001xx
```

  Header:

```
Content-Type: application/json
X-Auth-Token: "Token"
```

- Example response

```
{
 "id": "8abda8635e09d185015e09d188dd0001xx",
 "name": "MigrationTask",
 "type": "MIGRATE_FILE",
 "os_type": "LINUX",
 "state": "READY",
 "estimate_complete_time": 1600391014000,
 "create_date": 1600159831000,
 "start_date": 1600159831000,
 "finish_date": 1600343128000,
 "priority": 1,
 "speed_limit": 0,
 "migrate_speed": 0,
 "start_target_server": true,
```



```
"error_json": "",
"total_time": 935000,
"float_ip": "192.168.0.xxx",
"migration_ip": "192.168.0.xxx",
"vm_template_id": null,
"region_name": "region_name",
"region_id": "region_id",
"project_name": "project_name",
"project_id": "05825205120026802ff0c01721bc1xxx",
"sub_tasks": [
 {
  "id": 3514,
  "name": "SSL_CONFIG",
  "progress": 100,
  "start_date": 1600159847000,
  "end_date": 1600159851000,
  "user_op": "REPLICATE"
 },
 {
  "id": 3515,
  "name": "ATTACH_AGENT_IMAGE",
  "progress": 100,
  "start_date": 1600159851000,
  "end_date": 1600160027000,
  "user_op": "REPLICATE"
 },
 {
  "id": 3516,
  "name": "FORMAT_DISK_LINUX_FILE",
  "progress": 100,
  "start_date": 1600160027000,
  "end_date": 1600160030000,
  "user_op": "REPLICATE"
 },
 {
  "id": 3517,
  "name": "MIGRATE_LINUX_FILE",
  "progress": 100,
  "start_date": 1600160080000,
  "end_date": 1600160088000,
  "user_op": "REPLICATE"
 },
 {
  "id": 3582,
  "name": "CONFIGURE_LINUX_FILE",
  "progress": 100,
  "start_date": 1600333914000,
  "end_date": 1600334025000,
  "user_op": "CUTOVER0"
 },
 {
  "id": 3583,
  "name": "DETTACH_AGENT_IMAGE",
  "progress": 100,
  "start_date": 1600334029000,
  "end_date": 1600334103000,
  "user_op": "CUTOVER0"
 },
 {
  "id": 3584,
  "name": "SSL_CONFIG",
  "progress": 100,
  "start_date": 1600334185000,
  "end_date": 1600334188000,
  "user_op": "RESYNC0"
 },
 {
  "id": 3585,
  "name": "ATTACH_AGENT_IMAGE",
```



```
  "progress": 100,
  "start_date": 1600334189000,
  "end_date": 1600334375000,
  "user_op": "RESYNC0"
 },
 {
  "id": 3586,
  "name": "SYNC_LINUX_FILE",
  "progress": 100,
  "start_date": 1600334375000,
  "end_date": 1600334376000,
  "user_op": "RESYNC0"
 },
 {
  "id": 3587,
  "name": "CONFIGURE_LINUX_FILE",
  "progress": 100,
  "start_date": 1600342952000,
  "end_date": 1600343052000,
  "user_op": "CUTOVER1"
 },
 {
  "id": 3588,
  "name": "DETTACH_AGENT_IMAGE",
  "progress": 100,
  "start_date": 1600343056000,
  "end_date": 1600343128000,
  "user_op": "CUTOVER1"
 },
 {
  "id": 3589,
  "name": "SSL_CONFIG",
  "progress": 100,
  "start_date": 1600390756000,
  "end_date": 1600390760000,
  "user_op": "RESYNC1"
 },
 {
  "id": 3590,
  "name": "ATTACH_AGENT_IMAGE",
  "progress": 100,
  "start_date": 1600390760000,
  "end_date": 1600390953000,
  "user_op": "RESYNC1"
 },
 {
  "id": 3591,
  "name": "SYNC_LINUX_FILE",
  "progress": 100,
  "start_date": 1600390954000,
  "end_date": 1600390954000,
  "user_op": "RESYNC1"
 }
],
"source_server": {
 "id": "621f2cb5-ba4f-4819-b00d-ee48ca2c3xxx",
 "ip": "192.168.0.176",
 "name": "ecs-4bb4",
 "os_type": "LINUX",
 "os_version": "CENTOS_7_6_64BIT",
 "oem_system": false,
 "state": "syncing",
 "migration_cycle": "syncing"
},
"target_server": {
 "id": "de35f45d-b6d5-4769-9148-984c8fab4xxx",
 "vm_id": "6d51477f-0a02-4917-8d7d-b13969f4axxx",
 "name": "ecs-4bb4",
 "ip": null,
```



```
  "os_type": "LINUX",
  "os_version": null,
  "system_dir": null,
  "disks": [
   {
    "id": 86364,
    "name": "/dev/sda",
    "relation_name": "/dev/vda",
    "disk_id": "ed3c78d0-3166-48d1-bee0-d29e7d834xxx",
    "partition_style": "MBR",
    "size": 42949672960,
    "used_size": 42948624384,
    "device_use": "BOOT",
    "os_disk": false,
    "physical_volumes": [
     {
      "id": 132594,
      "uuid": null,
      "index": 0,
      "name": "/dev/sda1",
      "relation_name": "/dev/vda1",
      "device_use": "OS",
      "file_system": "ext4",
      "mount_point": "/",
      "size": 42948624384,
      "used_size": 2467393536,
      "free_size": 40481230848
     }
    ],
    "disk_index": "z"
   }
  ],
  "volume_groups": [],
  "image_disk_id": "9700f5ce-02ae-4a71-a057-deffe9b69xxx",
  "rollback_snapshot_ids": null
 },
 "clone_server": null,
 "target_snapshot_id": null,
 "remain_seconds": 60000
}
```

state indicates the task execution status. **READY** indicates that the task has been created.

----End

4.2 Starting a Migration Task

Scenarios

This section describes how to start a migration task by calling APIs. For details, see **Calling APIs**.

Before starting a migration task, you need to obtain a token. After starting the migration task, you need to query the task status. Only failed or paused tasks can be started.

Involved APIs

- **Obtaining a User Token Through Password Authentication**: This API is used to obtain a user token for authentication.
- **Managing Migration Tasks (Start)**: This API is used to start a failed or paused migration task using a target AK/SK pair.

- **Querying a Migration Task**: This API is used to query the details of a migration task with a specified ID to check whether the migration task is successfully started.

Procedure

Step 1 Obtain an IAM user token.

- API

  URI format: POST /v3/auth/tokens

  For details, see **Obtaining a User Token Through Password Authentication**.

- Example request

```
POST: https://{iam_endpoint}/v3/auth/tokens
```

  Obtain *{endpoint}* from **Regions and Endpoints**.

  Body:

```
{
    "auth": {
        "identity": {
            "methods": [
                "password"
            ],
            "password": {
                "user": {
                    "name": "testname",
                    "domain": {
                        "name": "testname"
                    },
                    "password": "Password"
                }
            }
        },
        "scope": {
            "project": {
                "id": "0215ef11e49d4743be23dd97a1561xxx"
            }
        }
    }
}
```

In the response header, the value of **X-Subject-Token** is the token.

```
X-Subject-Token: MIIDkgYJKoZIhvcNAQcCoIIDgzCCA38CAQExDTALBglghkgBZQMEAgEwgXXXXX...
```

Step 2 Start the desired migration task.

- API

  URI format:

```
POST /v3/tasks/{task_id}/action
```

  For details, see **Managing Migration Tasks (Start)**.

- Example request

```
POST  https://sms.ap-southeast-1.myhuaweicloud.com/v3/tasks/
8abda8635e09d185015e09d188dd0001xx/action
```

  Header:

```
Content-Type: application/json
X-Auth-Token: "Token"
```

  Body:

```
{
      "operation" : "start",
}
```


If the response code is 200, the API is successfully called.

Step 3 View the migration task status.

- API

  URI format:

  ```
  GET /v3/tasks/{task_id}
  ```

  For details, see **Querying a Migration Task**.

- Example request

  ```
  GET  https://sms.ap-southeast-1.myhuaweicloud.com/v3/tasks/
  8abda8635e09d185015e09d188dd0001xx
  ```

  Header:

  ```
  Content-Type: application/json
  X-Auth-Token: "Token"
  ```

- Example response


```
{
 "id": "8abda8635e09d185015e09d188dd0001xx",
 "name": "MigrationTask",
 "type": "MIGRATE_FILE",
 "os_type": "LINUX",
 "state": "RUNNING",
 "estimate_complete_time": 1600391014000,
 "create_date": 1600159831000,
 "start_date": 1600159831000,
 "finish_date": 1600343128000,
 "priority": 1,
 "speed_limit": 0,
 "migrate_speed": 0,
 "start_target_server": true,
 "error_json": "",
 "total_time": 935000,
 "float_ip": "192.168.0.xxx",
 "migration_ip": "192.168.0.xxx",
 "vm_template_id": null,
 "region_name": "region_name",
 "region_id": "region_id",
 "project_name": "project_name",
 "project_id": "05825205120026802ff0c01721bc1xxx",
 "sub_tasks": [
  {
   "id": 3514,
   "name": "SSL_CONFIG",
   "progress": 100,
   "start_date": 1600159847000,
   "end_date": 1600159851000,
   "user_op": "REPLICATE"
  },
  {
   "id": 3515,
   "name": "ATTACH_AGENT_IMAGE",
   "progress": 100,
   "start_date": 1600159851000,
   "end_date": 1600160027000,
   "user_op": "REPLICATE"
  },
  {
   "id": 3516,
   "name": "FORMAT_DISK_LINUX_FILE",
   "progress": 100,
   "start_date": 1600160027000,
   "end_date": 1600160030000,
   "user_op": "REPLICATE"
  },
  {
   "id": 3517,
```



```
  "name": "MIGRATE_LINUX_FILE",
  "progress": 100,
  "start_date": 1600160080000,
  "end_date": 1600160088000,
  "user_op": "REPLICATE"
 },
 {
  "id": 3582,
  "name": "CONFIGURE_LINUX_FILE",
  "progress": 100,
  "start_date": 1600333914000,
  "end_date": 1600334025000,
  "user_op": "CUTOVER0"
 },
 {
  "id": 3583,
  "name": "DETTACH_AGENT_IMAGE",
  "progress": 100,
  "start_date": 1600334029000,
  "end_date": 1600334103000,
  "user_op": "CUTOVER0"
 },
 {
  "id": 3584,
  "name": "SSL_CONFIG",
  "progress": 100,
  "start_date": 1600334185000,
  "end_date": 1600334188000,
  "user_op": "RESYNC0"
 },
 {
  "id": 3585,
  "name": "ATTACH_AGENT_IMAGE",
  "progress": 100,
  "start_date": 1600334189000,
  "end_date": 1600334375000,
  "user_op": "RESYNC0"
 },
 {
  "id": 3586,
  "name": "SYNC_LINUX_FILE",
  "progress": 100,
  "start_date": 1600334375000,
  "end_date": 1600334376000,
  "user_op": "RESYNC0"
 },
 {
  "id": 3587,
  "name": "CONFIGURE_LINUX_FILE",
  "progress": 100,
  "start_date": 1600342952000,
  "end_date": 1600343052000,
  "user_op": "CUTOVER1"
 },
 {
  "id": 3588,
  "name": "DETTACH_AGENT_IMAGE",
  "progress": 100,
  "start_date": 1600343056000,
  "end_date": 1600343128000,
  "user_op": "CUTOVER1"
 },
 {
  "id": 3589,
  "name": "SSL_CONFIG",
  "progress": 100,
  "start_date": 1600390756000,
  "end_date": 1600390760000,
  "user_op": "RESYNC1"
```



```
 },
 {
  "id": 3590,
  "name": "ATTACH_AGENT_IMAGE",
  "progress": 100,
  "start_date": 1600390760000,
  "end_date": 1600390953000,
  "user_op": "RESYNC1"
 },
 {
  "id": 3591,
  "name": "SYNC_LINUX_FILE",
  "progress": 100,
  "start_date": 1600390954000,
  "end_date": 1600390954000,
  "user_op": "RESYNC1"
 }
],
"source_server": {
 "id": "621f2cb5-ba4f-4819-b00d-ee48ca2c3xxx",
 "ip": "192.168.0.xxx",
 "name": "ecs-4bb4",
 "os_type": "LINUX",
 "os_version": "CENTOS_7_6_64BIT",
 "oem_system": false,
 "state": "syncing",
 "migration_cycle": "syncing"
},
"target_server": {
 "id": "de35f45d-b6d5-4769-9148-984c8fab4xxx",
 "vm_id": "6d51477f-0a02-4917-8d7d-b13969f4axxx",
 "name": "ecs-4bb4",
 "ip": null,
 "os_type": "LINUX",
 "os_version": null,
 "system_dir": null,
 "disks": [
  {
   "id": 86364,
   "name": "/dev/sda",
   "relation_name": "/dev/vda",
   "disk_id": "ed3c78d0-3166-48d1-bee0-d29e7d834xxx",
   "partition_style": "MBR",
   "size": 42949672960,
   "used_size": 42948624384,
   "device_use": "BOOT",
   "os_disk": false,
   "physical_volumes": [
    {
     "id": 132594,
     "uuid": null,
     "index": 0,
     "name": "/dev/sda1",
     "relation_name": "/dev/vda1",
     "device_use": "OS",
     "file_system": "ext4",
     "mount_point": "/",
     "size": 42948624384,
     "used_size": 2467393536,
     "free_size": 40481230848
    }
   ],
   "disk_index": "z"
  }
 ],
 "volume_groups": [],
 "image_disk_id": "9700f5ce-02ae-4a71-a057-deffe9b69xxx",
 "rollback_snapshot_ids": null
},
```

```
"clone_server": null,
"target_snapshot_id": null,
"remain_seconds": 60000
}
```

state indicates the task execution status. **RUNNING** indicates that the task is being executed, and **MIGRATE_SUCCESS** indicates that the task has been executed successfully.

----End

4.3 Pausing and Deleting a Migration Task

Scenarios

This section describes how to pause and delete a migration task by calling APIs. For details, see **Calling APIs**.

Before deleting a migration task, you need to obtain a token and pause the migration task. After the migration task is paused, you need to query the task status. The migration task can be deleted only after it is paused.

Involved APIs

- **Obtaining a User Token Through Password Authentication**: This API is used to obtain a user token for authentication.
- **Managing Migration Tasks (Pause)**: This API is used to pause a migration task with a specified ID.
- **Querying a Migration Task**: This API is used to query the details of a migration task with a specified ID to check whether the migration task is successfully paused.
- **Deleting a Migration Task**: This API is used to delete a migration task with a specified ID.

Procedure

Step 1 Obtain an IAM user token.

- API

  URI format: POST /v3/auth/tokens

  For details, see **Obtaining a User Token Through Password Authentication**.

- Example request

  ```
  POST: https://{iam_endpoint}/v3/auth/tokens
  ```

  Obtain *{endpoint}* from **Regions and Endpoints**.

  Body:

  ```
  {
      "auth": {
          "identity": {
              "methods": [
                  "password"
              ],
              "password": {
                  "user": {
                      "name": "testname",
                      "domain": {
  ```



```
                "name": "testname"
              },
              "password": "Password"
            }
          }
        },
        "scope": {
          "project": {
            "id": "0215ef11e49d4743be23dd97a1561xxx"
          }
        }
      }
}
```

In the response header, the value of **X-Subject-Token** is the token.

```
X-Subject-Token: MIIDkgYJKoZIhvcNAQcCoIIDgzCCA38CAQExDTALBglghkgBZQMEAgEwgXXXXX...
```

Step 2 Pause the desired migration task.

- API

  URI format:

  ```
  POST /v3/tasks/{task_id}/action
  ```

  For details, see **Managing Migration Tasks (Pause)**.

- Example request

  ```
  POST  https://sms.ap-southeast-1.myhuaweicloud.com/v3/tasks/
  8abda8635e09d185015e09d188dd0001xx/action
  ```

  Header:

  ```
  Content-Type: application/json
  X-Auth-Token: "Token"
  ```

  Body:

  ```
  {
        "operation" : "stop"
  }
  ```

  If the response code is 200, the API is successfully called.

Step 3 View the migration task status.

- API

  URI format:

  ```
  GET /v3/tasks/{task_id}
  ```

  For details, see **Querying a Migration Task**.

- Example request

  ```
  GET  https://sms.ap-southeast-1.myhuaweicloud.com/v3/tasks/
  8abda8635e09d185015e09d188dd0001xx
  ```

  Header:

  ```
  Content-Type: application/json
  X-Auth-Token: "Token"
  ```

- Example response

  ```
  {
   "id": "8abda8635e09d185015e09d188dd0001xx",
   "name": "MigrationTask",
   "type": "MIGRATE_FILE",
   "os_type": "LINUX",
   "state": "ABORT",
   "estimate_complete_time": 1600391014000,
   "create_date": 1600159831000,
   "start_date": 1600159831000,
   "finish_date": 1600343128000,
   "priority": 1,
  ```

```
"speed_limit": 0,
"migrate_speed": 0,
"start_target_server": true,
"error_json": "",
"total_time": 935000,
"float_ip": "192.168.0.xxx",
"migration_ip": "192.168.0.xxx",
"vm_template_id": null,
"region_name": "region_name",
"region_id": "region_id",
"project_name": ""project_name",
"project_id": "05825205120026802ff0c01721bc1xxx",
"sub_tasks": [
 {
  "id": 3514,
  "name": "SSL_CONFIG",
  "progress": 100,
  "start_date": 1600159847000,
  "end_date": 1600159851000,
  "user_op": "REPLICATE"
 },
 {
  "id": 3515,
  "name": "ATTACH_AGENT_IMAGE",
  "progress": 100,
  "start_date": 1600159851000,
  "end_date": 1600160027000,
  "user_op": "REPLICATE"
 },
 {
  "id": 3516,
  "name": "FORMAT_DISK_LINUX_FILE",
  "progress": 100,
  "start_date": 1600160027000,
  "end_date": 1600160030000,
  "user_op": "REPLICATE"
 },
 {
  "id": 3517,
  "name": "MIGRATE_LINUX_FILE",
  "progress": 100,
  "start_date": 1600160080000,
  "end_date": 1600160088000,
  "user_op": "REPLICATE"
 },
 {
  "id": 3582,
  "name": "CONFIGURE_LINUX_FILE",
  "progress": 100,
  "start_date": 1600333914000,
  "end_date": 1600334025000,
  "user_op": "CUTOVER0"
 },
 {
  "id": 3583,
  "name": "DETTACH_AGENT_IMAGE",
  "progress": 100,
  "start_date": 1600334029000,
  "end_date": 1600334103000,
  "user_op": "CUTOVER0"
 },
 {
  "id": 3584,
  "name": "SSL_CONFIG",
  "progress": 100,
  "start_date": 1600334185000,
  "end_date": 1600334188000,
  "user_op": "RESYNC0"
 },
```



```
 {
  "id": 3585,
  "name": "ATTACH_AGENT_IMAGE",
  "progress": 100,
  "start_date": 1600334189000,
  "end_date": 1600334375000,
  "user_op": "RESYNC0"
 },
 {
  "id": 3586,
  "name": "SYNC_LINUX_FILE",
  "progress": 100,
  "start_date": 1600334375000,
  "end_date": 1600334376000,
  "user_op": "RESYNC0"
 },
 {
  "id": 3587,
  "name": "CONFIGURE_LINUX_FILE",
  "progress": 100,
  "start_date": 1600342952000,
  "end_date": 1600343052000,
  "user_op": "CUTOVER1"
 },
 {
  "id": 3588,
  "name": "DETTACH_AGENT_IMAGE",
  "progress": 100,
  "start_date": 1600343056000,
  "end_date": 1600343128000,
  "user_op": "CUTOVER1"
 },
 {
  "id": 3589,
  "name": "SSL_CONFIG",
  "progress": 100,
  "start_date": 1600390756000,
  "end_date": 1600390760000,
  "user_op": "RESYNC1"
 },
 {
  "id": 3590,
  "name": "ATTACH_AGENT_IMAGE",
  "progress": 100,
  "start_date": 1600390760000,
  "end_date": 1600390953000,
  "user_op": "RESYNC1"
 },
 {
  "id": 3591,
  "name": "SYNC_LINUX_FILE",
  "progress": 100,
  "start_date": 1600390954000,
  "end_date": 1600390954000,
  "user_op": "RESYNC1"
 }
],
"source_server": {
 "id": "621f2cb5-ba4f-4819-b00d-ee48ca2c35ea",
 "ip": "192.168.0.176",
 "name": "ecs-4bb4",
 "os_type": "LINUX",
 "os_version": "CENTOS_7_6_64BIT",
 "oem_system": false,
 "state": "syncing",
 "migration_cycle": "syncing"
},
"target_server": {
 "id": "de35f45d-b6d5-4769-9148-984c8fab4xxx",
```

```
"vm_id": "6d51477f-0a02-4917-8d7d-b13969f4axxx",
"name": "ecs-4bb4",
"ip": null,
"os_type": "LINUX",
"os_version": null,
"system_dir": null,
"disks": [
 {
  "id": 86364,
  "name": "/dev/sda",
  "relation_name": "/dev/vda",
  "disk_id": "ed3c78d0-3166-48d1-bee0-d29e7d834xxx",
  "partition_style": "MBR",
  "size": 42949672960,
  "used_size": 42948624384,
  "device_use": "BOOT",
  "os_disk": false,
  "physical_volumes": [
   {
    "id": 132594,
    "uuid": null,
    "index": 0,
    "name": "/dev/sda1",
    "relation_name": "/dev/vda1",
    "device_use": "OS",
    "file_system": "ext4",
    "mount_point": "/",
    "size": 42948624384,
    "used_size": 2467393536,
    "free_size": 40481230848
   }
  ],
  "disk_index": "z"
 }
],
"volume_groups": [],
"image_disk_id": "9700f5ce-02ae-4a71-a057-deffe9b69xxx",
"rollback_snapshot_ids": null
},
"clone_server": null,
"target_snapshot_id": null,
"remain_seconds": 60000
}
```


state indicates the task execution status. **ABORT** indicates that the task has been paused.

Step 4 Delete the migration task.

- API

  URI format:

  ```
  DELETE /v3/tasks/{task_id}
  ```

  For details, see **Deleting a Migration Task**.

- Example request

  ```
  https://sms.ap-southeast-1.myhuaweicloud.com/v3/tasks/8abda8635e09d185015e09d188dd0001xx
  ```

  Header:

  ```
  Content-Type: application/json
  X-Auth-Token: "Token"
  ```

  If the response code is 200, the migration task has been deleted successfully.

----End

5 API

5.1 API Version Query

5.1.1 Listing SMS API Versions

Function

This API is used to list SMS API versions.

Calling Method

For details, see **Calling APIs**.

Authorization Information

Each account has all the permissions required to call all APIs, but IAM users must be assigned the required permissions.

- If you are using role/policy-based authorization, see **Permissions Policies and Supported Actions** for details on the required permissions.
- If you are using identity policy-based authorization, no identity policy-based permission required for calling this API.

URI

GET /

Request Parameters

Table 5-1 Request header parameters

Parameter	Mandatory	Type	Description
X-Auth-Token	Yes	String	User token. The token can be obtained by calling the IAM API for obtaining a user token. The value of **X-Subject-Token** in the response header is the user token. Minimum: **1** Maximum: **16384**

Response Parameters

Status code: 200

Table 5-2 Response body parameters

Parameter	Type	Description
versions	Array of **Version** objects	The list of all SMS API versions. Array Length: **0 - 1024**

Table 5-3 Version

Parameter	Type	Description
id	String	The API version. Minimum: **0** Maximum: **255**
links	Array of **Link** objects	The API link address. Array Length: **0 - 1024**
status	String	The version status. **SUPPORTED** indicates that the version is supported. Minimum: **0** Maximum: **255**

Parameter	Type	Description
updated	String	The time when the version was updated. The format is yyyy-mm-ddThh:mm:ssZ. **T** is the separator between the calendar and the hourly notation of time. **Z** indicates the Coordinated Universal Time (UTC). For example, 2018-09-30T00:00:00Z Minimum: **0** Maximum: **255**

Table 5-4 Link

Parameter	Type	Description
href	String	The API URL. Minimum: **0** Maximum: **1024**
rel	String	Its value is **self**, indicating that **href** is a local link. Minimum: **0** Maximum: **1024**

Example Requests

This example queries the list of supported API versions.

```
GET https://{endpoint}/
```

Example Responses

Status code: 200

The list of supported API versions was obtained.

```
{
  "versions" : [ {
    "links" : [ {
      "rel" : "self",
      "href" : "https://{endpoint}/"
    } ],
    "id" : "v3",
    "updated" : "2020-09-02T17:50:00Z",
    "status" : "SUPPORTED"
  } ]
}
```

SDK Sample Code

The SDK sample code is as follows.

Java

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;


public class ListApiVersionSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great 
security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or 
environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running 
this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        ListApiVersionRequest request = new ListApiVersionRequest();
        try {
            ListApiVersionResponse response = client.listApiVersion(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

Python

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security 
risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment 
variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this 
example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]
```

```
    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = ListApiVersionRequest()
        response = client.list_api_version(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

Go

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
        WithRegion(region.ValueOf("<YOUR REGION>")).
        WithCredential(auth).
        SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.ListApiVersionRequest{}
    response, err := client.ListApiVersion(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
```

```
  }
}
```

More

For SDK sample code of more programming languages, see the Sample Code tab in **API Explorer**. SDK sample code can be automatically generated.

Status Codes

Status Code	Description
200	The list of supported API versions was obtained.

Error Codes

See **Error Codes**.

5.1.2 Querying an SMS API Version

Function

This API is used to query a specified API version of SMS.

Calling Method

For details, see **Calling APIs**.

Authorization Information

Each account has all the permissions required to call all APIs, but IAM users must be assigned the required permissions.

- If you are using role/policy-based authorization, see **Permissions Policies and Supported Actions** for details on the required permissions.
- If you are using identity policy-based authorization, no identity policy-based permission required for calling this API.

URI

GET /{version}

Table 5-5 Path Parameters

Parameter	Mandatory	Type	Description
version	Yes	String	Version information, for example, v3. Minimum: **1** Maximum: **10**

Request Parameters

Table 5-6 Request header parameters

Parameter	Mandatory	Type	Description
X-Auth-Token	Yes	String	User token. Obtain it by calling the IAM API for obtaining a user token. The value of **X-Subject-Token** in the response header is the user token. Minimum: **1** Maximum: **16384**

Response Parameters

Status code: 200

Table 5-7 Response body parameters

Parameter	Type	Description
id	String	The API version. Minimum: **0** Maximum: **255**
links	Array of **Link** objects	The API link address. Array Length: **0 - 1024**
status	String	The version status. **SUPPORTED** indicates that the version is supported. Minimum: **0** Maximum: **255**

Parameter	Type	Description
updated	String	The time when the version was updated. The format is yyyy-mm-ddThh:mm:ssZ. **T** is the separator between the calendar and the hourly notation of time. **Z** indicates the Coordinated Universal Time (UTC). For example, 2018-09-30T00:00:00Z Minimum: **0** Maximum: **255**

Table 5-8 Link

Parameter	Type	Description
href	String	The API URL. Minimum: **0** Maximum: **1024**
rel	String	Its value is **self**, indicating that **href** is a local link. Minimum: **0** Maximum: **1024**

Example Requests

This example queries a specified SMS API version.

```
GET https://{endpoint}/v3
X-Auth-Token: XXXX
```

Example Responses

Status code: 200

Querying an SMS API Version Successfully

```
{
  "links" : [ {
    "rel" : "self",
    "href" : "https://{endpoint}/v3"
  } ],
  "id" : "v3",
  "updated" : "2020-09-02T17:50:00Z",
  "status" : "SUPPORTED"
}
```

SDK Sample Code

The SDK sample code is as follows.

Java

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;


public class ShowApiVersionSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        ShowApiVersionRequest request = new ShowApiVersionRequest();
        request.withVersion("{version}");
        try {
            ShowApiVersionResponse response = client.showApiVersion(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

Python

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
```

```
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = ShowApiVersionRequest()
        request.version = "{version}"
        response = client.show_api_version(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

Go

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
         WithRegion(region.ValueOf("<YOUR REGION>")).
         WithCredential(auth).
         SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.ShowApiVersionRequest{}
    request.Version = "{version}"
    response, err := client.ShowApiVersion(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
```

```
    } else {
        fmt.Println(err)
    }
}
```

More

For SDK sample code of more programming languages, see the Sample Code tab in **API Explorer**. SDK sample code can be automatically generated.

Status Codes

Status Code	Description
200	Querying an SMS API Version Successfully

Error Codes

See **Error Codes**.

5.2 Agent Running

5.2.1 Obtaining the Agent configuration information

Function

After the Agent installed on a source server is started, it calls this API to obtain configuration information.

Calling Method

For details, see **Calling APIs**.

Authorization Information

Each account has all the permissions required to call all APIs, but IAM users must be assigned the required permissions.

- If you are using role/policy-based authorization, see **Permissions Policies and Supported Actions** for details on the required permissions.
- If you are using identity policy-based authorization, the following identity policy-based permissions are required.

Action	Access Level	Resource Type (*: required)	Condition Key	Alias	Dependencies
sms::getConfig	Read	-	-	sms:server:queryServer	-

URI

GET /v3/config

Request Parameters

Table 5-9 Request header parameters

Parameter	Mandatory	Type	Description
X-Auth-Token	Yes	String	User token. The token can be obtained by calling the IAM API for obtaining a user token. The value of **X-Subject-Token** in the response header is the user token. Minimum: **1** Maximum: **16384**

Response Parameters

Status code: 200

Table 5-10 Response body parameters

Parameter	Type	Description
config	Map<String,String>	mainRegion, obs_domain, disktype, process_and_it, and information added.
regions	Array of Map<String,Object> objects	The region array. Array Length: **1 - 100**

Status code: 400

Table 5-11 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **1024**

Status code: 403

Table 5-12 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **255**
encoded_authorization_message	String	The encrypted authorization information. Minimum: **0** Maximum: **65535**
error_param	Array of strings	Error parameters. Minimum: **0** Maximum: **65535** Array Length: **1 - 20**
details	Array of **details** objects	The error details. Array Length: **1 - 20**

Table 5-13 details

Parameter	Type	Description
error_code	String	The SMS error code. Minimum: **0** Maximum: **65535**

Parameter	Type	Description
error_msg	String	The SMS error message. Minimum: **0** Maximum: **65535**

Example Requests

This example obtains the Agent configuration information for the current endpoint.

```
GET https://{endpoint}/v3/config
```

Example Responses

Status code: 200

The configuration information was obtained.


```
{
  "config" : {
    "mainRegion" : "region_name",
    "disktype" : "SATA"
  },
  "regions" : [ {
    "region_name" : "region_name",
    "project_name" : "project_name"
  }, {
    "region_name" : "region_name",
    "project_name" : "project_name"
  } ]
}
```


Status code: 403

Authentication failed.


```
{
  "error_code" : "SMS.9004",
  "error_msg" : "The current account does not have the permission to execute policy. You do not have permission to perform action XXX on resource XXX.",
  "encoded_authorization_message" : "XXXXXX",
  "error_param" : [ "You do not have permission to perform action XXX on resource XXX." ],
  "details" : [ {
    "error_code" : "SMS.9004",
    "error_msg" : "You do not have permission to perform action XXX on resource XXX."
  } ]
}
```


SDK Sample Code

The SDK sample code is as follows.

Java

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
```

```
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;


public class ShowConfigSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great
security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or
environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running
this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        ShowConfigRequest request = new ShowConfigRequest();
        try {
            ShowConfigResponse response = client.showConfig(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

Python

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security
risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment
variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this
example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()
```

```
try:
    request = ShowConfigRequest()
    response = client.show_config(request)
    print(response)
except exceptions.ClientRequestException as e:
    print(e.status_code)
    print(e.request_id)
    print(e.error_code)
    print(e.error_msg)
```

Go

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
         WithRegion(region.ValueOf("<YOUR REGION>")).
         WithCredential(auth).
         SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.ShowConfigRequest{}
    response, err := client.ShowConfig(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

More

For SDK sample code of more programming languages, see the Sample Code tab in **API Explorer**. SDK sample code can be automatically generated.

Status Codes

Status Code	Description
200	The configuration information was obtained.
400	Obtaining the configuration information failed.
403	Authentication failed.

Error Codes

See **Error Codes**.

5.3 Source Server Management

5.3.1 Listing Failed Source Servers

Function

This API is used to query the list of source servers that failed to be migrated and the reported error messages.

Calling Method

For details, see **Calling APIs**.

Authorization Information

Each account has all the permissions required to call all APIs, but IAM users must be assigned the required permissions.

- If you are using role/policy-based authorization, see **Permissions Policies and Supported Actions** for details on the required permissions.
- If you are using identity policy-based authorization, the following identity policy-based permissions are required.

Action	Access Level	Resource Type (*: required)	Condition Key	Alias	Dependencies
sms:server:listErrors	List	template *	-	server:queryServer	-

URI

GET /v3/errors

Table 5-14 Query Parameters

Parameter	Mandatory	Type	Description
limit	No	Integer	The number of failed source servers recorded on each page. Minimum: **0** Maximum: **100** Default: **50**
offset	Yes	Integer	The offset. Minimum: **0** Maximum: **65535** Default: **0**
enterprise_project_id	No	String	The ID of the enterprise project to be queried. Minimum: **0** Maximum: **255**

Request Parameters

Table 5-15 Request header parameters

Parameter	Mandatory	Type	Description
X-Auth-Token	Yes	String	User token. The token can be obtained by calling the IAM API for obtaining a user token. The value of **X-Subject-Token** in the response header is the user token. Minimum: **1** Maximum: **16384**

Response Parameters

Status code: 200

Table 5-16 Response body parameters

Parameter	Type	Description
count	Integer	The number of source servers that failed to be migrated due to errors. Minimum: **0** Maximum: **2147483647**
migration_errors	Array of **MigrationErrors** objects	The details of failed source servers. Array Length: **0 - 65535**

Table 5-17 MigrationErrors

Parameter	Type	Description
error_json	String	The error message in JSON format. Minimum: **0** Maximum: **255**
host_name	String	The hostname of the source server, which is obtained from the user system and may be empty. Minimum: **0** Maximum: **255**
name	String	The source server name in SMS. Minimum: **0** Maximum: **255**
source_id	String	The source server ID. Minimum: **0** Maximum: **255**
source_ip	String	The IP address of the source server. Minimum: **0** Maximum: **255**
target_ip	String	The IP address of the target server. Minimum: **0** Maximum: **255**

Status code: 403

Table 5-18 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **255**
encoded_authorization_message	String	The encrypted authorization information. Minimum: **0** Maximum: **65535**
error_param	Array of strings	Error parameters. Minimum: **0** Maximum: **65535** Array Length: **1 - 20**
details	Array of **details** objects	The error details. Array Length: **1 - 20**

Table 5-19 details

Parameter	Type	Description
error_code	String	The SMS error code. Minimum: **0** Maximum: **65535**
error_msg	String	The SMS error message. Minimum: **0** Maximum: **65535**

Example Requests

This example lists all failed source servers with 10 records on one page and navigates to page 0.

```
GET https://{endpoint}/v3/errors?limit=10&offset=0
```

Example Responses

Status code: 200

The list of failed source servers was obtained.


```
{
  "count" : 4,
  "migration_errors" : [ {
    "source_id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "source_ip" : "192.168.0.235",
    "target_ip" : null,
    "name" : "sms-ubuntu",
    "host_name" : null,
    "error_json" : "{\"error_code\":\"SMS.1302\",\"error_param\":\"[\\\"/\\\", \\\"/mnt/vdb1\\\"]\"}"
  }, {
    "source_id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "source_ip" : "192.168.0.163",
    "target_ip" : null,
    "name" : "sms-win08",
    "host_name" : "sms-win08",
    "error_json" : "{\"error_param\":\"[\\\"192.168.0.1\\\"]\",\"error_code\":\"SMS.2802\"}"
  }, {
    "source_id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "source_ip" : "192.168.0.154",
    "target_ip" : null,
    "name" : "sms-win16",
    "host_name" : "sms-win16",
    "error_json" : "{\"error_code\":\"SMS.1114\",\"error_param\":\"[]\"}"
  }, {
    "source_id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "source_ip" : "192.168.77.77",
    "target_ip" : null,
    "name" : "sms-centos",
    "host_name" : null,
    "error_json" : "{\"error_code\":\"SMS.3805\",\"error_param\":\"[]\"}"
  } ]
}
```


Status code: 403

Authentication failed.


```
{
  "error_code" : "SMS.9004",
  "error_msg" : "The current account does not have the permission to execute policy. You do not have permission to perform action XXX on resource XXX.",
  "encoded_authorization_message" : "XXXXXX",
  "error_param" : [ "You do not have permission to perform action XXX on resource XXX." ],
  "details" : [ {
    "error_code" : "SMS.9004",
    "error_msg" : "You do not have permission to perform action XXX on resource XXX."
  } ]
}
```


SDK Sample Code

The SDK sample code is as follows.

Java

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;


public class ListErrorServersSolution {
```

```
    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        ListErrorServersRequest request = new ListErrorServersRequest();
        try {
            ListErrorServersResponse response = client.listErrorServers(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

Python

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = ListErrorServersRequest()
        response = client.list_error_servers(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
```



```
        print(e.error_code)
        print(e.error_msg)
```

Go

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
         WithRegion(region.ValueOf("<YOUR REGION>")).
         WithCredential(auth).
         SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.ListErrorServersRequest{}
    response, err := client.ListErrorServers(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

More

For SDK sample code of more programming languages, see the Sample Code tab in **API Explorer**. SDK sample code can be automatically generated.

Status Codes

Status Code	Description
200	The list of failed source servers was obtained.
403	Authentication failed.

Error Codes

See **Error Codes**.

5.3.2 Registering a Source Server with SMS

Function

This API is automatically called by the Agent to report the basic information about a source server to SMS. After the source server is registered successfully, you can view the source server information on the SMS console.

Calling Method

For details, see **Calling APIs**.

Authorization Information

Each account has all the permissions required to call all APIs, but IAM users must be assigned the required permissions.

- If you are using role/policy-based authorization, see **Permissions Policies and Supported Actions** for details on the required permissions.
- If you are using identity policy-based authorization, the following identity policy-based permissions are required.

Action	Access Level	Resource Type (*: required)	Condition Key	Alias	Dependencies
sms:server:register	Write	server *	-	sms:server:registerServer	-
		-	g:EnterpriseProjectId		

URI

POST /v3/sources

Request Parameters

Table 5-20 Request header parameters

Parameter	Mandatory	Type	Description
X-Auth-Token	Yes	String	User token. The token can be obtained by calling the IAM API for obtaining a user token. The value of **X-Subject-Token** in the response header is the user token. Minimum: **1** Maximum: **16384**

Table 5-21 Request body parameters

Parameter	Mandatory	Type	Description
id	No	String	The ID of the source server in the SMS database. Minimum: **0** Maximum: **255**
ip	No	String	IP address of the source server. The value must be in the standard IP address format. Either the IP address or IPv6 address must be specified. Minimum: **0** Maximum: **255**
ipv6	No	String	IP address of the source server. The value must be in the standard IPv6 address format. Either the IP address or IPv6 address must be specified. Minimum: **0** Maximum: **255**
name	Yes	String	The source server name in SMS. Minimum: **1** Maximum: **64**

Parameter	Mandatory	Type	Description
hostname	No	String	Source server name. Minimum: **1** Maximum: **64**
os_type	Yes	String	The OS type of the source server, which can be Windows or Linux. This parameter is mandatory for source server registration and optional for source server updates. Minimum: **0** Maximum: **255** Enumeration values: ● **WINDOWS** ● **LINUX**
os_version	Yes	String	The OS version. This parameter is mandatory for registration and optional for updates. Minimum: **0** Maximum: **255**
linux_block_check	No	String	The Linux block-level check. Minimum: **0** Maximum: **255**
firmware	No	String	The boot mode of the source server, which can be BIOS or UEFI. Minimum: **0** Maximum: **255** Enumeration values: ● **BIOS** ● **UEFI**
cpu_quantity	No	Integer	The number of vCPUs. Minimum: **0** Maximum: **65535**
memory	No	Long	The memory size, in MB. Minimum: **0** Maximum: **9223372036854775807**

Parameter	Mandatory	Type	Description
disks	No	Array of **ServerDisk** objects	The information about source server disks. Array Length: **0 - 50**
btrfs_list	No	Array of **BtrfsFileSystem** objects	Btrfs information of the source end. If the source end does not have Btrfs, the value is **[]**. This parameter is mandatory in the Linux scenario. Otherwise, the subsequent environment check will fail. Array Length: **0 - 65535**
networks	Yes	Array of **NetWork** objects	The information about NICs on the source server. Array Length: **1 - 50**
domain_id	No	String	The tenant domain ID. Minimum: **0** Maximum: **255**
has_rsync	No	Boolean	Whether the rsync component is installed. This parameter is mandatory in the Linux scenario. Otherwise, the subsequent environment check will fail.
paravirtualization	No	Boolean	Whether the source end is paravirtualized. This parameter is mandatory in the Linux scenario. Otherwise, the subsequent environment check will fail.
raw_devices	No	String	Raw device list. This parameter is mandatory in the Linux scenario. Otherwise, the subsequent environment check will fail. Minimum: **0** Maximum: **255**
driver_files	No	Boolean	Whether the driver file is missing. This parameter is mandatory in the Windows scenario. Otherwise, the subsequent environment check will fail.

Parameter	Mandatory	Type	Description
system_services	No	Boolean	Whether there are abnormal services. This parameter is mandatory in the Windows scenario. Otherwise, the subsequent environment check will fail.
account_rights	No	Boolean	Whether the permission meets the requirements. This parameter is mandatory in the Windows scenario. Otherwise, the subsequent environment check will fail.
boot_loader	No	String	System boot type. Only the value **GRUB** is allowed. This parameter is mandatory in the Linux scenario. Otherwise, the subsequent environment check will fail. Minimum: **0** Maximum: **255**
system_dir	No	String	System directory. This parameter is mandatory in the Windows scenario. Otherwise, the subsequent environment check will fail. Minimum: **0** Maximum: **255**
volume_groups	No	Array of **VolumeGroups** objects	Volume group. If no volume group is available, enter **[]**. This parameter is mandatory in the Linux scenario. Otherwise, the subsequent environment check will fail. Array Length: **0 - 50**
agent_version	Yes	String	Agent version Minimum: **0** Maximum: **255**
kernel_version	No	String	The kernel version. Minimum: **0** Maximum: **255**

Parameter	Mandatory	Type	Description
migration_cycle	No	String	The migration stage. **cutovering** (target server being launched) **cutovered** (target server launched) **checking** **setting** **replicating** **syncing** Enumeration values: ● **cutovering** ● **cutovered** ● **checking** ● **setting** ● **replicating** ● **syncing**

Parameter	Mandatory	Type	Description
state	No	String	Status of the source server. **unavailable**: The environment verification fails. **waiting** **init**: Initializing. **replicate**: Replicating. **syncing** **stopping** **stopped** **skipping** **deleting** **clearing**: Clearing snapshot resources. **cleared**: Snapshot resources cleared. **clearfailed**: Failed to clear snapshot resources. **premigready**: Ready for migration drill. **premiging**: Performing migration drill. **premiged**: Migration drill completed. **premigfailed**: Migration drill failed. **cloning** **cutovering**: Starting the target server. **finished**: Target server started. **error** Minimum: **0** Maximum: **255** Enumeration values: ● **unavailable** ● **waiting** ● **initialize** ● **replicate** ● **syncing** ● **stopping** ● **stopped** ● **skipping**

Parameter	Mandatory	Type	Description
			• **deleting** • **clearing** • **cleared** • **clearfailed** • **premigready** • **premiged** • **premigfailed** • **cloning** • **cutovering** • **finished** • **error**
oem_system	No	Boolean	Indicates whether the OS is an OEM OS (Windows).
start_type	No	String	The start mode. The value can be **MANUAL**, **AUTO**, or empty. No verification is performed. The default value is **MANUAL**. Other values indicate that the server is started registered with the MgC platform. Minimum: **0** Maximum: **255**
io_read_wait	No	Double	The disk read latency (in ms). Minimum: **0.0** Maximum: **10000.0**
has_tc	No	Boolean	Whether the tc component is installed. This parameter is mandatory in the Linux scenario. Otherwise, the subsequent environment check will fail.

Parameter	Mandatory	Type	Description
platform	No	String	Platform information: **hw**: Huawei **ali**: Alibaba **aws**: Amazon **azure**: Microsoft Azure **gcp**: Google Cloud **tencent**: Tencent Cloud **vmware**: VMware **hyperv**: Hyper-V **other**: Others **default**: default Enumeration values: • **hw** • **ali** • **aws** • **azure** • **gcp** • **tencent** • **vmware** • **hyperv** • **other** • **default**

Table 5-22 ServerDisk

Parameter	Mandatory	Type	Description
name	No	String	The disk name. Minimum: **0** Maximum: **255**

Parameter	Mandatory	Type	Description
partition_style	No	String	Disk partition type. This parameter is mandatory when the source server is added. Otherwise, the subsequent environment check will fail. Options (required by EVS): **MBR**: Master Boot Record **GPT**: GUID Partition Table** **Partition Table** For details, see the description of differences between MBR and GPT in the EVS API document. Minimum: **0** Maximum: **255**
device_use	No	String	Disk type. This is not mandatory and can be left blank. The common values are as follows: **NORMAL** **OS**: system device **BOOT** **VOLUME_GROUP** **BTRFS** Minimum: **0** Maximum: **255**
size	No	Long	The disk size, in bytes. Minimum: **0** Maximum: **9223372036854775807**
used_size	No	Long	The used disk space, in bytes. Minimum: **0** Maximum: **9223372036854775807**
physical_volumes	No	Array of **PhysicalVolume** objects	The information about physical partitions on the disk. Array Length: **0 - 50**
os_disk	No	Boolean	Indicates whether the disk is the system disk.

Parameter	Mandatory	Type	Description
relation_name	No	String	The name of the paired disk on the Linux target ECS server. Minimum: **0** Maximum: **255**
inode_size	No	Integer	The inode size. Minimum: **0** Maximum: **2147483647**

Table 5-23 PhysicalVolume

Parameter	Mandatory	Type	Description
device_use	No	String	Disk type. This is not mandatory and can be left blank. The common values are as follows: **NORMAL** **OS**: system device **BOOT** **VOLUME_GROUP** **BTRFS** Minimum: **0** Maximum: **255**
file_system	No	String	The file system type. Minimum: **0** Maximum: **255**
index	No	Integer	The serial number. Minimum: **0** Maximum: **2147483647**
mount_point	No	String	The mount point. Minimum: **0** Maximum: **255**
name	No	String	The volume name. In Windows, it indicates the drive letter, and in Linux, it indicates the device ID. Minimum: **0** Maximum: **255**

Parameter	Mandatory	Type	Description
size	No	Long	The size. Minimum: **0** Maximum: **9223372036854775807**
used_size	No	Long	The used space. Minimum: **0** Maximum: **9223372036854775807**
inode_size	No	Integer	The number of inodes. Minimum: **0** Maximum: **2147483647**
inode_nums	No	Long	The number of inodes. Minimum: **0** Maximum: **9223372036854775807**
uuid	No	String	The GUID, which can be obtained from the source server. Minimum: **0** Maximum: **255**
size_per_cluster	No	Integer	The size of each cluster. Minimum: **0** Maximum: **2147483647**

Table 5-24 BtrfsFileSystem

Parameter	Mandatory	Type	Description
name	No	String	The file system name. Minimum: **0** Maximum: **255**
label	No	String	The file system tags. If no tag exists, the value is an empty string. Minimum: **0** Maximum: **255**

Parameter	Mandatory	Type	Description
uuid	No	String	The file system UUID. Minimum: **0** Maximum: **255**
device	No	String	The device names of the Btrfs file system. Minimum: **0** Maximum: **255**
size	No	Long	The space occupied by the file system data. Minimum: **0** Maximum: **9223372036854775807**
nodesize	No	Long	The Btrfs node size. Minimum: **0** Maximum: **9223372036854775807**
sectorsize	No	Integer	The sector size. Minimum: **0** Maximum: **2147483647**
data_profile	No	String	The data profile (RAD). Minimum: **0** Maximum: **255**
system_profile	No	String	The file system profile (RAD). Minimum: **0** Maximum: **255**
metadata_profile	No	String	The metadata profile (RAD). Minimum: **0** Maximum: **255**
global_reserve1	No	String	The Btrfs file system information. Minimum: **0** Maximum: **255**
g_vol_used_size	No	Long	The used space of the Btrfs volume. Minimum: **0** Maximum: **9223372036854775807**

Parameter	Mandatory	Type	Description
default_subvolid	No	String	The ID of the default subvolume. Minimum: **0** Maximum: **255**
default_subvol_name	No	String	The name of the default subvolume. Minimum: **0** Maximum: **255**
default_subvol_mountpath	No	String	The mount path of the default subvolume or Btrfs file system. Minimum: **0** Maximum: **255**
subvolumn	No	Array of **BtrfsSubvolumn** objects	The subvolume information. Array Length: **0 - 65535**

Table 5-25 BtrfsSubvolumn

Parameter	Mandatory	Type	Description
uuid	No	String	The UUID of the parent volume. Minimum: **0** Maximum: **255**
is_snapshot	No	String	Indicates whether the subvolume is a snapshot. Minimum: **0** Maximum: **255**
subvol_id	No	String	The subvolume ID. Minimum: **0** Maximum: **255**
parent_id	No	String	The parent volume ID. Minimum: **0** Maximum: **255**
subvol_name	No	String	The subvolume name. Minimum: **0** Maximum: **255**

Parameter	Mandatory	Type	Description
subvol_mount_path	No	String	The mount path of the subvolume. Minimum: **0** Maximum: **255**

Table 5-26 NetWork

Parameter	Mandatory	Type	Description
name	No	String	The NIC name. Minimum: **0** Maximum: **255**
ip	No	String	The IP address bound to the NIC. Minimum: **0** Maximum: **255**
ipv6	No	String	The IPv6 address. Minimum: **0** Maximum: **255**
netmask	No	String	The subnet mask. Minimum: **0** Maximum: **255**
gateway	No	String	Gateway. Minimum: **0** Maximum: **255**
mtu	No	Integer	The NIC MTU. This parameter is mandatory for Linux. Minimum: **0** Maximum: **2147483647**
mac	Yes	String	The first MAC address in the list must not be empty. Minimum: **0** Maximum: **255**
id	No	String	The database record ID. Minimum: **0** Maximum: **255**

Table 5-27 VolumeGroups

Parameter	Mandatory	Type	Description
components	No	String	The physical volume information. Minimum: **0** Maximum: **255**
free_size	No	Long	The available space. Minimum: **0** Maximum: **9223372036854775807**
logical_volumes	No	Array of **LogicalVolumes** objects	The logical volume information. Array Length: **0 - 50**
name	No	String	Name. Minimum: **0** Maximum: **255**
size	No	Long	Size. Minimum: **0** Maximum: **9223372036854775807**

Table 5-28 LogicalVolumes

Parameter	Mandatory	Type	Description
block_count	No	Integer	The number of blocks. Minimum: **0** Maximum: **2147483647** Default: **0**
block_size	No	Long	Block size. Minimum: **0** Maximum: **1048576** Default: **0**
file_system	No	String	The file system. Minimum: **0** Maximum: **255**
inode_size	No	Integer	The number of inodes. Minimum: **0** Maximum: **2147483647**

Parameter	Mandatory	Type	Description
inode_nums	No	Long	The number of inodes. Minimum: **0** Maximum: **9223372036854775807**
device_use	No	String	Partition type. This is not mandatory and can be left blank. The common values are as follows: **NORMAL** **OS**: system device **BOOT** **VOLUME_GROUP** **BTRFS** Minimum: **0** Maximum: **255**
mount_point	No	String	The mount point. Minimum: **0** Maximum: **256**
name	No	String	Name. Minimum: **0** Maximum: **1024**
size	No	Long	The size. Minimum: **0** Maximum: **9223372036854775807**
used_size	No	Long	The used space. Minimum: **0** Maximum: **9223372036854775807**
free_size	No	Long	The available space. Minimum: **0** Maximum: **9223372036854775807**

Response Parameters

Status code: 200

Table 5-29 Response body parameters

Parameter	Type	Description
id	String	The source server ID. Minimum: **0** Maximum: **255**

Status code: 403

Table 5-30 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **255**
encoded_authorization_message	String	The encrypted authorization information. Minimum: **0** Maximum: **65535**
error_param	Array of strings	Error parameters. Minimum: **0** Maximum: **65535** Array Length: **1 - 20**
details	Array of **details** objects	The error details. Array Length: **1 - 20**

Table 5-31 details

Parameter	Type	Description
error_code	String	The SMS error code. Minimum: **0** Maximum: **65535**
error_msg	String	The SMS error message. Minimum: **0** Maximum: **65535**

Example Requests

This example reports the information about a source server to SMS. The source server runs Linux SUSE12_64BIT_SP2, with the name of host-192-168-136-xxx, the IP address of 192.168.136.xxx, the BOOT system disk named /dev/vda, and the system disk size of 42,949,672,960 bytes. After the source server is reported, you can view the source server information on the SMS console.

```
POST https://{endpoint}/v3/sources

{
  "os_type" : "LINUX",
  "name" : "host-192-168-136-xxx",
  "os_version" : "SUSE12_64BIT_SP2",
  "linux_block_check" : "{\"release_type\": \"SUSE\", \"release_version\": \"12.2\", \"kernel_simplification\": \"4.4.21\", \"architecture\": \"x86_64\", \"kernel_version\": \"4.4.21-69-default\"}",
  "kernel_version" : "4.4.21-69-default",
  "paravirtualization" : true,
  "firmware" : "BIOS",
  "has_rsync" : true,
  "io_read_wait" : 3.4,
  "boot_loader" : "GRUB",
  "disks" : [ {
    "name" : "/dev/vda",
    "device_use" : "BOOT",
    "size" : 42949672960,
    "partition_style" : "MBR",
    "used_size" : 42948624384,
    "physical_volumes" : [ {
      "name" : "/dev/vda1",
      "size" : 2153775104,
      "device_use" : "NORMAL",
      "used_size" : 2153775104,
      "inode_size" : 0,
      "inode_nums" : 0,
      "file_system" : "swap",
      "mount_point" : ""
    }, {
      "name" : "/dev/vda2",
      "size" : 16862150656,
      "device_use" : "BTRFS",
      "used_size" : 16862150656,
      "inode_size" : 0,
      "inode_nums" : 0,
      "file_system" : "btrfs",
      "mount_point" : ""
    }, {
      "name" : "/dev/vda3",
      "size" : 23932698624,
      "device_use" : "NORMAL",
      "used_size" : 33988608,
      "inode_size" : 0,
      "inode_nums" : 12345,
      "file_system" : "xfs",
      "mount_point" : "/home"
    } ]
  }, {
    "name" : "/dev/vdb",
    "device_use" : "NORMAL",
    "size" : 21474836480,
    "partition_style" : "MBR",
    "used_size" : 21473787904,
    "physical_volumes" : [ {
      "name" : "/dev/vdb1",
      "size" : 21473787904,
      "device_use" : "VOLUME_GROUP",
      "used_size" : 21473787904,
      "inode_size" : 0,
```



```
      "inode_nums" : 0,
      "file_system" : "LVM2_member",
      "mount_point" : ""
    } ]
  }, {
    "name" : "/dev/vdc",
    "device_use" : "VOLUME_GROUP",
    "size" : 21474836480,
    "partition_style" : "MBR",
    "used_size" : 0,
    "physical_volumes" : [ ]
  } ],
  "volume_groups" : [ {
    "name" : "vg1",
    "size" : 42948624384,
    "components" : "/dev/vdb1;/dev/vdc",
    "logical_volumes" : [ {
      "name" : "/dev/mapper/vg1-lv1",
      "device_use" : "NORMAL",
      "size" : 10737418240,
      "free_size" : 10713837568,
      "used_size" : 23580672,
      "file_system" : "ext4",
      "mount_point" : "/mnt/lv1",
      "inode_nums" : 12345,
      "inode_size" : "256"
    } ]
  } ],
  "btrfs_list" : [ {
    "name" : "/dev/vda2",
    "label" : "none",
    "uuid" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "device" : "/dev/vda2",
    "size" : "3.30GiB",
    "nodesize" : "16384",
    "sectorsize" : "4096",
    "data_profile" : "single",
    "system_profile" : "single",
    "metadata_profile" : "single",
    "global_reserve1" : "single",
    "g_vol_used_size" : "3894038528",
    "default_subvolid" : "259",
    "default_subvol_name" : "@/.snapshots/1/snapshot",
    "default_subvol_mountpath" : "/",
    "subvolumn" : [ {
      "uuid" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
      "is_snapshot" : "false",
      "subvol_id" : "257",
      "parent_id" : "5",
      "subvol_name" : "@",
      "subvol_mount_path" : "null"
    }, {
      "uuid" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
      "is_snapshot" : "false",
      "subvol_id" : "258",
      "parent_id" : "257",
      "subvol_name" : "@/.snapshots",
      "subvol_mount_path" : "/.snapshots"
    }, {
      "uuid" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
      "is_snapshot" : "true",
      "subvol_id" : "259",
      "parent_id" : "258",
      "subvol_name" : "@/.snapshots/1/snapshot",
      "subvol_mount_path" : "/"
    }, {
      "uuid" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
      "is_snapshot" : "false",
      "subvol_id" : "260",
```

```
    "parent_id" : "257",
    "subvol_name" : "@/boot/grub2/i386-pc",
    "subvol_mount_path" : "/boot/grub2/i386-pc"
  }, {
    "uuid" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "is_snapshot" : "false",
    "subvol_id" : "261",
    "parent_id" : "257",
    "subvol_name" : "@/boot/grub2/x86_64-efi",
    "subvol_mount_path" : "/boot/grub2/x86_64-efi"
  }, {
    "uuid" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "is_snapshot" : "false",
    "subvol_id" : "262",
    "parent_id" : "257",
    "subvol_name" : "@/opt",
    "subvol_mount_path" : "/opt"
  }, {
    "uuid" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "is_snapshot" : "false",
    "subvol_id" : "263",
    "parent_id" : "257",
    "subvol_name" : "@/srv",
    "subvol_mount_path" : "/srv"
  }, {
    "uuid" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "is_snapshot" : "false",
    "subvol_id" : "264",
    "parent_id" : "257",
    "subvol_name" : "@/tmp",
    "subvol_mount_path" : "/tmp"
  }, {
    "uuid" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "is_snapshot" : "false",
    "subvol_id" : "265",
    "parent_id" : "257",
    "subvol_name" : "@/usr/local",
    "subvol_mount_path" : "/usr/local"
  }, {
    "uuid" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "is_snapshot" : "false",
    "subvol_id" : "266",
    "parent_id" : "257",
    "subvol_name" : "@/var/cache",
    "subvol_mount_path" : "/var/cache"
  }, {
    "uuid" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "is_snapshot" : "false",
    "subvol_id" : "267",
    "parent_id" : "257",
    "subvol_name" : "@/var/crash",
    "subvol_mount_path" : "/var/crash"
  }, {
    "uuid" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "is_snapshot" : "false",
    "subvol_id" : "268",
    "parent_id" : "257",
    "subvol_name" : "@/var/lib/libvirt/images",
    "subvol_mount_path" : "/var/lib/libvirt/images"
  }, {
    "uuid" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "is_snapshot" : "false",
    "subvol_id" : "269",
    "parent_id" : "257",
    "subvol_name" : "@/var/lib/machines",
    "subvol_mount_path" : "/var/lib/machines"
  }, {
    "uuid" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "is_snapshot" : "false",
```

```
    "subvol_id" : "270",
    "parent_id" : "257",
    "subvol_name" : "@/var/lib/mailman",
    "subvol_mount_path" : "/var/lib/mailman"
  }, {
    "uuid" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "is_snapshot" : "false",
    "subvol_id" : "271",
    "parent_id" : "257",
    "subvol_name" : "@/var/lib/mariadb",
    "subvol_mount_path" : "/var/lib/mariadb"
  }, {
    "uuid" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "is_snapshot" : "false",
    "subvol_id" : "272",
    "parent_id" : "257",
    "subvol_name" : "@/var/lib/mysql",
    "subvol_mount_path" : "/var/lib/mysql"
  }, {
    "uuid" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "is_snapshot" : "false",
    "subvol_id" : "273",
    "parent_id" : "257",
    "subvol_name" : "@/var/lib/named",
    "subvol_mount_path" : "/var/lib/named"
  }, {
    "uuid" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "is_snapshot" : "false",
    "subvol_id" : "274",
    "parent_id" : "257",
    "subvol_name" : "@/var/lib/pgsql",
    "subvol_mount_path" : "/var/lib/pgsql"
  }, {
    "uuid" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "is_snapshot" : "false",
    "subvol_id" : "275",
    "parent_id" : "257",
    "subvol_name" : "@/var/log",
    "subvol_mount_path" : "/var/log"
  }, {
    "uuid" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "is_snapshot" : "false",
    "subvol_id" : "276",
    "parent_id" : "257",
    "subvol_name" : "@/var/opt",
    "subvol_mount_path" : "/var/opt"
  }, {
    "uuid" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "is_snapshot" : "false",
    "subvol_id" : "277",
    "parent_id" : "257",
    "subvol_name" : "@/var/spool",
    "subvol_mount_path" : "/var/spool"
  }, {
    "uuid" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "is_snapshot" : "false",
    "subvol_id" : "278",
    "parent_id" : "257",
    "subvol_name" : "@/var/tmp",
    "subvol_mount_path" : "/var/tmp"
  }, {
    "uuid" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "is_snapshot" : "true",
    "subvol_id" : "282",
    "parent_id" : "258",
    "subvol_name" : "@/.snapshots/2/snapshot",
    "subvol_mount_path" : "null"
  } ]
} ],
```

```
  "cpu_quantity" : 1,
  "memory" : 934752256,
  "networks" : [ {
    "name" : "eth0",
    "ip" : "192.168.136.xxx",
    "netmask" : "netmask",
    "mtu" : 1,
    "gateway" : "gateway",
    "mac" : "1a9660eb8a3ffcf4df6d7865b52eb54f7b0cd194029e0eadd8e2c7f1267d80c0"
  } ],
  "raw_devices" : "string",
  "has_tc" : true,
  "ip" : "192.168.136.xxx",
  "agent_version" : "2.2.1",
  "platform" : "hw"
}
```

Example Responses

Status code: 200

The source server was registered with SMS.

```
{
  "id" : "xxxxxxxxxxxxxxxxxxxxxxxx00000001"
}
```

Status code: 403

Authentication failed.

```
{
  "error_code" : "SMS.9004",
  "error_msg" : "The current account does not have the permission to execute policy. You do not have permission to perform action XXX on resource XXX.",
  "encoded_authorization_message" : "XXXXXX",
  "error_param" : [ "You do not have permission to perform action XXX on resource XXX." ],
  "details" : [ {
    "error_code" : "SMS.9004",
    "error_msg" : "You do not have permission to perform action XXX on resource XXX."
  } ]
}
```

SDK Sample Code

The SDK sample code is as follows.

Java

This example reports the information about a source server to SMS. The source server runs Linux SUSE12_64BIT_SP2, with the name of host-192-168-136-xxx, the IP address of 192.168.136.xxx, the BOOT system disk named /dev/vda, and the system disk size of 42,949,672,960 bytes. After the source server is reported, you can view the source server information on the SMS console.

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;
```

```
import java.util.List;
import java.util.ArrayList;

public class RegisterServerSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great
security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or
environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running
this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        RegisterServerRequest request = new RegisterServerRequest();
        PostSourceServerBody body = new PostSourceServerBody();
        List<LogicalVolumes> listVolumeGroupsLogicalVolumes = new ArrayList<>();
        listVolumeGroupsLogicalVolumes.add(
            new LogicalVolumes()
                .withFileSystem("ext4")
                .withInodeSize(256)
                .withInodeNums(12345L)
                .withDeviceUse("NORMAL")
                .withMountPoint("/mnt/lv1")
                .withName("/dev/mapper/vg1-lv1")
                .withSize(10737418240L)
                .withUsedSize(23580672L)
                .withFreeSize(10713837568L)
        );
        List<VolumeGroups> listbodyVolumeGroups = new ArrayList<>();
        listbodyVolumeGroups.add(
            new VolumeGroups()
                .withComponents("/dev/vdb1;/dev/vdc")
                .withLogicalVolumes(listVolumeGroupsLogicalVolumes)
                .withName("vg1")
                .withSize(42948624384L)
        );
        List<NetWork> listbodyNetworks = new ArrayList<>();
        listbodyNetworks.add(
            new NetWork()
                .withName("eth0")
                .withIp("192.168.136.xxx")
                .withNetmask("netmask")
                .withGateway("gateway")
                .withMtu(1)
                .withMac("1a9660eb8a3ffcf4df6d7865b52eb54f7b0cd194029e0eadd8e2c7f1267d80c0")
        );
        List<BtrfsSubvolumn> listBtrfsListSubvolumn = new ArrayList<>();
        listBtrfsListSubvolumn.add(
            new BtrfsSubvolumn()
                .withUuid("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
                .withIsSnapshot("false")
                .withSubvolId("257")
                .withParentId("5")
                .withSubvolName("@")
                .withSubvolMountPath("null")
        );
        listBtrfsListSubvolumn.add(
            new BtrfsSubvolumn()
                .withUuid("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
```

```
            .withIsSnapshot("false")
            .withSubvolId("258")
            .withParentId("257")
            .withSubvolName("@/.snapshots")
            .withSubvolMountPath("/.snapshots")
    );
    listBtrfsListSubvolumn.add(
        new BtrfsSubvolumn()
            .withUuid("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
            .withIsSnapshot("true")
            .withSubvolId("259")
            .withParentId("258")
            .withSubvolName("@/.snapshots/1/snapshot")
            .withSubvolMountPath("/")
    );
    listBtrfsListSubvolumn.add(
        new BtrfsSubvolumn()
            .withUuid("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
            .withIsSnapshot("false")
            .withSubvolId("260")
            .withParentId("257")
            .withSubvolName("@/boot/grub2/i386-pc")
            .withSubvolMountPath("/boot/grub2/i386-pc")
    );
    listBtrfsListSubvolumn.add(
        new BtrfsSubvolumn()
            .withUuid("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
            .withIsSnapshot("false")
            .withSubvolId("261")
            .withParentId("257")
            .withSubvolName("@/boot/grub2/x86_64-efi")
            .withSubvolMountPath("/boot/grub2/x86_64-efi")
    );
    listBtrfsListSubvolumn.add(
        new BtrfsSubvolumn()
            .withUuid("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
            .withIsSnapshot("false")
            .withSubvolId("262")
            .withParentId("257")
            .withSubvolName("@/opt")
            .withSubvolMountPath("/opt")
    );
    listBtrfsListSubvolumn.add(
        new BtrfsSubvolumn()
            .withUuid("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
            .withIsSnapshot("false")
            .withSubvolId("263")
            .withParentId("257")
            .withSubvolName("@/srv")
            .withSubvolMountPath("/srv")
    );
    listBtrfsListSubvolumn.add(
        new BtrfsSubvolumn()
            .withUuid("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
            .withIsSnapshot("false")
            .withSubvolId("264")
            .withParentId("257")
            .withSubvolName("@/tmp")
            .withSubvolMountPath("/tmp")
    );
    listBtrfsListSubvolumn.add(
        new BtrfsSubvolumn()
            .withUuid("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
            .withIsSnapshot("false")
            .withSubvolId("265")
            .withParentId("257")
            .withSubvolName("@/usr/local")
            .withSubvolMountPath("/usr/local")
    );
```

```
listBtrfsListSubvolumn.add(
    new BtrfsSubvolumn()
        .withUuid("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
        .withIsSnapshot("false")
        .withSubvolId("266")
        .withParentId("257")
        .withSubvolName("@/var/cache")
        .withSubvolMountPath("/var/cache")
);
listBtrfsListSubvolumn.add(
    new BtrfsSubvolumn()
        .withUuid("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
        .withIsSnapshot("false")
        .withSubvolId("267")
        .withParentId("257")
        .withSubvolName("@/var/crash")
        .withSubvolMountPath("/var/crash")
);
listBtrfsListSubvolumn.add(
    new BtrfsSubvolumn()
        .withUuid("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
        .withIsSnapshot("false")
        .withSubvolId("268")
        .withParentId("257")
        .withSubvolName("@/var/lib/libvirt/images")
        .withSubvolMountPath("/var/lib/libvirt/images")
);
listBtrfsListSubvolumn.add(
    new BtrfsSubvolumn()
        .withUuid("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
        .withIsSnapshot("false")
        .withSubvolId("269")
        .withParentId("257")
        .withSubvolName("@/var/lib/machines")
        .withSubvolMountPath("/var/lib/machines")
);
listBtrfsListSubvolumn.add(
    new BtrfsSubvolumn()
        .withUuid("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
        .withIsSnapshot("false")
        .withSubvolId("270")
        .withParentId("257")
        .withSubvolName("@/var/lib/mailman")
        .withSubvolMountPath("/var/lib/mailman")
);
listBtrfsListSubvolumn.add(
    new BtrfsSubvolumn()
        .withUuid("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
        .withIsSnapshot("false")
        .withSubvolId("271")
        .withParentId("257")
        .withSubvolName("@/var/lib/mariadb")
        .withSubvolMountPath("/var/lib/mariadb")
);
listBtrfsListSubvolumn.add(
    new BtrfsSubvolumn()
        .withUuid("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
        .withIsSnapshot("false")
        .withSubvolId("272")
        .withParentId("257")
        .withSubvolName("@/var/lib/mysql")
        .withSubvolMountPath("/var/lib/mysql")
);
listBtrfsListSubvolumn.add(
    new BtrfsSubvolumn()
        .withUuid("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
        .withIsSnapshot("false")
        .withSubvolId("273")
        .withParentId("257")
```



```
                .withSubvolName("@/var/lib/named")
                .withSubvolMountPath("/var/lib/named")
        );
        listBtrfsListSubvolumn.add(
            new BtrfsSubvolumn()
                .withUuid("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
                .withIsSnapshot("false")
                .withSubvolId("274")
                .withParentId("257")
                .withSubvolName("@/var/lib/pgsql")
                .withSubvolMountPath("/var/lib/pgsql")
        );
        listBtrfsListSubvolumn.add(
            new BtrfsSubvolumn()
                .withUuid("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
                .withIsSnapshot("false")
                .withSubvolId("275")
                .withParentId("257")
                .withSubvolName("@/var/log")
                .withSubvolMountPath("/var/log")
        );
        listBtrfsListSubvolumn.add(
            new BtrfsSubvolumn()
                .withUuid("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
                .withIsSnapshot("false")
                .withSubvolId("276")
                .withParentId("257")
                .withSubvolName("@/var/opt")
                .withSubvolMountPath("/var/opt")
        );
        listBtrfsListSubvolumn.add(
            new BtrfsSubvolumn()
                .withUuid("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
                .withIsSnapshot("false")
                .withSubvolId("277")
                .withParentId("257")
                .withSubvolName("@/var/spool")
                .withSubvolMountPath("/var/spool")
        );
        listBtrfsListSubvolumn.add(
            new BtrfsSubvolumn()
                .withUuid("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
                .withIsSnapshot("false")
                .withSubvolId("278")
                .withParentId("257")
                .withSubvolName("@/var/tmp")
                .withSubvolMountPath("/var/tmp")
        );
        listBtrfsListSubvolumn.add(
            new BtrfsSubvolumn()
                .withUuid("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
                .withIsSnapshot("true")
                .withSubvolId("282")
                .withParentId("258")
                .withSubvolName("@/.snapshots/2/snapshot")
                .withSubvolMountPath("null")
        );
        List<BtrfsFileSystem> listbodyBtrfsList = new ArrayList<>();
        listbodyBtrfsList.add(
            new BtrfsFileSystem()
                .withName("/dev/vda2")
                .withLabel("none")
                .withUuid("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
                .withDevice("/dev/vda2")
                .withSize(3.30GiBL)
                .withNodesize(16384L)
                .withSectorsize(4096)
                .withDataProfile("single")
                .withSystemProfile("single")
```

```
            .withMetadataProfile("single")
            .withGlobalReserve1("single")
            .withGVolUsedSize(3894038528L)
            .withDefaultSubvolid("259")
            .withDefaultSubvolName("@/.snapshots/1/snapshot")
            .withDefaultSubvolMountpath("/")
            .withSubvolumn(listBtrfsListSubvolumn)
    );
    List<PhysicalVolume> listDisksPhysicalVolumes = new ArrayList<>();
    listDisksPhysicalVolumes.add(
        new PhysicalVolume()
            .withDeviceUse("VOLUME_GROUP")
            .withFileSystem("LVM2_member")
            .withMountPoint("")
            .withName("/dev/vdb1")
            .withSize(21473787904L)
            .withUsedSize(21473787904L)
            .withInodeSize(0)
            .withInodeNums(0L)
    );
    List<PhysicalVolume> listDisksPhysicalVolumes1 = new ArrayList<>();
    listDisksPhysicalVolumes1.add(
        new PhysicalVolume()
            .withDeviceUse("NORMAL")
            .withFileSystem("swap")
            .withMountPoint("")
            .withName("/dev/vda1")
            .withSize(2153775104L)
            .withUsedSize(2153775104L)
            .withInodeSize(0)
            .withInodeNums(0L)
    );
    listDisksPhysicalVolumes1.add(
        new PhysicalVolume()
            .withDeviceUse("BTRFS")
            .withFileSystem("btrfs")
            .withMountPoint("")
            .withName("/dev/vda2")
            .withSize(16862150656L)
            .withUsedSize(16862150656L)
            .withInodeSize(0)
            .withInodeNums(0L)
    );
    listDisksPhysicalVolumes1.add(
        new PhysicalVolume()
            .withDeviceUse("NORMAL")
            .withFileSystem("xfs")
            .withMountPoint("/home")
            .withName("/dev/vda3")
            .withSize(23932698624L)
            .withUsedSize(33988608L)
            .withInodeSize(0)
            .withInodeNums(12345L)
    );
    List<ServerDisk> listbodyDisks = new ArrayList<>();
    listbodyDisks.add(
        new ServerDisk()
            .withName("/dev/vda")
            .withPartitionStyle("MBR")
            .withDeviceUse("BOOT")
            .withSize(42949672960L)
            .withUsedSize(42948624384L)
            .withPhysicalVolumes(listDisksPhysicalVolumes1)
    );
    listbodyDisks.add(
        new ServerDisk()
            .withName("/dev/vdb")
            .withPartitionStyle("MBR")
            .withDeviceUse("NORMAL")
```

```
                .withSize(21474836480L)
                .withUsedSize(21473787904L)
                .withPhysicalVolumes(listDisksPhysicalVolumes)
        );
        listbodyDisks.add(
            new ServerDisk()
                .withName("/dev/vdc")
                .withPartitionStyle("MBR")
                .withDeviceUse("VOLUME_GROUP")
                .withSize(21474836480L)
                .withUsedSize(0L)
                .withPhysicalVolumes()
        );
        body.withPlatform(PostSourceServerBody.PlatformEnum.fromValue("hw"));
        body.withHasTc(true);
        body.withIoReadWait((double)3.4);
        body.withKernelVersion("4.4.21-69-default");
        body.withAgentVersion("2.2.1");
        body.withVolumeGroups(listbodyVolumeGroups);
        body.withBootLoader("GRUB");
        body.withRawDevices("string");
        body.withParavirtualization(true);
        body.withHasRsync(true);
        body.withNetworks(listbodyNetworks);
        body.withBtrfsList(listbodyBtrfsList);
        body.withDisks(listbodyDisks);
        body.withMemory(934752256L);
        body.withCpuQuantity(1);
        body.withFirmware(PostSourceServerBody.FirmwareEnum.fromValue("BIOS"));
        body.withLinuxBlockCheck("{"release_type": "SUSE", "release_version": "12.2", "kernel_simplification":
"4.4.21", "architecture": "x86_64", "kernel_version": "4.4.21-69-default"}");
        body.withOsVersion("SUSE12_64BIT_SP2");
        body.withOsType(PostSourceServerBody.OsTypeEnum.fromValue("LINUX"));
        body.withName("host-192-168-136-xxx");
        body.withIp("192.168.136.xxx");
        request.withBody(body);
        try {
            RegisterServerResponse response = client.registerServer(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

Python

This example reports the information about a source server to SMS. The source server runs Linux SUSE12_64BIT_SP2, with the name of host-192-168-136-xxx, the IP address of 192.168.136.xxx, the BOOT system disk named /dev/vda, and the system disk size of 42,949,672,960 bytes. After the source server is reported, you can view the source server information on the SMS console.

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *
```

```
if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = RegisterServerRequest()
        listLogicalVolumesVolumeGroups = [
            LogicalVolumes(
                file_system="ext4",
                inode_size=256,
                inode_nums=12345,
                device_use="NORMAL",
                mount_point="/mnt/lv1",
                name="/dev/mapper/vg1-lv1",
                size=10737418240,
                used_size=23580672,
                free_size=10713837568
            )
        ]
        listVolumeGroupsbody = [
            VolumeGroups(
                components="/dev/vdb1;/dev/vdc",
                logical_volumes=listLogicalVolumesVolumeGroups,
                name="vg1",
                size=42948624384
            )
        ]
        listNetworksbody = [
            NetWork(
                name="eth0",
                ip="192.168.136.xxx",
                netmask="netmask",
                gateway="gateway",
                mtu=1,
                mac="1a9660eb8a3ffcf4df6d7865b52eb54f7b0cd194029e0eadd8e2c7f1267d80c0"
            )
        ]
        listSubvolumnBtrfsList = [
            BtrfsSubvolumn(
                uuid="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
                is_snapshot="false",
                subvol_id="257",
                parent_id="5",
                subvol_name="@",
                subvol_mount_path="null"
            ),
            BtrfsSubvolumn(
                uuid="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
                is_snapshot="false",
                subvol_id="258",
                parent_id="257",
                subvol_name="@/.snapshots",
                subvol_mount_path="/.snapshots"
            ),
            BtrfsSubvolumn(
                uuid="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
```

```
        is_snapshot="true",
        subvol_id="259",
        parent_id="258",
        subvol_name="@/.snapshots/1/snapshot",
        subvol_mount_path="/"
    ),
    BtrfsSubvolumn(
        uuid="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
        is_snapshot="false",
        subvol_id="260",
        parent_id="257",
        subvol_name="@/boot/grub2/i386-pc",
        subvol_mount_path="/boot/grub2/i386-pc"
    ),
    BtrfsSubvolumn(
        uuid="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
        is_snapshot="false",
        subvol_id="261",
        parent_id="257",
        subvol_name="@/boot/grub2/x86_64-efi",
        subvol_mount_path="/boot/grub2/x86_64-efi"
    ),
    BtrfsSubvolumn(
        uuid="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
        is_snapshot="false",
        subvol_id="262",
        parent_id="257",
        subvol_name="@/opt",
        subvol_mount_path="/opt"
    ),
    BtrfsSubvolumn(
        uuid="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
        is_snapshot="false",
        subvol_id="263",
        parent_id="257",
        subvol_name="@/srv",
        subvol_mount_path="/srv"
    ),
    BtrfsSubvolumn(
        uuid="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
        is_snapshot="false",
        subvol_id="264",
        parent_id="257",
        subvol_name="@/tmp",
        subvol_mount_path="/tmp"
    ),
    BtrfsSubvolumn(
        uuid="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
        is_snapshot="false",
        subvol_id="265",
        parent_id="257",
        subvol_name="@/usr/local",
        subvol_mount_path="/usr/local"
    ),
    BtrfsSubvolumn(
        uuid="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
        is_snapshot="false",
        subvol_id="266",
        parent_id="257",
        subvol_name="@/var/cache",
        subvol_mount_path="/var/cache"
    ),
    BtrfsSubvolumn(
        uuid="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
        is_snapshot="false",
        subvol_id="267",
        parent_id="257",
        subvol_name="@/var/crash",
        subvol_mount_path="/var/crash"
```



```
        ),
        BtrfsSubvolumn(
            uuid="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
            is_snapshot="false",
            subvol_id="268",
            parent_id="257",
            subvol_name="@/var/lib/libvirt/images",
            subvol_mount_path="/var/lib/libvirt/images"
        ),
        BtrfsSubvolumn(
            uuid="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
            is_snapshot="false",
            subvol_id="269",
            parent_id="257",
            subvol_name="@/var/lib/machines",
            subvol_mount_path="/var/lib/machines"
        ),
        BtrfsSubvolumn(
            uuid="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
            is_snapshot="false",
            subvol_id="270",
            parent_id="257",
            subvol_name="@/var/lib/mailman",
            subvol_mount_path="/var/lib/mailman"
        ),
        BtrfsSubvolumn(
            uuid="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
            is_snapshot="false",
            subvol_id="271",
            parent_id="257",
            subvol_name="@/var/lib/mariadb",
            subvol_mount_path="/var/lib/mariadb"
        ),
        BtrfsSubvolumn(
            uuid="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
            is_snapshot="false",
            subvol_id="272",
            parent_id="257",
            subvol_name="@/var/lib/mysql",
            subvol_mount_path="/var/lib/mysql"
        ),
        BtrfsSubvolumn(
            uuid="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
            is_snapshot="false",
            subvol_id="273",
            parent_id="257",
            subvol_name="@/var/lib/named",
            subvol_mount_path="/var/lib/named"
        ),
        BtrfsSubvolumn(
            uuid="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
            is_snapshot="false",
            subvol_id="274",
            parent_id="257",
            subvol_name="@/var/lib/pgsql",
            subvol_mount_path="/var/lib/pgsql"
        ),
        BtrfsSubvolumn(
            uuid="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
            is_snapshot="false",
            subvol_id="275",
            parent_id="257",
            subvol_name="@/var/log",
            subvol_mount_path="/var/log"
        ),
        BtrfsSubvolumn(
            uuid="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
            is_snapshot="false",
            subvol_id="276",
```

```
            parent_id="257",
            subvol_name="@/var/opt",
            subvol_mount_path="/var/opt"
        ),
        BtrfsSubvolumn(
            uuid="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
            is_snapshot="false",
            subvol_id="277",
            parent_id="257",
            subvol_name="@/var/spool",
            subvol_mount_path="/var/spool"
        ),
        BtrfsSubvolumn(
            uuid="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
            is_snapshot="false",
            subvol_id="278",
            parent_id="257",
            subvol_name="@/var/tmp",
            subvol_mount_path="/var/tmp"
        ),
        BtrfsSubvolumn(
            uuid="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
            is_snapshot="true",
            subvol_id="282",
            parent_id="258",
            subvol_name="@/.snapshots/2/snapshot",
            subvol_mount_path="null"
        )
    ]
    listBtrfsListbody = [
        BtrfsFileSystem(
            name="/dev/vda2",
            label="none",
            uuid="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
            device="/dev/vda2",
            size=3.30GiB,
            nodesize=16384,
            sectorsize=4096,
            data_profile="single",
            system_profile="single",
            metadata_profile="single",
            global_reserve1="single",
            g_vol_used_size=3894038528,
            default_subvolid="259",
            default_subvol_name="@/.snapshots/1/snapshot",
            default_subvol_mountpath="/",
            subvolumn=listSubvolumnBtrfsList
        )
    ]
    listPhysicalVolumesDisks = [
        PhysicalVolume(
            device_use="VOLUME_GROUP",
            file_system="LVM2_member",
            mount_point="",
            name="/dev/vdb1",
            size=21473787904,
            used_size=21473787904,
            inode_size=0,
            inode_nums=0
        )
    ]
    listPhysicalVolumesDisks1 = [
        PhysicalVolume(
            device_use="NORMAL",
            file_system="swap",
            mount_point="",
            name="/dev/vda1",
            size=2153775104,
            used_size=2153775104,
```

```
            inode_size=0,
            inode_nums=0
        ),
        PhysicalVolume(
            device_use="BTRFS",
            file_system="btrfs",
            mount_point="",
            name="/dev/vda2",
            size=16862150656,
            used_size=16862150656,
            inode_size=0,
            inode_nums=0
        ),
        PhysicalVolume(
            device_use="NORMAL",
            file_system="xfs",
            mount_point="/home",
            name="/dev/vda3",
            size=23932698624,
            used_size=33988608,
            inode_size=0,
            inode_nums=12345
        )
    ]
    listDisksbody = [
        ServerDisk(
            name="/dev/vda",
            partition_style="MBR",
            device_use="BOOT",
            size=42949672960,
            used_size=42948624384,
            physical_volumes=listPhysicalVolumesDisks1
        ),
        ServerDisk(
            name="/dev/vdb",
            partition_style="MBR",
            device_use="NORMAL",
            size=21474836480,
            used_size=21473787904,
            physical_volumes=listPhysicalVolumesDisks
        ),
        ServerDisk(
            name="/dev/vdc",
            partition_style="MBR",
            device_use="VOLUME_GROUP",
            size=21474836480,
            used_size=0,
        )
    ]
    request.body = PostSourceServerBody(
        platform="hw",
        has_tc=True,
        io_read_wait=3.4,
        kernel_version="4.4.21-69-default",
        agent_version="2.2.1",
        volume_groups=listVolumeGroupsbody,
        boot_loader="GRUB",
        raw_devices="string",
        paravirtualization=True,
        has_rsync=True,
        networks=listNetworksbody,
        btrfs_list=listBtrfsListbody,
        disks=listDisksbody,
        memory=934752256,
        cpu_quantity=1,
        firmware="BIOS",
        linux_block_check="{"release_type": "SUSE", "release_version": "12.2", "kernel_simplification":
"4.4.21", "architecture": "x86_64", "kernel_version": "4.4.21-69-default"}",
        os_version="SUSE12_64BIT_SP2",
```

```
            os_type="LINUX",
            name="host-192-168-136-xxx",
            ip="192.168.136.xxx"
        )
        response = client.register_server(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

Go

This example reports the information about a source server to SMS. The source server runs Linux SUSE12_64BIT_SP2, with the name of host-192-168-136-xxx, the IP address of 192.168.136.xxx, the BOOT system disk named /dev/vda, and the system disk size of 42,949,672,960 bytes. After the source server is reported, you can view the source server information on the SMS console.

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security 
risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment 
variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before running this 
example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
         WithRegion(region.ValueOf("<YOUR REGION>")).
         WithCredential(auth).
         SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.RegisterServerRequest{}
    fileSystemLogicalVolumes:= "ext4"
    inodeSizeLogicalVolumes:= int32(256)
    inodeNumsLogicalVolumes:= int64(12345)
    deviceUseLogicalVolumes:= "NORMAL"
    mountPointLogicalVolumes:= "/mnt/lv1"
```

```
nameLogicalVolumes:= "/dev/mapper/vg1-lv1"
sizeLogicalVolumes:= int64(10737418240)
usedSizeLogicalVolumes:= int64(23580672)
freeSizeLogicalVolumes:= int64(10713837568)
var listLogicalVolumesVolumeGroups = []model.LogicalVolumes{
    {
        FileSystem: &fileSystemLogicalVolumes,
        InodeSize: &inodeSizeLogicalVolumes,
        InodeNums: &inodeNumsLogicalVolumes,
        DeviceUse: &deviceUseLogicalVolumes,
        MountPoint: &mountPointLogicalVolumes,
        Name: &nameLogicalVolumes,
        Size: &sizeLogicalVolumes,
        UsedSize: &usedSizeLogicalVolumes,
        FreeSize: &freeSizeLogicalVolumes,
    },
}
componentsVolumeGroups:= "/dev/vdb1;/dev/vdc"
nameVolumeGroups:= "vg1"
sizeVolumeGroups:= int64(42948624384)
var listVolumeGroupsbody = []model.VolumeGroups{
    {
        Components: &componentsVolumeGroups,
        LogicalVolumes: &listLogicalVolumesVolumeGroups,
        Name: &nameVolumeGroups,
        Size: &sizeVolumeGroups,
    },
}
nameNetworks:= "eth0"
ipNetworks:= "192.168.136.xxx"
netmaskNetworks:= "netmask"
gatewayNetworks:= "gateway"
mtuNetworks:= int32(1)
var listNetworksbody = []model.NetWork{
    {
        Name: &nameNetworks,
        Ip: &ipNetworks,
        Netmask: &netmaskNetworks,
        Gateway: &gatewayNetworks,
        Mtu: &mtuNetworks,
        Mac: "1a9660eb8a3ffcf4df6d7865b52eb54f7b0cd194029e0eadd8e2c7f1267d80c0",
    },
}
uuidSubvolumn:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
isSnapshotSubvolumn:= "false"
subvolIdSubvolumn:= "257"
parentIdSubvolumn:= "5"
subvolNameSubvolumn:= "@"
subvolMountPathSubvolumn:= "null"
uuidSubvolumn1:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
isSnapshotSubvolumn1:= "false"
subvolIdSubvolumn1:= "258"
parentIdSubvolumn1:= "257"
subvolNameSubvolumn1:= "@/.snapshots"
subvolMountPathSubvolumn1:= "/.snapshots"
uuidSubvolumn2:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
isSnapshotSubvolumn2:= "true"
subvolIdSubvolumn2:= "259"
parentIdSubvolumn2:= "258"
subvolNameSubvolumn2:= "@/.snapshots/1/snapshot"
subvolMountPathSubvolumn2:= "/"
uuidSubvolumn3:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
isSnapshotSubvolumn3:= "false"
subvolIdSubvolumn3:= "260"
parentIdSubvolumn3:= "257"
subvolNameSubvolumn3:= "@/boot/grub2/i386-pc"
subvolMountPathSubvolumn3:= "/boot/grub2/i386-pc"
uuidSubvolumn4:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
isSnapshotSubvolumn4:= "false"
```

```
subvolIdSubvolumn4:= "261"
parentIdSubvolumn4:= "257"
subvolNameSubvolumn4:= "@/boot/grub2/x86_64-efi"
subvolMountPathSubvolumn4:= "/boot/grub2/x86_64-efi"
uuidSubvolumn5:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
isSnapshotSubvolumn5:= "false"
subvolIdSubvolumn5:= "262"
parentIdSubvolumn5:= "257"
subvolNameSubvolumn5:= "@/opt"
subvolMountPathSubvolumn5:= "/opt"
uuidSubvolumn6:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
isSnapshotSubvolumn6:= "false"
subvolIdSubvolumn6:= "263"
parentIdSubvolumn6:= "257"
subvolNameSubvolumn6:= "@/srv"
subvolMountPathSubvolumn6:= "/srv"
uuidSubvolumn7:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
isSnapshotSubvolumn7:= "false"
subvolIdSubvolumn7:= "264"
parentIdSubvolumn7:= "257"
subvolNameSubvolumn7:= "@/tmp"
subvolMountPathSubvolumn7:= "/tmp"
uuidSubvolumn8:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
isSnapshotSubvolumn8:= "false"
subvolIdSubvolumn8:= "265"
parentIdSubvolumn8:= "257"
subvolNameSubvolumn8:= "@/usr/local"
subvolMountPathSubvolumn8:= "/usr/local"
uuidSubvolumn9:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
isSnapshotSubvolumn9:= "false"
subvolIdSubvolumn9:= "266"
parentIdSubvolumn9:= "257"
subvolNameSubvolumn9:= "@/var/cache"
subvolMountPathSubvolumn9:= "/var/cache"
uuidSubvolumn10:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
isSnapshotSubvolumn10:= "false"
subvolIdSubvolumn10:= "267"
parentIdSubvolumn10:= "257"
subvolNameSubvolumn10:= "@/var/crash"
subvolMountPathSubvolumn10:= "/var/crash"
uuidSubvolumn11:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
isSnapshotSubvolumn11:= "false"
subvolIdSubvolumn11:= "268"
parentIdSubvolumn11:= "257"
subvolNameSubvolumn11:= "@/var/lib/libvirt/images"
subvolMountPathSubvolumn11:= "/var/lib/libvirt/images"
uuidSubvolumn12:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
isSnapshotSubvolumn12:= "false"
subvolIdSubvolumn12:= "269"
parentIdSubvolumn12:= "257"
subvolNameSubvolumn12:= "@/var/lib/machines"
subvolMountPathSubvolumn12:= "/var/lib/machines"
uuidSubvolumn13:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
isSnapshotSubvolumn13:= "false"
subvolIdSubvolumn13:= "270"
parentIdSubvolumn13:= "257"
subvolNameSubvolumn13:= "@/var/lib/mailman"
subvolMountPathSubvolumn13:= "/var/lib/mailman"
uuidSubvolumn14:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
isSnapshotSubvolumn14:= "false"
subvolIdSubvolumn14:= "271"
parentIdSubvolumn14:= "257"
subvolNameSubvolumn14:= "@/var/lib/mariadb"
subvolMountPathSubvolumn14:= "/var/lib/mariadb"
uuidSubvolumn15:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
isSnapshotSubvolumn15:= "false"
subvolIdSubvolumn15:= "272"
parentIdSubvolumn15:= "257"
subvolNameSubvolumn15:= "@/var/lib/mysql"
```

```
subvolMountPathSubvolumn15:= "/var/lib/mysql"
uuidSubvolumn16:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
isSnapshotSubvolumn16:= "false"
subvolIdSubvolumn16:= "273"
parentIdSubvolumn16:= "257"
subvolNameSubvolumn16:= "@/var/lib/named"
subvolMountPathSubvolumn16:= "/var/lib/named"
uuidSubvolumn17:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
isSnapshotSubvolumn17:= "false"
subvolIdSubvolumn17:= "274"
parentIdSubvolumn17:= "257"
subvolNameSubvolumn17:= "@/var/lib/pgsql"
subvolMountPathSubvolumn17:= "/var/lib/pgsql"
uuidSubvolumn18:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
isSnapshotSubvolumn18:= "false"
subvolIdSubvolumn18:= "275"
parentIdSubvolumn18:= "257"
subvolNameSubvolumn18:= "@/var/log"
subvolMountPathSubvolumn18:= "/var/log"
uuidSubvolumn19:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
isSnapshotSubvolumn19:= "false"
subvolIdSubvolumn19:= "276"
parentIdSubvolumn19:= "257"
subvolNameSubvolumn19:= "@/var/opt"
subvolMountPathSubvolumn19:= "/var/opt"
uuidSubvolumn20:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
isSnapshotSubvolumn20:= "false"
subvolIdSubvolumn20:= "277"
parentIdSubvolumn20:= "257"
subvolNameSubvolumn20:= "@/var/spool"
subvolMountPathSubvolumn20:= "/var/spool"
uuidSubvolumn21:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
isSnapshotSubvolumn21:= "false"
subvolIdSubvolumn21:= "278"
parentIdSubvolumn21:= "257"
subvolNameSubvolumn21:= "@/var/tmp"
subvolMountPathSubvolumn21:= "/var/tmp"
uuidSubvolumn22:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
isSnapshotSubvolumn22:= "true"
subvolIdSubvolumn22:= "282"
parentIdSubvolumn22:= "258"
subvolNameSubvolumn22:= "@/.snapshots/2/snapshot"
subvolMountPathSubvolumn22:= "null"
var listSubvolumnBtrfsList = []model.BtrfsSubvolumn{
    {
        Uuid: &uuidSubvolumn,
        IsSnapshot: &isSnapshotSubvolumn,
        SubvolId: &subvolIdSubvolumn,
        ParentId: &parentIdSubvolumn,
        SubvolName: &subvolNameSubvolumn,
        SubvolMountPath: &subvolMountPathSubvolumn,
    },
    {
        Uuid: &uuidSubvolumn1,
        IsSnapshot: &isSnapshotSubvolumn1,
        SubvolId: &subvolIdSubvolumn1,
        ParentId: &parentIdSubvolumn1,
        SubvolName: &subvolNameSubvolumn1,
        SubvolMountPath: &subvolMountPathSubvolumn1,
    },
    {
        Uuid: &uuidSubvolumn2,
        IsSnapshot: &isSnapshotSubvolumn2,
        SubvolId: &subvolIdSubvolumn2,
        ParentId: &parentIdSubvolumn2,
        SubvolName: &subvolNameSubvolumn2,
        SubvolMountPath: &subvolMountPathSubvolumn2,
    },
    {
```

```
        Uuid: &uuidSubvolumn3,
        IsSnapshot: &isSnapshotSubvolumn3,
        SubvolId: &subvolIdSubvolumn3,
        ParentId: &parentIdSubvolumn3,
        SubvolName: &subvolNameSubvolumn3,
        SubvolMountPath: &subvolMountPathSubvolumn3,
    },
    {
        Uuid: &uuidSubvolumn4,
        IsSnapshot: &isSnapshotSubvolumn4,
        SubvolId: &subvolIdSubvolumn4,
        ParentId: &parentIdSubvolumn4,
        SubvolName: &subvolNameSubvolumn4,
        SubvolMountPath: &subvolMountPathSubvolumn4,
    },
    {
        Uuid: &uuidSubvolumn5,
        IsSnapshot: &isSnapshotSubvolumn5,
        SubvolId: &subvolIdSubvolumn5,
        ParentId: &parentIdSubvolumn5,
        SubvolName: &subvolNameSubvolumn5,
        SubvolMountPath: &subvolMountPathSubvolumn5,
    },
    {
        Uuid: &uuidSubvolumn6,
        IsSnapshot: &isSnapshotSubvolumn6,
        SubvolId: &subvolIdSubvolumn6,
        ParentId: &parentIdSubvolumn6,
        SubvolName: &subvolNameSubvolumn6,
        SubvolMountPath: &subvolMountPathSubvolumn6,
    },
    {
        Uuid: &uuidSubvolumn7,
        IsSnapshot: &isSnapshotSubvolumn7,
        SubvolId: &subvolIdSubvolumn7,
        ParentId: &parentIdSubvolumn7,
        SubvolName: &subvolNameSubvolumn7,
        SubvolMountPath: &subvolMountPathSubvolumn7,
    },
    {
        Uuid: &uuidSubvolumn8,
        IsSnapshot: &isSnapshotSubvolumn8,
        SubvolId: &subvolIdSubvolumn8,
        ParentId: &parentIdSubvolumn8,
        SubvolName: &subvolNameSubvolumn8,
        SubvolMountPath: &subvolMountPathSubvolumn8,
    },
    {
        Uuid: &uuidSubvolumn9,
        IsSnapshot: &isSnapshotSubvolumn9,
        SubvolId: &subvolIdSubvolumn9,
        ParentId: &parentIdSubvolumn9,
        SubvolName: &subvolNameSubvolumn9,
        SubvolMountPath: &subvolMountPathSubvolumn9,
    },
    {
        Uuid: &uuidSubvolumn10,
        IsSnapshot: &isSnapshotSubvolumn10,
        SubvolId: &subvolIdSubvolumn10,
        ParentId: &parentIdSubvolumn10,
        SubvolName: &subvolNameSubvolumn10,
        SubvolMountPath: &subvolMountPathSubvolumn10,
    },
    {
        Uuid: &uuidSubvolumn11,
        IsSnapshot: &isSnapshotSubvolumn11,
        SubvolId: &subvolIdSubvolumn11,
        ParentId: &parentIdSubvolumn11,
        SubvolName: &subvolNameSubvolumn11,
```

```
        SubvolMountPath: &subvolMountPathSubvolumn11,
    },
    {
        Uuid: &uuidSubvolumn12,
        IsSnapshot: &isSnapshotSubvolumn12,
        SubvolId: &subvolIdSubvolumn12,
        ParentId: &parentIdSubvolumn12,
        SubvolName: &subvolNameSubvolumn12,
        SubvolMountPath: &subvolMountPathSubvolumn12,
    },
    {
        Uuid: &uuidSubvolumn13,
        IsSnapshot: &isSnapshotSubvolumn13,
        SubvolId: &subvolIdSubvolumn13,
        ParentId: &parentIdSubvolumn13,
        SubvolName: &subvolNameSubvolumn13,
        SubvolMountPath: &subvolMountPathSubvolumn13,
    },
    {
        Uuid: &uuidSubvolumn14,
        IsSnapshot: &isSnapshotSubvolumn14,
        SubvolId: &subvolIdSubvolumn14,
        ParentId: &parentIdSubvolumn14,
        SubvolName: &subvolNameSubvolumn14,
        SubvolMountPath: &subvolMountPathSubvolumn14,
    },
    {
        Uuid: &uuidSubvolumn15,
        IsSnapshot: &isSnapshotSubvolumn15,
        SubvolId: &subvolIdSubvolumn15,
        ParentId: &parentIdSubvolumn15,
        SubvolName: &subvolNameSubvolumn15,
        SubvolMountPath: &subvolMountPathSubvolumn15,
    },
    {
        Uuid: &uuidSubvolumn16,
        IsSnapshot: &isSnapshotSubvolumn16,
        SubvolId: &subvolIdSubvolumn16,
        ParentId: &parentIdSubvolumn16,
        SubvolName: &subvolNameSubvolumn16,
        SubvolMountPath: &subvolMountPathSubvolumn16,
    },
    {
        Uuid: &uuidSubvolumn17,
        IsSnapshot: &isSnapshotSubvolumn17,
        SubvolId: &subvolIdSubvolumn17,
        ParentId: &parentIdSubvolumn17,
        SubvolName: &subvolNameSubvolumn17,
        SubvolMountPath: &subvolMountPathSubvolumn17,
    },
    {
        Uuid: &uuidSubvolumn18,
        IsSnapshot: &isSnapshotSubvolumn18,
        SubvolId: &subvolIdSubvolumn18,
        ParentId: &parentIdSubvolumn18,
        SubvolName: &subvolNameSubvolumn18,
        SubvolMountPath: &subvolMountPathSubvolumn18,
    },
    {
        Uuid: &uuidSubvolumn19,
        IsSnapshot: &isSnapshotSubvolumn19,
        SubvolId: &subvolIdSubvolumn19,
        ParentId: &parentIdSubvolumn19,
        SubvolName: &subvolNameSubvolumn19,
        SubvolMountPath: &subvolMountPathSubvolumn19,
    },
    {
        Uuid: &uuidSubvolumn20,
        IsSnapshot: &isSnapshotSubvolumn20,
```



```
            SubvolId: &subvolIdSubvolumn20,
            ParentId: &parentIdSubvolumn20,
            SubvolName: &subvolNameSubvolumn20,
            SubvolMountPath: &subvolMountPathSubvolumn20,
        },
        {
            Uuid: &uuidSubvolumn21,
            IsSnapshot: &isSnapshotSubvolumn21,
            SubvolId: &subvolIdSubvolumn21,
            ParentId: &parentIdSubvolumn21,
            SubvolName: &subvolNameSubvolumn21,
            SubvolMountPath: &subvolMountPathSubvolumn21,
        },
        {
            Uuid: &uuidSubvolumn22,
            IsSnapshot: &isSnapshotSubvolumn22,
            SubvolId: &subvolIdSubvolumn22,
            ParentId: &parentIdSubvolumn22,
            SubvolName: &subvolNameSubvolumn22,
            SubvolMountPath: &subvolMountPathSubvolumn22,
        },
    }
    nameBtrfsList:= "/dev/vda2"
    labelBtrfsList:= "none"
    uuidBtrfsList:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
    deviceBtrfsList:= "/dev/vda2"
    sizeBtrfsList:= int64(3.30GiB)
    nodesizeBtrfsList:= int64(16384)
    sectorsizeBtrfsList:= int32(4096)
    dataProfileBtrfsList:= "single"
    systemProfileBtrfsList:= "single"
    metadataProfileBtrfsList:= "single"
    globalReserve1BtrfsList:= "single"
    gVolUsedSizeBtrfsList:= int64(3894038528)
    defaultSubvolidBtrfsList:= "259"
    defaultSubvolNameBtrfsList:= "@/.snapshots/1/snapshot"
    defaultSubvolMountpathBtrfsList:= "/"
    var listBtrfsListbody = []model.BtrfsFileSystem{
        {
            Name: &nameBtrfsList,
            Label: &labelBtrfsList,
            Uuid: &uuidBtrfsList,
            Device: &deviceBtrfsList,
            Size: &sizeBtrfsList,
            Nodesize: &nodesizeBtrfsList,
            Sectorsize: &sectorsizeBtrfsList,
            DataProfile: &dataProfileBtrfsList,
            SystemProfile: &systemProfileBtrfsList,
            MetadataProfile: &metadataProfileBtrfsList,
            GlobalReserve1: &globalReserve1BtrfsList,
            GVolUsedSize: &gVolUsedSizeBtrfsList,
            DefaultSubvolid: &defaultSubvolidBtrfsList,
            DefaultSubvolName: &defaultSubvolNameBtrfsList,
            DefaultSubvolMountpath: &defaultSubvolMountpathBtrfsList,
            Subvolumn: &listSubvolumnBtrfsList,
        },
    }
    deviceUsePhysicalVolumes:= "VOLUME_GROUP"
    fileSystemPhysicalVolumes:= "LVM2_member"
    mountPointPhysicalVolumes:= ""
    namePhysicalVolumes:= "/dev/vdb1"
    sizePhysicalVolumes:= int64(21473787904)
    usedSizePhysicalVolumes:= int64(21473787904)
    inodeSizePhysicalVolumes:= int32(0)
    inodeNumsPhysicalVolumes:= int64(0)
    var listPhysicalVolumesDisks = []model.PhysicalVolume{
        {
            DeviceUse: &deviceUsePhysicalVolumes,
            FileSystem: &fileSystemPhysicalVolumes,
```

```
            MountPoint: &mountPointPhysicalVolumes,
            Name: &namePhysicalVolumes,
            Size: &sizePhysicalVolumes,
            UsedSize: &usedSizePhysicalVolumes,
            InodeSize: &inodeSizePhysicalVolumes,
            InodeNums: &inodeNumsPhysicalVolumes,
        },
    }
    deviceUsePhysicalVolumes1:= "NORMAL"
    fileSystemPhysicalVolumes1:= "swap"
    mountPointPhysicalVolumes1:= ""
    namePhysicalVolumes1:= "/dev/vda1"
    sizePhysicalVolumes1:= int64(2153775104)
    usedSizePhysicalVolumes1:= int64(2153775104)
    inodeSizePhysicalVolumes1:= int32(0)
    inodeNumsPhysicalVolumes1:= int64(0)
    deviceUsePhysicalVolumes2:= "BTRFS"
    fileSystemPhysicalVolumes2:= "btrfs"
    mountPointPhysicalVolumes2:= ""
    namePhysicalVolumes2:= "/dev/vda2"
    sizePhysicalVolumes2:= int64(16862150656)
    usedSizePhysicalVolumes2:= int64(16862150656)
    inodeSizePhysicalVolumes2:= int32(0)
    inodeNumsPhysicalVolumes2:= int64(0)
    deviceUsePhysicalVolumes3:= "NORMAL"
    fileSystemPhysicalVolumes3:= "xfs"
    mountPointPhysicalVolumes3:= "/home"
    namePhysicalVolumes3:= "/dev/vda3"
    sizePhysicalVolumes3:= int64(23932698624)
    usedSizePhysicalVolumes3:= int64(33988608)
    inodeSizePhysicalVolumes3:= int32(0)
    inodeNumsPhysicalVolumes3:= int64(12345)
    var listPhysicalVolumesDisks1 = []model.PhysicalVolume{
        {
            DeviceUse: &deviceUsePhysicalVolumes1,
            FileSystem: &fileSystemPhysicalVolumes1,
            MountPoint: &mountPointPhysicalVolumes1,
            Name: &namePhysicalVolumes1,
            Size: &sizePhysicalVolumes1,
            UsedSize: &usedSizePhysicalVolumes1,
            InodeSize: &inodeSizePhysicalVolumes1,
            InodeNums: &inodeNumsPhysicalVolumes1,
        },
        {
            DeviceUse: &deviceUsePhysicalVolumes2,
            FileSystem: &fileSystemPhysicalVolumes2,
            MountPoint: &mountPointPhysicalVolumes2,
            Name: &namePhysicalVolumes2,
            Size: &sizePhysicalVolumes2,
            UsedSize: &usedSizePhysicalVolumes2,
            InodeSize: &inodeSizePhysicalVolumes2,
            InodeNums: &inodeNumsPhysicalVolumes2,
        },
        {
            DeviceUse: &deviceUsePhysicalVolumes3,
            FileSystem: &fileSystemPhysicalVolumes3,
            MountPoint: &mountPointPhysicalVolumes3,
            Name: &namePhysicalVolumes3,
            Size: &sizePhysicalVolumes3,
            UsedSize: &usedSizePhysicalVolumes3,
            InodeSize: &inodeSizePhysicalVolumes3,
            InodeNums: &inodeNumsPhysicalVolumes3,
        },
    }
    nameDisks:= "/dev/vda"
    partitionStyleDisks:= "MBR"
    deviceUseDisks:= "BOOT"
    sizeDisks:= int64(42949672960)
    usedSizeDisks:= int64(42948624384)
```

```
    nameDisks1:= "/dev/vdb"
    partitionStyleDisks1:= "MBR"
    deviceUseDisks1:= "NORMAL"
    sizeDisks1:= int64(21474836480)
    usedSizeDisks1:= int64(21473787904)
    nameDisks2:= "/dev/vdc"
    partitionStyleDisks2:= "MBR"
    deviceUseDisks2:= "VOLUME_GROUP"
    sizeDisks2:= int64(21474836480)
    usedSizeDisks2:= int64(0)
    var listDisksbody = []model.ServerDisk{
        {
            Name: &nameDisks,
            PartitionStyle: &partitionStyleDisks,
            DeviceUse: &deviceUseDisks,
            Size: &sizeDisks,
            UsedSize: &usedSizeDisks,
            PhysicalVolumes: &listPhysicalVolumesDisks1,
        },
        {
            Name: &nameDisks1,
            PartitionStyle: &partitionStyleDisks1,
            DeviceUse: &deviceUseDisks1,
            Size: &sizeDisks1,
            UsedSize: &usedSizeDisks1,
            PhysicalVolumes: &listPhysicalVolumesDisks,
        },
        {
            Name: &nameDisks2,
            PartitionStyle: &partitionStyleDisks2,
            DeviceUse: &deviceUseDisks2,
            Size: &sizeDisks2,
            UsedSize: &usedSizeDisks2,
        },
    }
    platformPostSourceServerBody:= model.GetPostSourceServerBodyPlatformEnum().HW
    hasTcPostSourceServerBody:= true
    ioReadWaitPostSourceServerBody:= float64(3.4)
    kernelVersionPostSourceServerBody:= "4.4.21-69-default"
    bootLoaderPostSourceServerBody:= "GRUB"
    rawDevicesPostSourceServerBody:= "string"
    paravirtualizationPostSourceServerBody:= true
    hasRsyncPostSourceServerBody:= true
    memoryPostSourceServerBody:= int64(934752256)
    cpuQuantityPostSourceServerBody:= int32(1)
    firmwarePostSourceServerBody:= model.GetPostSourceServerBodyFirmwareEnum().BIOS
    linuxBlockCheckPostSourceServerBody:= "{"release_type": "SUSE", "release_version": "12.2",
"kernel_simplification": "4.4.21", "architecture": "x86_64", "kernel_version": "4.4.21-69-default"}"
    ipPostSourceServerBody:= "192.168.136.xxx"
    request.Body = &model.PostSourceServerBody{
        Platform: &platformPostSourceServerBody,
        HasTc: &hasTcPostSourceServerBody,
        IoReadWait: &ioReadWaitPostSourceServerBody,
        KernelVersion: &kernelVersionPostSourceServerBody,
        AgentVersion: "2.2.1",
        VolumeGroups: &listVolumeGroupsbody,
        BootLoader: &bootLoaderPostSourceServerBody,
        RawDevices: &rawDevicesPostSourceServerBody,
        Paravirtualization: &paravirtualizationPostSourceServerBody,
        HasRsync: &hasRsyncPostSourceServerBody,
        Networks: listNetworksbody,
        BtrfsList: &listBtrfsListbody,
        Disks: &listDisksbody,
        Memory: &memoryPostSourceServerBody,
        CpuQuantity: &cpuQuantityPostSourceServerBody,
        Firmware: &firmwarePostSourceServerBody,
        LinuxBlockCheck: &linuxBlockCheckPostSourceServerBody,
        OsVersion: "SUSE12_64BIT_SP2",
        OsType: model.GetPostSourceServerBodyOsTypeEnum().LINUX,
```

```
        Name: "host-192-168-136-xxx",
        Ip: &ipPostSourceServerBody,
    }
    response, err := client.RegisterServer(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

More

For SDK sample code of more programming languages, see the Sample Code tab in **API Explorer**. SDK sample code can be automatically generated.

Status Codes

Status Code	Description
200	The source server was registered with SMS.
403	Authentication failed.

Error Codes

See **Error Codes**.

5.3.3 Listing Source Servers

Function

After the Agent installed on a source server is started, the Agent registers the source server information with SMS. This API is used to list registered source servers.

Calling Method

For details, see **Calling APIs**.

Authorization Information

Each account has all the permissions required to call all APIs, but IAM users must be assigned the required permissions.

- If you are using role/policy-based authorization, see **Permissions Policies and Supported Actions** for details on the required permissions.
- If you are using identity policy-based authorization, the following identity policy-based permissions are required.

Action	Access Level	Resource Type (*: required)	Condition Key	Alias	Dependencies
sms:server:list	List	server *	-	sms:server:queryServer	-
		-	g:EnterpriseProjectId		

URI

GET /v3/sources

Table 5-32 Query Parameters

Parameter	Mandatory	Type	Description
state	No	String	The source server status. **unavailable**: The source server fails the environment check. **waiting**: indicates that the source server is waiting for migration. **initialize**: indicates that the migration of the source server is being initialized. **replicate**: indicates that the source server is being replicated. **syncing**: indicates that the source server is being synchronized. **stopping**: indicates that the migration of source server is being stopped. **stopped** indicates that the migration of source server is stopped. **skipping**: indicates that the migration of source server is being skipped. **deleting**: indicates that the migration task of source server is being deleted. **clearing**: The snapshot resources are being cleared. **cleared**: The snapshot resources have been cleared. **clearfailed**: The snapshot resources fail to be cleared. **premigready**: The migration drill is ready. **premiged**: The migration drill has been completed. **premigfailed**: The migration drill fails. **cloning**: indicates that the target server for the source server is being cloned.

Parameter	Mandatory	Type	Description
			cutovering: indicates that the target server for the source server is being launched. **finished**: indicates that the target server for the source server is launched. **error**: indicated that an error occurs. Enumeration values: • **unavailable** • **waiting** • **initialize** • **replicate** • **syncing** • **stopping** • **stopped** • **skipping** • **deleting** • **clearing** • **cleared** • **clearfailed** • **premigready** • **premiged** • **premigfailed** • **cloning** • **cutovering** • **finished** • **error**
name	No	String	The source server name. Minimum: **0** Maximum: **255**
id	No	String	The source server ID. Minimum: **0** Maximum: **255**
ip	No	String	The IP address of the source server. Minimum: **0** Maximum: **255**

Parameter	Mandatory	Type	Description
ipv6	No	String	IPv6 address of the source server. The IP address is preferentially used for query. Minimum: **0** Maximum: **255**
migproject	No	String	The migration project ID. If this parameter is specified, only the source servers in the project are queried. Minimum: **0** Maximum: **255**
limit	No	Integer	The number of source servers recorded on each page. **0** indicates that the default value **200** is used. Minimum: **0** Maximum: **200** Default: **200**
offset	No	Integer	The offset. The default value is **0**. Minimum: **0** Maximum: **65535** Default: **0**

Parameter	Mandatory	Type	Description
migration_cycle	No	String	**checking**: indicates that the check is in progress. **setting**: indicates that the configuration is in progress. **replicating**: indicates that the data is being replicated. **syncing**: indicates that the data is being synchronized. **cutovering**: indicates that the target server for the source server is being launched. **cutovered**: indicates that the target server for the source server is launched. Minimum: **0** Maximum: **255** Enumeration values: ● **checking** ● **setting** ● **replicating** ● **syncing** ● **cutovering** ● **cutovered**
connected	No	Boolean	Indicates whether to query source servers that are disconnected from SMS.
enterprise_project_id	No	String	The ID of the enterprise project to be queried. Minimum: **0** Maximum: **255**
is_consistency_result_exist	No	Boolean	Indicates whether there are consistency verification results.
vm_id	No	String	ID of the clone server on the platform. Minimum: **0** Maximum: **255**

Request Parameters

Table 5-33 Request header parameters

Parameter	Mandatory	Type	Description
X-Auth-Token	Yes	String	User token. The token can be obtained by calling the IAM API for obtaining a user token. The value of **X-Subject-Token** in the response header is the user token. Minimum: **1** Maximum: **16384**

Response Parameters

Status code: 200

Table 5-34 Response body parameters

Parameter	Type	Description
count	Integer	The total number of source servers that meet the query criteria, which is not affected by **limit** and **offset**. Minimum: **0** Maximum: **2147483647**
source_servers	Array of **SourceServersResponseBody** objects	The list of queried source servers. Array Length: **0 - 65535**

Table 5-35 SourceServersResponseBody

Parameter	Type	Description
id	String	The source server ID. Minimum: **0** Maximum: **255**
ip	String	The IP address of the source server. Minimum: **0** Maximum: **255**

Parameter	Type	Description
name	String	The source server name. Minimum: **0** Maximum: **255**
enterprise_project_id	String	Enterprise project ID. Minimum: **0** Maximum: **255**
add_date	Long	The time when the source server was registered. Minimum: **0** Maximum: **9223372036854775807**
os_type	String	The OS type of the source server, which can be Windows or Linux. Minimum: **0** Maximum: **255** Enumeration values: ● **WINDOWS** ● **LINUX**
os_version	String	The OS version, for example, **CENTOS7.6**. Minimum: **0** Maximum: **255**
oem_system	Boolean	Indicates whether the OS is an OEM OS (Windows).

Parameter	Type	Description
state	String	The source server status. **unavailable** (The source server fails the environment check.) **waiting** **initialize** **replicate** **syncing** **stopping** **stopped** **skipping** **deleting** **clearing** **cleared** (snapshot resources cleared) **clearfailed** (snapshot resource clearing failed) **premigready** (migration drill ready) **premiged** (migration drill completed) **premigfailed** (migration drill failed) **cloning** **cutovering** (target server being launched) **finished** (target server launched) **error** Minimum: **0** Maximum: **255** Enumeration values: • **unavailable** • **waiting** • **initialize** • **replicate** • **syncing** • **stopping** • **stopped** • **skipping** • **deleting** • **clearing** • **cleared** • **clearfailed** • **premigready**

Parameter	Type	Description
		• **premiged** • **premigfailed** • **cloning** • **cutovering** • **finished** • **error**
connected	Boolean	Indicates whether the source server is properly connected to SMS.
cpu_quantity	Integer	The number of CPUs on the source server. Minimum: **0** Maximum: **2147483647**
memory	Long	The physical memory size of the source server, in bytes. Minimum: **0** Maximum: **9223372036854775807**
current_task	**TaskByServerSources** object	The migration task associated with the source server.
checks	Array of **EnvironmentCheck** objects	The check items of the source server. Array Length: **0 - 65535**
init_target_server	**InitTargetServer** object	The recommended target server configuration.
replicatesize	Long	The volume of migrated data, in bytes. Minimum: **0** Maximum: **9223372036854775807**
stage_action_time	Long	The time when the migration stage of the source server last changed. The migration stage is defined by **migration_cycle**. Minimum: **0** Maximum: **9223372036854775807**
totalsize	Long	The volume of data to be migrated, in bytes. Minimum: **0** Maximum: **9223372036854775807**

Parameter	Type	Description
last_visit_time	Long	The time when the Agent connection status last changed. Minimum: **0** Maximum: **9223372036854775807**
migration_cycle	String	The migration stage. **cutovering** (target server being launched) **cutovered** (target server launched) **checking** **setting** **replicating** **syncing** Minimum: **0** Maximum: **255** Enumeration values: ● **cutovering** ● **cutovered** ● **checking** ● **setting** ● **replicating** ● **syncing**
state_action_time	Long	The time when the status of the source server last changed. The source server status is defined by **state**. Minimum: **0** Maximum: **9223372036854775807**
is_consistency_result_exist	Boolean	Indicates whether there are consistency verification results. Default: **false**
has_tc	Boolean	Indicates whether TC is installed. This parameter is mandatory for Linux.

Parameter	Type	Description
start_type	String	Start mode. The value can be **MANUAL**, **AUTO**, or empty. No verification is performed. The default value is **MANUAL**. Other values indicate that the server is started registered with the MgC platform. Minimum: **0** Maximum: **255**

Table 5-36 TaskByServerSources

Parameter	Type	Description
id	String	The task ID. Minimum: **1** Maximum: **255**
name	String	The task name. Minimum: **0** Maximum: **255**
type	String	The migration project type. **MIGRATE_BLOCK** **MIGRATE_FILE** Enumeration values: • **MIGRATE_BLOCK** • **MIGRATE_FILE**
state	String	The task status. Minimum: **0** Maximum: **255**
estimate_complete_time	Long	The estimated end time. Minimum: **0** Maximum: **9223372036854775807**
start_date	Long	The start time. Minimum: **0** Maximum: **9223372036854775807**
speed_limit	Integer	The migration rate limit. Minimum: **0** Maximum: **10000**

Parameter	Type	Description
migrate_speed	Double	The migration speed. Minimum: **0** Maximum: **10000**
compress_rate	Double	The compression rate. Minimum: **0** Maximum: **10000**
start_target_server	Boolean	Indicates whether the target server is started.
vm_template_id	String	The server template ID. Minimum: **0** Maximum: **255**
region_id	String	region_id Minimum: **0** Maximum: **255**
project_name	String	The project name. Minimum: **0** Maximum: **255**
project_id	String	The project ID. Minimum: **0** Maximum: **255**
target_server	**TargetServerById** object	Destination.
log_collect_status	String	The log collection status. **INIT** (ready) **UPLOADING** **UPLOAD_FAIL** **UPLOADED** Enumeration values: ● **INIT** ● **UPLOADING** ● **UPLOAD_FAIL** ● **UPLOADED**
exist_server	Boolean	Indicates whether an existing server is used as the target server.
use_public_ip	Boolean	Indicates whether a public IP address is used for migration.

Parameter	Type	Description
clone_server	**CloneServer** object	The information about the cloned server.
remain_seconds	Long	The remaining migration time in seconds. Minimum: **0** Maximum: **9223372036854775807**
log_bucket	String	The name of the bucket to which logs are uploaded. Minimum: **0** Maximum: **255**
log_expire	Long	The validity period of the share link. Minimum: **300** Maximum: **64800**
log_upload_time	Long	The log upload time. Minimum: **0** Maximum: **9223372036854775807**
log_share_url	String	The share URL. Minimum: **0** Maximum: **65535**
subtask_info	String	The current subtask and progress. Minimum: **0** Maximum: **255**

Table 5-37 TargetServerById

Parameter	Type	Description
vm_id	String	The target server ID. Minimum: **0** Maximum: **255**
name	String	The name of the target server. Minimum: **0** Maximum: **255**

Table 5-38 CloneServer

Parameter	Type	Description
vm_id	String	The ID of the cloned server. Minimum: **0** Maximum: **255**
name	String	The name of the cloned server. Minimum: **0** Maximum: **255**
clone_error	String	The error code returned for a clone failure. Minimum: **0** Maximum: **255**
clone_state	String	The clone status. NOT_READY READY PREPARING CREATING ERROR FINISHED Enumeration values: ● **NOT_READY** ● **READY** ● **PREPARING** ● **CREATING** ● **ERROR** ● **FINISHED**
error_msg	String	The error message returned for a clone failure. Minimum: **0** Maximum: **1024**

Table 5-39 EnvironmentCheck

Parameter	Type	Description
id	Long	The check item ID. Minimum: **0** Maximum: **9223372036854775807**

Parameter	Type	Description
params	Array of strings	Parameters. Minimum: **0** Maximum: **255** Array Length: **0 - 65535**
name	String	The check item name. Minimum: **0** Maximum: **255**
result	String	The check result. **OK**: The check is passed. **WARN**: A warning is generated. **ERROR**: The check fails. Minimum: **0** Maximum: **255** Enumeration values: ● **OK** ● **WARN** ● **ERROR**
error_code	String	The returned error code. Minimum: **0** Maximum: **255**
error_or_warn	String	The error or warning. Minimum: **0** Maximum: **255**
error_params	String	The parameters that failed the check. Minimum: **0** Maximum: **255**

Table 5-40 InitTargetServer

Parameter	Type	Description
disks	Array of **DiskIntargetServer** objects	The information about the recommended target server disks. Array Length: **0 - 65535**
volume_groups	Array of **VolumeGroups** objects	This parameter is mandatory for Linux. If there are no volume groups, the value is an empty array []. Array Length: **0 - 65535**

Table 5-41 DiskIntargetServer

Parameter	Type	Description
name	String	The disk name. Minimum: **0** Maximum: **255**
size	Long	The disk size, in bytes. Minimum: **0** Maximum: **9223372036854775807**
device_use	String	The disk function. **BOOT**: boot device **OS**: system device **NORMAL**: general device Minimum: **0** Maximum: **255** Enumeration values: • **BOOT** • **OS** • **NORMAL**
used_size	Long	The used disk space, in bytes. Minimum: **0** Maximum: **9223372036854775807**
physical_volumes	Array of **PhysicalVolumes** objects	The physical volume information. Array Length: **0 - 65535**

Table 5-42 PhysicalVolumes

Parameter	Type	Description
device_use	String	The partition function. The partition can be a general, boot, or OS partition. Minimum: **0** Maximum: **255**
file_system	String	The file system type. Minimum: **0** Maximum: **255**

Parameter	Type	Description
index	Integer	The serial number. Minimum: **0** Maximum: **2147483647**
mount_point	String	The mount point. Minimum: **0** Maximum: **255**
name	String	The volume name. In Windows, it indicates the drive letter, and in Linux, it indicates the device ID. Minimum: **0** Maximum: **255**
size	Long	The size. Minimum: **0** Maximum: **9223372036854775807**
inode_size	Long	The number of inodes. Minimum: **0** Maximum: **9223372036854775807**
used_size	Long	The used space. Minimum: **0** Maximum: **9223372036854775807**
uuid	String	The GUID, which can be obtained from the source server. Minimum: **0** Maximum: **255**

Table 5-43 VolumeGroups

Parameter	Type	Description
components	String	The physical volume information. Minimum: **0** Maximum: **255**
free_size	Long	The available space. Minimum: **0** Maximum: **9223372036854775807**
logical_volumes	Array of **LogicalVolumes** objects	The logical volume information. Array Length: **0 - 50**

Parameter	Type	Description
name	String	Name. Minimum: **0** Maximum: **255**
size	Long	Size. Minimum: **0** Maximum: **9223372036854775807**

Table 5-44 LogicalVolumes

Parameter	Type	Description
block_count	Integer	The number of blocks. Minimum: **0** Maximum: **2147483647** Default: **0**
block_size	Long	Block size. Minimum: **0** Maximum: **1048576** Default: **0**
file_system	String	The file system. Minimum: **0** Maximum: **255**
inode_size	Integer	The number of inodes. Minimum: **0** Maximum: **2147483647**
inode_nums	Long	The number of inodes. Minimum: **0** Maximum: **9223372036854775807**

Parameter	Type	Description
device_use	String	Partition type. This is not mandatory and can be left blank. The common values are as follows: **NORMAL** **OS**: system device **BOOT** **VOLUME_GROUP** **BTRFS** Minimum: **0** Maximum: **255**
mount_point	String	The mount point. Minimum: **0** Maximum: **256**
name	String	Name. Minimum: **0** Maximum: **1024**
size	Long	The size. Minimum: **0** Maximum: **9223372036854775807**
used_size	Long	The used space. Minimum: **0** Maximum: **9223372036854775807**
free_size	Long	The available space. Minimum: **0** Maximum: **9223372036854775807**

Status code: 403

Table 5-45 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**

Parameter	Type	Description
error_msg	String	The error message. Minimum: **0** Maximum: **255**
encoded_authorization_message	String	The encrypted authorization information. Minimum: **0** Maximum: **65535**
error_param	Array of strings	Error parameters. Minimum: **0** Maximum: **65535** Array Length: **1 - 20**
details	Array of **details** objects	The error details. Array Length: **1 - 20**

Table 5-46 details

Parameter	Type	Description
error_code	String	The SMS error code. Minimum: **0** Maximum: **65535**
error_msg	String	The SMS error message. Minimum: **0** Maximum: **65535**

Status code: 500

Table 5-47 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **1024**

Example Requests

This example lists all registered source servers with 10 records on one page and navigates to page 0.

```
GET https://{endpoint}/v3/sources?limit=10&offset=0
X-Auth-Token: XXXX
```

Example Responses

Status code: 200

The list of source servers was obtained.


```
{
  "count" : 10,
  "source_servers" : [ {
    "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "ip" : "192.168.0.1",
    "name" : "sms-test",
    "enterprise_project_id" : "0",
    "add_date" : 1598417717000,
    "os_type" : "WINDOWS",
    "os_version" : "WINDOWS2008_R2_64BIT",
    "oem_system" : false,
    "state" : "finished",
    "connected" : true,
    "cpu_quantity" : 1,
    "memory" : 2146557952,
    "current_task" : {
      "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
      "name" : "MigrationTask",
      "type" : "MIGRATE_BLOCK",
      "state" : "MIGRATE_SUCCESS",
      "estimate_complete_time" : null,
      "start_date" : 1598417771000,
      "speed_limit" : 0,
      "migrate_speed" : 0,
      "start_target_server" : true,
      "vm_template_id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
      "region_id" : "region_id",
      "project_name" : "project_name",
      "project_id" : "xxxxxxxxxxxxxxxxxxxxxxxx00000001",
      "target_server" : {
        "vm_id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
        "name" : "sms-test"
      },
      "log_collect_status" : "INIT",
      "exist_server" : false,
      "use_public_ip" : true,
      "clone_server" : null,
      "remain_seconds" : null
    },
    "checks" : [ {
      "id" : 524062,
      "params" : [ "" ],
      "name" : "OS_VERSION",
      "result" : "OK",
      "error_code" : null,
      "error_params" : ""
    }, {
      "id" : 524063,
      "params" : [ "" ],
      "name" : "FIRMWARE",
      "result" : "OK",
      "error_code" : null,
      "error_params" : ""
    }, {
```

```
    "id" : 524064,
    "params" : [ "" ],
    "name" : "CPU",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 524065,
    "params" : [ "" ],
    "name" : "MEMORY",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 524066,
    "params" : [ "" ],
    "name" : "SYSTEM_ROOT",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 524067,
    "params" : [ "" ],
    "name" : "PARTITION_STYLE",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 524068,
    "params" : [ "" ],
    "name" : "FILE_SYSTEM",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 524069,
    "params" : [ "" ],
    "name" : "FREE_SPACE",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 524070,
    "params" : [ "" ],
    "name" : "OEM_SYSTEM",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 524071,
    "params" : [ "" ],
    "name" : "DRIVER_FILE",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 524072,
    "params" : [ "" ],
    "name" : "SERVICE",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 524073,
    "params" : [ "" ],
    "name" : "ACCOUNT_RIGHTS",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
```

```
    } ],
    "init_target_server" : {
      "disks" : [ {
        "name" : "Disk 0",
        "size" : 42949672960,
        "device_use" : "OS"
      } ]
    },
    "replicatesize" : 0,
    "stage_action_time" : 1598419352959,
    "totalsize" : 0,
    "last_visit_time" : 1598434312002,
    "migration_cycle" : "cutovered",
    "state_action_time" : 1598419352959,
    "start_type" : "MANUAL"
  }, {
    "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "ip" : "192.168.0.154",
    "name" : "sms-win16",
    "add_date" : 1598417612000,
    "os_type" : "WINDOWS",
    "os_version" : "WINDOWS2016_64BIT",
    "oem_system" : false,
    "state" : "finished",
    "connected" : true,
    "cpu_quantity" : 1,
    "memory" : 2146553856,
    "current_task" : {
      "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
      "name" : "MigrationTask",
      "type" : "MIGRATE_BLOCK",
      "state" : "MIGRATE_SUCCESS",
      "estimate_complete_time" : null,
      "start_date" : 1598417627000,
      "speed_limit" : 0,
      "migrate_speed" : 0,
      "start_target_server" : true,
      "vm_template_id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
      "region_id" : "region_id",
      "project_name" : "project_name",
      "project_id" : "xxxxxxxxxxxxxxxxxxxxxxxx00000001",
      "target_server" : {
        "vm_id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
        "name" : "e2e-sms-win16"
      },
      "log_collect_status" : "INIT",
      "exist_server" : false,
      "use_public_ip" : true,
      "clone_server" : null,
      "remain_seconds" : null
    },
    "checks" : [ {
      "id" : 524050,
      "params" : [ "" ],
      "name" : "OS_VERSION",
      "result" : "OK",
      "error_code" : null,
      "error_params" : ""
    }, {
      "id" : 524051,
      "params" : [ "" ],
      "name" : "FIRMWARE",
      "result" : "OK",
      "error_code" : null,
      "error_params" : ""
    }, {
      "id" : 524052,
      "params" : [ "" ],
      "name" : "CPU",
```

```
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 524053,
    "params" : [ "" ],
    "name" : "MEMORY",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 524054,
    "params" : [ "" ],
    "name" : "SYSTEM_ROOT",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 524055,
    "params" : [ "" ],
    "name" : "PARTITION_STYLE",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 524056,
    "params" : [ "" ],
    "name" : "FILE_SYSTEM",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 524057,
    "params" : [ "" ],
    "name" : "FREE_SPACE",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 524058,
    "params" : [ "" ],
    "name" : "OEM_SYSTEM",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 524059,
    "params" : [ "" ],
    "name" : "DRIVER_FILE",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 524060,
    "params" : [ "" ],
    "name" : "SERVICE",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 524061,
    "params" : [ "" ],
    "name" : "ACCOUNT_RIGHTS",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  } ],
  "init_target_server" : {
    "disks" : [ {
```

```
        "name" : "Disk 0",
        "size" : 42949672960,
        "device_use" : "OS"
      } ]
    },
    "replicatesize" : 0,
    "stage_action_time" : 1598419339661,
    "totalsize" : 0,
    "last_visit_time" : 1598434316810,
    "migration_cycle" : "cutovered",
    "state_action_time" : 1598419339661,
    "start_type" : "AUTO"
  }, {
    "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "ip" : "192.168.77.77",
    "name" : "sms-centos",
    "add_date" : 1598417551000,
    "os_type" : "LINUX",
    "os_version" : "CENTOS_7_4_64BIT",
    "oem_system" : false,
    "state" : "error",
    "connected" : true,
    "cpu_quantity" : 1,
    "memory" : 1038716928,
    "current_task" : {
      "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
      "name" : "MigrationTask",
      "type" : "MIGRATE_BLOCK",
      "state" : "MIGRATE_FAIL",
      "estimate_complete_time" : null,
      "start_date" : 1598417588000,
      "speed_limit" : 0,
      "migrate_speed" : 0,
      "start_target_server" : true,
      "vm_template_id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
      "region_id" : "region_id",
      "project_name" : "project_name",
      "project_id" : "xxxxxxxxxxxxxxxxxxxxxxxx00000001",
      "target_server" : {
        "vm_id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
        "name" : "e2e-sms-centos"
      },
      "log_collect_status" : "INIT",
      "exist_server" : false,
      "use_public_ip" : true,
      "clone_server" : null,
      "remain_seconds" : null
    },
    "checks" : [ {
      "id" : 524038,
      "params" : [ "" ],
      "name" : "OS_VERSION",
      "result" : "OK",
      "error_code" : null,
      "error_params" : ""
    }, {
      "id" : 524039,
      "params" : [ "" ],
      "name" : "CPU",
      "result" : "OK",
      "error_code" : null,
      "error_params" : ""
    }, {
      "id" : 524040,
      "params" : [ "" ],
      "name" : "MEMORY",
      "result" : "OK",
      "error_code" : null,
      "error_params" : ""
```

```
  }, {
    "id" : 524041,
    "params" : [ "" ],
    "name" : "PARAVIRTUALIZATION",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 524042,
    "params" : [ "" ],
    "name" : "FIRMWARE",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 524043,
    "params" : [ "" ],
    "name" : "BOOT_LOADER",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 524044,
    "params" : [ "" ],
    "name" : "RSYNC",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 524045,
    "params" : [ "" ],
    "name" : "RAW_DEVICES",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 524046,
    "params" : [ "" ],
    "name" : "DISK_INFO",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 524047,
    "params" : [ "" ],
    "name" : "PARTITION_STYLE",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 524048,
    "params" : [ "" ],
    "name" : "FILE_SYSTEM",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 524049,
    "params" : [ "" ],
    "name" : "LINUX_BLOCK_SUPPORT",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  } ],
  "init_target_server" : {
    "disks" : [ {
      "name" : "/dev/vda",
      "size" : 42949672960,
      "device_use" : "BOOT"
```

```
    } ]
  },
  "replicatesize" : 42949672960,
  "stage_action_time" : 1598428182454,
  "totalsize" : 42949672960,
  "last_visit_time" : 1598434308889,
  "migration_cycle" : "syncing",
  "state_action_time" : 1598428182454,
  "start_type" : ""
}, {
  "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
  "ip" : "192.168.0.235",
  "name" : "sms-ubuntu",
  "add_date" : 1598417522000,
  "os_type" : "LINUX",
  "os_version" : "UBUNTU_18_4_64BIT",
  "oem_system" : false,
  "state" : "unavailable",
  "connected" : false,
  "cpu_quantity" : 1,
  "memory" : 1032556544,
  "current_task" : null,
  "checks" : [ ],
  "init_target_server" : {
    "disks" : [ {
      "name" : "/dev/vda",
      "size" : 42949672960,
      "device_use" : "BOOT"
    }, {
      "name" : "/dev/vdb",
      "size" : 21474836480,
      "device_use" : "NORMAL"
    } ]
  },
  "replicatesize" : 0,
  "stage_action_time" : 1598417521797,
  "totalsize" : 0,
  "last_visit_time" : 1598417521795,
  "migration_cycle" : "checking",
  "state_action_time" : null,
  "start_type" : "MANUAL"
}, {
  "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
  "ip" : "192.168.178.214",
  "name" : "sms-sms2",
  "add_date" : 1598403465000,
  "os_type" : "WINDOWS",
  "os_version" : "WINDOWS2012_R2_64BIT",
  "oem_system" : false,
  "state" : "waiting",
  "connected" : false,
  "cpu_quantity" : 1,
  "memory" : 2146553856,
  "current_task" : null,
  "checks" : [ {
    "id" : 523970,
    "params" : [ "" ],
    "name" : "OS_VERSION",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 523971,
    "params" : [ "" ],
    "name" : "FIRMWARE",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
```

```
    "id" : 523972,
    "params" : [ "" ],
    "name" : "CPU",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 523973,
    "params" : [ "" ],
    "name" : "MEMORY",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 523974,
    "params" : [ "" ],
    "name" : "SYSTEM_ROOT",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 523975,
    "params" : [ "" ],
    "name" : "PARTITION_STYLE",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 523976,
    "params" : [ "" ],
    "name" : "FILE_SYSTEM",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 523977,
    "params" : [ "" ],
    "name" : "FREE_SPACE",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 523978,
    "params" : [ "" ],
    "name" : "OEM_SYSTEM",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 523979,
    "params" : [ "" ],
    "name" : "DRIVER_FILE",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 523980,
    "params" : [ "" ],
    "name" : "SERVICE",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 523981,
    "params" : [ "" ],
    "name" : "ACCOUNT_RIGHTS",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
```

```
  } ],
  "init_target_server" : {
    "disks" : [ {
      "name" : "Disk 0",
      "size" : 42949672960,
      "device_use" : "OS"
    } ]
  },
  "replicatesize" : 0,
  "stage_action_time" : 1598403465315,
  "totalsize" : 0,
  "last_visit_time" : 1598403588140,
  "migration_cycle" : "checking",
  "state_action_time" : 1598403465414,
  "start_type" : "MANUAL"
}, {
  "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
  "ip" : "192.168.0.1",
  "name" : "linux sources",
  "add_date" : 1598369476000,
  "os_type" : "LINUX",
  "os_version" : "REDHAT_7_3_64BIT",
  "oem_system" : false,
  "state" : "unavailable",
  "connected" : false,
  "cpu_quantity" : 4,
  "memory" : 8581140480,
  "current_task" : null,
  "checks" : [ ],
  "init_target_server" : {
    "disks" : [ {
      "name" : "sda",
      "size" : 85899345920,
      "device_use" : "BOOT"
    }, {
      "name" : "sdb",
      "size" : 214748364800,
      "device_use" : "NORMAL"
    } ]
  },
  "replicatesize" : 0,
  "stage_action_time" : 1598369475726,
  "totalsize" : 0,
  "last_visit_time" : 1598369475725,
  "migration_cycle" : "checking",
  "state_action_time" : null,
  "start_type" : "MANUAL"
}, {
  "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
  "ip" : "192.168.0.1",
  "name" : "linux sources",
  "add_date" : 1598351694000,
  "os_type" : "LINUX",
  "os_version" : "REDHAT_7_3_64BIT",
  "oem_system" : false,
  "state" : "unavailable",
  "connected" : false,
  "cpu_quantity" : 4,
  "memory" : 8581140480,
  "current_task" : null,
  "checks" : [ ],
  "init_target_server" : {
    "disks" : [ {
      "name" : "sda",
      "size" : 85899345920,
      "device_use" : "BOOT"
    }, {
      "name" : "sdb",
      "size" : 214748364800,
```

```
      "device_use" : "NORMAL"
    } ]
  },
  "replicatesize" : 0,
  "stage_action_time" : 1598351693858,
  "totalsize" : 0,
  "last_visit_time" : 1598351693857,
  "migration_cycle" : "checking",
  "state_action_time" : null,
  "start_type" : "MANUAL"
}, {
  "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
  "ip" : "192.168.0.1",
  "name" : "linux sources",
  "add_date" : 1598348080000,
  "os_type" : "LINUX",
  "os_version" : "REDHAT_7_3_64BIT",
  "oem_system" : false,
  "state" : "unavailable",
  "connected" : false,
  "cpu_quantity" : 4,
  "memory" : 8581140480,
  "current_task" : null,
  "checks" : [ ],
  "init_target_server" : {
    "disks" : [ {
      "name" : "sda",
      "size" : 85899345920,
      "device_use" : "BOOT"
    }, {
      "name" : "sdb",
      "size" : 214748364800,
      "device_use" : "NORMAL"
    } ]
  },
  "replicatesize" : 0,
  "stage_action_time" : 1598348079782,
  "totalsize" : 0,
  "last_visit_time" : 1598348079781,
  "migration_cycle" : "checking",
  "state_action_time" : null,
  "start_type" : "MANUAL"
}, {
  "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
  "ip" : "192.168.0.239",
  "name" : "sms-centos7",
  "add_date" : 1598326505000,
  "os_type" : "LINUX",
  "os_version" : "CENTOS_8_5_64BIT",
  "oem_system" : false,
  "state" : "unavailable",
  "connected" : false,
  "cpu_quantity" : 1,
  "memory" : 1926860800,
  "current_task" : null,
  "checks" : [ {
    "id" : 523794,
    "params" : [ "" ],
    "name" : "OS_VERSION",
    "result" : "ERROR",
    "error_code" : "SMS.6504",
    "error_params" : ""
  }, {
    "id" : 523795,
    "params" : [ "" ],
    "name" : "CPU",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
```

```
  }, {
    "id" : 523796,
    "params" : [ "" ],
    "name" : "MEMORY",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 523797,
    "params" : [ "" ],
    "name" : "PARAVIRTUALIZATION",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 523798,
    "params" : [ "" ],
    "name" : "FIRMWARE",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 523799,
    "params" : [ "" ],
    "name" : "BOOT_LOADER",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 523800,
    "params" : [ "" ],
    "name" : "RSYNC",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 523801,
    "params" : [ "" ],
    "name" : "RAW_DEVICES",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 523802,
    "params" : [ "" ],
    "name" : "DISK_INFO",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 523803,
    "params" : [ "" ],
    "name" : "PARTITION_STYLE",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 523804,
    "params" : [ "" ],
    "name" : "FILE_SYSTEM",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 523805,
    "params" : [ "" ],
    "name" : "LINUX_BLOCK_SUPPORT",
    "result" : "OK",
    "error_code" : null,
```



```
    "error_params" : ""
  } ],
  "init_target_server" : {
    "disks" : [ {
      "name" : "/dev/vda",
      "size" : 42949672960,
      "device_use" : "BOOT"
    }, {
      "name" : "/dev/vdb",
      "size" : 42949672960,
      "device_use" : "NORMAL"
    } ]
  },
  "replicatesize" : 0,
  "stage_action_time" : 1598326505378,
  "totalsize" : 0,
  "last_visit_time" : 1598423828868,
  "migration_cycle" : "checking",
  "state_action_time" : 1598326505459,
  "start_type" : "MANUAL"
}, {
  "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
  "ip" : "192.168.0.65",
  "name" : "smsc-test",
  "add_date" : 1598238727000,
  "os_type" : "LINUX",
  "os_version" : "CENTOS_6_5_64BIT",
  "oem_system" : false,
  "state" : "finished",
  "connected" : true,
  "cpu_quantity" : 1,
  "memory" : 1043931136,
  "current_task" : {
    "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "name" : "MigrationTask",
    "type" : "MIGRATE_FILE",
    "state" : "MIGRATE_SUCCESS",
    "estimate_complete_time" : null,
    "start_date" : 1598239243000,
    "speed_limit" : 0,
    "migrate_speed" : 0,
    "start_target_server" : true,
    "vm_template_id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "region_id" : "region_id",
    "project_name" : "project_name",
    "project_id" : "xxxxxxxxxxxxxxxxxxxxxxxx00000001",
    "target_server" : {
      "vm_id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
      "name" : "smsc-test"
    },
    "log_collect_status" : "INIT",
    "exist_server" : false,
    "use_public_ip" : true,
    "clone_server" : null,
    "remain_seconds" : null
  },
  "checks" : [ {
    "id" : 523686,
    "params" : [ "" ],
    "name" : "OS_VERSION",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 523687,
    "params" : [ "" ],
    "name" : "CPU",
    "result" : "OK",
    "error_code" : null,
```

```
    "error_params" : ""
  }, {
    "id" : 523688,
    "params" : [ "" ],
    "name" : "MEMORY",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 523689,
    "params" : [ "" ],
    "name" : "PARAVIRTUALIZATION",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 523690,
    "params" : [ "" ],
    "name" : "FIRMWARE",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 523691,
    "params" : [ "" ],
    "name" : "BOOT_LOADER",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 523692,
    "params" : [ "" ],
    "name" : "RSYNC",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 523693,
    "params" : [ "" ],
    "name" : "RAW_DEVICES",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 523694,
    "params" : [ "" ],
    "name" : "DISK_INFO",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 523695,
    "params" : [ "" ],
    "name" : "PARTITION_STYLE",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 523696,
    "params" : [ "" ],
    "name" : "FILE_SYSTEM",
    "result" : "OK",
    "error_code" : null,
    "error_params" : ""
  }, {
    "id" : 523697,
    "params" : [ "" ],
    "name" : "LINUX_BLOCK_SUPPORT",
    "result" : "WARN",
```

```
      "error_code" : "SMS.6617",
      "error_params" : ""
    } ],
    "init_target_server" : {
      "disks" : [ {
        "name" : "/dev/vda",
        "size" : 42949672960,
        "device_use" : "BOOT"
      }, {
        "name" : "/dev/vdb",
        "size" : 10737418240,
        "device_use" : "NORMAL"
      } ]
    },
    "replicatesize" : 0,
    "stage_action_time" : 1598240178677,
    "totalsize" : 0,
    "last_visit_time" : 1598434314748,
    "migration_cycle" : "cutovered",
    "state_action_time" : 1598240178677,
    "start_type" : "MANUAL"
  } ]
}
```


Status code: 403

Authentication failed.


```
{
  "error_code" : "SMS.9004",
  "error_msg" : "The current account does not have the permission to execute policy. You do not have permission to perform action XXX on resource XXX.",
  "encoded_authorization_message" : "XXXXXX",
  "error_param" : [ "You do not have permission to perform action XXX on resource XXX." ],
  "details" : [ {
    "error_code" : "SMS.9004",
    "error_msg" : "You do not have permission to perform action XXX on resource XXX."
  } ]
}
```


SDK Sample Code

The SDK sample code is as follows.

Java

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;


public class ListServersSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");
```

```
        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        ListServersRequest request = new ListServersRequest();
        try {
            ListServersResponse response = client.listServers(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

Python

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = ListServersRequest()
        response = client.list_servers(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

Go

```
package main

import (
    "fmt"
```

```
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
         WithRegion(region.ValueOf("<YOUR REGION>")).
         WithCredential(auth).
         SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.ListServersRequest{}
    response, err := client.ListServers(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

More

For SDK sample code of more programming languages, see the Sample Code tab in **API Explorer**. SDK sample code can be automatically generated.

Status Codes

Status Code	Description
200	The list of source servers was obtained.
403	Authentication failed.
500	Internal Server Error

Error Codes

See **Error Codes**.

5.3.4 Batch Deleting Source Server Records

Function

This API is used to delete source server records in batches. Once the information about a source server is deleted, you can add the information about the source server to the SMS console again only by restarting the migration Agent on the source server.

Calling Method

For details, see **Calling APIs**.

Authorization Information

Each account has all the permissions required to call all APIs, but IAM users must be assigned the required permissions.

- If you are using role/policy-based authorization, see **Permissions Policies and Supported Actions** for details on the required permissions.
- If you are using identity policy-based authorization, no identity policy-based permission required for calling this API.

URI

POST /v3/sources/delete

Request Parameters

Table 5-48 Request header parameters

Parameter	Mandatory	Type	Description
X-Auth-Token	Yes	String	User token. The token can be obtained by calling the IAM API for obtaining a user token. The value of **X-Subject-Token** in the response header is the user token. Minimum: **1** Maximum: **16384**

Table 5-49 Request body parameters

Parameter	Mandatory	Type	Description
ids	Yes	Array of strings	The IDs of all objects to be deleted. Minimum: **1** Maximum: **36** Array Length: **1 - 50**

Response Parameters

Status code: 200

Table 5-50 Response body parameters

Parameter	Type	Description
-	String	The response for deleting source servers in batches.

Status code: 403

Table 5-51 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **255**
encoded_authorization_message	String	The encrypted authorization information. Minimum: **0** Maximum: **65535**
error_param	Array of strings	Error parameters. Minimum: **0** Maximum: **65535** Array Length: **1 - 20**
details	Array of **details** objects	The error details. Array Length: **1 - 20**

Table 5-52 details

Parameter	Type	Description
error_code	String	The SMS error code. Minimum: **0** Maximum: **65535**
error_msg	String	The SMS error message. Minimum: **0** Maximum: **65535**

Example Requests

This example deletes source servers **ec2a894f-0d92-47c5-ac22-168ef61dxxxx** and **5f13089f-799f-4f33-b3e1-c499397dxxxx** in a batch.

```
POST https://{endpoint}/v3/sources/delete

{
  "ids" : [ "ec2a894f-0d92-47c5-ac22-168ef61dxxxx", "5f13089f-799f-4f33-b3e1-c499397dxxxx" ]
}
```

Example Responses

Status code: 200

Batch deleting source server records succeeded.

```
{ }
```

Status code: 403

Authentication failed.

```
{
  "error_code" : "SMS.9004",
  "error_msg" : "The current account does not have the permission to execute policy. You do not have permission to perform action XXX on resource XXX.",
  "encoded_authorization_message" : "XXXXXX",
  "error_param" : [ "You do not have permission to perform action XXX on resource XXX." ],
  "details" : [ {
    "error_code" : "SMS.9004",
    "error_msg" : "You do not have permission to perform action XXX on resource XXX."
  } ]
}
```

SDK Sample Code

The SDK sample code is as follows.

Java

This example deletes source servers **ec2a894f-0d92-47c5-ac22-168ef61dxxxx** and **5f13089f-799f-4f33-b3e1-c499397dxxxx** in a batch.


```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;

import java.util.List;
import java.util.ArrayList;

public class DeleteServersSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        DeleteServersRequest request = new DeleteServersRequest();
        DeleteIds body = new DeleteIds();
        List<String> listbodyIds = new ArrayList<>();
        listbodyIds.add("ec2a894f-0d92-47c5-ac22-168ef61dxxxx");
        listbodyIds.add("5f13089f-799f-4f33-b3e1-c499397dxxxx");
        body.withIds(listbodyIds);
        request.withBody(body);
        try {
            DeleteServersResponse response = client.deleteServers(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

Python

This example deletes source servers **ec2a894f-0d92-47c5-ac22-168ef61dxxxx** and **5f13089f-799f-4f33-b3e1-c499397dxxxx** in a batch.

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *
```

```
if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = DeleteServersRequest()
        listIdsbody = [
            "ec2a894f-0d92-47c5-ac22-168ef61dxxxx",
            "5f13089f-799f-4f33-b3e1-c499397dxxxx"
        ]
        request.body = DeleteIds(
            ids=listIdsbody
        )
        response = client.delete_servers(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

Go

This example deletes source servers **ec2a894f-0d92-47c5-ac22-168ef61dxxxx** and **5f13089f-799f-4f33-b3e1-c499397dxxxx** in a batch.

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
```

```
        WithRegion(region.ValueOf("<YOUR REGION>")).
        WithCredential(auth).
        SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.DeleteServersRequest{}
    var listIdsbody = []string{
        "ec2a894f-0d92-47c5-ac22-168ef61dxxxx",
        "5f13089f-799f-4f33-b3e1-c499397dxxxx",
    }
    request.Body = &model.DeleteIds{
        Ids: listIdsbody,
    }
    response, err := client.DeleteServers(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

More

For SDK sample code of more programming languages, see the Sample Code tab in **API Explorer**. SDK sample code can be automatically generated.

Status Codes

Status Code	Description
200	Batch deleting source server records succeeded.
403	Authentication failed.

Error Codes

See **Error Codes**.

5.3.5 Modifying Source Server Information

Function

This API is used to modify the information of a source server in SMS to facilitate server management.

Calling Method

For details, see **Calling APIs**.

Authorization Information

Each account has all the permissions required to call all APIs, but IAM users must be assigned the required permissions.

- If you are using role/policy-based authorization, see **Permissions Policies and Supported Actions** for details on the required permissions.
- If you are using identity policy-based authorization, the following identity policy-based permissions are required.

Action	Access Level	Resource Type (*: required)	Condition Key	Alias	Dependencies
sms:server:update	Write	server *	-	sms:server:migrationServer	-
		-	g:EnterpriseProjectId		

URI

PUT /v3/sources/{source_id}

Table 5-53 Path Parameters

Parameter	Mandatory	Type	Description
source_id	Yes	String	The ID of the source server in SMS. Minimum: **1** Maximum: **36**

Request Parameters

Table 5-54 Request header parameters

Parameter	Mandatory	Type	Description
X-Auth-Token	Yes	String	User token. The token can be obtained by calling the IAM API for obtaining a user token. The value of **X-Subject-Token** in the response header is the user token. Minimum: **1** Maximum: **16384**

Table 5-55 Request body parameters

Parameter	Mandatory	Type	Description
name	No	String	The new name of the source server. Minimum: **0** Maximum: **255**
migprojectid	No	String	The ID of the migration project to which the source server belongs after the modification. Minimum: **0** Maximum: **255**
disks	No	Array of **PutDisk** objects	The modification on the disk takes effect only when the target end is in the **waiting** state. Array Length: **0 - 65535**
volume_groups	No	Array of **PutVolumeGroups** objects	The modification on the volume group takes effect only when the target end is in the **waiting** state. Array Length: **0 - 65535**

Table 5-56 PutDisk

Parameter	Mandatory	Type	Description
need_migration	No	Boolean	The disk name. Default: **true**
id	No	String	The disk ID. Minimum: **0** Maximum: **255**
adjust_size	No	Long	The new size. Minimum: **0** Maximum: **9223372036854775807** Default: **0**
physical_volumes	No	Array of **PutVolume** objects	The updated volume information. Array Length: **0 - 65535**

Table 5-57 PutVolume

Parameter	Mandatory	Type	Description
id	No	String	The database record ID. Minimum: **0** Maximum: **255**
need_migration	No	Boolean	Indicates whether it needs to be migrated. Default: **true**
adjust_size	No	Long	The new size. Minimum: **0** Maximum: **9223372036854775807** Default: **0**

Table 5-58 PutVolumeGroups

Parameter	Mandatory	Type	Description
logical_volumes	No	Array of **PutLogicalVolume** objects	The logical volume information. Array Length: **0 - 65535**
id	No	String	The volume group ID. Minimum: **0** Maximum: **255**
need_migration	No	Boolean	Indicates whether it needs to be migrated. Default: **true**
adjust_size	No	Long	The new size. Minimum: **0** Maximum: **9223372036854775807** Default: **0**

Table 5-59 PutLogicalVolume

Parameter	Mandatory	Type	Description
id	No	String	The logical volume ID. Minimum: **0** Maximum: **255**

Parameter	Mandatory	Type	Description
need_migration	No	Boolean	Indicates whether it needs to be migrated. Default: **true**
adjust_size	No	Long	The new size. Minimum: **0** Maximum: **9223372036854775807** Default: **0**

Response Parameters

Status code: 200

Table 5-60 Response body parameters

Parameter	Type	Description
-	String	The information about the source server with a specified ID was modified successfully.

Status code: 403

Table 5-61 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **255**
encoded_authorization_message	String	The encrypted authorization information. Minimum: **0** Maximum: **65535**

Parameter	Type	Description
error_param	Array of strings	Error parameters. Minimum: **0** Maximum: **65535** Array Length: **1 - 20**
details	Array of **details** objects	The error details. Array Length: **1 - 20**

Table 5-62 details

Parameter	Type	Description
error_code	String	The SMS error code. Minimum: **0** Maximum: **65535**
error_msg	String	The SMS error message. Minimum: **0** Maximum: **65535**

Example Requests

This example changes the name of the source server whose ID is **dcdbe339-b02d-4578-95a1-9c9c547dxxxx** to **abcd**.

```
PUT https://{endpoint}/v3/sources/dcdbe339-b02d-4578-95a1-9c9c547dxxxx

{
  "name" : "abcd"
}
```

Example Responses

Status code: 200

The information about the source server with a specified ID was modified successfully.

```
{ }
```

Status code: 403

Authentication failed.

```
{
  "error_code" : "SMS.9004",
  "error_msg" : "The current account does not have the permission to execute policy. You do not have permission to perform action XXX on resource XXX.",
  "encoded_authorization_message" : "XXXXXX",
  "error_param" : [ "You do not have permission to perform action XXX on resource XXX." ],
  "details" : [ {
```



```
    "error_code" : "SMS.9004",
    "error_msg" : "You do not have permission to perform action XXX on resource XXX."
  } ]
}
```

SDK Sample Code

The SDK sample code is as follows.

Java

This example changes the name of the source server whose ID is **dcdbe339-b02d-4578-95a1-9c9c547dxxxx** to **abcd**.

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;


public class UpdateServerNameSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        UpdateServerNameRequest request = new UpdateServerNameRequest();
        request.withSourceId("{source_id}");
        PutSourceServerBody body = new PutSourceServerBody();
        body.withName("abcd");
        request.withBody(body);
        try {
            UpdateServerNameResponse response = client.updateServerName(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

Python

This example changes the name of the source server whose ID is **dcdbe339-b02d-4578-95a1-9c9c547dxxxx** to **abcd**.

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = UpdateServerNameRequest()
        request.source_id = "{source_id}"
        request.body = PutSourceServerBody(
            name="abcd"
        )
        response = client.update_server_name(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

Go

This example changes the name of the source server whose ID is **dcdbe339-b02d-4578-95a1-9c9c547dxxxx** to **abcd**.

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
```

```
            WithAk(ak).
            WithSk(sk).
            SafeBuild()

        if err != nil {
            fmt.Println(err)
            return
        }

        hcClient, err := sms.SmsClientBuilder().
             WithRegion(region.ValueOf("<YOUR REGION>")).
             WithCredential(auth).
             SafeBuild()


        if err != nil {
            fmt.Println(err)
            return
        }

        client := sms.NewSmsClient(hcClient)

        request := &model.UpdateServerNameRequest{}
        request.SourceId = "{source_id}"
        namePutSourceServerBody:= "abcd"
        request.Body = &model.PutSourceServerBody{
            Name: &namePutSourceServerBody,
        }
        response, err := client.UpdateServerName(request)
        if err == nil {
            fmt.Printf("%+v\n", response)
        } else {
            fmt.Println(err)
        }
}
```

More

For SDK sample code of more programming languages, see the Sample Code tab in **API Explorer**. SDK sample code can be automatically generated.

Status Codes

Status Code	Description
200	The information about the source server with a specified ID was modified successfully.
403	Authentication failed.

Error Codes

See **Error Codes**.

5.3.6 Updating the Migration Status of a Source Server

Function

This API is used to update the migration status of a source server.

The request parameters are not verified and the update does not take effect if the source server is in the **unavailable**, **stopping**, **stopped**, **deleting**, **finished**, **clearing**, or **clearfailed** state.

Calling Method

For details, see **Calling APIs**.

Authorization Information

Each account has all the permissions required to call all APIs, but IAM users must be assigned the required permissions.

- If you are using role/policy-based authorization, see **Permissions Policies and Supported Actions** for details on the required permissions.
- If you are using identity policy-based authorization, the following identity policy-based permissions are required.

Action	Access Level	Resource Type (*: required)	Condition Key	Alias	Dependencies
sms:server:updateState	Write	server *	-	sms:server:migrationServer	-
		-	g:EnterpriseProjectId		

URI

PUT /v3/sources/{source_id}/changestate

Table 5-63 Path Parameters

Parameter	Mandatory	Type	Description
source_id	Yes	String	The ID of the source server in SMS. Minimum: **1** Maximum: **36**

Request Parameters

Table 5-64 Request header parameters

Parameter	Mandatory	Type	Description
X-Auth-Token	Yes	String	User token. The token can be obtained by calling the IAM API for obtaining a user token. The value of **X-Subject-Token** in the response header is the user token. Minimum: **1** Maximum: **16384**

Table 5-65 Request body parameters

Parameter	Mandatory	Type	Description
copystate	Yes	String	The source server status. **unavailable** (environment check failed) **waiting** **init** (initialization) **replicate** **syncing** **stopping** **stopped** **skipping** **deleting** **clearing** (clearing snapshot resources) **cleared** (snapshot resources cleared) **clearfailed** (snapshot resource clearing failed) **premigready** (migration drill ready) **premiging** (migration drill in progress) **premiged** (migration drill completed) **premigfailed** (migration drill failed) **cloning** (waiting for the clone to complete) **cutovering** (target server being launched) **finished** (target server launched) **error** (error occurred) Enumeration values: • **unavailable** • **waiting** • **init** • **replicate** • **syncing** • **stopping** • **stopped**

Parameter	Mandatory	Type	Description
			• **skipping** • **deleting** • **clearing** • **cleared** • **clearfailed** • **premigready** • **premiging** • **premiged** • **premigfailed** • **cloning** • **cutovering** • **finished** • **error**
migrationcycle	Yes	String	Migration period. **no_ready**: not ready. **ready_for_test** **testing** **tested** **cutovering**: starting the target server. **cutovered**: target server started. **checking** **setting** **replicating** **syncing** Enumeration values: • **no_ready** • **ready_for_test** • **testing** • **tested** • **cutovering** • **cutovered** • **checking** • **setting** • **replicating** • **syncing**

Response Parameters

Status code: 200

Table 5-66 Response body parameters

Parameter	Type	Description
-	String	The migration task status of a source server was updated.

Status code: 403

Table 5-67 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **255**
encoded_authorization_message	String	The encrypted authorization information. Minimum: **0** Maximum: **65535**
error_param	Array of strings	Error parameters. Minimum: **0** Maximum: **65535** Array Length: **1 - 20**
details	Array of **details** objects	The error details. Array Length: **1 - 20**

Table 5-68 details

Parameter	Type	Description
error_code	String	The SMS error code. Minimum: **0** Maximum: **65535**

Parameter	Type	Description
error_msg	String	The SMS error message. Minimum: **0** Maximum: **65535**

Example Requests

This example updates the replication status to **WAITING** and the migration stage to **cutovered** for source server **dcdbe339-b02d-4578-95a1-9c9c547dxxxx**.

```
PUT https://{endpoint}/v3/sources/dcdbe339-b02d-4578-95a1-9c9c547dxxxx/changestate

{
  "copystate" : "waiting",
  "migrationcycle" : "cutovered"
}
```

Example Responses

Status code: 200

The migration task status of a source server was updated.

```
{ }
```

Status code: 403

Authentication failed.

```
{
  "error_code" : "SMS.9004",
  "error_msg" : "The current account does not have the permission to execute policy. You do not have 
permission to perform action XXX on resource XXX.",
  "encoded_authorization_message" : "XXXXXX",
  "error_param" : [ "You do not have permission to perform action XXX on resource XXX." ],
  "details" : [ {
    "error_code" : "SMS.9004",
    "error_msg" : "You do not have permission to perform action XXX on resource XXX."
  } ]
}
```

SDK Sample Code

The SDK sample code is as follows.

Java

This example updates the replication status to **WAITING** and the migration stage to **cutovered** for source server **dcdbe339-b02d-4578-95a1-9c9c547dxxxx**.

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
```

```
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;


public class UpdateCopyStateSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        UpdateCopyStateRequest request = new UpdateCopyStateRequest();
        request.withSourceId("{source_id}");
        PutCopyStateReq body = new PutCopyStateReq();
        body.withMigrationcycle(PutCopyStateReq.MigrationcycleEnum.fromValue("cutovered"));
        body.withCopystate(PutCopyStateReq.CopystateEnum.fromValue("waiting"));
        request.withBody(body);
        try {
            UpdateCopyStateResponse response = client.updateCopyState(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

Python

This example updates the replication status to **WAITING** and the migration stage to **cutovered** for source server **dcdbe339-b02d-4578-95a1-9c9c547dxxxx**.

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)
```

```
client = SmsClient.new_builder() \
    .with_credentials(credentials) \
    .with_region(SmsRegion.value_of("<YOUR REGION>")) \
    .build()

try:
    request = UpdateCopyStateRequest()
    request.source_id = "{source_id}"
    request.body = PutCopyStateReq(
        migrationcycle="cutovered",
        copystate="waiting"
    )
    response = client.update_copy_state(request)
    print(response)
except exceptions.ClientRequestException as e:
    print(e.status_code)
    print(e.request_id)
    print(e.error_code)
    print(e.error_msg)
```

Go

This example updates the replication status to **WAITING** and the migration stage to **cutovered** for source server **dcdbe339-b02d-4578-95a1-9c9c547dxxxx**.

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security
risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment
variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before running this
example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
         WithRegion(region.ValueOf("<YOUR REGION>")).
         WithCredential(auth).
         SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.UpdateCopyStateRequest{}
```

```
    request.SourceId = "{source_id}"
    request.Body = &model.PutCopyStateReq{
        Migrationcycle: model.GetPutCopyStateReqMigrationcycleEnum().CUTOVERED,
        Copystate: model.GetPutCopyStateReqCopystateEnum().WAITING,
    }
    response, err := client.UpdateCopyState(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

More

For SDK sample code of more programming languages, see the Sample Code tab in **API Explorer**. SDK sample code can be automatically generated.

Status Codes

Status Code	Description
200	The migration task status of a source server was updated.
403	Authentication failed.

Error Codes

See **Error Codes**.

5.3.7 Deleting a Source Server Record

Function

This API is used to delete the record of a source server with a specified ID from the SMS console. Once the information about a source server is deleted, you can add the information about the source server to the SMS console again only by restarting the migration Agent on the source server.

Calling Method

For details, see **Calling APIs**.

Authorization Information

Each account has all the permissions required to call all APIs, but IAM users must be assigned the required permissions.

- If you are using role/policy-based authorization, see **Permissions Policies and Supported Actions** for details on the required permissions.
- If you are using identity policy-based authorization, the following identity policy-based permissions are required.

Action	Access Level	Resource Type (*: required)	Condition Key	Alias	Dependencies
sms:server:delete	Write	server *	-	sms:server:migrationServer	-
		-	g:EnterpriseProjectId		

URI

DELETE /v3/sources/{source_id}

Table 5-69 Path Parameters

Parameter	Mandatory	Type	Description
source_id	Yes	String	The ID of the source server in SMS. Minimum: **1** Maximum: **36**

Request Parameters

Table 5-70 Request header parameters

Parameter	Mandatory	Type	Description
X-Auth-Token	Yes	String	User token. The token can be obtained by calling the IAM API for obtaining a user token. The value of **X-Subject-Token** in the response header is the user token. Minimum: **1** Maximum: **16384**

Response Parameters

Status code: 200

Table 5-71 Response body parameters

Parameter	Type	Description
-	String	The information about the source server with a specified ID was deleted.

Status code: 403

Table 5-72 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **255**
encoded_authorization_message	String	The encrypted authorization information. Minimum: **0** Maximum: **65535**
error_param	Array of strings	Error parameters. Minimum: **0** Maximum: **65535** Array Length: **1 - 20**
details	Array of **details** objects	The error details. Array Length: **1 - 20**

Table 5-73 details

Parameter	Type	Description
error_code	String	The SMS error code. Minimum: **0** Maximum: **65535**
error_msg	String	The SMS error message. Minimum: **0** Maximum: **65535**

Example Requests

This example deletes the record of source server with ID **1a6d1e0b-62e5-4376-b59f-ff2fd569xxxx**.

```
DELETE https://{endpoint}/v3/sources/1a6d1e0b-62e5-4376-b59f-ff2fd569xxxx
```

Example Responses

Status code: 403

Authentication failed.


```
{
  "error_code" : "SMS.9004",
  "error_msg" : "The current account does not have the permission to execute policy. You do not have permission to perform action XXX on resource XXX.",
  "encoded_authorization_message" : "XXXXXX",
  "error_param" : [ "You do not have permission to perform action XXX on resource XXX." ],
  "details" : [ {
    "error_code" : "SMS.9004",
    "error_msg" : "You do not have permission to perform action XXX on resource XXX."
  } ]
}
```


SDK Sample Code

The SDK sample code is as follows.

Java

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;


public class DeleteServerSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        DeleteServerRequest request = new DeleteServerRequest();
        request.withSourceId("{source_id}");
        try {
            DeleteServerResponse response = client.deleteServer(request);
```

```
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

Python

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = DeleteServerRequest()
        request.source_id = "{source_id}"
        response = client.delete_server(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

Go

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before running this
```

```
example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
         WithRegion(region.ValueOf("<YOUR REGION>")).
         WithCredential(auth).
         SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.DeleteServerRequest{}
    request.SourceId = "{source_id}"
    response, err := client.DeleteServer(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

More

For SDK sample code of more programming languages, see the Sample Code tab in **API Explorer**. SDK sample code can be automatically generated.

Status Codes

Status Code	Description
200	The information about the source server with a specified ID was deleted.
403	Authentication failed.

Error Codes

See **Error Codes**.

5.3.8 Querying a Source Server

Function

After the migration Agent reports the source server information to SMS, SMS checks the migration feasibility. This API returns the basic information and check results of the source server.

Calling Method

For details, see **Calling APIs**.

Authorization Information

Each account has all the permissions required to call all APIs, but IAM users must be assigned the required permissions.

- If you are using role/policy-based authorization, see **Permissions Policies and Supported Actions** for details on the required permissions.
- If you are using identity policy-based authorization, the following identity policy-based permissions are required.

Action	Access Level	Resource Type (*: required)	Condition Key	Alias	Dependencies
sms:server:get	Read	server *	-	sms:server:queryServer	-
		-	g:EnterpriseProjectId		

URI

GET /v3/sources/{source_id}

Table 5-74 Path Parameters

Parameter	Mandatory	Type	Description
source_id	Yes	String	The ID of the source server in SMS. Minimum: **1** Maximum: **36**

Request Parameters

Table 5-75 Request header parameters

Parameter	Mandatory	Type	Description
X-Auth-Token	Yes	String	User token. The token can be obtained by calling the IAM API for obtaining a user token. The value of **X-Subject-Token** in the response header is the user token. Minimum: **1** Maximum: **16384**

Response Parameters

Status code: 200

Table 5-76 Response body parameters

Parameter	Type	Description
id	String	The source server ID. Minimum: **0** Maximum: **255**
ip	String	The IP address of the source server. Minimum: **0** Maximum: **255**
name	String	The source server name in SMS. Minimum: **0** Maximum: **255**
hostname	String	The hostname of the source server. This parameter is mandatory for source server registration and optional for source server updates. Minimum: **0** Maximum: **255**
enterprise_project_id	String	Enterprise project ID. Minimum: **1** Maximum: **255**

Parameter	Type	Description
add_date	Long	The time when the source server was registered. Minimum: **0** Maximum: **9223372036854775807**
os_type	String	The OS type of the source server, which can be Windows or Linux. This parameter is mandatory for source server registration and optional for source server updates. Minimum: **0** Maximum: **255**
os_version	String	The OS version. This parameter is mandatory for registration and optional for updates. Minimum: **0** Maximum: **255**
oem_system	Boolean	Indicates whether the OS is an OEM OS (Windows).

Parameter	Type	Description
state	String	The source server status. **unavailable** (The source server fails the environment check.) **waiting** **initialize** **replicate** **syncing** **stopping** **stopped** **skipping** **deleting** **clearing** **cleared** (snapshot resources cleared) **clearfailed** (snapshot resource clearing failed) **premigready** (migration drill ready) **premiged** (migration drill completed) **premigfailed** (migration drill failed) **cloning** **cutovering** (target server being launched) **finished** (target server launched) **error** Enumeration values: ● **unavailable** ● **waiting** ● **initialize** ● **replicate** ● **syncing** ● **stopping** ● **stopped** ● **skipping** ● **deleting** ● **clearing** ● **cleared** ● **clearfailed** ● **premigready** ● **premiged** ● **premigfailed**

Parameter	Type	Description
		• **cloning** • **cutovering** • **finished** • **error**
connected	Boolean	Indicate whether the Agent installed on the source server is connected to SMS.
firmware	String	The boot mode of the source server, which can be BIOS or UEFI. Enumeration values: • **BIOS** • **UEFI**
init_target_server	**InitTargetServer** object	The recommended target server configuration.
cpu_quantity	Integer	The number of CPUs on the source server. Minimum: **0** Maximum: **65535**
memory	Long	The physical memory size (MB) of the source server. Minimum: **0** Maximum: **9223372036854775807**
current_task	**TaskByServerSource** object	The migration task associated with the source server.
disks	Array of **ServerDisk** objects	The information about source server disks. Array Length: **0 - 65535**
volume_groups	Array of **VolumeGroups** objects	The volume group information of the source server. This parameter is mandatory for Linux. If there are no volume groups, the value is an empty array []. Array Length: **0 - 65535**
btrfs_list	Array of **BtrfsFileSystem** objects	The information about the Btrfs file systems on the source server. This parameter is mandatory for Linux. If there are no Btrfs file systems on the source server, the value is an empty array []. Array Length: **0 - 65535**

Parameter	Type	Description
networks	Array of **NetWork** objects	The information about NICs on the source server. Array Length: **0 - 65535**
checks	Array of **EnvironmentCheck** objects	The environment check information for the source server. Array Length: **0 - 65535**
migration_cycle	String	The migration stage. **cutovering** (target server being launched) **cutovered** (target server launched) **checking** **setting** **replicating** **syncing** Minimum: **0** Maximum: **255** Enumeration values: ● **cutovering** ● **cutovered** ● **checking** ● **setting** ● **replicating** ● **syncing**
state_action_time	Long	The timestamp when the status of the source server last changed. The source server status is defined by **state**. Minimum: **0** Maximum: **9223372036854775807**
replicatesize	Long	The amount of data that has been migrated, in bytes. Minimum: **0** Maximum: **9223372036854775807**
totalsize	Long	The total amount of data to be migrated, in bytes. Minimum: **0** Maximum: **9223372036854775807**

Parameter	Type	Description
last_visit_time	Long	The timestamp when the Agent connection status last changed. Minimum: **0** Maximum: **9223372036854775807**
stage_action_time	Long	The timestamp when the migration stage of the source server last changed. The migration stage is defined by **migration_cycle**. Minimum: **0** Maximum: **9223372036854775807**
agent_version	String	The Agent version. Minimum: **0** Maximum: **255**
has_tc	Boolean	Indicates whether TC is installed. This parameter is mandatory for Linux.
start_type	String	Start mode. The value can be **MANUAL**, **AUTO**, or empty. No verification is performed. The default value is **MANUAL**. Other values indicate that the server is started registered with the MgC platform. Minimum: **0** Maximum: **255**

Table 5-77 InitTargetServer

Parameter	Type	Description
disks	Array of **DiskIntargetServer** objects	The information about the recommended target server disks. Array Length: **0 - 65535**
volume_groups	Array of **VolumeGroups** objects	This parameter is mandatory for Linux. If there are no volume groups, the value is an empty array []. Array Length: **0 - 65535**

Table 5-78 DiskIntargetServer

Parameter	Type	Description
name	String	The disk name. Minimum: **0** Maximum: **255**
size	Long	The disk size, in bytes. Minimum: **0** Maximum: **9223372036854775807**
device_use	String	The disk function. **BOOT**: boot device **OS**: system device **NORMAL**: general device Minimum: **0** Maximum: **255** Enumeration values: ● **BOOT** ● **OS** ● **NORMAL**
used_size	Long	The used disk space, in bytes. Minimum: **0** Maximum: **9223372036854775807**
physical_volumes	Array of **PhysicalVolumes** objects	The physical volume information. Array Length: **0 - 65535**

Table 5-79 PhysicalVolumes

Parameter	Type	Description
device_use	String	The partition function. The partition can be a general, boot, or OS partition. Minimum: **0** Maximum: **255**
file_system	String	The file system type. Minimum: **0** Maximum: **255**
index	Integer	The serial number. Minimum: **0** Maximum: **2147483647**

Parameter	Type	Description
mount_point	String	The mount point. Minimum: **0** Maximum: **255**
name	String	The volume name. In Windows, it indicates the drive letter, and in Linux, it indicates the device ID. Minimum: **0** Maximum: **255**
size	Long	The size. Minimum: **0** Maximum: **9223372036854775807**
inode_size	Long	The number of inodes. Minimum: **0** Maximum: **9223372036854775807**
used_size	Long	The used space. Minimum: **0** Maximum: **9223372036854775807**
uuid	String	The GUID, which can be obtained from the source server. Minimum: **0** Maximum: **255**

Table 5-80 TaskByServerSource

Parameter	Type	Description
id	String	The task ID. Minimum: **0** Maximum: **255**
name	String	The task name. Minimum: **0** Maximum: **255**

Parameter	Type	Description
type	String	The migration project type. **MIGRATE_BLOCK** **MIGRATE_FILE** Enumeration values: • **MIGRATE_BLOCK** • **MIGRATE_FILE**
state	String	The task status. Minimum: **0** Maximum: **255**
start_date	Long	The start time. Minimum: **0** Maximum: **9223372036854775807**
speed_limit	Integer	The migration rate limit. Minimum: **0** Maximum: **10000**
migrate_speed	Double	The migration speed. Minimum: **0** Maximum: **10000**
start_target_server	Boolean	Indicates whether the target server is started.
vm_template_id	String	The server template ID. Minimum: **0** Maximum: **255**
region_id	String	region_id Minimum: **0** Maximum: **255**
project_name	String	The project name. Minimum: **0** Maximum: **255**
project_id	String	The project ID. Minimum: **0** Maximum: **255**
target_server	**TargetServerById** object	Destination.

Parameter	Type	Description
log_collect_status	String	The log collection status. **INIT** (ready) **UPLOADING** **UPLOAD_FAIL** **UPLOADED** Enumeration values: • **INIT** • **UPLOADING** • **UPLOAD_FAIL** • **UPLOADED**
exist_server	Boolean	Indicates whether an existing server is used as the target server.
use_public_ip	Boolean	Indicates whether a public IP address is used for migration.
clone_server	**CloneServer** object	The information about the cloned server.
subtask_info	String	The current subtask and progress. Minimum: **0** Maximum: **255**

Table 5-81 TargetServerById

Parameter	Type	Description
vm_id	String	The target server ID. Minimum: **0** Maximum: **255**
name	String	The name of the target server. Minimum: **0** Maximum: **255**

Table 5-82 CloneServer

Parameter	Type	Description
vm_id	String	The ID of the cloned server. Minimum: **0** Maximum: **255**

Parameter	Type	Description
name	String	The name of the cloned server. Minimum: **0** Maximum: **255**
clone_error	String	The error code returned for a clone failure. Minimum: **0** Maximum: **255**
clone_state	String	The clone status. NOT_READY READY PREPARING CREATING ERROR FINISHED Enumeration values: ● **NOT_READY** ● **READY** ● **PREPARING** ● **CREATING** ● **ERROR** ● **FINISHED**
error_msg	String	The error message returned for a clone failure. Minimum: **0** Maximum: **1024**

Table 5-83 ServerDisk

Parameter	Type	Description
name	String	The disk name. Minimum: **0** Maximum: **255**

Parameter	Type	Description
partition_style	String	Disk partition type. This parameter is mandatory when the source server is added. Otherwise, the subsequent environment check will fail. Options (required by EVS): **MBR**: Master Boot Record **GPT**: GUID Partition Table** **Partition Table** For details, see the description of differences between MBR and GPT in the EVS API document. Minimum: **0** Maximum: **255**
device_use	String	Disk type. This is not mandatory and can be left blank. The common values are as follows: **NORMAL** **OS**: system device **BOOT** **VOLUME_GROUP** **BTRFS** Minimum: **0** Maximum: **255**
size	Long	The disk size, in bytes. Minimum: **0** Maximum: **9223372036854775807**
used_size	Long	The used disk space, in bytes. Minimum: **0** Maximum: **9223372036854775807**
physical_volumes	Array of **PhysicalVolume** objects	The information about physical partitions on the disk. Array Length: **0 - 50**
os_disk	Boolean	Indicates whether the disk is the system disk.
relation_name	String	The name of the paired disk on the Linux target ECS server. Minimum: **0** Maximum: **255**

Parameter	Type	Description
inode_size	Integer	The inode size. Minimum: **0** Maximum: **2147483647**

Table 5-84 PhysicalVolume

Parameter	Type	Description
device_use	String	Disk type. This is not mandatory and can be left blank. The common values are as follows: **NORMAL** **OS**: system device **BOOT** **VOLUME_GROUP** **BTRFS** Minimum: **0** Maximum: **255**
file_system	String	The file system type. Minimum: **0** Maximum: **255**
index	Integer	The serial number. Minimum: **0** Maximum: **2147483647**
mount_point	String	The mount point. Minimum: **0** Maximum: **255**
name	String	The volume name. In Windows, it indicates the drive letter, and in Linux, it indicates the device ID. Minimum: **0** Maximum: **255**
size	Long	The size. Minimum: **0** Maximum: **9223372036854775807**

Parameter	Type	Description
used_size	Long	The used space. Minimum: **0** Maximum: **9223372036854775807**
inode_size	Integer	The number of inodes. Minimum: **0** Maximum: **2147483647**
inode_nums	Long	The number of inodes. Minimum: **0** Maximum: **9223372036854775807**
uuid	String	The GUID, which can be obtained from the source server. Minimum: **0** Maximum: **255**
size_per_cluster	Integer	The size of each cluster. Minimum: **0** Maximum: **2147483647**

Table 5-85 VolumeGroups

Parameter	Type	Description
components	String	The physical volume information. Minimum: **0** Maximum: **255**
free_size	Long	The available space. Minimum: **0** Maximum: **9223372036854775807**
logical_volumes	Array of **LogicalVolumes** objects	The logical volume information. Array Length: **0 - 50**
name	String	Name. Minimum: **0** Maximum: **255**
size	Long	Size. Minimum: **0** Maximum: **9223372036854775807**

Table 5-86 LogicalVolumes

Parameter	Type	Description
block_count	Integer	The number of blocks. Minimum: **0** Maximum: **2147483647** Default: **0**
block_size	Long	Block size. Minimum: **0** Maximum: **1048576** Default: **0**
file_system	String	The file system. Minimum: **0** Maximum: **255**
inode_size	Integer	The number of inodes. Minimum: **0** Maximum: **2147483647**
inode_nums	Long	The number of inodes. Minimum: **0** Maximum: **9223372036854775807**
device_use	String	Partition type. This is not mandatory and can be left blank. The common values are as follows: **NORMAL** **OS**: system device **BOOT** **VOLUME_GROUP** **BTRFS** Minimum: **0** Maximum: **255**
mount_point	String	The mount point. Minimum: **0** Maximum: **256**
name	String	Name. Minimum: **0** Maximum: **1024**

Parameter	Type	Description
size	Long	The size. Minimum: **0** Maximum: **9223372036854775807**
used_size	Long	The used space. Minimum: **0** Maximum: **9223372036854775807**
free_size	Long	The available space. Minimum: **0** Maximum: **9223372036854775807**

Table 5-87 BtrfsFileSystem

Parameter	Type	Description
name	String	The file system name. Minimum: **0** Maximum: **255**
label	String	The file system tags. If no tag exists, the value is an empty string. Minimum: **0** Maximum: **255**
uuid	String	The file system UUID. Minimum: **0** Maximum: **255**
device	String	The device names of the Btrfs file system. Minimum: **0** Maximum: **255**
size	Long	The space occupied by the file system data. Minimum: **0** Maximum: **9223372036854775807**
nodesize	Long	The Btrfs node size. Minimum: **0** Maximum: **9223372036854775807**

Parameter	Type	Description
sectorsize	Integer	The sector size. Minimum: **0** Maximum: **2147483647**
data_profile	String	The data profile (RAD). Minimum: **0** Maximum: **255**
system_profile	String	The file system profile (RAD). Minimum: **0** Maximum: **255**
metadata_profile	String	The metadata profile (RAD). Minimum: **0** Maximum: **255**
global_reserve1	String	The Btrfs file system information. Minimum: **0** Maximum: **255**
g_vol_used_size	Long	The used space of the Btrfs volume. Minimum: **0** Maximum: **9223372036854775807**
default_subvolid	String	The ID of the default subvolume. Minimum: **0** Maximum: **255**
default_subvol_name	String	The name of the default subvolume. Minimum: **0** Maximum: **255**
default_subvol_mountpath	String	The mount path of the default subvolume or Btrfs file system. Minimum: **0** Maximum: **255**
subvolumn	Array of **BtrfsSubvolumn** objects	The subvolume information. Array Length: **0 - 65535**

Table 5-88 BtrfsSubvolumn

Parameter	Type	Description
uuid	String	The UUID of the parent volume. Minimum: **0** Maximum: **255**
is_snapshot	String	Indicates whether the subvolume is a snapshot. Minimum: **0** Maximum: **255**
subvol_id	String	The subvolume ID. Minimum: **0** Maximum: **255**
parent_id	String	The parent volume ID. Minimum: **0** Maximum: **255**
subvol_name	String	The subvolume name. Minimum: **0** Maximum: **255**
subvol_mount_path	String	The mount path of the subvolume. Minimum: **0** Maximum: **255**

Table 5-89 NetWork

Parameter	Type	Description
name	String	The NIC name. Minimum: **0** Maximum: **255**
ip	String	The IP address bound to the NIC. Minimum: **0** Maximum: **255**
ipv6	String	The IPv6 address. Minimum: **0** Maximum: **255**
netmask	String	The subnet mask. Minimum: **0** Maximum: **255**

Parameter	Type	Description
gateway	String	Gateway. Minimum: **0** Maximum: **255**
mtu	Integer	The NIC MTU. This parameter is mandatory for Linux. Minimum: **0** Maximum: **2147483647**
mac	String	The first MAC address in the list must not be empty. Minimum: **0** Maximum: **255**
id	String	The database record ID. Minimum: **0** Maximum: **255**

Table 5-90 EnvironmentCheck

Parameter	Type	Description
id	Long	The check item ID. Minimum: **0** Maximum: **9223372036854775807**
params	Array of strings	Parameters. Minimum: **0** Maximum: **255** Array Length: **0 - 65535**
name	String	The check item name. Minimum: **0** Maximum: **255**

Parameter	Type	Description
result	String	The check result. **OK**: The check is passed. **WARN**: A warning is generated. **ERROR**: The check fails. Minimum: **0** Maximum: **255** Enumeration values: ● **OK** ● **WARN** ● **ERROR**
error_code	String	The returned error code. Minimum: **0** Maximum: **255**
error_or_warn	String	The error or warning. Minimum: **0** Maximum: **255**
error_params	String	The parameters that failed the check. Minimum: **0** Maximum: **255**

Status code: 403

Table 5-91 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **255**
encoded_authorization_message	String	The encrypted authorization information. Minimum: **0** Maximum: **65535**

Parameter	Type	Description
error_param	Array of strings	Error parameters. Minimum: **0** Maximum: **65535** Array Length: **1 - 20**
details	Array of **details** objects	The error details. Array Length: **1 - 20**

Table 5-92 details

Parameter	Type	Description
error_code	String	The SMS error code. Minimum: **0** Maximum: **65535**
error_msg	String	The SMS error message. Minimum: **0** Maximum: **65535**

Example Requests

This example queries the information about the source server with the ID **211d7878-d7ba-4cac-acf1-a02ccfb8xxxx**.

```
GET https://{endpoint}/v3/sources/211d7878-d7ba-4cac-acf1-a02ccfb8xxxx
X-Auth-Token: XXXX
```

Example Responses

Status code: 200

The details about a source server were obtained.


```
{
  "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
  "ip" : "192.168.0.154",
  "name" : "sms-win16",
  "hostname" : "sms-win16",
  "add_date" : 1598435769000,
  "os_type" : "WINDOWS",
  "os_version" : "WINDOWS2016_64BIT",
  "oem_system" : false,
  "state" : "initialize",
  "connected" : true,
  "firmware" : "BIOS",
  "cpu_quantity" : 1,
  "memory" : 2146553856,
  "current_task" : {
    "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "name" : "MigrationTask",
```

```
  "type" : "MIGRATE_BLOCK",
  "state" : "RUNNING",
  "speed_limit" : 0,
  "start_target_server" : true,
  "vm_template_id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
  "region_id" : "region_id",
  "project_name" : "project_name",
  "project_id" : "xxxxxxxxxxxxxxxxxxxxxxxx00000001",
  "target_server" : {
    "vm_id" : "",
    "name" : ""
  },
  "log_collect_status" : "INIT",
  "exist_server" : false,
  "use_public_ip" : true,
  "clone_server" : null
},
"disks" : [ {
  "name" : "Disk 0",
  "relation_name" : null,
  "partition_style" : "MBR",
  "size" : 42949672960,
  "used_size" : 42947575808,
  "device_use" : "BOOT",
  "os_disk" : false,
  "physical_volumes" : [ {
    "uuid" : "\\\\?\\Volume{586b7157-0000-0000-0000-100000000000}\\",
    "index" : 1,
    "name" : "(Reserved)",
    "device_use" : "BOOT",
    "file_system" : "NTFS",
    "mount_point" : null,
    "size" : 524288000,
    "used_size" : 410275840
  }, {
    "uuid" : "\\\\?\\Volume{586b7157-0000-0000-0000-501f00000000}\\",
    "index" : 2,
    "name" : "C:\\",
    "device_use" : "OS",
    "file_system" : "NTFS",
    "mount_point" : null,
    "size" : 42423287808,
    "used_size" : 23170301952
  } ]
} ],
"volume_groups" : [ ],
"networks" : [ {
  "name" : null,
  "ip" : null,
  "netmask" : null,
  "gateway" : null,
  "mtu" : 0,
  "mac" : "dac20cd4f6318ca6458673b0046ddcc89e936df292d0806cb868ba63a817853c"
} ],
"checks" : [ {
  "id" : 524146,
  "params" : [ "" ],
  "name" : "OS_VERSION",
  "result" : "OK",
  "error_code" : null,
  "error_or_warn" : null,
  "error_params" : ""
}, {
  "id" : 524147,
  "params" : [ "" ],
  "name" : "FIRMWARE",
  "result" : "OK",
  "error_code" : null,
  "error_or_warn" : null,
```

```
  "error_params" : ""
}, {
  "id" : 524148,
  "params" : [ "" ],
  "name" : "CPU",
  "result" : "OK",
  "error_code" : null,
  "error_or_warn" : null,
  "error_params" : ""
}, {
  "id" : 524149,
  "params" : [ "" ],
  "name" : "MEMORY",
  "result" : "OK",
  "error_code" : null,
  "error_or_warn" : null,
  "error_params" : ""
}, {
  "id" : 524150,
  "params" : [ "" ],
  "name" : "SYSTEM_ROOT",
  "result" : "OK",
  "error_code" : null,
  "error_or_warn" : null,
  "error_params" : ""
}, {
  "id" : 524151,
  "params" : [ "" ],
  "name" : "PARTITION_STYLE",
  "result" : "OK",
  "error_code" : null,
  "error_or_warn" : null,
  "error_params" : ""
}, {
  "id" : 524152,
  "params" : [ "" ],
  "name" : "FILE_SYSTEM",
  "result" : "OK",
  "error_code" : null,
  "error_or_warn" : null,
  "error_params" : ""
}, {
  "id" : 524153,
  "params" : [ "" ],
  "name" : "FREE_SPACE",
  "result" : "OK",
  "error_code" : null,
  "error_or_warn" : null,
  "error_params" : ""
}, {
  "id" : 524154,
  "params" : [ "" ],
  "name" : "OEM_SYSTEM",
  "result" : "OK",
  "error_code" : null,
  "error_or_warn" : null,
  "error_params" : ""
}, {
  "id" : 524155,
  "params" : [ "" ],
  "name" : "DRIVER_FILE",
  "result" : "OK",
  "error_code" : null,
  "error_or_warn" : null,
  "error_params" : ""
}, {
  "id" : 524156,
  "params" : [ "" ],
  "name" : "SERVICE",
```

```
    "result" : "OK",
    "error_code" : null,
    "error_or_warn" : null,
    "error_params" : ""
  }, {
    "id" : 524157,
    "params" : [ "" ],
    "name" : "ACCOUNT_RIGHTS",
    "result" : "OK",
    "error_code" : null,
    "error_or_warn" : null,
    "error_params" : ""
  } ],
  "init_target_server" : {
    "disks" : [ {
      "name" : "Disk 0",
      "size" : 42949672960,
      "used_size" : 42947575808,
      "device_use" : "OS",
      "physical_volumes" : [ {
        "uuid" : "\\\\?\\Volume{586b7157-0000-0000-0000-100000000000}\\",
        "index" : 1,
        "name" : "(Reserved)",
        "device_use" : "BOOT",
        "file_system" : "NTFS",
        "mount_point" : null,
        "size" : 524288000,
        "used_size" : 410275840
      }, {
        "uuid" : "\\\\?\\Volume{586b7157-0000-0000-0000-501f00000000}\\",
        "index" : 2,
        "name" : "C:\\",
        "device_use" : "OS",
        "file_system" : "NTFS",
        "mount_point" : null,
        "size" : 42423287808,
        "used_size" : 23170301952
      } ]
    } ],
    "volume_groups" : [ ]
  },
  "replicatesize" : 0,
  "stage_action_time" : 1598435768945,
  "totalsize" : 0,
  "last_visit_time" : 1598435801422,
  "agent_version" : "6.1.8",
  "migration_cycle" : "replicating",
  "state_action_time" : 1598435783569
}
```


Status code: 403

Authentication failed.


```
{
  "error_code" : "SMS.9004",
  "error_msg" : "The current account does not have the permission to execute policy. You do not have permission to perform action XXX on resource XXX.",
  "encoded_authorization_message" : "XXXXXX",
  "error_param" : [ "You do not have permission to perform action XXX on resource XXX." ],
  "details" : [ {
    "error_code" : "SMS.9004",
    "error_msg" : "You do not have permission to perform action XXX on resource XXX."
  } ]
}
```


SDK Sample Code

The SDK sample code is as follows.

Java

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;


public class ShowServerSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great
security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or
environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running
this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        ShowServerRequest request = new ShowServerRequest();
        request.withSourceId("{source_id}");
        try {
            ShowServerResponse response = client.showServer(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

Python

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security
risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment
variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this
example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
```

```
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = ShowServerRequest()
        request.source_id = "{source_id}"
        response = client.show_server(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

Go

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
         WithRegion(region.ValueOf("<YOUR REGION>")).
         WithCredential(auth).
         SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.ShowServerRequest{}
    request.SourceId = "{source_id}"
    response, err := client.ShowServer(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
```

```
    } else {
        fmt.Println(err)
    }
}
```

More

For SDK sample code of more programming languages, see the Sample Code tab in **API Explorer**. SDK sample code can be automatically generated.

Status Codes

Status Code	Description
200	The details about a source server were obtained.
403	Authentication failed.

Error Codes

See **Error Codes**.

5.3.9 Obtaining an Overview of Source Servers

Function

Obtaining an Overview of Source Servers

Calling Method

For details, see **Calling APIs**.

Authorization Information

Each account has all the permissions required to call all APIs, but IAM users must be assigned the required permissions.

- If you are using role/policy-based authorization, see **Permissions Policies and Supported Actions** for details on the required permissions.
- If you are using identity policy-based authorization, the following identity policy-based permissions are required.

Action	Access Level	Resource Type (*: required)	Condition Key	Alias	Dependencies
sms:server:overview	Read	server *	-	sms:server:queryServer	-

URI

GET /v3/sources/overview

Request Parameters

Table 5-93 Request header parameters

Parameter	Mandatory	Type	Description
X-Auth-Token	Yes	String	User token. The token can be obtained by calling the IAM API for obtaining a user token. The value of **X-Subject-Token** in the response header is the user token. Minimum: **1** Maximum: **16384**

Response Parameters

Status code: 200

Table 5-94 Response body parameters

Parameter	Type	Description
waiting	Integer	The number of servers that are in a waiting migration status. Minimum: **0** Maximum: **1000**
replicate	Integer	The number of servers that are in a replicating migration status. Minimum: **0** Maximum: **1000**
syncing	Integer	The number of servers that are in a synchronizing migration status. Minimum: **0** Maximum: **1000**
stopped	Integer	The number of servers that are in a paused migration status. Minimum: **0** Maximum: **1000**

Parameter	Type	Description
deleting	Integer	The number of servers that are in a deleting migration status. Minimum: **0** Maximum: **1000**
cutovering	Integer	The number of servers whose paired target servers are being launched. Minimum: **0** Maximum: **1000**
unavailable	Integer	The number of servers that fail the environment check. Minimum: **0** Maximum: **1000**
stopping	Integer	The number of servers that are in a **stopping** migration status. Minimum: **0** Maximum: **1000**
skipping	Integer	The number of servers that are being skipped. Minimum: **0** Maximum: **1000**
finished	Integer	The number of servers whose paired target servers have been launched. Minimum: **0** Maximum: **1000**
initialize	Integer	The number of servers that are in an initializing migration status. Minimum: **0** Maximum: **1000**
error	Integer	Error. Minimum: **0** Maximum: **1000**
cloning	Integer	The number of servers whose paired target servers are being cloned. Minimum: **0** Maximum: **1000**

Parameter	Type	Description
unconfigured	Integer	The number of servers that do not have target server configurations. Minimum: **0** Maximum: **1000**

Status code: 403

Table 5-95 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **255**
encoded_authorization_message	String	The encrypted authorization information. Minimum: **0** Maximum: **65535**
error_param	Array of strings	Error parameters. Minimum: **0** Maximum: **65535** Array Length: **1 - 20**
details	Array of **details** objects	The error details. Array Length: **1 - 20**

Table 5-96 details

Parameter	Type	Description
error_code	String	The SMS error code. Minimum: **0** Maximum: **65535**
error_msg	String	The SMS error message. Minimum: **0** Maximum: **65535**

Example Requests

This example obtains an overview of source servers.

```
GET https://{endpoint}/v3/sources/overview
X-Auth-Token: XXXX
```

Example Responses

Status code: 200

An overview of source servers was obtained.


```
{
  "replicate" : 0,
  "stopped" : 0,
  "unconfigured" : 13,
  "waiting" : 0,
  "syncing" : 0,
  "unavailable" : 0,
  "cutovering" : 0,
  "finished" : 6,
  "error" : 3,
  "deleting" : 1,
  "stopping" : 1,
  "skipping" : 1,
  "initialize" : 0,
  "cloning" : 0
}
```


Status code: 403

Authentication failed.


```
{
  "error_code" : "SMS.9004",
  "error_msg" : "The current account does not have the permission to execute policy. You do not have permission to perform action XXX on resource XXX.",
  "encoded_authorization_message" : "XXXXXX",
  "error_param" : [ "You do not have permission to perform action XXX on resource XXX." ],
  "details" : [ {
    "error_code" : "SMS.9004",
    "error_msg" : "You do not have permission to perform action XXX on resource XXX."
  } ]
}
```


SDK Sample Code

The SDK sample code is as follows.

Java

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;


public class ShowOverviewSolution {
```

```
    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        ShowOverviewRequest request = new ShowOverviewRequest();
        try {
            ShowOverviewResponse response = client.showOverview(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

Python

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = ShowOverviewRequest()
        response = client.show_overview(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
```

```
        print(e.error_code)
        print(e.error_msg)
```

Go

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
        WithRegion(region.ValueOf("<YOUR REGION>")).
        WithCredential(auth).
        SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.ShowOverviewRequest{}
    response, err := client.ShowOverview(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

More

For SDK sample code of more programming languages, see the Sample Code tab in **API Explorer**. SDK sample code can be automatically generated.

Status Codes

Status Code	Description
200	An overview of source servers was obtained.
403	Authentication failed.

Error Codes

See **Error Codes**.

5.3.10 Updating Disk Information

Function

This API is used to overwrite the original disk information of a server with the new disk information.

This API takes effect only when the source server is in the **waiting** state. After the migration starts, the modification on the disk information does not take effect.

Calling Method

For details, see **Calling APIs**.

Authorization Information

Each account has all the permissions required to call all APIs, but IAM users must be assigned the required permissions.

- If you are using role/policy-based authorization, see **Permissions Policies and Supported Actions** for details on the required permissions.
- If you are using identity policy-based authorization, the following identity policy-based permissions are required.

Action	Access Level	Resource Type (*: required)	Condition Key	Alias	Dependencies
sms:server:updateDiskInfo	Write	server *	-	sms:server:registerServer	-
		-	g:EnterpriseProjectId		

URI

PUT /v3/sources/{source_id}/diskinfo

Table 5-97 Path Parameters

Parameter	Mandatory	Type	Description
source_id	Yes	String	The source server ID. Minimum: **1** Maximum: **36**

Request Parameters

Table 5-98 Request header parameters

Parameter	Mandatory	Type	Description
X-Auth-Token	Yes	String	User token. The token can be obtained by calling the IAM API for obtaining a user token. The value of **X-Subject-Token** in the response header is the user token. Minimum: **1** Maximum: **16384**

Table 5-99 Request body parameters

Parameter	Mandatory	Type	Description
disks	No	Array of **ServerDisk** objects	The updated disk information. Array Length: **0 - 50**
volumegroups	No	Array of **VolumeGroups** objects	The updated volume information. Array Length: **0 - 50**
btrfs_list	No	Array of **BtrfsFileSystem** objects	The updated Btrfs file system information. Array Length: **0 - 65535**

Table 5-100 ServerDisk

Parameter	Mandatory	Type	Description
name	No	String	The disk name. Minimum: **0** Maximum: **255**
partition_style	No	String	Disk partition type. This parameter is mandatory when the source server is added. Otherwise, the subsequent environment check will fail. Options (required by EVS): **MBR**: Master Boot Record **GPT**: GUID Partition Table** **Partition Table** For details, see the description of differences between MBR and GPT in the EVS API document. Minimum: **0** Maximum: **255**
device_use	No	String	Disk type. This is not mandatory and can be left blank. The common values are as follows: **NORMAL** **OS**: system device **BOOT** **VOLUME_GROUP** **BTRFS** Minimum: **0** Maximum: **255**
size	No	Long	The disk size, in bytes. Minimum: **0** Maximum: **9223372036854775807**
used_size	No	Long	The used disk space, in bytes. Minimum: **0** Maximum: **9223372036854775807**

Parameter	Mandatory	Type	Description
physical_volumes	No	Array of **PhysicalVolume** objects	The information about physical partitions on the disk. Array Length: **0 - 50**
os_disk	No	Boolean	Indicates whether the disk is the system disk.
relation_name	No	String	The name of the paired disk on the Linux target ECS server. Minimum: **0** Maximum: **255**
inode_size	No	Integer	The inode size. Minimum: **0** Maximum: **2147483647**

Table 5-101 PhysicalVolume

Parameter	Mandatory	Type	Description
device_use	No	String	Disk type. This is not mandatory and can be left blank. The common values are as follows: **NORMAL** **OS**: system device **BOOT** **VOLUME_GROUP** **BTRFS** Minimum: **0** Maximum: **255**
file_system	No	String	The file system type. Minimum: **0** Maximum: **255**
index	No	Integer	The serial number. Minimum: **0** Maximum: **2147483647**
mount_point	No	String	The mount point. Minimum: **0** Maximum: **255**

Parameter	Mandatory	Type	Description
name	No	String	The volume name. In Windows, it indicates the drive letter, and in Linux, it indicates the device ID. Minimum: **0** Maximum: **255**
size	No	Long	The size. Minimum: **0** Maximum: **9223372036854775807**
used_size	No	Long	The used space. Minimum: **0** Maximum: **9223372036854775807**
inode_size	No	Integer	The number of inodes. Minimum: **0** Maximum: **2147483647**
inode_nums	No	Long	The number of inodes. Minimum: **0** Maximum: **9223372036854775807**
uuid	No	String	The GUID, which can be obtained from the source server. Minimum: **0** Maximum: **255**
size_per_cluster	No	Integer	The size of each cluster. Minimum: **0** Maximum: **2147483647**

Table 5-102 VolumeGroups

Parameter	Mandatory	Type	Description
components	No	String	The physical volume information. Minimum: **0** Maximum: **255**

Parameter	Mandatory	Type	Description
free_size	No	Long	The available space. Minimum: **0** Maximum: **9223372036854775807**
logical_volumes	No	Array of **LogicalVolumes** objects	The logical volume information. Array Length: **0 - 50**
name	No	String	Name. Minimum: **0** Maximum: **255**
size	No	Long	Size. Minimum: **0** Maximum: **9223372036854775807**

Table 5-103 LogicalVolumes

Parameter	Mandatory	Type	Description
block_count	No	Integer	The number of blocks. Minimum: **0** Maximum: **2147483647** Default: **0**
block_size	No	Long	Block size. Minimum: **0** Maximum: **1048576** Default: **0**
file_system	No	String	The file system. Minimum: **0** Maximum: **255**
inode_size	No	Integer	The number of inodes. Minimum: **0** Maximum: **2147483647**
inode_nums	No	Long	The number of inodes. Minimum: **0** Maximum: **9223372036854775807**

Parameter	Mandatory	Type	Description
device_use	No	String	Partition type. This is not mandatory and can be left blank. The common values are as follows: **NORMAL** **OS**: system device **BOOT** **VOLUME_GROUP** **BTRFS** Minimum: **0** Maximum: **255**
mount_point	No	String	The mount point. Minimum: **0** Maximum: **256**
name	No	String	Name. Minimum: **0** Maximum: **1024**
size	No	Long	The size. Minimum: **0** Maximum: **9223372036854775807**
used_size	No	Long	The used space. Minimum: **0** Maximum: **9223372036854775807**
free_size	No	Long	The available space. Minimum: **0** Maximum: **9223372036854775807**

Table 5-104 BtrfsFileSystem

Parameter	Mandatory	Type	Description
name	No	String	The file system name. Minimum: **0** Maximum: **255**

Parameter	Mandatory	Type	Description
label	No	String	The file system tags. If no tag exists, the value is an empty string. Minimum: **0** Maximum: **255**
uuid	No	String	The file system UUID. Minimum: **0** Maximum: **255**
device	No	String	The device names of the Btrfs file system. Minimum: **0** Maximum: **255**
size	No	Long	The space occupied by the file system data. Minimum: **0** Maximum: **9223372036854775807**
nodesize	No	Long	The Btrfs node size. Minimum: **0** Maximum: **9223372036854775807**
sectorsize	No	Integer	The sector size. Minimum: **0** Maximum: **2147483647**
data_profile	No	String	The data profile (RAD). Minimum: **0** Maximum: **255**
system_profile	No	String	The file system profile (RAD). Minimum: **0** Maximum: **255**
metadata_profile	No	String	The metadata profile (RAD). Minimum: **0** Maximum: **255**
global_reserve1	No	String	The Btrfs file system information. Minimum: **0** Maximum: **255**

Parameter	Mandatory	Type	Description
g_vol_used_size	No	Long	The used space of the Btrfs volume. Minimum: **0** Maximum: **9223372036854775807**
default_subvolid	No	String	The ID of the default subvolume. Minimum: **0** Maximum: **255**
default_subvol_name	No	String	The name of the default subvolume. Minimum: **0** Maximum: **255**
default_subvol_mountpath	No	String	The mount path of the default subvolume or Btrfs file system. Minimum: **0** Maximum: **255**
subvolumn	No	Array of **BtrfsSubvolumn** objects	The subvolume information. Array Length: **0 - 65535**

Table 5-105 BtrfsSubvolumn

Parameter	Mandatory	Type	Description
uuid	No	String	The UUID of the parent volume. Minimum: **0** Maximum: **255**
is_snapshot	No	String	Indicates whether the subvolume is a snapshot. Minimum: **0** Maximum: **255**
subvol_id	No	String	The subvolume ID. Minimum: **0** Maximum: **255**
parent_id	No	String	The parent volume ID. Minimum: **0** Maximum: **255**

Parameter	Mandatory	Type	Description
subvol_name	No	String	The subvolume name. Minimum: **0** Maximum: **255**
subvol_mount_path	No	String	The mount path of the subvolume. Minimum: **0** Maximum: **255**

Response Parameters

Status code: 200

Table 5-106 Response body parameters

Parameter	Type	Description
-	String	The disk information was updated.

Status code: 403

Table 5-107 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **255**
encoded_authorization_message	String	The encrypted authorization information. Minimum: **0** Maximum: **65535**
error_param	Array of strings	Error parameters. Minimum: **0** Maximum: **65535** Array Length: **1 - 20**

Parameter	Type	Description
details	Array of **details** objects	The error details. Array Length: **1 - 20**

Table 5-108 details

Parameter	Type	Description
error_code	String	The SMS error code. Minimum: **0** Maximum: **65535**
error_msg	String	The SMS error message. Minimum: **0** Maximum: **65535**

Example Requests

This example updates the disk information of a server. The new disk name is /dev/vda, the disk function is BOOT, and the disk size is 42,949,672,960 bytes.

```
PUT https://{endpoint}/v3/sources/{source_id}/diskinfo

{
  "disks" : [ {
    "name" : "/dev/vda",
    "device_use" : "BOOT",
    "size" : 42949672960,
    "partition_style" : "MBR",
    "used_size" : 42948624384,
    "physical_volumes" : [ {
      "name" : "/dev/vda1",
      "size" : 2153775104,
      "device_use" : "NORMAL",
      "used_size" : 2153775104,
      "inode_size" : 0,
      "file_system" : "swap",
      "mount_point" : ""
    }, {
      "name" : "/dev/vda2",
      "size" : 16862150656,
      "device_use" : "BTRFS",
      "used_size" : 16862150656,
      "inode_size" : 0,
      "file_system" : "btrfs",
      "mount_point" : ""
    }, {
      "name" : "/dev/vda3",
      "size" : 23932698624,
      "device_use" : "NORMAL",
      "used_size" : 33988608,
      "inode_size" : 0,
      "file_system" : "xfs",
      "mount_point" : "/home"
    } ]
  }, {
```



```
  "name" : "/dev/vdb",
  "device_use" : "NORMAL",
  "size" : 21474836480,
  "partition_style" : "MBR",
  "used_size" : 21473787904,
  "physical_volumes" : [ {
    "name" : "/dev/vdb1",
    "size" : 21473787904,
    "device_use" : "VOLUME_GROUP",
    "used_size" : 21473787904,
    "inode_size" : 0,
    "file_system" : "LVM2_member",
    "mount_point" : ""
  } ]
}, {
  "name" : "/dev/vdc",
  "device_use" : "VOLUME_GROUP",
  "size" : 21474836480,
  "partition_style" : "MBR",
  "used_size" : 0,
  "physical_volumes" : [ ]
} ],
"volumegroups" : [ {
  "name" : "vg1",
  "size" : 42948624384,
  "components" : "/dev/vdb1;/dev/vdc",
  "logical_volumes" : [ {
    "name" : "/dev/mapper/vg1-lv1",
    "device_use" : "NORMAL",
    "size" : 10737418240,
    "free_size" : 10713837568,
    "used_size" : 23580672,
    "file_system" : "ext4",
    "mount_point" : "/mnt/lv1",
    "inode_size" : 256
  } ]
} ],
"btrfs_list" : [ {
  "name" : "/dev/vda2",
  "label" : "none",
  "uuid" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
  "device" : "/dev/vda2",
  "size" : 3538944000,
  "nodesize" : 16384,
  "sectorsize" : 4096,
  "data_profile" : "single",
  "system_profile" : "single",
  "metadata_profile" : "single",
  "global_reserve1" : "single",
  "g_vol_used_size" : 3894038528,
  "default_subvolid" : "259",
  "default_subvol_name" : "@/.snapshots/1/snapshot",
  "default_subvol_mountpath" : "/",
  "subvolumn" : [ {
    "uuid" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "is_snapshot" : "false",
    "subvol_id" : "257",
    "parent_id" : "5",
    "subvol_name" : "@",
    "subvol_mount_path" : null
  }, {
    "uuid" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "is_snapshot" : "false",
    "subvol_id" : "258",
    "parent_id" : "257",
    "subvol_name" : "@/.snapshots",
    "subvol_mount_path" : "/.snapshots"
  }, {
    "uuid" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
```

```
  "is_snapshot" : "true",
  "subvol_id" : "259",
  "parent_id" : "258",
  "subvol_name" : "@/.snapshots/1/snapshot",
  "subvol_mount_path" : "/"
}, {
  "uuid" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
  "is_snapshot" : "false",
  "subvol_id" : "260",
  "parent_id" : "257",
  "subvol_name" : "@/boot/grub2/i386-pc",
  "subvol_mount_path" : "/boot/grub2/i386-pc"
}, {
  "uuid" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
  "is_snapshot" : "false",
  "subvol_id" : "261",
  "parent_id" : "257",
  "subvol_name" : "@/boot/grub2/x86_64-efi",
  "subvol_mount_path" : "/boot/grub2/x86_64-efi"
}, {
  "uuid" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
  "is_snapshot" : "false",
  "subvol_id" : "262",
  "parent_id" : "257",
  "subvol_name" : "@/opt",
  "subvol_mount_path" : "/opt"
}, {
  "uuid" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
  "is_snapshot" : "false",
  "subvol_id" : "263",
  "parent_id" : "257",
  "subvol_name" : "@/srv",
  "subvol_mount_path" : "/srv"
}, {
  "uuid" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
  "is_snapshot" : "false",
  "subvol_id" : "264",
  "parent_id" : "257",
  "subvol_name" : "@/tmp",
  "subvol_mount_path" : "/tmp"
}, {
  "uuid" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
  "is_snapshot" : "false",
  "subvol_id" : "265",
  "parent_id" : "257",
  "subvol_name" : "@/usr/local",
  "subvol_mount_path" : "/usr/local"
}, {
  "uuid" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
  "is_snapshot" : "false",
  "subvol_id" : "266",
  "parent_id" : "257",
  "subvol_name" : "@/var/cache",
  "subvol_mount_path" : "/var/cache"
}, {
  "uuid" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
  "is_snapshot" : "false",
  "subvol_id" : "267",
  "parent_id" : "257",
  "subvol_name" : "@/var/crash",
  "subvol_mount_path" : "/var/crash"
}, {
  "uuid" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
  "is_snapshot" : "false",
  "subvol_id" : "268",
  "parent_id" : "257",
  "subvol_name" : "@/var/lib/libvirt/images",
  "subvol_mount_path" : "/var/lib/libvirt/images"
}, {
```



```
    "uuid" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "is_snapshot" : "false",
    "subvol_id" : "269",
    "parent_id" : "257",
    "subvol_name" : "@/var/lib/machines",
    "subvol_mount_path" : "/var/lib/machines"
  }, {
    "uuid" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "is_snapshot" : "false",
    "subvol_id" : "270",
    "parent_id" : "257",
    "subvol_name" : "@/var/lib/mailman",
    "subvol_mount_path" : "/var/lib/mailman"
  }, {
    "uuid" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "is_snapshot" : "false",
    "subvol_id" : "271",
    "parent_id" : "257",
    "subvol_name" : "@/var/lib/mariadb",
    "subvol_mount_path" : "/var/lib/mariadb"
  }, {
    "uuid" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "is_snapshot" : "false",
    "subvol_id" : "272",
    "parent_id" : "257",
    "subvol_name" : "@/var/lib/mysql",
    "subvol_mount_path" : "/var/lib/mysql"
  }, {
    "uuid" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "is_snapshot" : "false",
    "subvol_id" : "273",
    "parent_id" : "257",
    "subvol_name" : "@/var/lib/named",
    "subvol_mount_path" : "/var/lib/named"
  }, {
    "uuid" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "is_snapshot" : "false",
    "subvol_id" : "274",
    "parent_id" : "257",
    "subvol_name" : "@/var/lib/pgsql",
    "subvol_mount_path" : "/var/lib/pgsql"
  }, {
    "uuid" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "is_snapshot" : "false",
    "subvol_id" : "275",
    "parent_id" : "257",
    "subvol_name" : "@/var/log",
    "subvol_mount_path" : "/var/log"
  }, {
    "uuid" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "is_snapshot" : "false",
    "subvol_id" : "276",
    "parent_id" : "257",
    "subvol_name" : "@/var/opt",
    "subvol_mount_path" : "/var/opt"
  }, {
    "uuid" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "is_snapshot" : "false",
    "subvol_id" : "277",
    "parent_id" : "257",
    "subvol_name" : "@/var/spool",
    "subvol_mount_path" : "/var/spool"
  }, {
    "uuid" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "is_snapshot" : "false",
    "subvol_id" : "278",
    "parent_id" : "257",
    "subvol_name" : "@/var/tmp",
    "subvol_mount_path" : "/var/tmp"
```

```
    }, {
      "uuid" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
      "is_snapshot" : "true",
      "subvol_id" : "282",
      "parent_id" : "258",
      "subvol_name" : "@/.snapshots/2/snapshot",
      "subvol_mount_path" : null
    } ]
  } ]
}
```

Example Responses

Status code: 200

The disk information was updated.

```
{ }
```

Status code: 403

Authentication failed.

```
{
  "error_code" : "SMS.9004",
  "error_msg" : "The current account does not have the permission to execute policy. You do not have permission to perform action XXX on resource XXX.",
  "encoded_authorization_message" : "XXXXXX",
  "error_param" : [ "You do not have permission to perform action XXX on resource XXX." ],
  "details" : [ {
    "error_code" : "SMS.9004",
    "error_msg" : "You do not have permission to perform action XXX on resource XXX."
  } ]
}
```

SDK Sample Code

The SDK sample code is as follows.

Java

This example updates the disk information of a server. The new disk name is /dev/vda, the disk function is BOOT, and the disk size is 42,949,672,960 bytes.

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;

import java.util.List;
import java.util.ArrayList;

public class UpdateDiskInfoSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
```

```
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        UpdateDiskInfoRequest request = new UpdateDiskInfoRequest();
        request.withSourceId("{source_id}");
        PutDiskInfoReq body = new PutDiskInfoReq();
        List<BtrfsSubvolumn> listBtrfsListSubvolumn = new ArrayList<>();
        listBtrfsListSubvolumn.add(
            new BtrfsSubvolumn()
                .withUuid("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
                .withIsSnapshot("false")
                .withSubvolId("257")
                .withParentId("5")
                .withSubvolName("@")
        );
        listBtrfsListSubvolumn.add(
            new BtrfsSubvolumn()
                .withUuid("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
                .withIsSnapshot("false")
                .withSubvolId("258")
                .withParentId("257")
                .withSubvolName("@/.snapshots")
                .withSubvolMountPath("/.snapshots")
        );
        listBtrfsListSubvolumn.add(
            new BtrfsSubvolumn()
                .withUuid("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
                .withIsSnapshot("true")
                .withSubvolId("259")
                .withParentId("258")
                .withSubvolName("@/.snapshots/1/snapshot")
                .withSubvolMountPath("/")
        );
        listBtrfsListSubvolumn.add(
            new BtrfsSubvolumn()
                .withUuid("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
                .withIsSnapshot("false")
                .withSubvolId("260")
                .withParentId("257")
                .withSubvolName("@/boot/grub2/i386-pc")
                .withSubvolMountPath("/boot/grub2/i386-pc")
        );
        listBtrfsListSubvolumn.add(
            new BtrfsSubvolumn()
                .withUuid("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
                .withIsSnapshot("false")
                .withSubvolId("261")
                .withParentId("257")
                .withSubvolName("@/boot/grub2/x86_64-efi")
                .withSubvolMountPath("/boot/grub2/x86_64-efi")
        );
        listBtrfsListSubvolumn.add(
            new BtrfsSubvolumn()
                .withUuid("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
                .withIsSnapshot("false")
                .withSubvolId("262")
                .withParentId("257")
                .withSubvolName("@/opt")
                .withSubvolMountPath("/opt")
        );
        listBtrfsListSubvolumn.add(
```

```
        new BtrfsSubvolumn()
            .withUuid("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
            .withIsSnapshot("false")
            .withSubvolId("263")
            .withParentId("257")
            .withSubvolName("@/srv")
            .withSubvolMountPath("/srv")
    );
    listBtrfsListSubvolumn.add(
        new BtrfsSubvolumn()
            .withUuid("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
            .withIsSnapshot("false")
            .withSubvolId("264")
            .withParentId("257")
            .withSubvolName("@/tmp")
            .withSubvolMountPath("/tmp")
    );
    listBtrfsListSubvolumn.add(
        new BtrfsSubvolumn()
            .withUuid("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
            .withIsSnapshot("false")
            .withSubvolId("265")
            .withParentId("257")
            .withSubvolName("@/usr/local")
            .withSubvolMountPath("/usr/local")
    );
    listBtrfsListSubvolumn.add(
        new BtrfsSubvolumn()
            .withUuid("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
            .withIsSnapshot("false")
            .withSubvolId("266")
            .withParentId("257")
            .withSubvolName("@/var/cache")
            .withSubvolMountPath("/var/cache")
    );
    listBtrfsListSubvolumn.add(
        new BtrfsSubvolumn()
            .withUuid("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
            .withIsSnapshot("false")
            .withSubvolId("267")
            .withParentId("257")
            .withSubvolName("@/var/crash")
            .withSubvolMountPath("/var/crash")
    );
    listBtrfsListSubvolumn.add(
        new BtrfsSubvolumn()
            .withUuid("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
            .withIsSnapshot("false")
            .withSubvolId("268")
            .withParentId("257")
            .withSubvolName("@/var/lib/libvirt/images")
            .withSubvolMountPath("/var/lib/libvirt/images")
    );
    listBtrfsListSubvolumn.add(
        new BtrfsSubvolumn()
            .withUuid("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
            .withIsSnapshot("false")
            .withSubvolId("269")
            .withParentId("257")
            .withSubvolName("@/var/lib/machines")
            .withSubvolMountPath("/var/lib/machines")
    );
    listBtrfsListSubvolumn.add(
        new BtrfsSubvolumn()
            .withUuid("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
            .withIsSnapshot("false")
            .withSubvolId("270")
            .withParentId("257")
            .withSubvolName("@/var/lib/mailman")
```

```
            .withSubvolMountPath("/var/lib/mailman")
    );
    listBtrfsListSubvolumn.add(
        new BtrfsSubvolumn()
            .withUuid("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
            .withIsSnapshot("false")
            .withSubvolId("271")
            .withParentId("257")
            .withSubvolName("@/var/lib/mariadb")
            .withSubvolMountPath("/var/lib/mariadb")
    );
    listBtrfsListSubvolumn.add(
        new BtrfsSubvolumn()
            .withUuid("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
            .withIsSnapshot("false")
            .withSubvolId("272")
            .withParentId("257")
            .withSubvolName("@/var/lib/mysql")
            .withSubvolMountPath("/var/lib/mysql")
    );
    listBtrfsListSubvolumn.add(
        new BtrfsSubvolumn()
            .withUuid("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
            .withIsSnapshot("false")
            .withSubvolId("273")
            .withParentId("257")
            .withSubvolName("@/var/lib/named")
            .withSubvolMountPath("/var/lib/named")
    );
    listBtrfsListSubvolumn.add(
        new BtrfsSubvolumn()
            .withUuid("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
            .withIsSnapshot("false")
            .withSubvolId("274")
            .withParentId("257")
            .withSubvolName("@/var/lib/pgsql")
            .withSubvolMountPath("/var/lib/pgsql")
    );
    listBtrfsListSubvolumn.add(
        new BtrfsSubvolumn()
            .withUuid("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
            .withIsSnapshot("false")
            .withSubvolId("275")
            .withParentId("257")
            .withSubvolName("@/var/log")
            .withSubvolMountPath("/var/log")
    );
    listBtrfsListSubvolumn.add(
        new BtrfsSubvolumn()
            .withUuid("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
            .withIsSnapshot("false")
            .withSubvolId("276")
            .withParentId("257")
            .withSubvolName("@/var/opt")
            .withSubvolMountPath("/var/opt")
    );
    listBtrfsListSubvolumn.add(
        new BtrfsSubvolumn()
            .withUuid("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
            .withIsSnapshot("false")
            .withSubvolId("277")
            .withParentId("257")
            .withSubvolName("@/var/spool")
            .withSubvolMountPath("/var/spool")
    );
    listBtrfsListSubvolumn.add(
        new BtrfsSubvolumn()
            .withUuid("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
            .withIsSnapshot("false")
```

```
            .withSubvolId("278")
            .withParentId("257")
            .withSubvolName("@/var/tmp")
            .withSubvolMountPath("/var/tmp")
    );
    listBtrfsListSubvolumn.add(
        new BtrfsSubvolumn()
            .withUuid("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
            .withIsSnapshot("true")
            .withSubvolId("282")
            .withParentId("258")
            .withSubvolName("@/.snapshots/2/snapshot")
    );
    List<BtrfsFileSystem> listbodyBtrfsList = new ArrayList<>();
    listbodyBtrfsList.add(
        new BtrfsFileSystem()
            .withName("/dev/vda2")
            .withLabel("none")
            .withUuid("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
            .withDevice("/dev/vda2")
            .withSize(3538944000L)
            .withNodesize(16384L)
            .withSectorsize(4096)
            .withDataProfile("single")
            .withSystemProfile("single")
            .withMetadataProfile("single")
            .withGlobalReserve1("single")
            .withGVolUsedSize(3894038528L)
            .withDefaultSubvolid("259")
            .withDefaultSubvolName("@/.snapshots/1/snapshot")
            .withDefaultSubvolMountpath("/")
            .withSubvolumn(listBtrfsListSubvolumn)
    );
    List<LogicalVolumes> listVolumegroupsLogicalVolumes = new ArrayList<>();
    listVolumegroupsLogicalVolumes.add(
        new LogicalVolumes()
            .withFileSystem("ext4")
            .withInodeSize(256)
            .withDeviceUse("NORMAL")
            .withMountPoint("/mnt/lv1")
            .withName("/dev/mapper/vg1-lv1")
            .withSize(10737418240L)
            .withUsedSize(23580672L)
            .withFreeSize(10713837568L)
    );
    List<VolumeGroups> listbodyVolumegroups = new ArrayList<>();
    listbodyVolumegroups.add(
        new VolumeGroups()
            .withComponents("/dev/vdb1;/dev/vdc")
            .withLogicalVolumes(listVolumegroupsLogicalVolumes)
            .withName("vg1")
            .withSize(42948624384L)
    );
    List<PhysicalVolume> listDisksPhysicalVolumes = new ArrayList<>();
    listDisksPhysicalVolumes.add(
        new PhysicalVolume()
            .withDeviceUse("VOLUME_GROUP")
            .withFileSystem("LVM2_member")
            .withMountPoint("")
            .withName("/dev/vdb1")
            .withSize(21473787904L)
            .withUsedSize(21473787904L)
            .withInodeSize(0)
    );
    List<PhysicalVolume> listDisksPhysicalVolumes1 = new ArrayList<>();
    listDisksPhysicalVolumes1.add(
        new PhysicalVolume()
            .withDeviceUse("NORMAL")
            .withFileSystem("swap")
```

```
            .withMountPoint("")
            .withName("/dev/vda1")
            .withSize(2153775104L)
            .withUsedSize(2153775104L)
            .withInodeSize(0)
    );
    listDisksPhysicalVolumes1.add(
        new PhysicalVolume()
            .withDeviceUse("BTRFS")
            .withFileSystem("btrfs")
            .withMountPoint("")
            .withName("/dev/vda2")
            .withSize(16862150656L)
            .withUsedSize(16862150656L)
            .withInodeSize(0)
    );
    listDisksPhysicalVolumes1.add(
        new PhysicalVolume()
            .withDeviceUse("NORMAL")
            .withFileSystem("xfs")
            .withMountPoint("/home")
            .withName("/dev/vda3")
            .withSize(23932698624L)
            .withUsedSize(33988608L)
            .withInodeSize(0)
    );
    List<ServerDisk> listbodyDisks = new ArrayList<>();
    listbodyDisks.add(
        new ServerDisk()
            .withName("/dev/vda")
            .withPartitionStyle("MBR")
            .withDeviceUse("BOOT")
            .withSize(42949672960L)
            .withUsedSize(42948624384L)
            .withPhysicalVolumes(listDisksPhysicalVolumes1)
    );
    listbodyDisks.add(
        new ServerDisk()
            .withName("/dev/vdb")
            .withPartitionStyle("MBR")
            .withDeviceUse("NORMAL")
            .withSize(21474836480L)
            .withUsedSize(21473787904L)
            .withPhysicalVolumes(listDisksPhysicalVolumes)
    );
    listbodyDisks.add(
        new ServerDisk()
            .withName("/dev/vdc")
            .withPartitionStyle("MBR")
            .withDeviceUse("VOLUME_GROUP")
            .withSize(21474836480L)
            .withUsedSize(0L)
            .withPhysicalVolumes()
    );
    body.withBtrfsList(listbodyBtrfsList);
    body.withVolumegroups(listbodyVolumegroups);
    body.withDisks(listbodyDisks);
    request.withBody(body);
    try {
        UpdateDiskInfoResponse response = client.updateDiskInfo(request);
        System.out.println(response.toString());
    } catch (ConnectionException e) {
        e.printStackTrace();
    } catch (RequestTimeoutException e) {
        e.printStackTrace();
    } catch (ServiceResponseException e) {
        e.printStackTrace();
        System.out.println(e.getHttpStatusCode());
        System.out.println(e.getRequestId());
```

```
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

Python

This example updates the disk information of a server. The new disk name is /dev/vda, the disk function is BOOT, and the disk size is 42,949,672,960 bytes.

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = UpdateDiskInfoRequest()
        request.source_id = "{source_id}"
        listSubvolumnBtrfsList = [
            BtrfsSubvolumn(
                uuid="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
                is_snapshot="false",
                subvol_id="257",
                parent_id="5",
                subvol_name="@"
            ),
            BtrfsSubvolumn(
                uuid="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
                is_snapshot="false",
                subvol_id="258",
                parent_id="257",
                subvol_name="@/.snapshots",
                subvol_mount_path="/.snapshots"
            ),
            BtrfsSubvolumn(
                uuid="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
                is_snapshot="true",
                subvol_id="259",
                parent_id="258",
                subvol_name="@/.snapshots/1/snapshot",
                subvol_mount_path="/"
            ),
            BtrfsSubvolumn(
                uuid="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
                is_snapshot="false",
                subvol_id="260",
                parent_id="257",
                subvol_name="@/boot/grub2/i386-pc",
                subvol_mount_path="/boot/grub2/i386-pc"
```

```
    ),
    BtrfsSubvolumn(
        uuid="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
        is_snapshot="false",
        subvol_id="261",
        parent_id="257",
        subvol_name="@/boot/grub2/x86_64-efi",
        subvol_mount_path="/boot/grub2/x86_64-efi"
    ),
    BtrfsSubvolumn(
        uuid="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
        is_snapshot="false",
        subvol_id="262",
        parent_id="257",
        subvol_name="@/opt",
        subvol_mount_path="/opt"
    ),
    BtrfsSubvolumn(
        uuid="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
        is_snapshot="false",
        subvol_id="263",
        parent_id="257",
        subvol_name="@/srv",
        subvol_mount_path="/srv"
    ),
    BtrfsSubvolumn(
        uuid="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
        is_snapshot="false",
        subvol_id="264",
        parent_id="257",
        subvol_name="@/tmp",
        subvol_mount_path="/tmp"
    ),
    BtrfsSubvolumn(
        uuid="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
        is_snapshot="false",
        subvol_id="265",
        parent_id="257",
        subvol_name="@/usr/local",
        subvol_mount_path="/usr/local"
    ),
    BtrfsSubvolumn(
        uuid="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
        is_snapshot="false",
        subvol_id="266",
        parent_id="257",
        subvol_name="@/var/cache",
        subvol_mount_path="/var/cache"
    ),
    BtrfsSubvolumn(
        uuid="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
        is_snapshot="false",
        subvol_id="267",
        parent_id="257",
        subvol_name="@/var/crash",
        subvol_mount_path="/var/crash"
    ),
    BtrfsSubvolumn(
        uuid="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
        is_snapshot="false",
        subvol_id="268",
        parent_id="257",
        subvol_name="@/var/lib/libvirt/images",
        subvol_mount_path="/var/lib/libvirt/images"
    ),
    BtrfsSubvolumn(
        uuid="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
        is_snapshot="false",
        subvol_id="269",
```

```
        parent_id="257",
        subvol_name="@/var/lib/machines",
        subvol_mount_path="/var/lib/machines"
    ),
    BtrfsSubvolumn(
        uuid="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
        is_snapshot="false",
        subvol_id="270",
        parent_id="257",
        subvol_name="@/var/lib/mailman",
        subvol_mount_path="/var/lib/mailman"
    ),
    BtrfsSubvolumn(
        uuid="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
        is_snapshot="false",
        subvol_id="271",
        parent_id="257",
        subvol_name="@/var/lib/mariadb",
        subvol_mount_path="/var/lib/mariadb"
    ),
    BtrfsSubvolumn(
        uuid="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
        is_snapshot="false",
        subvol_id="272",
        parent_id="257",
        subvol_name="@/var/lib/mysql",
        subvol_mount_path="/var/lib/mysql"
    ),
    BtrfsSubvolumn(
        uuid="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
        is_snapshot="false",
        subvol_id="273",
        parent_id="257",
        subvol_name="@/var/lib/named",
        subvol_mount_path="/var/lib/named"
    ),
    BtrfsSubvolumn(
        uuid="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
        is_snapshot="false",
        subvol_id="274",
        parent_id="257",
        subvol_name="@/var/lib/pgsql",
        subvol_mount_path="/var/lib/pgsql"
    ),
    BtrfsSubvolumn(
        uuid="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
        is_snapshot="false",
        subvol_id="275",
        parent_id="257",
        subvol_name="@/var/log",
        subvol_mount_path="/var/log"
    ),
    BtrfsSubvolumn(
        uuid="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
        is_snapshot="false",
        subvol_id="276",
        parent_id="257",
        subvol_name="@/var/opt",
        subvol_mount_path="/var/opt"
    ),
    BtrfsSubvolumn(
        uuid="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
        is_snapshot="false",
        subvol_id="277",
        parent_id="257",
        subvol_name="@/var/spool",
        subvol_mount_path="/var/spool"
    ),
    BtrfsSubvolumn(
```

```
            uuid="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
            is_snapshot="false",
            subvol_id="278",
            parent_id="257",
            subvol_name="@/var/tmp",
            subvol_mount_path="/var/tmp"
        ),
        BtrfsSubvolumn(
            uuid="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
            is_snapshot="true",
            subvol_id="282",
            parent_id="258",
            subvol_name="@/.snapshots/2/snapshot"
        )
    ]
    listBtrfsListbody = [
        BtrfsFileSystem(
            name="/dev/vda2",
            label="none",
            uuid="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
            device="/dev/vda2",
            size=3538944000,
            nodesize=16384,
            sectorsize=4096,
            data_profile="single",
            system_profile="single",
            metadata_profile="single",
            global_reserve1="single",
            g_vol_used_size=3894038528,
            default_subvolid="259",
            default_subvol_name="@/.snapshots/1/snapshot",
            default_subvol_mountpath="/",
            subvolumn=listSubvolumnBtrfsList
        )
    ]
    listLogicalVolumesVolumegroups = [
        LogicalVolumes(
            file_system="ext4",
            inode_size=256,
            device_use="NORMAL",
            mount_point="/mnt/lv1",
            name="/dev/mapper/vg1-lv1",
            size=10737418240,
            used_size=23580672,
            free_size=10713837568
        )
    ]
    listVolumegroupsbody = [
        VolumeGroups(
            components="/dev/vdb1;/dev/vdc",
            logical_volumes=listLogicalVolumesVolumegroups,
            name="vg1",
            size=42948624384
        )
    ]
    listPhysicalVolumesDisks = [
        PhysicalVolume(
            device_use="VOLUME_GROUP",
            file_system="LVM2_member",
            mount_point="",
            name="/dev/vdb1",
            size=21473787904,
            used_size=21473787904,
            inode_size=0
        )
    ]
    listPhysicalVolumesDisks1 = [
        PhysicalVolume(
            device_use="NORMAL",
```



```
                file_system="swap",
                mount_point="",
                name="/dev/vda1",
                size=2153775104,
                used_size=2153775104,
                inode_size=0
            ),
            PhysicalVolume(
                device_use="BTRFS",
                file_system="btrfs",
                mount_point="",
                name="/dev/vda2",
                size=16862150656,
                used_size=16862150656,
                inode_size=0
            ),
            PhysicalVolume(
                device_use="NORMAL",
                file_system="xfs",
                mount_point="/home",
                name="/dev/vda3",
                size=23932698624,
                used_size=33988608,
                inode_size=0
            )
        ]
        listDisksbody = [
            ServerDisk(
                name="/dev/vda",
                partition_style="MBR",
                device_use="BOOT",
                size=42949672960,
                used_size=42948624384,
                physical_volumes=listPhysicalVolumesDisks1
            ),
            ServerDisk(
                name="/dev/vdb",
                partition_style="MBR",
                device_use="NORMAL",
                size=21474836480,
                used_size=21473787904,
                physical_volumes=listPhysicalVolumesDisks
            ),
            ServerDisk(
                name="/dev/vdc",
                partition_style="MBR",
                device_use="VOLUME_GROUP",
                size=21474836480,
                used_size=0,
            )
        ]
        request.body = PutDiskInfoReq(
            btrfs_list=listBtrfsListbody,
            volumegroups=listVolumegroupsbody,
            disks=listDisksbody
        )
        response = client.update_disk_info(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

Go

This example updates the disk information of a server. The new disk name is /dev/vda, the disk function is BOOT, and the disk size is 42,949,672,960 bytes.

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
         WithRegion(region.ValueOf("<YOUR REGION>")).
         WithCredential(auth).
         SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.UpdateDiskInfoRequest{}
    request.SourceId = "{source_id}"
    uuidSubvolumn:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
    isSnapshotSubvolumn:= "false"
    subvolIdSubvolumn:= "257"
    parentIdSubvolumn:= "5"
    subvolNameSubvolumn:= "@"
    uuidSubvolumn1:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
    isSnapshotSubvolumn1:= "false"
    subvolIdSubvolumn1:= "258"
    parentIdSubvolumn1:= "257"
    subvolNameSubvolumn1:= "@/.snapshots"
    subvolMountPathSubvolumn:= "/.snapshots"
    uuidSubvolumn2:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
    isSnapshotSubvolumn2:= "true"
    subvolIdSubvolumn2:= "259"
    parentIdSubvolumn2:= "258"
    subvolNameSubvolumn2:= "@/.snapshots/1/snapshot"
    subvolMountPathSubvolumn1:= "/"
    uuidSubvolumn3:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
    isSnapshotSubvolumn3:= "false"
    subvolIdSubvolumn3:= "260"
    parentIdSubvolumn3:= "257"
    subvolNameSubvolumn3:= "@/boot/grub2/i386-pc"
    subvolMountPathSubvolumn2:= "/boot/grub2/i386-pc"
    uuidSubvolumn4:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
    isSnapshotSubvolumn4:= "false"
```

```
subvolIdSubvolumn4:= "261"
parentIdSubvolumn4:= "257"
subvolNameSubvolumn4:= "@/boot/grub2/x86_64-efi"
subvolMountPathSubvolumn3:= "/boot/grub2/x86_64-efi"
uuidSubvolumn5:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
isSnapshotSubvolumn5:= "false"
subvolIdSubvolumn5:= "262"
parentIdSubvolumn5:= "257"
subvolNameSubvolumn5:= "@/opt"
subvolMountPathSubvolumn4:= "/opt"
uuidSubvolumn6:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
isSnapshotSubvolumn6:= "false"
subvolIdSubvolumn6:= "263"
parentIdSubvolumn6:= "257"
subvolNameSubvolumn6:= "@/srv"
subvolMountPathSubvolumn5:= "/srv"
uuidSubvolumn7:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
isSnapshotSubvolumn7:= "false"
subvolIdSubvolumn7:= "264"
parentIdSubvolumn7:= "257"
subvolNameSubvolumn7:= "@/tmp"
subvolMountPathSubvolumn6:= "/tmp"
uuidSubvolumn8:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
isSnapshotSubvolumn8:= "false"
subvolIdSubvolumn8:= "265"
parentIdSubvolumn8:= "257"
subvolNameSubvolumn8:= "@/usr/local"
subvolMountPathSubvolumn7:= "/usr/local"
uuidSubvolumn9:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
isSnapshotSubvolumn9:= "false"
subvolIdSubvolumn9:= "266"
parentIdSubvolumn9:= "257"
subvolNameSubvolumn9:= "@/var/cache"
subvolMountPathSubvolumn8:= "/var/cache"
uuidSubvolumn10:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
isSnapshotSubvolumn10:= "false"
subvolIdSubvolumn10:= "267"
parentIdSubvolumn10:= "257"
subvolNameSubvolumn10:= "@/var/crash"
subvolMountPathSubvolumn9:= "/var/crash"
uuidSubvolumn11:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
isSnapshotSubvolumn11:= "false"
subvolIdSubvolumn11:= "268"
parentIdSubvolumn11:= "257"
subvolNameSubvolumn11:= "@/var/lib/libvirt/images"
subvolMountPathSubvolumn10:= "/var/lib/libvirt/images"
uuidSubvolumn12:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
isSnapshotSubvolumn12:= "false"
subvolIdSubvolumn12:= "269"
parentIdSubvolumn12:= "257"
subvolNameSubvolumn12:= "@/var/lib/machines"
subvolMountPathSubvolumn11:= "/var/lib/machines"
uuidSubvolumn13:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
isSnapshotSubvolumn13:= "false"
subvolIdSubvolumn13:= "270"
parentIdSubvolumn13:= "257"
subvolNameSubvolumn13:= "@/var/lib/mailman"
subvolMountPathSubvolumn12:= "/var/lib/mailman"
uuidSubvolumn14:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
isSnapshotSubvolumn14:= "false"
subvolIdSubvolumn14:= "271"
parentIdSubvolumn14:= "257"
subvolNameSubvolumn14:= "@/var/lib/mariadb"
subvolMountPathSubvolumn13:= "/var/lib/mariadb"
uuidSubvolumn15:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
isSnapshotSubvolumn15:= "false"
subvolIdSubvolumn15:= "272"
parentIdSubvolumn15:= "257"
subvolNameSubvolumn15:= "@/var/lib/mysql"
```



```
subvolMountPathSubvolumn14:= "/var/lib/mysql"
uuidSubvolumn16:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
isSnapshotSubvolumn16:= "false"
subvolIdSubvolumn16:= "273"
parentIdSubvolumn16:= "257"
subvolNameSubvolumn16:= "@/var/lib/named"
subvolMountPathSubvolumn15:= "/var/lib/named"
uuidSubvolumn17:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
isSnapshotSubvolumn17:= "false"
subvolIdSubvolumn17:= "274"
parentIdSubvolumn17:= "257"
subvolNameSubvolumn17:= "@/var/lib/pgsql"
subvolMountPathSubvolumn16:= "/var/lib/pgsql"
uuidSubvolumn18:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
isSnapshotSubvolumn18:= "false"
subvolIdSubvolumn18:= "275"
parentIdSubvolumn18:= "257"
subvolNameSubvolumn18:= "@/var/log"
subvolMountPathSubvolumn17:= "/var/log"
uuidSubvolumn19:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
isSnapshotSubvolumn19:= "false"
subvolIdSubvolumn19:= "276"
parentIdSubvolumn19:= "257"
subvolNameSubvolumn19:= "@/var/opt"
subvolMountPathSubvolumn18:= "/var/opt"
uuidSubvolumn20:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
isSnapshotSubvolumn20:= "false"
subvolIdSubvolumn20:= "277"
parentIdSubvolumn20:= "257"
subvolNameSubvolumn20:= "@/var/spool"
subvolMountPathSubvolumn19:= "/var/spool"
uuidSubvolumn21:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
isSnapshotSubvolumn21:= "false"
subvolIdSubvolumn21:= "278"
parentIdSubvolumn21:= "257"
subvolNameSubvolumn21:= "@/var/tmp"
subvolMountPathSubvolumn20:= "/var/tmp"
uuidSubvolumn22:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
isSnapshotSubvolumn22:= "true"
subvolIdSubvolumn22:= "282"
parentIdSubvolumn22:= "258"
subvolNameSubvolumn22:= "@/.snapshots/2/snapshot"
var listSubvolumnBtrfsList = []model.BtrfsSubvolumn{
    {
        Uuid: &uuidSubvolumn,
        IsSnapshot: &isSnapshotSubvolumn,
        SubvolId: &subvolIdSubvolumn,
        ParentId: &parentIdSubvolumn,
        SubvolName: &subvolNameSubvolumn,
    },
    {
        Uuid: &uuidSubvolumn1,
        IsSnapshot: &isSnapshotSubvolumn1,
        SubvolId: &subvolIdSubvolumn1,
        ParentId: &parentIdSubvolumn1,
        SubvolName: &subvolNameSubvolumn1,
        SubvolMountPath: &subvolMountPathSubvolumn,
    },
    {
        Uuid: &uuidSubvolumn2,
        IsSnapshot: &isSnapshotSubvolumn2,
        SubvolId: &subvolIdSubvolumn2,
        ParentId: &parentIdSubvolumn2,
        SubvolName: &subvolNameSubvolumn2,
        SubvolMountPath: &subvolMountPathSubvolumn1,
    },
    {
        Uuid: &uuidSubvolumn3,
        IsSnapshot: &isSnapshotSubvolumn3,
```

```
        SubvolId: &subvolIdSubvolumn3,
        ParentId: &parentIdSubvolumn3,
        SubvolName: &subvolNameSubvolumn3,
        SubvolMountPath: &subvolMountPathSubvolumn2,
    },
    {
        Uuid: &uuidSubvolumn4,
        IsSnapshot: &isSnapshotSubvolumn4,
        SubvolId: &subvolIdSubvolumn4,
        ParentId: &parentIdSubvolumn4,
        SubvolName: &subvolNameSubvolumn4,
        SubvolMountPath: &subvolMountPathSubvolumn3,
    },
    {
        Uuid: &uuidSubvolumn5,
        IsSnapshot: &isSnapshotSubvolumn5,
        SubvolId: &subvolIdSubvolumn5,
        ParentId: &parentIdSubvolumn5,
        SubvolName: &subvolNameSubvolumn5,
        SubvolMountPath: &subvolMountPathSubvolumn4,
    },
    {
        Uuid: &uuidSubvolumn6,
        IsSnapshot: &isSnapshotSubvolumn6,
        SubvolId: &subvolIdSubvolumn6,
        ParentId: &parentIdSubvolumn6,
        SubvolName: &subvolNameSubvolumn6,
        SubvolMountPath: &subvolMountPathSubvolumn5,
    },
    {
        Uuid: &uuidSubvolumn7,
        IsSnapshot: &isSnapshotSubvolumn7,
        SubvolId: &subvolIdSubvolumn7,
        ParentId: &parentIdSubvolumn7,
        SubvolName: &subvolNameSubvolumn7,
        SubvolMountPath: &subvolMountPathSubvolumn6,
    },
    {
        Uuid: &uuidSubvolumn8,
        IsSnapshot: &isSnapshotSubvolumn8,
        SubvolId: &subvolIdSubvolumn8,
        ParentId: &parentIdSubvolumn8,
        SubvolName: &subvolNameSubvolumn8,
        SubvolMountPath: &subvolMountPathSubvolumn7,
    },
    {
        Uuid: &uuidSubvolumn9,
        IsSnapshot: &isSnapshotSubvolumn9,
        SubvolId: &subvolIdSubvolumn9,
        ParentId: &parentIdSubvolumn9,
        SubvolName: &subvolNameSubvolumn9,
        SubvolMountPath: &subvolMountPathSubvolumn8,
    },
    {
        Uuid: &uuidSubvolumn10,
        IsSnapshot: &isSnapshotSubvolumn10,
        SubvolId: &subvolIdSubvolumn10,
        ParentId: &parentIdSubvolumn10,
        SubvolName: &subvolNameSubvolumn10,
        SubvolMountPath: &subvolMountPathSubvolumn9,
    },
    {
        Uuid: &uuidSubvolumn11,
        IsSnapshot: &isSnapshotSubvolumn11,
        SubvolId: &subvolIdSubvolumn11,
        ParentId: &parentIdSubvolumn11,
        SubvolName: &subvolNameSubvolumn11,
        SubvolMountPath: &subvolMountPathSubvolumn10,
    },
```

```
    {
        Uuid: &uuidSubvolumn12,
        IsSnapshot: &isSnapshotSubvolumn12,
        SubvolId: &subvolIdSubvolumn12,
        ParentId: &parentIdSubvolumn12,
        SubvolName: &subvolNameSubvolumn12,
        SubvolMountPath: &subvolMountPathSubvolumn11,
    },
    {
        Uuid: &uuidSubvolumn13,
        IsSnapshot: &isSnapshotSubvolumn13,
        SubvolId: &subvolIdSubvolumn13,
        ParentId: &parentIdSubvolumn13,
        SubvolName: &subvolNameSubvolumn13,
        SubvolMountPath: &subvolMountPathSubvolumn12,
    },
    {
        Uuid: &uuidSubvolumn14,
        IsSnapshot: &isSnapshotSubvolumn14,
        SubvolId: &subvolIdSubvolumn14,
        ParentId: &parentIdSubvolumn14,
        SubvolName: &subvolNameSubvolumn14,
        SubvolMountPath: &subvolMountPathSubvolumn13,
    },
    {
        Uuid: &uuidSubvolumn15,
        IsSnapshot: &isSnapshotSubvolumn15,
        SubvolId: &subvolIdSubvolumn15,
        ParentId: &parentIdSubvolumn15,
        SubvolName: &subvolNameSubvolumn15,
        SubvolMountPath: &subvolMountPathSubvolumn14,
    },
    {
        Uuid: &uuidSubvolumn16,
        IsSnapshot: &isSnapshotSubvolumn16,
        SubvolId: &subvolIdSubvolumn16,
        ParentId: &parentIdSubvolumn16,
        SubvolName: &subvolNameSubvolumn16,
        SubvolMountPath: &subvolMountPathSubvolumn15,
    },
    {
        Uuid: &uuidSubvolumn17,
        IsSnapshot: &isSnapshotSubvolumn17,
        SubvolId: &subvolIdSubvolumn17,
        ParentId: &parentIdSubvolumn17,
        SubvolName: &subvolNameSubvolumn17,
        SubvolMountPath: &subvolMountPathSubvolumn16,
    },
    {
        Uuid: &uuidSubvolumn18,
        IsSnapshot: &isSnapshotSubvolumn18,
        SubvolId: &subvolIdSubvolumn18,
        ParentId: &parentIdSubvolumn18,
        SubvolName: &subvolNameSubvolumn18,
        SubvolMountPath: &subvolMountPathSubvolumn17,
    },
    {
        Uuid: &uuidSubvolumn19,
        IsSnapshot: &isSnapshotSubvolumn19,
        SubvolId: &subvolIdSubvolumn19,
        ParentId: &parentIdSubvolumn19,
        SubvolName: &subvolNameSubvolumn19,
        SubvolMountPath: &subvolMountPathSubvolumn18,
    },
    {
        Uuid: &uuidSubvolumn20,
        IsSnapshot: &isSnapshotSubvolumn20,
        SubvolId: &subvolIdSubvolumn20,
        ParentId: &parentIdSubvolumn20,
```

```
            SubvolName: &subvolNameSubvolumn20,
            SubvolMountPath: &subvolMountPathSubvolumn19,
        },
        {
            Uuid: &uuidSubvolumn21,
            IsSnapshot: &isSnapshotSubvolumn21,
            SubvolId: &subvolIdSubvolumn21,
            ParentId: &parentIdSubvolumn21,
            SubvolName: &subvolNameSubvolumn21,
            SubvolMountPath: &subvolMountPathSubvolumn20,
        },
        {
            Uuid: &uuidSubvolumn22,
            IsSnapshot: &isSnapshotSubvolumn22,
            SubvolId: &subvolIdSubvolumn22,
            ParentId: &parentIdSubvolumn22,
            SubvolName: &subvolNameSubvolumn22,
        },
    }
    nameBtrfsList:= "/dev/vda2"
    labelBtrfsList:= "none"
    uuidBtrfsList:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
    deviceBtrfsList:= "/dev/vda2"
    sizeBtrfsList:= int64(3538944000)
    nodesizeBtrfsList:= int64(16384)
    sectorsizeBtrfsList:= int32(4096)
    dataProfileBtrfsList:= "single"
    systemProfileBtrfsList:= "single"
    metadataProfileBtrfsList:= "single"
    globalReserve1BtrfsList:= "single"
    gVolUsedSizeBtrfsList:= int64(3894038528)
    defaultSubvolidBtrfsList:= "259"
    defaultSubvolNameBtrfsList:= "@/.snapshots/1/snapshot"
    defaultSubvolMountpathBtrfsList:= "/"
    var listBtrfsListbody = []model.BtrfsFileSystem{
        {
            Name: &nameBtrfsList,
            Label: &labelBtrfsList,
            Uuid: &uuidBtrfsList,
            Device: &deviceBtrfsList,
            Size: &sizeBtrfsList,
            Nodesize: &nodesizeBtrfsList,
            Sectorsize: &sectorsizeBtrfsList,
            DataProfile: &dataProfileBtrfsList,
            SystemProfile: &systemProfileBtrfsList,
            MetadataProfile: &metadataProfileBtrfsList,
            GlobalReserve1: &globalReserve1BtrfsList,
            GVolUsedSize: &gVolUsedSizeBtrfsList,
            DefaultSubvolid: &defaultSubvolidBtrfsList,
            DefaultSubvolName: &defaultSubvolNameBtrfsList,
            DefaultSubvolMountpath: &defaultSubvolMountpathBtrfsList,
            Subvolumn: &listSubvolumnBtrfsList,
        },
    }
    fileSystemLogicalVolumes:= "ext4"
    inodeSizeLogicalVolumes:= int32(256)
    deviceUseLogicalVolumes:= "NORMAL"
    mountPointLogicalVolumes:= "/mnt/lv1"
    nameLogicalVolumes:= "/dev/mapper/vg1-lv1"
    sizeLogicalVolumes:= int64(10737418240)
    usedSizeLogicalVolumes:= int64(23580672)
    freeSizeLogicalVolumes:= int64(10713837568)
    var listLogicalVolumesVolumegroups = []model.LogicalVolumes{
        {
            FileSystem: &fileSystemLogicalVolumes,
            InodeSize: &inodeSizeLogicalVolumes,
            DeviceUse: &deviceUseLogicalVolumes,
            MountPoint: &mountPointLogicalVolumes,
            Name: &nameLogicalVolumes,
```

```
            Size: &sizeLogicalVolumes,
            UsedSize: &usedSizeLogicalVolumes,
            FreeSize: &freeSizeLogicalVolumes,
        },
    }
    componentsVolumegroups:= "/dev/vdb1;/dev/vdc"
    nameVolumegroups:= "vg1"
    sizeVolumegroups:= int64(42948624384)
    var listVolumegroupsbody = []model.VolumeGroups{
        {
            Components: &componentsVolumegroups,
            LogicalVolumes: &listLogicalVolumesVolumegroups,
            Name: &nameVolumegroups,
            Size: &sizeVolumegroups,
        },
    }
    deviceUsePhysicalVolumes:= "VOLUME_GROUP"
    fileSystemPhysicalVolumes:= "LVM2_member"
    mountPointPhysicalVolumes:= ""
    namePhysicalVolumes:= "/dev/vdb1"
    sizePhysicalVolumes:= int64(21473787904)
    usedSizePhysicalVolumes:= int64(21473787904)
    inodeSizePhysicalVolumes:= int32(0)
    var listPhysicalVolumesDisks = []model.PhysicalVolume{
        {
            DeviceUse: &deviceUsePhysicalVolumes,
            FileSystem: &fileSystemPhysicalVolumes,
            MountPoint: &mountPointPhysicalVolumes,
            Name: &namePhysicalVolumes,
            Size: &sizePhysicalVolumes,
            UsedSize: &usedSizePhysicalVolumes,
            InodeSize: &inodeSizePhysicalVolumes,
        },
    }
    deviceUsePhysicalVolumes1:= "NORMAL"
    fileSystemPhysicalVolumes1:= "swap"
    mountPointPhysicalVolumes1:= ""
    namePhysicalVolumes1:= "/dev/vda1"
    sizePhysicalVolumes1:= int64(2153775104)
    usedSizePhysicalVolumes1:= int64(2153775104)
    inodeSizePhysicalVolumes1:= int32(0)
    deviceUsePhysicalVolumes2:= "BTRFS"
    fileSystemPhysicalVolumes2:= "btrfs"
    mountPointPhysicalVolumes2:= ""
    namePhysicalVolumes2:= "/dev/vda2"
    sizePhysicalVolumes2:= int64(16862150656)
    usedSizePhysicalVolumes2:= int64(16862150656)
    inodeSizePhysicalVolumes2:= int32(0)
    deviceUsePhysicalVolumes3:= "NORMAL"
    fileSystemPhysicalVolumes3:= "xfs"
    mountPointPhysicalVolumes3:= "/home"
    namePhysicalVolumes3:= "/dev/vda3"
    sizePhysicalVolumes3:= int64(23932698624)
    usedSizePhysicalVolumes3:= int64(33988608)
    inodeSizePhysicalVolumes3:= int32(0)
    var listPhysicalVolumesDisks1 = []model.PhysicalVolume{
        {
            DeviceUse: &deviceUsePhysicalVolumes1,
            FileSystem: &fileSystemPhysicalVolumes1,
            MountPoint: &mountPointPhysicalVolumes1,
            Name: &namePhysicalVolumes1,
            Size: &sizePhysicalVolumes1,
            UsedSize: &usedSizePhysicalVolumes1,
            InodeSize: &inodeSizePhysicalVolumes1,
        },
        {
            DeviceUse: &deviceUsePhysicalVolumes2,
            FileSystem: &fileSystemPhysicalVolumes2,
            MountPoint: &mountPointPhysicalVolumes2,
```

```
            Name: &namePhysicalVolumes2,
            Size: &sizePhysicalVolumes2,
            UsedSize: &usedSizePhysicalVolumes2,
            InodeSize: &inodeSizePhysicalVolumes2,
        },
        {
            DeviceUse: &deviceUsePhysicalVolumes3,
            FileSystem: &fileSystemPhysicalVolumes3,
            MountPoint: &mountPointPhysicalVolumes3,
            Name: &namePhysicalVolumes3,
            Size: &sizePhysicalVolumes3,
            UsedSize: &usedSizePhysicalVolumes3,
            InodeSize: &inodeSizePhysicalVolumes3,
        },
    }
    nameDisks:= "/dev/vda"
    partitionStyleDisks:= "MBR"
    deviceUseDisks:= "BOOT"
    sizeDisks:= int64(42949672960)
    usedSizeDisks:= int64(42948624384)
    nameDisks1:= "/dev/vdb"
    partitionStyleDisks1:= "MBR"
    deviceUseDisks1:= "NORMAL"
    sizeDisks1:= int64(21474836480)
    usedSizeDisks1:= int64(21473787904)
    nameDisks2:= "/dev/vdc"
    partitionStyleDisks2:= "MBR"
    deviceUseDisks2:= "VOLUME_GROUP"
    sizeDisks2:= int64(21474836480)
    usedSizeDisks2:= int64(0)
    var listDisksbody = []model.ServerDisk{
        {
            Name: &nameDisks,
            PartitionStyle: &partitionStyleDisks,
            DeviceUse: &deviceUseDisks,
            Size: &sizeDisks,
            UsedSize: &usedSizeDisks,
            PhysicalVolumes: &listPhysicalVolumesDisks1,
        },
        {
            Name: &nameDisks1,
            PartitionStyle: &partitionStyleDisks1,
            DeviceUse: &deviceUseDisks1,
            Size: &sizeDisks1,
            UsedSize: &usedSizeDisks1,
            PhysicalVolumes: &listPhysicalVolumesDisks,
        },
        {
            Name: &nameDisks2,
            PartitionStyle: &partitionStyleDisks2,
            DeviceUse: &deviceUseDisks2,
            Size: &sizeDisks2,
            UsedSize: &usedSizeDisks2,
        },
    }
    request.Body = &model.PutDiskInfoReq{
        BtrfsList: &listBtrfsListbody,
        Volumegroups: &listVolumegroupsbody,
        Disks: &listDisksbody,
    }
    response, err := client.UpdateDiskInfo(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

More

For SDK sample code of more programming languages, see the Sample Code tab in **API Explorer**. SDK sample code can be automatically generated.

Status Codes

Status Code	Description
200	The disk information was updated.
403	Authentication failed.

Error Codes

See **Error Codes**.

5.4 Task Management

5.4.1 Creating a Migration Task

Function

This API is used to create a migration task for a source server.

Calling Method

For details, see **Calling APIs**.

Authorization Information

Each account has all the permissions required to call all APIs, but IAM users must be assigned the required permissions.

- If you are using role/policy-based authorization, see **Permissions Policies and Supported Actions** for details on the required permissions.
- If you are using identity policy-based authorization, the following identity policy-based permissions are required.

Action	Access Level	Resource Type (*: required)	Condition Key	Alias	Dependencies
sms:server:createTask	Write	server *	-	sms:server:migrationServer	-
		-	g:EnterpriseProjectId		

URI

POST /v3/tasks

Request Parameters

Table 5-109 Request header parameters

Parameter	Mandatory	Type	Description
X-Auth-Token	Yes	String	User token. The token can be obtained by calling the IAM API for obtaining a user token. The value of **X-Subject-Token** in the response header is the user token. Minimum: **1** Maximum: **16384**

Table 5-110 Request body parameters

Parameter	Mandatory	Type	Description
name	Yes	String	Task name. The value can contain only letters, digits, underscores (_), and hyphens (-). Minimum: **1** Maximum: **64**
type	Yes	String	The task type. **MIGRATE_FILE** **MIGRATE_BLOCK** Enumeration values: • **MIGRATE_FILE** • **MIGRATE_BLOCK**
start_target_server	No	Boolean	Indicates whether the target server is started after the migration is complete. Default: **true**
auto_start	No	Boolean	Indicates whether to automatically start the migration.

Parameter	Mandatory	Type	Description
os_type	No	String	The OS type. **WINDOWS** **LINUX** Enumeration values: • **WINDOWS** • **LINUX**
source_server	Yes	**SourceServerByTask** object	The source server must exist, be connected, and does not have migration tasks.
target_server	Yes	**TargetServerByTask** object	The target server information.
migration_ip	No	String	The IP address used for migration. This parameter is not required if the target server is automatically created. If **use_ipv6** is set to **false**, **migration_ip** must be in the IPv4 format. If **use_ipv6** is set to **true**, **migration_ip** must be in the IPv6 format. Minimum: **0** Maximum: **255**
region_name	Yes	String	The region name. Minimum: **0** Maximum: **255**
region_id	Yes	String	region ID Minimum: **0** Maximum: **255**
project_name	Yes	String	The project name. Minimum: **0** Maximum: **255**
project_id	Yes	String	The project ID. Minimum: **0** Maximum: **255**
priority	No	Integer	Priority. The default value is **1**. Minimum: **0** Maximum: **65535**

Parameter	Mandatory	Type	Description
vm_template_id	No	String	The template used to create target servers automatically. Minimum: **0** Maximum: **100**
clonevm_template_id	No	String	The ID of the template for cloning the target server. Minimum: **0** Maximum: **100**
use_public_ip	No	Boolean	Indicates whether a public IP address is used for migration. Default: **true**
use_ipv6	No	Boolean	Indicates whether IPv6 is used. Default: **false**
syncing	No	Boolean	Indicates whether to perform a continuous synchronization after the first replication. If this parameter is not specified, the default value **false** is used. Default: **false**
exist_server	No	Boolean	Checks whether the service exists. If yes, create the task.
start_network_check	No	Boolean	Indicates whether network performance measurement is enabled.
speed_limit	No	Integer	The migration speed limit. Minimum: **0** Maximum: **1000** Default: **0**

Parameter	Mandatory	Type	Description
over_speed_threshold	No	Double	The overspeed threshold for stopping migration. This is a protection measure. If the migration speed exceeds the threshold, the task is stopped. It is used to control the consumption of resources (especially network bandwidth) during the migration to ensure that the overall system performance is not affected by a single migration task. The unit is percentage. Minimum: **10** Maximum: **100** Default: **10**
is_need_consistency_check	No	Boolean	Indicates whether consistency verification is enabled.
need_migration_test	No	Boolean	Indicates whether migration drilling is enabled.

Table 5-111 SourceServerByTask

Parameter	Mandatory	Type	Description
id	Yes	String	The source server ID. Minimum: **0** Maximum: **255**

Table 5-112 TargetServerByTask

Parameter	Mandatory	Type	Description
btrfs_list	No	Array of **BtrfsFileSystem** objects	The Btrfs information, which is obtained from the source server. Array Length: **0 - 65535**
disks	No	Array of **TargetDisks** objects	The disk information. Array Length: **0 - 50**

Parameter	Mandatory	Type	Description
name	Yes	String	Name. Minimum: **0** Maximum: **255**
vm_id	Yes	String	VM ID Minimum: **0** Maximum: **255**
volume_groups	No	Array of **VolumeGroups** objects	The volume group information, which is obtained from the source server. Array Length: **0 - 50**

Table 5-113 BtrfsFileSystem

Parameter	Mandatory	Type	Description
name	No	String	The file system name. Minimum: **0** Maximum: **255**
label	No	String	The file system tags. If no tag exists, the value is an empty string. Minimum: **0** Maximum: **255**
uuid	No	String	The file system UUID. Minimum: **0** Maximum: **255**
device	No	String	The device names of the Btrfs file system. Minimum: **0** Maximum: **255**
size	No	Long	The space occupied by the file system data. Minimum: **0** Maximum: **9223372036854775807**
nodesize	No	Long	The Btrfs node size. Minimum: **0** Maximum: **9223372036854775807**

Parameter	Mandatory	Type	Description
sectorsize	No	Integer	The sector size. Minimum: **0** Maximum: **2147483647**
data_profile	No	String	The data profile (RAD). Minimum: **0** Maximum: **255**
system_profile	No	String	The file system profile (RAD). Minimum: **0** Maximum: **255**
metadata_profile	No	String	The metadata profile (RAD). Minimum: **0** Maximum: **255**
global_reserve1	No	String	The Btrfs file system information. Minimum: **0** Maximum: **255**
g_vol_used_size	No	Long	The used space of the Btrfs volume. Minimum: **0** Maximum: **9223372036854775807**
default_subvolid	No	String	The ID of the default subvolume. Minimum: **0** Maximum: **255**
default_subvol_name	No	String	The name of the default subvolume. Minimum: **0** Maximum: **255**
default_subvol_mountpath	No	String	The mount path of the default subvolume or Btrfs file system. Minimum: **0** Maximum: **255**
subvolumn	No	Array of **BtrfsSubvolumn** objects	The subvolume information. Array Length: **0 - 65535**

Table 5-114 BtrfsSubvolumn

Parameter	Mandatory	Type	Description
uuid	No	String	The UUID of the parent volume. Minimum: **0** Maximum: **255**
is_snapshot	No	String	Indicates whether the subvolume is a snapshot. Minimum: **0** Maximum: **255**
subvol_id	No	String	The subvolume ID. Minimum: **0** Maximum: **255**
parent_id	No	String	The parent volume ID. Minimum: **0** Maximum: **255**
subvol_name	No	String	The subvolume name. Minimum: **0** Maximum: **255**
subvol_mount_path	No	String	The mount path of the subvolume. Minimum: **0** Maximum: **255**

Table 5-115 TargetDisks

Parameter	Mandatory	Type	Description
device_use	No	String	Disk type, which is only used for information display and is not verified. **BOOT**: disk where the boot partition is located. **OS**: disk where the OS is located. **NORMAL**: common disk. Default: **NORMAL** Enumeration values: - **NORMAL** - **OS** - **BOOT**
disk_id	No	String	The disk ID. This parameter is not required if the target server is automatically created. Minimum: **0** Maximum: **255**
name	No	String	The disk name. Set this parameter to **Disk X** based on the disk sequence. Minimum: **0** Maximum: **255**
physical_volumes	No	Array of **PhysicalVolumes** objects	The physical volume information. Array Length: **0 - 50**
size	No	Long	The size. Minimum: **0** Maximum: **9223372036854775807**
used_size	No	Long	The used space. Minimum: **0** Maximum: **9223372036854775807**

Table 5-116 PhysicalVolumes

Parameter	Mandatory	Type	Description
device_use	No	String	The partition function. The partition can be a general, boot, or OS partition. Minimum: **0** Maximum: **255**
file_system	No	String	The file system type. Minimum: **0** Maximum: **255**
index	No	Integer	The serial number. Minimum: **0** Maximum: **2147483647**
mount_point	No	String	The mount point. Minimum: **0** Maximum: **255**
name	No	String	The volume name. In Windows, it indicates the drive letter, and in Linux, it indicates the device ID. Minimum: **0** Maximum: **255**
size	No	Long	The size. Minimum: **0** Maximum: **9223372036854775807**
inode_size	No	Long	The number of inodes. Minimum: **0** Maximum: **9223372036854775807**
used_size	No	Long	The used space. Minimum: **0** Maximum: **9223372036854775807**
uuid	No	String	The GUID, which can be obtained from the source server. Minimum: **0** Maximum: **255**

Table 5-117 VolumeGroups

Parameter	Mandatory	Type	Description
components	No	String	The physical volume information. Minimum: **0** Maximum: **255**
free_size	No	Long	The available space. Minimum: **0** Maximum: **9223372036854775807**
logical_volumes	No	Array of **LogicalVolumes** objects	The logical volume information. Array Length: **0 - 50**
name	No	String	Name. Minimum: **0** Maximum: **255**
size	No	Long	Size. Minimum: **0** Maximum: **9223372036854775807**

Table 5-118 LogicalVolumes

Parameter	Mandatory	Type	Description
block_count	No	Integer	The number of blocks. Minimum: **0** Maximum: **2147483647** Default: **0**
block_size	No	Long	Block size. Minimum: **0** Maximum: **1048576** Default: **0**
file_system	No	String	The file system. Minimum: **0** Maximum: **255**
inode_size	No	Integer	The number of inodes. Minimum: **0** Maximum: **2147483647**

Parameter	Mandatory	Type	Description
inode_nums	No	Long	The number of inodes. Minimum: **0** Maximum: **9223372036854775807**
device_use	No	String	Partition type. This is not mandatory and can be left blank. The common values are as follows: **NORMAL** **OS**: system device **BOOT** **VOLUME_GROUP** **BTRFS** Minimum: **0** Maximum: **255**
mount_point	No	String	The mount point. Minimum: **0** Maximum: **256**
name	No	String	Name. Minimum: **0** Maximum: **1024**
size	No	Long	The size. Minimum: **0** Maximum: **9223372036854775807**
used_size	No	Long	The used space. Minimum: **0** Maximum: **9223372036854775807**
free_size	No	Long	The available space. Minimum: **0** Maximum: **9223372036854775807**

Response Parameters

Status code: 200

Table 5-119 Response body parameters

Parameter	Type	Description
id	String	The task ID returned when the task is created successfully. Minimum: **0** Maximum: **255**

Status code: 403

Table 5-120 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **255**
encoded_authorization_message	String	The encrypted authorization information. Minimum: **0** Maximum: **65535**
error_param	Array of strings	Error parameters. Minimum: **0** Maximum: **65535** Array Length: **1 - 20**
details	Array of **details** objects	The error details. Array Length: **1 - 20**

Table 5-121 details

Parameter	Type	Description
error_code	String	The SMS error code. Minimum: **0** Maximum: **65535**

Parameter	Type	Description
error_msg	String	The SMS error message. Minimum: **0** Maximum: **65535**

Example Requests

This example creates a migration task whose name is **MigrationTask**. The migration type is **MIGRATE_FILE**, the source server runs Linux, the migration uses the public network, the migration region name is **region_name**, and the migration region ID is **region_id**.

```
POST https://{endpoint}/v3/tasks

{
  "name" : "MigrationTask",
  "type" : "MIGRATE_FILE",
  "os_type" : "LINUX",
  "start_target_server" : true,
  "use_public_ip" : true,
  "migration_ip" : "192.168.0.1",
  "region_name" : "region_name",
  "region_id" : "region_id",
  "project_name" : "project_name",
  "project_id" : "xxxxxxxxxxxxxxxxxxxxxxxx00000001",
  "source_server" : {
    "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
  },
  "target_server" : {
    "vm_id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "name" : "Auto-tar-name",
    "disks" : [ {
      "name" : "/dev/vda",
      "disk_id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
      "size" : 42949672960,
      "used_size" : 429496,
      "device_use" : "BOOT",
      "physical_volumes" : [ {
        "uuid" : null,
        "index" : 0,
        "name" : "/dev/vda1",
        "device_use" : "OS",
        "file_system" : "ext4",
        "mount_point" : "/",
        "size" : 42947575808,
        "used_size" : 5346484224
      } ]
    } ],
    "volume_groups" : [ ]
  },
  "is_need_consistency_check" : true
}
```

Example Responses

Status code: 200

The migration task was successfully created.

```
{
  "id" : "xxxxxxxxxxxxxxxxxxxxxxxx00000001"
}
```

Status code: 403

Authentication failed.

```
{
  "error_code" : "SMS.9004",
  "error_msg" : "The current account does not have the permission to execute policy. You do not have permission to perform action XXX on resource XXX.",
  "encoded_authorization_message" : "XXXXXX",
  "error_param" : [ "You do not have permission to perform action XXX on resource XXX." ],
  "details" : [ {
    "error_code" : "SMS.9004",
    "error_msg" : "You do not have permission to perform action XXX on resource XXX."
  } ]
}
```

SDK Sample Code

The SDK sample code is as follows.

Java

This example creates a migration task whose name is **MigrationTask**. The migration type is **MIGRATE_FILE**, the source server runs Linux, the migration uses the public network, the migration region name is **region_name**, and the migration region ID is **region_id**.

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;

import java.util.List;
import java.util.ArrayList;

public class CreateTaskSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        CreateTaskRequest request = new CreateTaskRequest();
        PostTask body = new PostTask();
```



```
        List<PhysicalVolumes> listDisksPhysicalVolumes = new ArrayList<>();
        listDisksPhysicalVolumes.add(
            new PhysicalVolumes()
                .withDeviceUse("OS")
                .withFileSystem("ext4")
                .withIndex(0)
                .withMountPoint("/")
                .withName("/dev/vda1")
                .withSize(42947575808L)
                .withUsedSize(5346484224L)
        );
        List<TargetDisks> listTargetServerDisks = new ArrayList<>();
        listTargetServerDisks.add(
            new TargetDisks()
                .withDeviceUse(TargetDisks.DeviceUseEnum.fromValue("BOOT"))
                .withDiskId("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
                .withName("/dev/vda")
                .withPhysicalVolumes(listDisksPhysicalVolumes)
                .withSize(42949672960L)
                .withUsedSize(429496L)
        );
        TargetServerByTask targetServerbody = new TargetServerByTask();
        targetServerbody.withDisks(listTargetServerDisks)
            .withName("Auto-tar-name")
            .withVmId("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001");
        SourceServerByTask sourceServerbody = new SourceServerByTask();
        sourceServerbody.withId("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001");
        body.withIsNeedConsistencyCheck(true);
        body.withUsePublicIp(true);
        body.withProjectId("xxxxxxxxxxxxxxxxxxxxxxxx00000001");
        body.withProjectName("project_name");
        body.withRegionId("region_id");
        body.withRegionName("region_name");
        body.withMigrationIp("192.168.0.1");
        body.withTargetServer(targetServerbody);
        body.withSourceServer(sourceServerbody);
        body.withOsType(PostTask.OsTypeEnum.fromValue("LINUX"));
        body.withStartTargetServer(true);
        body.withType(PostTask.TypeEnum.fromValue("MIGRATE_FILE"));
        body.withName("MigrationTask");
        request.withBody(body);
        try {
            CreateTaskResponse response = client.createTask(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

Python

This example creates a migration task whose name is **MigrationTask**. The migration type is **MIGRATE_FILE**, the source server runs Linux, the migration uses the public network, the migration region name is **region_name**, and the migration region ID is **region_id**.

```
# coding: utf-8

import os
```

```
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = CreateTaskRequest()
        listPhysicalVolumesDisks = [
            PhysicalVolumes(
                device_use="OS",
                file_system="ext4",
                index=0,
                mount_point="/",
                name="/dev/vda1",
                size=42947575808,
                used_size=5346484224
            )
        ]
        listDisksTargetServer = [
            TargetDisks(
                device_use="BOOT",
                disk_id="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
                name="/dev/vda",
                physical_volumes=listPhysicalVolumesDisks,
                size=42949672960,
                used_size=429496
            )
        ]
        targetServerbody = TargetServerByTask(
            disks=listDisksTargetServer,
            name="Auto-tar-name",
            vm_id="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
        )
        sourceServerbody = SourceServerByTask(
            id="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
        )
        request.body = PostTask(
            is_need_consistency_check=True,
            use_public_ip=True,
            project_id="xxxxxxxxxxxxxxxxxxxxxxxx00000001",
            project_name="project_name",
            region_id="region_id",
            region_name="region_name",
            migration_ip="192.168.0.1",
            target_server=targetServerbody,
            source_server=sourceServerbody,
            os_type="LINUX",
            start_target_server=True,
            type="MIGRATE_FILE",
            name="MigrationTask"
        )
        response = client.create_task(request)
        print(response)
```

```
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

Go

This example creates a migration task whose name is **MigrationTask**. The migration type is **MIGRATE_FILE**, the source server runs Linux, the migration uses the public network, the migration region name is **region_name**, and the migration region ID is **region_id**.

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
         WithRegion(region.ValueOf("<YOUR REGION>")).
         WithCredential(auth).
         SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.CreateTaskRequest{}
    deviceUsePhysicalVolumes:= "OS"
    fileSystemPhysicalVolumes:= "ext4"
    indexPhysicalVolumes:= int32(0)
    mountPointPhysicalVolumes:= "/"
    namePhysicalVolumes:= "/dev/vda1"
    sizePhysicalVolumes:= int64(42947575808)
    usedSizePhysicalVolumes:= int64(5346484224)
    var listPhysicalVolumesDisks = []model.PhysicalVolumes{
        {
            DeviceUse: &deviceUsePhysicalVolumes,
            FileSystem: &fileSystemPhysicalVolumes,
            Index: &indexPhysicalVolumes,
```

```
            MountPoint: &mountPointPhysicalVolumes,
            Name: &namePhysicalVolumes,
            Size: &sizePhysicalVolumes,
            UsedSize: &usedSizePhysicalVolumes,
        },
    }
    deviceUseDisks:= model.GetTargetDisksDeviceUseEnum().BOOT
    diskIdDisks:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
    nameDisks:= "/dev/vda"
    sizeDisks:= int64(42949672960)
    usedSizeDisks:= int64(429496)
    var listDisksTargetServer = []model.TargetDisks{
        {
            DeviceUse: &deviceUseDisks,
            DiskId: &diskIdDisks,
            Name: &nameDisks,
            PhysicalVolumes: &listPhysicalVolumesDisks,
            Size: &sizeDisks,
            UsedSize: &usedSizeDisks,
        },
    }
    targetServerbody := &model.TargetServerByTask{
        Disks: &listDisksTargetServer,
        Name: "Auto-tar-name",
        VmId: "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    }
    sourceServerbody := &model.SourceServerByTask{
        Id: "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    }
    isNeedConsistencyCheckPostTask:= true
    usePublicIpPostTask:= true
    migrationIpPostTask:= "192.168.0.1"
    osTypePostTask:= model.GetPostTaskOsTypeEnum().LINUX
    startTargetServerPostTask:= true
    request.Body = &model.PostTask{
        IsNeedConsistencyCheck: &isNeedConsistencyCheckPostTask,
        UsePublicIp: &usePublicIpPostTask,
        ProjectId: "xxxxxxxxxxxxxxxxxxxxxxxx00000001",
        ProjectName: "project_name",
        RegionId: "region_id",
        RegionName: "region_name",
        MigrationIp: &migrationIpPostTask,
        TargetServer: targetServerbody,
        SourceServer: sourceServerbody,
        OsType: &osTypePostTask,
        StartTargetServer: &startTargetServerPostTask,
        Type: model.GetPostTaskTypeEnum().MIGRATE_FILE,
        Name: "MigrationTask",
    }
    response, err := client.CreateTask(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

More

For SDK sample code of more programming languages, see the Sample Code tab in **API Explorer**. SDK sample code can be automatically generated.

Status Codes

Status Code	Description
200	The migration task was successfully created.
403	Authentication failed.

Error Codes

See **Error Codes**.

5.4.2 Listing Migration Tasks

Function

After the target server is configured for a source server, SMS automatically creates a migration task. This API is used to list all your migration tasks.

Calling Method

For details, see **Calling APIs**.

Authorization Information

Each account has all the permissions required to call all APIs, but IAM users must be assigned the required permissions.

- If you are using role/policy-based authorization, see **Permissions Policies and Supported Actions** for details on the required permissions.
- If you are using identity policy-based authorization, the following identity policy-based permissions are required.

Action	Access Level	Resource Type (*: required)	Condition Key	Alias	Dependencies
sms:server:listTask	List	server *	-	sms:server:queryServer	-
		-	g:EnterpriseProjectId		

URI

GET /v3/tasks

Table 5-122 Query Parameters

Parameter	Mandatory	Type	Description
state	No	String	The migration task status. **READY**: indicates that the migration task is ready for execution. **RUNNING**: indicates that the migration task is being executed. **SYNCING**: indicates that the incremental data is being synchronized. **MIGRATE_SUCCESS**: indicates that the migration succeeds. **MIGRATE_FAIL**: indicates that the migration fails. **ABORTING**: The migration task is being stopped. **ABORT**: The migration task is stopped. **DELETING**: indicates that the task is being deleted. **SYNC_F_ROLLBACKING**: indicates that the synchronization fails and the task is being rolled back. **SYNC_F_ROLLBACK_SUCCESS**: indicates that the synchronization fails and the rollback is successful. Enumeration values: • **READY** • **RUNNING** • **SYNCING** • **MIGRATE_SUCCESS** • **MIGRATE_FAIL** • **ABORTING** • **ABORT** • **DELETING** • **SYNC_F_ROLLBACKING** • **SYNC_F_ROLLBACK_SUCCESS**

Parameter	Mandatory	Type	Description
name	No	String	The task name. Minimum: **0** Maximum: **255**
id	No	String	The task ID. Minimum: **1** Maximum: **36**
source_server_id	No	String	The source server ID. Minimum: **1** Maximum: **36**
limit	No	Integer	The number of tasks recorded on each page. Minimum: **0** Maximum: **200** Default: **100**
offset	No	Integer	The offset. Minimum: **0** Maximum: **65535** Default: **0**
enterprise_project_id	No	String	The ID of the enterprise project to be queried. Minimum: **0** Maximum: **255**

Request Parameters

Table 5-123 Request header parameters

Parameter	Mandatory	Type	Description
X-Auth-Token	Yes	String	User token. The token can be obtained by calling the IAM API for obtaining a user token. The value of **X-Subject-Token** in the response header is the user token. Minimum: **1** Maximum: **16384**

Response Parameters

Status code: 200

Table 5-124 Response body parameters

Parameter	Type	Description
count	Integer	The number of tasks that meet the search criteria, which is not affected by pagination. Minimum: **0** Maximum: **2147483647**
tasks	Array of **TasksResponseBody** objects	The list of queried tasks. Array Length: **0 - 65535**

Table 5-125 TasksResponseBody

Parameter	Type	Description
id	String	The migration task ID. Minimum: **0** Maximum: **255**
name	String	The task name, which is defined by the user. Minimum: **0** Maximum: **255**
type	String	The task type. This parameter is mandatory for creating a task and optional for updating a task. **MIGRATE_FILE** **MIGRATE_BLOCK** Minimum: **0** Maximum: **255** Enumeration values: ● **MIGRATE_FILE** ● **MIGRATE_BLOCK**

Parameter	Type	Description
os_type	String	The OS type, which can be Windows or Linux. This parameter is mandatory for creating a task and optional for updating a task. Minimum: **0** Maximum: **255** Enumeration values: ● **WINDOWS** ● **LINUX**
state	String	The migration task status. **READY** **RUNNING** **SYNCING** **MIGRATE_SUCCESS** **SYNC_SUCCESS** **MIGRATE_FAIL** **SYNC_FAIL** **ABORTING** **ABORT** **SKIPPING** **DELETING** **RESETING** Enumeration values: ● **READY** ● **RUNNING** ● **SYNCING** ● **MIGRATE_SUCCESS** ● **SYNC_SUCCESS** ● **MIGRATE_FAIL** ● **SYNC_FAIL** ● **ABORTING** ● **ABORT** ● **SKIPPING** ● **DELETING** ● **RESETING**
estimate_complete_time	Long	The estimated completion time. Minimum: **0** Maximum: **9223372036854775807**

Parameter	Type	Description
create_date	Long	The task creation time. Minimum: **0** Maximum: **9223372036854775807**
priority	Integer	The migration process priority. **0**: low **1**: standard **2**: high Minimum: **0** Maximum: **2** Enumeration values: • **0** • **1** • **2**
speed_limit	Integer	The migration rate limit. Minimum: **0** Maximum: **65535**
migrate_speed	Double	The migration rate, in Mbit/s. Minimum: **0** Maximum: **10000**
compress_rate	Double	The compression rate. Minimum: **0** Maximum: **10000**
start_target_server	Boolean	Indicates whether the target server is started after the migration is complete. **true**: yes **false**: no Default: **false**
error_json	String	The error message. Minimum: **0** Maximum: **1024**
total_time	Long	The task duration. Minimum: **0** Maximum: **9223372036854775807**

Parameter	Type	Description
migration_ip	String	The IP address of the target server. If you use a public network for migration, enter the EIP bound to the target server. If you use a private line for migration, enter a private IP address of the target server. Minimum: **0** Maximum: **255**
sub_tasks	Array of **SubTaskAssociatedWithTask** objects	The information about subtasks associated with the migration task. Array Length: **0 - 65535**
source_server	**SourceServerAssociatedWithTask** object	The information about the source server associated with the migration task.
enterprise_project_id	String	The migration project ID. Minimum: **0** Maximum: **255**
target_server	**TargetServerAssociatedWithTask** object	The information about the target server associated with the migration task.
log_collect_status	String	The log collection status. **INIT** (ready) **UPLOADING** **UPLOAD_FAIL** **UPLOADED** Enumeration values: • **INIT** • **UPLOADING** • **UPLOAD_FAIL** • **UPLOADED**
clone_server	**CloneServerBrief** object	The information about the cloned server.
syncing	Boolean	Indicates whether synchronization is enabled.
network_check_info	**NetworkCheckInfoRequestBody** object	The network measurement results.

Parameter	Type	Description
special_config	Array of **ConfigBody** objects	The configuration information of advanced migration options. Array Length: **0 - 1000**
total_cpu_usage	Double	The CPU usage of the server, in percentage. Minimum: **0** Maximum: **100**
agent_cpu_usage	Double	The CPU usage of the Agent, in percentage. Minimum: **0** Maximum: **100**
total_mem_usage	Double	The memory usage of the server, in MB. Minimum: **0** Maximum: **1048576.0**
agent_mem_usage	Double	The memory usage of the Agent, in MB. Minimum: **0** Maximum: **1048576.0**
total_disk_io	Double	The disk I/O of the server, in Mbit/s. Minimum: **0** Maximum: **10000.0**
agent_disk_io	Double	The disk I/O of the Agent, in Mbit/s. Minimum: **0** Maximum: **10000.0**
need_migration_test	Boolean	Indicates whether migration drilling is enabled.
subtask_info	String	The current subtask and progress. Minimum: **0** Maximum: **255**

Table 5-126 SubTaskAssociatedWithTask

Parameter	Type	Description
id	Long	The subtask ID. Minimum: **0** Maximum: **9223372036854775807**

Parameter	Type	Description
name	String	The subtask name. Minimum: **0** Maximum: **255**
progress	Integer	The progress of the subtask. The value is an integer ranging from 0 to 100. Minimum: **0** Maximum: **100**
start_date	Long	The start time of the subtask. Minimum: **0** Maximum: **9223372036854775807**
end_date	Long	The end time of the subtask. If the subtask is not complete, this parameter is left blank. Minimum: **0** Maximum: **9223372036854775807**
process_trace	String	The detailed progress of the migration or synchronization. Minimum: **0** Maximum: **2048**

Table 5-127 SourceServerAssociatedWithTask

Parameter	Type	Description
id	String	The ID of the source server in the SMS database. Minimum: **0** Maximum: **255**
ip	String	The IP address of the source server. This parameter is mandatory for source server registration and optional for source server updates. Minimum: **0** Maximum: **255**
name	String	The source server name in SMS. Minimum: **0** Maximum: **255**

Parameter	Type	Description
os_type	String	The OS type of the source server, which can be Windows or Linux. This parameter is mandatory for source server registration and optional for source server updates. Minimum: **0** Maximum: **255** Enumeration values: • **WINDOWS** • **LINUX**
os_version	String	The OS version. This parameter is mandatory for registration and optional for updates. Minimum: **0** Maximum: **255**
oem_system	Boolean	Indicates whether the OS is an OEM OS (Windows).

Parameter	Type	Description
state	String	The source server status. **unavailable** (The source server fails the environment check.) **waiting** **initialize** **replicate** **syncing** **stopping** **stopped** **skipping** **deleting** **clearing** (clearing snapshot resources) **cleared** (snapshot resources cleared) **clearfailed** (snapshot resource clearing failed) **premigready** (migration drill ready) **premiged** (migration drill completed) **premigfailed** (migration drill failed) **cloning** **cutovering** (target server being launched) **finished** (target server launched) **error** Enumeration values: • **unavailable** • **waiting** • **initialize** • **replicate** • **syncing** • **stopping** • **stopped** • **skipping** • **deleting** • **clearing** • **cleared** • **clearfailed** • **premigready** • **premiged** • **premigfailed**

Parameter	Type	Description
		• **cloning** • **cutovering** • **finished** • **error**

Table 5-128 TargetServerAssociatedWithTask

Parameter	Type	Description
id	String	The ID of the target server in the SMS database. Minimum: **0** Maximum: **255**
vm_id	String	The ID of the target server. Minimum: **0** Maximum: **255**
name	String	The name of the target server. Minimum: **0** Maximum: **255**
ip	String	The IP address of the target server. Minimum: **0** Maximum: **255**
os_type	String	The OS type of the target server. **WINDOWS**: Windows system **LINUX**: Linux system Minimum: **0** Maximum: **255** Enumeration values: • **WINDOWS** • **LINUX**
os_version	String	The OS version. Minimum: **0** Maximum: **255**

Table 5-129 CloneServerBrief

Parameter	Type	Description
vm_id	String	The ID of the cloned server. Minimum: **0** Maximum: **255**
name	String	The name of the cloned server. Minimum: **0** Maximum: **255**

Table 5-130 NetworkCheckInfoRequestBody

Parameter	Type	Description
domain_connectivity	Boolean	The connectivity to domain names.
destination_connectivity	Boolean	The connectivity to the target server.
network_delay	Double	Network latency (ms). Minimum: **0** Maximum: **10000.0**
network_jitter	Double	Network jitter (ms). Minimum: **0** Maximum: **10000**
migration_speed	Double	Bandwidth (Mbit/s). Minimum: **0** Maximum: **10000**
loss_percentage	Double	Packet loss rate (%). Minimum: **0** Maximum: **100**
cpu_usage	Double	CPU usage (%). Minimum: **0** Maximum: **100**
mem_usage	Double	Memory usage (%). Minimum: **0** Maximum: **100**
evaluation_result	String	The network evaluation result. Minimum: **5** Maximum: **9**

Table 5-131 ConfigBody

Parameter	Type	Description
config_key	String	The advanced migration options to be configured. **MIGRATE_EXCLUDE_DIR**: indicates the directories to be excluded from the migration. **SYNC_EXCLUDE_DIR**: indicates the directories to be excluded from the synchronization. **ONLY_SYNC_DIR**: indicates the directories to be included in the synchronization. **CONSISTENCY_DIR**: indicates the directories for consistency verification. **CONSISTENCY_DIR_ILLEGAL**: indicates illegal directories found after consistency verification. **LINUX_BLOCK_COMPRESS_THREAD_NUM**: indicates the number of compression threads for Linux block-level migration. **MIGRATE_DST_IP**: indicates the target IP address used for the migration. **LINUX_BLOCK_CACHE_SIZE**: indicates the cache size for Linux block-level migration. **LINUX_CPU_LIMIT**: indicates the CPU limit for Linux migration. **LINUX_MEM_LIMIT**: indicates the memory limit for Linux migration. **LINUX_IO_LIMIT**: indicates the I/O limit for Linux migration. **NUM_PROCESS_MIGRATE**: indicates the number of migration processes. **NUM_PROCESS_SYNC**: indicates the number of synchronization processes. **CONSISTENCY_RECHECK**: indicates the re-verification of inconsistencies found. **CONSISTENCY_MODE**: indicates the consistency verification mode. **DYNAMIC_PORT**: indicates a dynamic port. **INSTALL_PWD_AGENT**: indicates whether to install the password reset

Parameter	Type	Description
		plug-in. The value **true** indicates that the plug-in will be installed. The default value is **true**. Enumeration values: ● **MIGRATE_EXCLUDE_DIR** ● **SYNC_EXCLUDE_DIR** ● **ONLY_SYNC_DIR** ● **CONSISTENCY_DIR** ● **CONSISTENCY_DIR_ILLEGAL** ● **LINUX_BLOCK_COMPRESS_THREAD_NUM** ● **MIGRATE_DST_IP** ● **LINUX_BLOCK_CACHE_SIZE** ● **LINUX_CPU_LIMIT** ● **LINUX_MEM_LIMIT** ● **LINUX_IO_LIMIT** ● **NUM_PROCESS_MIGRATE** ● **NUM_PROCESS_SYNC** ● **CONSISTENCY_RECHECK** ● **CONSISTENCY_MODE** ● **DYNAMIC_PORT**
config_value	String	The value specified for the advanced migration option. It is stored in the database and parsed on the Agent. Minimum: **0** Maximum: **1024**
config_status	String	The reserved field that describes the configuration status. Minimum: **0** Maximum: **255**

Status code: 403

Table 5-132 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**

Parameter	Type	Description
error_msg	String	The error message. Minimum: **0** Maximum: **255**
encoded_authorization_message	String	The encrypted authorization information. Minimum: **0** Maximum: **65535**
error_param	Array of strings	Error parameters. Minimum: **0** Maximum: **65535** Array Length: **1 - 20**
details	Array of **details** objects	The error details. Array Length: **1 - 20**

Table 5-133 details

Parameter	Type	Description
error_code	String	The SMS error code. Minimum: **0** Maximum: **65535**
error_msg	String	The SMS error message. Minimum: **0** Maximum: **65535**

Example Requests

This example lists all migration tasks.

```
GET https://{endpoint}/v3/tasks
```

Example Responses

Status code: 200

The list of migration tasks was obtained.


```
{
  "count" : 3,
  "tasks" : [ {
    "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "name" : "sms_task_lxxxx11",
    "type" : "MIGRATE_FILE",
    "os_type" : "LINUX",
```

```
  "state" : "MIGRATE_SUCCESS",
  "estimate_complete_time" : null,
  "create_date" : 1585139506000,
  "priority" : 1,
  "speed_limit" : 0,
  "migrate_speed" : 0,
  "start_target_server" : true,
  "error_json" : "",
  "total_time" : 3878000,
  "migration_ip" : "",
  "source_server" : {
    "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "ip" : "192.168.*.107",
    "name" : "xxx-linux-1",
    "os_type" : "LINUX",
    "os_version" : "CENTOS_7_6_64BIT",
    "oem_system" : false,
    "state" : "waiting"
  },
  "target_server" : {
    "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "vm_id" : "",
    "name" : "",
    "ip" : null,
    "os_type" : "LINUX",
    "os_version" : null
  },
  "log_collect_status" : "INIT"
}, {
  "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
  "name" : "sms_task_xxx22",
  "type" : "MIGRATE_BLOCK",
  "os_type" : "WINDOWS",
  "state" : "MIGRATE_SUCCESS",
  "estimate_complete_time" : null,
  "create_date" : 1585138569000,
  "priority" : 1,
  "speed_limit" : 0,
  "migrate_speed" : 0,
  "start_target_server" : true,
  "error_json" : "",
  "total_time" : 10824000,
  "migration_ip" : "",
  "source_server" : {
    "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "ip" : "192.168.*.245",
    "name" : "xxx-windows-2",
    "os_type" : "WINDOWS",
    "os_version" : "WINDOWS2012_R2_64BIT",
    "oem_system" : false,
    "state" : "waiting"
  },
  "target_server" : {
    "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "vm_id" : "",
    "name" : "",
    "ip" : null,
    "os_type" : "WINDOWS",
    "os_version" : "WINDOWS2012_R2_64BIT"
  },
  "log_collect_status" : "INIT"
}, {
  "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
  "name" : "sms_task_leddesktop",
  "type" : "MIGRATE_BLOCK",
  "os_type" : "WINDOWS",
  "state" : "MIGRATE_SUCCESS",
  "estimate_complete_time" : null,
  "create_date" : 1566130392000,
```

```
    "priority" : 1,
    "speed_limit" : 200,
    "migrate_speed" : 0,
    "start_target_server" : true,
    "error_json" : "",
    "total_time" : 882000,
    "migration_ip" : "192.168.1.201",
    "source_server" : null,
    "target_server" : {
      "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
      "vm_id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
      "name" : "xxx-sms-target",
      "ip" : null,
      "os_type" : "WINDOWS",
      "os_version" : "WINDOWS2008_R2_64BIT"
    },
    "log_collect_status" : "INIT"
  } ]
}
```

Status code: 403

Authentication failed.

```
{
  "error_code" : "SMS.9004",
  "error_msg" : "The current account does not have the permission to execute policy. You do not have permission to perform action XXX on resource XXX.",
  "encoded_authorization_message" : "XXXXXX",
  "error_param" : [ "You do not have permission to perform action XXX on resource XXX." ],
  "details" : [ {
    "error_code" : "SMS.9004",
    "error_msg" : "You do not have permission to perform action XXX on resource XXX."
  } ]
}
```

SDK Sample Code

The SDK sample code is as follows.

Java

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;


public class ListTasksSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);
```

```
        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        ListTasksRequest request = new ListTasksRequest();
        try {
            ListTasksResponse response = client.listTasks(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

Python

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = ListTasksRequest()
        response = client.list_tasks(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

Go

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
```



```
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
         WithRegion(region.ValueOf("<YOUR REGION>")).
         WithCredential(auth).
         SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.ListTasksRequest{}
    response, err := client.ListTasks(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

More

For SDK sample code of more programming languages, see the Sample Code tab in **API Explorer**. SDK sample code can be automatically generated.

Status Codes

Status Code	Description
200	The list of migration tasks was obtained.
403	Authentication failed.

Error Codes

See **Error Codes**.

5.4.3 Batch Deleting Migration Tasks

Function

This API is used to delete multiple migration tasks in a batch.

Constraints

Migration tasks cannot be deleted if the source server is connected to SMS, and the source server is in one of the following statuses: initialize, stopping, deleting, cloning, replicate, syncing, cutovering, skipping, clearing, premiging.

Calling Method

For details, see **Calling APIs**.

Authorization Information

Each account has all the permissions required to call all APIs, but IAM users must be assigned the required permissions.

- If you are using role/policy-based authorization, see **Permissions Policies and Supported Actions** for details on the required permissions.
- If you are using identity policy-based authorization, no identity policy-based permission required for calling this API.

URI

POST /v3/tasks/delete

Request Parameters

Table 5-134 Request header parameters

Parameter	Mandatory	Type	Description
X-Auth-Token	Yes	String	User token. The token can be obtained by calling the IAM API for obtaining a user token. The value of **X-Subject-Token** in the response header is the user token. Minimum: **1** Maximum: **16384**

Table 5-135 Request body parameters

Parameter	Mandatory	Type	Description
ids	Yes	Array of strings	The IDs of the tasks to be deleted. Minimum: **0** Maximum: **255** Array Length: **1 - 50**

Response Parameters

Status code: 200

Table 5-136 Response body parameters

Parameter	Type	Description
-	String	Deleting migration tasks in batches succeeded.

Status code: 403

Table 5-137 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **255**
encoded_authorization_message	String	The encrypted authorization information. Minimum: **0** Maximum: **65535**
error_param	Array of strings	Error parameters. Minimum: **0** Maximum: **65535** Array Length: **1 - 20**
details	Array of **details** objects	The error details. Array Length: **1 - 20**

Table 5-138 details

Parameter	Type	Description
error_code	String	The SMS error code. Minimum: **0** Maximum: **65535**
error_msg	String	The SMS error message. Minimum: **0** Maximum: **65535**

Example Requests

This example deletes migration tasks with the ID set of 1, 2, and 3 in a batch.

```
POST https://{endpoint}/v3/tasks/delete

{
  "ids" : [ "1", "2", "3" ]
}
```

Example Responses

Status code: 200

Deleting migration tasks in batches succeeded.

```
{ }
```

Status code: 403

Authentication failed.

```
{
  "error_code" : "SMS.9004",
  "error_msg" : "The current account does not have the permission to execute policy. You do not have permission to perform action XXX on resource XXX.",
  "encoded_authorization_message" : "XXXXXX",
  "error_param" : [ "You do not have permission to perform action XXX on resource XXX." ],
  "details" : [ {
    "error_code" : "SMS.9004",
    "error_msg" : "You do not have permission to perform action XXX on resource XXX."
  } ]
}
```

SDK Sample Code

The SDK sample code is as follows.

Java

This example deletes migration tasks with the ID set of 1, 2, and 3 in a batch.

```
package com.huaweicloud.sdk.test;
```

```
import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;

import java.util.List;
import java.util.ArrayList;

public class DeleteTasksSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        DeleteTasksRequest request = new DeleteTasksRequest();
        DeleteTasksReq body = new DeleteTasksReq();
        List<String> listbodyIds = new ArrayList<>();
        listbodyIds.add("1");
        listbodyIds.add("2");
        listbodyIds.add("3");
        body.withIds(listbodyIds);
        request.withBody(body);
        try {
            DeleteTasksResponse response = client.deleteTasks(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

Python

This example deletes migration tasks with the ID set of 1, 2, and 3 in a batch.

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
```



```
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = DeleteTasksRequest()
        listIdsbody = [
            "1",
            "2",
            "3"
        ]
        request.body = DeleteTasksReq(
            ids=listIdsbody
        )
        response = client.delete_tasks(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

Go

This example deletes migration tasks with the ID set of 1, 2, and 3 in a batch.

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
         WithRegion(region.ValueOf("<YOUR REGION>")).
         WithCredential(auth).
```

```
        SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.DeleteTasksRequest{}
    var listIdsbody = []string{
        "1",
        "2",
        "3",
    }
    request.Body = &model.DeleteTasksReq{
        Ids: listIdsbody,
    }
    response, err := client.DeleteTasks(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

More

For SDK sample code of more programming languages, see the Sample Code tab in **API Explorer**. SDK sample code can be automatically generated.

Status Codes

Status Code	Description
200	Deleting migration tasks in batches succeeded.
403	Authentication failed.

Error Codes

See **Error Codes**.

5.4.4 Deleting a Migration Task

Function

This API is used to delete a migration task with a specified ID.

Constraints

Migration tasks cannot be deleted if the source server is connected to SMS, and the source server is in one of the following statuses: initialize, stopping, deleting, cloning, replicate, syncing, cutovering, skipping, clearing, premiging.

Calling Method

For details, see **Calling APIs**.

Authorization Information

Each account has all the permissions required to call all APIs, but IAM users must be assigned the required permissions.

- If you are using role/policy-based authorization, see **Permissions Policies and Supported Actions** for details on the required permissions.
- If you are using identity policy-based authorization, the following identity policy-based permissions are required.

Action	Access Level	Resource Type (*: required)	Condition Key	Alias	Dependencies
sms:server:deleteTask	Write	server *	-	sms:server:migrationServer	-
		-	g:EnterpriseProjectId		

URI

DELETE /v3/tasks/{task_id}

Table 5-139 Path Parameters

Parameter	Mandatory	Type	Description
task_id	Yes	String	The ID of the migration task to be deleted. Minimum: **1** Maximum: **36**

Request Parameters

Table 5-140 Request header parameters

Parameter	Mandatory	Type	Description
X-Auth-Token	Yes	String	User token. The token can be obtained by calling the IAM API for obtaining a user token. The value of **X-Subject-Token** in the response header is the user token. Minimum: **1** Maximum: **16384**

Response Parameters

Status code: 200

Table 5-141 Response body parameters

Parameter	Type	Description
-	String	The migration task with a specified ID was deleted.

Status code: 403

Table 5-142 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **255**
encoded_authorization_message	String	The encrypted authorization information. Minimum: **0** Maximum: **65535**

Parameter	Type	Description
error_param	Array of strings	Error parameters. Minimum: **0** Maximum: **65535** Array Length: **1 - 20**
details	Array of **details** objects	The error details. Array Length: **1 - 20**

Table 5-143 details

Parameter	Type	Description
error_code	String	The SMS error code. Minimum: **0** Maximum: **65535**
error_msg	String	The SMS error message. Minimum: **0** Maximum: **65535**

Example Requests

This example deletes the migration task with ID **7a9a9540-ff28-4869-b9e4-855fbe12xxxx**.

```
DELETE https://{endpoint}/v3/tasks/7a9a9540-ff28-4869-b9e4-855fbe12xxxx
```

Example Responses

Status code: 403

Authentication failed.

```
{
  "error_code" : "SMS.9004",
  "error_msg" : "The current account does not have the permission to execute policy. You do not have permission to perform action XXX on resource XXX.",
  "encoded_authorization_message" : "XXXXXX",
  "error_param" : [ "You do not have permission to perform action XXX on resource XXX." ],
  "details" : [ {
    "error_code" : "SMS.9004",
    "error_msg" : "You do not have permission to perform action XXX on resource XXX."
  } ]
}
```

SDK Sample Code

The SDK sample code is as follows.

Java

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;


public class DeleteTaskSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great
security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or
environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running
this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        DeleteTaskRequest request = new DeleteTaskRequest();
        request.withTaskId("{task_id}");
        try {
            DeleteTaskResponse response = client.deleteTask(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

Python

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security
risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment
variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this
example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
```



```
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = DeleteTaskRequest()
        request.task_id = "{task_id}"
        response = client.delete_task(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

Go

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
         WithRegion(region.ValueOf("<YOUR REGION>")).
         WithCredential(auth).
         SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.DeleteTaskRequest{}
    request.TaskId = "{task_id}"
    response, err := client.DeleteTask(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
```

```
    } else {
        fmt.Println(err)
    }
}
```

More

For SDK sample code of more programming languages, see the Sample Code tab in **API Explorer**. SDK sample code can be automatically generated.

Status Codes

Status Code	Description
200	The migration task with a specified ID was deleted.
403	Authentication failed.

Error Codes

See **Error Codes**.

5.4.5 Querying a Migration Task with a Specified ID

Function

This API is used to query a migration task with a specified ID.

Calling Method

For details, see **Calling APIs**.

Authorization Information

Each account has all the permissions required to call all APIs, but IAM users must be assigned the required permissions.

- If you are using role/policy-based authorization, see **Permissions Policies and Supported Actions** for details on the required permissions.
- If you are using identity policy-based authorization, the following identity policy-based permissions are required.

Action	Access Level	Resource Type (*: required)	Condition Key	Alias	Dependencies
sms:server:getTask	Read	server *	-	sms:server:queryServer	-
		-	g:EnterpriseProjectId		

URI

GET /v3/tasks/{task_id}

Table 5-144 Path Parameters

Parameter	Mandatory	Type	Description
task_id	Yes	String	The migration task ID. Minimum: **1** Maximum: **36**

Request Parameters

Table 5-145 Request header parameters

Parameter	Mandatory	Type	Description
X-Auth-Token	Yes	String	User token. The token can be obtained by calling the IAM API for obtaining a user token. The value of **X-Subject-Token** in the response header is the user token. Minimum: **1** Maximum: **16384**

Response Parameters

Status code: 200

Table 5-146 Response body parameters

Parameter	Type	Description
name	String	The task name, which is defined by the user. Minimum: **0** Maximum: **255**

Parameter	Type	Description
type	String	The task type. This parameter is mandatory for creating a task and optional for updating a task. **MIGRATE_FILE** **MIGRATE_BLOCK** Minimum: **0** Maximum: **255** Enumeration values: • **MIGRATE_FILE** • **MIGRATE_BLOCK**
os_type	String	The OS type, which can be Windows or Linux. This parameter is mandatory for creating a task and optional for updating a task. Enumeration values: • **WINDOWS** • **LINUX**
id	String	The migration task ID. Minimum: **0** Maximum: **255**
priority	Integer	The migration process priority. **0**: low **1**: standard (default) **2**: high Minimum: **0** Maximum: **2** Enumeration values: • **0** • **1** • **2**
speed_limit	Integer	The migration rate limit. Minimum: **0** Maximum: **65535**
region_id	String	The ID of the region where the target server is located. Minimum: **0** Maximum: **255**

Parameter	Type	Description
start_target_server	Boolean	Indicates whether the target server is started after the migration is complete. **true**: yes **false**: no Default: **true**
enterprise_project_id	String	Enterprise project ID. Minimum: **1** Maximum: **255**
migration_ip	String	The IP address of the target server. If you use a public network for migration, enter the EIP bound to the target server. If you use a private line for migration, enter a private IP address of the target server. Minimum: **0** Maximum: **255**
region_name	String	The name of the region where the target server is located. Minimum: **0** Maximum: **255**
project_name	String	The name of the project to which the target server belongs. Minimum: **0** Maximum: **255**
project_id	String	The ID of the project to which the target server belongs. Minimum: **0** Maximum: **255**
vm_template_id	String	The template ID. Minimum: **0** Maximum: **255**
source_server	**SourceServerResponse** object	The returned source server information.
target_server	**TaskTargetServer** object	The target server information.

Parameter	Type	Description
state	String	The migration task status. **READY** **RUNNING** **SYNCING** **MIGRATE_SUCCESS** **SYNC_SUCCESS** **MIGRATE_FAIL** **SYNC_FAIL** **ABORTING** **ABORT** **SKIPPING** **DELETING** **RESETING** Enumeration values: ● **READY** ● **RUNNING** ● **SYNCING** ● **MIGRATE_SUCCESS** ● **SYNC_SUCCESS** ● **MIGRATE_FAIL** ● **SYNC_FAIL** ● **ABORTING** ● **ABORT** ● **SKIPPING** ● **DELETING** ● **RESETING**
estimate_complete_time	Long	The estimated completion time. Minimum: **0** Maximum: **9223372036854775807**
connected	Boolean	The connection status.
create_date	Long	The task creation time. Minimum: **0** Maximum: **9223372036854775807**
start_date	Long	The task start time. Minimum: **0** Maximum: **9223372036854775807**

Parameter	Type	Description
finish_date	Long	The task end time. Minimum: **0** Maximum: **9223372036854775807**
migrate_speed	Double	The migration rate, in Mbit/s. Minimum: **0** Maximum: **10000**
compress_rate	Double	The compression rate. Minimum: **0** Maximum: **10000**
error_json	String	The error message. Minimum: **0** Maximum: **1024**
total_time	Long	The task duration. Minimum: **0** Maximum: **9223372036854775807**
float_ip	String	Retain this parameter temporarily to ensure the compatibility with the earlier versions of SMS-Agent. Minimum: **0** Maximum: **255**
remain_seconds	Long	The remaining migration time in seconds. Minimum: **0** Maximum: **9223372036854775807**
target_snapshot_id	String	The ID of the snapshot for the target server. Minimum: **0** Maximum: **255**
clone_server	**CloneServer** object	The information about the cloned server.
sub_tasks	Array of **SubTask** objects	The subtasks contained in the migration task. Array Length: **0 - 65535**
network_check_info	**NetworkCheckInfoRequestBody** object	The network measurement results.

Parameter	Type	Description
total_cpu_usage	Double	The CPU usage of the server, in percentage. Minimum: **0** Maximum: **100**
agent_cpu_usage	Double	The CPU usage of the Agent, in percentage. Minimum: **0** Maximum: **100**
total_mem_usage	Double	The memory usage of the server, in MB. Minimum: **0** Maximum: **1048576.0**
agent_mem_usage	Double	The memory usage of the Agent, in MB. Minimum: **0** Maximum: **1048576.0**
total_disk_io	Double	The disk I/O of the server, in Mbit/s. Minimum: **0** Maximum: **10000.0**
agent_disk_io	Double	The disk I/O of the Agent, in Mbit/s. Minimum: **0** Maximum: **10000.0**
need_migration_test	Boolean	Indicates whether migration drilling is enabled.
subtask_info	String	The current subtask and progress. Minimum: **0** Maximum: **255**

Table 5-147 SourceServerResponse

Parameter	Type	Description
id	String	The ID of the source server in the SMS database. Minimum: **0** Maximum: **255**

Parameter	Type	Description
ip	String	The IP address of the source server. This parameter is mandatory for source server registration and optional for source server updates. Minimum: **0** Maximum: **255**
name	String	The source server name in SMS. Minimum: **0** Maximum: **255**
os_type	String	The OS type of the source server, which can be Windows or Linux. This parameter is mandatory for source server registration and optional for source server updates. Minimum: **0** Maximum: **255** Enumeration values: ● **WINDOWS** ● **LINUX**
os_version	String	The OS version. This parameter is mandatory for registration and optional for updates. Minimum: **0** Maximum: **255**
oem_system	Boolean	Indicates whether the OS is an OEM OS (Windows).

Parameter	Type	Description
state	String	The source server status. **unavailable** (The source server fails the environment check.) **waiting** **initialize** **replicate** **syncing** **stopping** **stopped** **skipping** **deleting** **clearing** **cleared** (snapshot resources cleared) **clearfailed** (snapshot resource clearing failed) **premigready** (migration drill ready) **premiged** (migration drill completed) **premigfailed** (migration drill failed) **cloning** **cutovering** (target server being launched) **finished** (target server launched) **error** Enumeration values: • **unavailable** • **waiting** • **initialize** • **replicate** • **syncing** • **stopping** • **stopped** • **skipping** • **deleting** • **clearing** • **cleared** • **clearfailed** • **premigready** • **premiged** • **premigfailed**

Parameter	Type	Description
		• **cloning** • **cutovering** • **finished** • **error**
migration_cycle	String	The migration stage. **cutovering** (target server being launched) **cutovered** (target server launched) **checking** **setting** **replicating** **syncing** Minimum: **0** Maximum: **255** Enumeration values: • **cutovering** • **cutovered** • **checking** • **setting** • **replicating** • **syncing**

Table 5-148 TaskTargetServer

Parameter	Type	Description
id	String	The ID of the target server in the SMS database. Minimum: **0** Maximum: **255**
vm_id	String	The ID of the target server. This parameter is not required for automatically created target servers. Minimum: **0** Maximum: **255**
name	String	The name of the target server. Minimum: **0** Maximum: **255**

Parameter	Type	Description
ip	String	The IP address of the target server. Minimum: **0** Maximum: **255**
os_type	String	The OS type of the source server, which can be Windows or Linux. This parameter is mandatory for source server registration and optional for source server updates. Minimum: **0** Maximum: **255** Enumeration values: ● **WINDOWS** ● **LINUX**
os_version	String	The OS version. This parameter is mandatory for registration and optional for updates. Minimum: **0** Maximum: **255**
system_dir	String	The system directory. This parameter is mandatory for Windows. Minimum: **0** Maximum: **255**
disks	Array of **TargetDisk** objects	The information about target server disks, which is generally the same as that of the source server. Array Length: **0 - 65535**
volume_groups	Array of **VolumeGroups** objects	The information about logical volumes on the target server, which is generally the same as that on the source server. Array Length: **0 - 65535**
btrfs_list	Array of strings	The information about the Btrfs file systems on the source server. This parameter is mandatory for Linux. If there are no Btrfs file systems on the source server, the value is an empty array []. Minimum: **0** Maximum: **255** Array Length: **0 - 65535**

Parameter	Type	Description
image_disk_id	String	The ID of the disk that hosts the agent image on the target server. Minimum: **0** Maximum: **255**
cutovered_snapshot_ids	String	The ID of the snapshot used for rollback on the target server. Minimum: **0** Maximum: **255**

Table 5-149 TargetDisk

Parameter	Type	Description
id	Long	The disk ID. Minimum: **0** Maximum: **9223372036854775807**
device_use	String	The partition function. The partition can be a general, boot, or OS partition. **BOOT**: boot device **OS**: system device **NORMAL**: general device Default: **NORMAL** Enumeration values: ● **NORMAL** ● **OS** ● **BOOT**
disk_id	String	The disk ID. Minimum: **0** Maximum: **255**
name	String	The disk name. Minimum: **0** Maximum: **255**
physical_volumes	Array of **TargetPhysicalVolumes** objects	The logical volume information. Array Length: **0 - 65535**
size	Long	The size. Minimum: **0** Maximum: **9223372036854775807**

Parameter	Type	Description
used_size	Long	The used space. Minimum: **0** Maximum: **9223372036854775807**
disk_index	String	The disk index. Minimum: **0** Maximum: **255**
os_disk	Boolean	Indicates whether the disk is the system disk.
partition_style	String	The disk partition type. This parameter is mandatory for source server registration. **MBR**: Master Boot Record **GPT**: GUID Partition Table Enumeration values: ● **MBR** ● **GPT**
relation_name	String	The name of the paired disk on the Linux target ECS server. Minimum: **0** Maximum: **255**

Table 5-150 TargetPhysicalVolumes

Parameter	Type	Description
id	Long	The logical volume ID. Minimum: **0** Maximum: **9223372036854775807**
device_use	String	The partition function. **NORMAL**: general device **OS**: system device **BOOT**: boot device Default: **NORMAL** Enumeration values: ● **NORMAL** ● **OS** ● **BOOT**

Parameter	Type	Description
file_system	String	The file system. Minimum: **0** Maximum: **255**
index	Integer	No. Minimum: **0** Maximum: **2147483647**
mount_point	String	The mount point. Minimum: **0** Maximum: **255**
name	String	Name. Minimum: **0** Maximum: **255**
size	Long	The size. Minimum: **0** Maximum: **9223372036854775807**
used_size	Long	The used space. Minimum: **0** Maximum: **9223372036854775807**
uuid	String	uuid Minimum: **0** Maximum: **255**
relation_name	String	The name of the paired disk on the Linux target ECS server. Minimum: **0** Maximum: **255**
free_size	Long	The idle partition space. Minimum: **0** Maximum: **9223372036854775807**

Table 5-151 VolumeGroups

Parameter	Type	Description
components	String	The physical volume information. Minimum: **0** Maximum: **255**

Parameter	Type	Description
free_size	Long	The available space. Minimum: **0** Maximum: **9223372036854775807**
logical_volumes	Array of **LogicalVolumes** objects	The logical volume information. Array Length: **0 - 50**
name	String	Name. Minimum: **0** Maximum: **255**
size	Long	Size. Minimum: **0** Maximum: **9223372036854775807**

Table 5-152 LogicalVolumes

Parameter	Type	Description
block_count	Integer	The number of blocks. Minimum: **0** Maximum: **2147483647** Default: **0**
block_size	Long	Block size. Minimum: **0** Maximum: **1048576** Default: **0**
file_system	String	The file system. Minimum: **0** Maximum: **255**
inode_size	Integer	The number of inodes. Minimum: **0** Maximum: **2147483647**
inode_nums	Long	The number of inodes. Minimum: **0** Maximum: **9223372036854775807**

Parameter	Type	Description
device_use	String	Partition type. This is not mandatory and can be left blank. The common values are as follows: **NORMAL** **OS**: system device **BOOT** **VOLUME_GROUP** **BTRFS** Minimum: **0** Maximum: **255**
mount_point	String	The mount point. Minimum: **0** Maximum: **256**
name	String	Name. Minimum: **0** Maximum: **1024**
size	Long	The size. Minimum: **0** Maximum: **9223372036854775807**
used_size	Long	The used space. Minimum: **0** Maximum: **9223372036854775807**
free_size	Long	The available space. Minimum: **0** Maximum: **9223372036854775807**

Table 5-153 CloneServer

Parameter	Type	Description
vm_id	String	The ID of the cloned server. Minimum: **0** Maximum: **255**
name	String	The name of the cloned server. Minimum: **0** Maximum: **255**

Parameter	Type	Description
clone_error	String	The error code returned for a clone failure. Minimum: **0** Maximum: **255**
clone_state	String	The clone status. NOT_READY READY PREPARING CREATING ERROR FINISHED Enumeration values: ● **NOT_READY** ● **READY** ● **PREPARING** ● **CREATING** ● **ERROR** ● **FINISHED**
error_msg	String	The error message returned for a clone failure. Minimum: **0** Maximum: **1024**

Table 5-154 SubTask

Parameter	Type	Description
id	Long	The subtask ID. Minimum: **0** Maximum: **9223372036854775807**
name	String	The subtask name. Minimum: **0** Maximum: **255**
progress	Integer	The progress of the subtask. The value is an integer ranging from 0 to 100. Minimum: **0** Maximum: **100**

Parameter	Type	Description
start_date	Long	The start time of the subtask. Minimum: **0** Maximum: **9223372036854775807**
end_date	Long	The end time of the subtask. If the subtask is not complete, this parameter is left blank. Minimum: **0** Maximum: **9223372036854775807**
migrate_speed	Double	The migration speed, in Mbit/s. Minimum: **0** Maximum: **10000**
user_op	String	The user operation that triggers the subtask. Minimum: **0** Maximum: **50**
process_trace	String	The detailed progress of the migration or synchronization. Minimum: **0** Maximum: **2048**

Table 5-155 NetworkCheckInfoRequestBody

Parameter	Type	Description
domain_connectivity	Boolean	The connectivity to domain names.
destination_connectivity	Boolean	The connectivity to the target server.
network_delay	Double	Network latency (ms). Minimum: **0** Maximum: **10000.0**
network_jitter	Double	Network jitter (ms). Minimum: **0** Maximum: **10000**
migration_speed	Double	Bandwidth (Mbit/s). Minimum: **0** Maximum: **10000**

Parameter	Type	Description
loss_percentage	Double	Packet loss rate (%). Minimum: **0** Maximum: **100**
cpu_usage	Double	CPU usage (%). Minimum: **0** Maximum: **100**
mem_usage	Double	Memory usage (%). Minimum: **0** Maximum: **100**
evaluation_result	String	The network evaluation result. Minimum: **5** Maximum: **9**

Status code: 403

Table 5-156 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **255**
encoded_authorization_message	String	The encrypted authorization information. Minimum: **0** Maximum: **65535**
error_param	Array of strings	Error parameters. Minimum: **0** Maximum: **65535** Array Length: **1 - 20**
details	Array of **details** objects	The error details. Array Length: **1 - 20**

Table 5-157 details

Parameter	Type	Description
error_code	String	The SMS error code. Minimum: **0** Maximum: **65535**
error_msg	String	The SMS error message. Minimum: **0** Maximum: **65535**

Example Requests

This example queries a migration task with a specified ID.

```
GET https://{endpoint}/v3/tasks/ef3b9722-07a0-40ae-89b0-889ee96dfc56
```

Example Responses

Status code: 200

The migration task with a specified ID was queried successfully.


```
{
  "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
  "name" : "MigrationTask",
  "type" : "MIGRATE_BLOCK",
  "os_type" : "WINDOWS",
  "state" : "RUNNING",
  "estimate_complete_time" : null,
  "create_date" : 1598435778000,
  "start_date" : 1598435784000,
  "finish_date" : null,
  "priority" : 1,
  "speed_limit" : 0,
  "migrate_speed" : 0.0,
  "start_target_server" : true,
  "error_json" : "",
  "total_time" : 115,
  "float_ip" : null,
  "migration_ip" : null,
  "vm_template_id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
  "region_name" : "region name",
  "region_id" : "region id",
  "project_name" : "project name",
  "project_id" : "xxxxxxxxxxxxxxxxxxxxxxxx00000001",
  "sub_tasks" : [ {
    "id" : 7278,
    "name" : "CREATE_CLOUD_SERVER",
    "progress" : 0,
    "start_date" : 1598435802000,
    "end_date" : null,
    "user_op" : "REPLICATE",
    "process_trace" : null
  }, {
    "id" : 7279,
    "name" : "SSL_CONFIG",
    "progress" : 0,
    "start_date" : null,
    "end_date" : null,
```



```
  "user_op" : "REPLICATE",
  "process_trace" : null
}, {
  "id" : 7280,
  "name" : "ATTACH_AGENT_IMAGE",
  "progress" : 0,
  "start_date" : null,
  "end_date" : null,
  "user_op" : "REPLICATE",
  "process_trace" : null
}, {
  "id" : 7281,
  "name" : "FORMAT_DISK_WINDOWS",
  "progress" : 0,
  "start_date" : null,
  "end_date" : null,
  "user_op" : "REPLICATE",
  "process_trace" : null
}, {
  "id" : 7282,
  "name" : "MIGRATE_WINDOWS_BLOCK",
  "progress" : 0,
  "start_date" : null,
  "end_date" : null,
  "user_op" : "REPLICATE",
  "process_trace" : null
} ],
"source_server" : {
  "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
  "ip" : "192.168.0.154",
  "name" : "name-win16",
  "os_type" : "WINDOWS",
  "os_version" : "WINDOWS2016_64BIT",
  "oem_system" : false,
  "state" : "initialize",
  "migration_cycle" : "replicating"
},
"target_server" : {
  "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
  "vm_id" : "",
  "name" : "",
  "ip" : null,
  "os_type" : "WINDOWS",
  "os_version" : "WINDOWS2016_64BIT",
  "system_dir" : "Y:\\Windows\\System32",
  "disks" : [ {
    "id" : 88008,
    "name" : "Disk 1",
    "relation_name" : null,
    "disk_id" : "0",
    "partition_style" : "MBR",
    "size" : 42949672960,
    "used_size" : 42947575808,
    "device_use" : "OS",
    "os_disk" : true,
    "physical_volumes" : [ {
      "id" : 135055,
      "uuid" : "\\\\?\\Volume{586b7157-0000-0000-0000-100000000000}\\",
      "index" : 1,
      "name" : "Z:",
      "relation_name" : null,
      "device_use" : "BOOT",
      "file_system" : "NTFS",
      "mount_point" : null,
      "size" : 524288000,
      "used_size" : 410275840,
      "free_size" : 114012160
    }, {
      "id" : 135056,
```

```
        "uuid" : "\\\\?\\Volume{586b7157-0000-0000-0000-501f00000000}\\",
        "index" : 2,
        "name" : "Y:",
        "relation_name" : null,
        "device_use" : "OS",
        "file_system" : "NTFS",
        "mount_point" : null,
        "size" : 42423287808,
        "used_size" : 23170301952,
        "free_size" : 19252985856
      } ],
      "disk_index" : "0"
    } ],
    "volume_groups" : [ ],
    "image_disk_id" : null,
    "cutovered_snapshot_ids" : null
  },
  "clone_server" : null
}
```


Status code: 403

Authentication failed.


```
{
  "error_code" : "SMS.9004",
  "error_msg" : "The current account does not have the permission to execute policy. You do not have permission to perform action XXX on resource XXX.",
  "encoded_authorization_message" : "XXXXXX",
  "error_param" : [ "You do not have permission to perform action XXX on resource XXX." ],
  "details" : [ {
    "error_code" : "SMS.9004",
    "error_msg" : "You do not have permission to perform action XXX on resource XXX."
  } ]
}
```


SDK Sample Code

The SDK sample code is as follows.

Java

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;


public class ShowTaskSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);
```

```
        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        ShowTaskRequest request = new ShowTaskRequest();
        request.withTaskId("{task_id}");
        try {
            ShowTaskResponse response = client.showTask(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

Python

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = ShowTaskRequest()
        request.task_id = "{task_id}"
        response = client.show_task(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

Go

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
```

```
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
         WithRegion(region.ValueOf("<YOUR REGION>")).
         WithCredential(auth).
         SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.ShowTaskRequest{}
    request.TaskId = "{task_id}"
    response, err := client.ShowTask(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

More

For SDK sample code of more programming languages, see the Sample Code tab in **API Explorer**. SDK sample code can be automatically generated.

Status Codes

Status Code	Description
200	The migration task with a specified ID was queried successfully.
403	Authentication failed.

Error Codes

See **Error Codes**.

5.4.6 Updating a Migration Task

Function

This API is used to update a migration task with a specified ID.

Calling Method

For details, see **Calling APIs**.

Authorization Information

Each account has all the permissions required to call all APIs, but IAM users must be assigned the required permissions.

- If you are using role/policy-based authorization, see **Permissions Policies and Supported Actions** for details on the required permissions.
- If you are using identity policy-based authorization, the following identity policy-based permissions are required.

Action	Access Level	Resource Type (*: required)	Condition Key	Alias	Dependencies
sms:server:updateTask	Write	server *	-	sms:server:migrationServer	-
		-	g:EnterpriseProjectId		

URI

PUT /v3/tasks/{task_id}

Table 5-158 Path Parameters

Parameter	Mandatory	Type	Description
task_id	Yes	String	The migration task ID. Minimum: **1** Maximum: **36**

Request Parameters

Table 5-159 Request header parameters

Parameter	Mandatory	Type	Description
X-Auth-Token	Yes	String	User token. The token can be obtained by calling the IAM API for obtaining a user token. The value of **X-Subject-Token** in the response header is the user token. Minimum: **1** Maximum: **16384**

Table 5-160 Request body parameters

Parameter	Mandatory	Type	Description
name	No	String	Task name (customizable). The value can contain only letters, digits, underscores (_), and hyphens (-). Minimum: **1** Maximum: **64**
type	No	String	The task type. This parameter is mandatory for creating a task and optional for updating a task. **MIGRATE_FILE** **MIGRATE_BLOCK** Minimum: **0** Maximum: **255** Enumeration values: ● **MIGRATE_FILE** ● **MIGRATE_BLOCK**
region_id	No	String	The ID of the region where the target server is located. Minimum: **0** Maximum: **255**

Parameter	Mandatory	Type	Description
region_name	No	String	The name of the region where the target server is located. Minimum: **0** Maximum: **255**
exist_server	No	Boolean	Indicates whether an existing server is used as the target server. **true** indicates that an existing server is used as the target server. **false** indicates that the target server is newly created.
migration_ip	No	String	The IP address of the target server. For public network migration, enter the EIP. For Direct Connect migration, enter the private IP address. If **use_ipv6** is set to **true**, the value must be in the IPv6 format. If **use_ipv6** is set to **false**, the value must be in the IPv4 format. Minimum: **0** Maximum: **255**
use_ipv6	No	Boolean	Whether the IP address of the destination server uses IPv6. Default: **false**
use_public_ip	No	Boolean	Public network or not.
speed_limit	No	Integer	The migration rate limit, in Mbit/s. Minimum: **0** Maximum: **1000**
project_name	No	String	The name of the project to which the target server belongs. Minimum: **0** Maximum: **255**
project_id	No	String	The ID of the project to which the target server belongs. Minimum: **0** Maximum: **255**

Parameter	Mandatory	Type	Description
enterprise_project	No	String	Enterprise project ID. Minimum: **0** Maximum: **36**
image_disk_id	No	String	Disk ID of the image agent on the target server. Minimum: **0** Maximum: **255**
vm_template_id	No	String	The template ID. Minimum: **0** Maximum: **100**
target_disk_ids	No	String	Disk ID array of the target server. Separate IDs with spaces. The length of a single ID cannot exceed 255 characters. Minimum: **0** Maximum: **16384**
snapshot_ids	No	String	Snapshot ID of the target server. Separate IDs with semicolons (;). Minimum: **0** Maximum: **1024**
cutovered_snapshot_ids	No	String	ID of the snapshot for cutover. Minimum: **0** Maximum: **1024**
target_server	No	**TargetServer** object	The target server information.
clone_server	No	**CloneServer** object	Clone server. The modification on the clone server takes effect only when the source server is in the **syncing**, **stopping**, **stopped**, **cutovering**, **finished**, or **error** state.

Table 5-161 TargetServer

Parameter	Mandatory	Type	Description
id	No	String	The ID of the server in the SMS database. Minimum: **0** Maximum: **255**
ip	No	String	The IP address of the server. This parameter is mandatory for registration and optional for updates. Minimum: **0** Maximum: **255**
name	No	String	The server name. Minimum: **0** Maximum: **255**
hostname	No	String	The hostname of the server. This parameter is mandatory for registration and optional for updates. Minimum: **0** Maximum: **255**
os_type	No	String	The OS type of the server, which can be Windows or Linux. This parameter is mandatory for source server registration and optional for source server updates. Minimum: **0** Maximum: **255** Enumeration values: • **WINDOWS** • **LINUX**
os_version	No	String	The OS version. This parameter is mandatory for registration and optional for updates. Minimum: **0** Maximum: **255**

Parameter	Mandatory	Type	Description
firmware	No	String	The boot mode of the server, which can be BIOS or UEFI. Minimum: **0** Maximum: **255** Enumeration values: ● **BIOS** ● **UEFI**
cpu_quantity	No	Integer	The number of vCPUs. Minimum: **0** Maximum: **65535**
memory	No	Long	The memory size, in MB. Minimum: **0** Maximum: **9223372036854775807**
btrfs_list	No	Array of **BtrfsFileSystem** objects	The Btrfs information of the server, which is mandatory for Linux. If there are no Btrfs file systems on the server, the value is an empty array []. Array Length: **0 - 65535**
networks	No	Array of **NetWork** objects	The information about NICs on the server. Array Length: **0 - 65535**
domain_id	No	String	The tenant domain ID. Minimum: **0** Maximum: **255**
has_rsync	No	Boolean	Indicates whether rsync is installed. This parameter is mandatory for Linux.
paravirtualization	No	Boolean	Indicates whether the source server is paravirtualized. This parameter is mandatory for Linux.
raw_devices	No	String	The list of raw devices. This parameter is mandatory for Linux. Minimum: **0** Maximum: **255**

Parameter	Mandatory	Type	Description
driver_files	No	Boolean	Indicates whether any driver files are missing. This parameter is mandatory for Windows.
system_services	No	Boolean	Indicates whether there are abnormal services. This parameter is mandatory for Windows.
account_rights	No	Boolean	Indicates whether the account has the required permissions. This parameter is mandatory for Windows.
boot_loader	No	String	The system boot loader, which can be GRUB or LILO. This parameter is mandatory for Linux. Enumeration values: • **GRUB** • **LILO**
system_dir	No	String	The system directory. This parameter is mandatory for Windows. Minimum: **0** Maximum: **255**
volume_groups	No	Array of **VolumeGroups** objects	This parameter is mandatory for Linux. If there are no volume groups, the value is an empty array []. Array Length: **0 - 65535**
vm_id	No	String	The ID of the target server. This parameter is not required for automatically created target servers. Minimum: **0** Maximum: **255**
flavor	No	String	The flavor of the target server. Minimum: **0** Maximum: **255**

Parameter	Mandatory	Type	Description
disks	No	Array of **TargetDisk** objects	The information about target server disks, which is generally the same as that of the source server. Array Length: **0 - 65535**
image_disk_id	No	String	The ID of the disk that hosts the agent image on the target server. Minimum: **0** Maximum: **255**
snapshot_ids	No	String	The ID of the snapshot for the target server. Minimum: **0** Maximum: **255**
cutovered_snapshot_ids	No	String	The ID of the snapshot used for rollback on the target server. Minimum: **0** Maximum: **255**

Table 5-162 BtrfsFileSystem

Parameter	Mandatory	Type	Description
name	No	String	The file system name. Minimum: **0** Maximum: **255**
label	No	String	The file system tags. If no tag exists, the value is an empty string. Minimum: **0** Maximum: **255**
uuid	No	String	The file system UUID. Minimum: **0** Maximum: **255**
device	No	String	The device names of the Btrfs file system. Minimum: **0** Maximum: **255**

Parameter	Mandatory	Type	Description
size	No	Long	The space occupied by the file system data. Minimum: **0** Maximum: **9223372036854775807**
nodesize	No	Long	The Btrfs node size. Minimum: **0** Maximum: **9223372036854775807**
sectorsize	No	Integer	The sector size. Minimum: **0** Maximum: **2147483647**
data_profile	No	String	The data profile (RAD). Minimum: **0** Maximum: **255**
system_profile	No	String	The file system profile (RAD). Minimum: **0** Maximum: **255**
metadata_profile	No	String	The metadata profile (RAD). Minimum: **0** Maximum: **255**
global_reserve1	No	String	The Btrfs file system information. Minimum: **0** Maximum: **255**
g_vol_used_size	No	Long	The used space of the Btrfs volume. Minimum: **0** Maximum: **9223372036854775807**
default_subvolid	No	String	The ID of the default subvolume. Minimum: **0** Maximum: **255**
default_subvol_name	No	String	The name of the default subvolume. Minimum: **0** Maximum: **255**

Parameter	Mandatory	Type	Description
default_subvol_mountpath	No	String	The mount path of the default subvolume or Btrfs file system. Minimum: **0** Maximum: **255**
subvolumn	No	Array of **BtrfsSubvolumn** objects	The subvolume information. Array Length: **0 - 65535**

Table 5-163 BtrfsSubvolumn

Parameter	Mandatory	Type	Description
uuid	No	String	The UUID of the parent volume. Minimum: **0** Maximum: **255**
is_snapshot	No	String	Indicates whether the subvolume is a snapshot. Minimum: **0** Maximum: **255**
subvol_id	No	String	The subvolume ID. Minimum: **0** Maximum: **255**
parent_id	No	String	The parent volume ID. Minimum: **0** Maximum: **255**
subvol_name	No	String	The subvolume name. Minimum: **0** Maximum: **255**
subvol_mount_path	No	String	The mount path of the subvolume. Minimum: **0** Maximum: **255**

Table 5-164 NetWork

Parameter	Mandatory	Type	Description
name	No	String	The NIC name. Minimum: **0** Maximum: **255**
ip	No	String	The IP address bound to the NIC. Minimum: **0** Maximum: **255**
ipv6	No	String	The IPv6 address. Minimum: **0** Maximum: **255**
netmask	No	String	The subnet mask. Minimum: **0** Maximum: **255**
gateway	No	String	Gateway. Minimum: **0** Maximum: **255**
mtu	No	Integer	The NIC MTU. This parameter is mandatory for Linux. Minimum: **0** Maximum: **2147483647**
mac	Yes	String	The first MAC address in the list must not be empty. Minimum: **0** Maximum: **255**
id	No	String	The database record ID. Minimum: **0** Maximum: **255**

Table 5-165 VolumeGroups

Parameter	Mandatory	Type	Description
components	No	String	The physical volume information. Minimum: **0** Maximum: **255**

Parameter	Mandatory	Type	Description
free_size	No	Long	The available space. Minimum: **0** Maximum: **9223372036854775807**
logical_volumes	No	Array of **LogicalVolumes** objects	The logical volume information. Array Length: **0 - 50**
name	No	String	Name. Minimum: **0** Maximum: **255**
size	No	Long	Size. Minimum: **0** Maximum: **9223372036854775807**

Table 5-166 LogicalVolumes

Parameter	Mandatory	Type	Description
block_count	No	Integer	The number of blocks. Minimum: **0** Maximum: **2147483647** Default: **0**
block_size	No	Long	Block size. Minimum: **0** Maximum: **1048576** Default: **0**
file_system	No	String	The file system. Minimum: **0** Maximum: **255**
inode_size	No	Integer	The number of inodes. Minimum: **0** Maximum: **2147483647**
inode_nums	No	Long	The number of inodes. Minimum: **0** Maximum: **9223372036854775807**

Parameter	Mandatory	Type	Description
device_use	No	String	Partition type. This is not mandatory and can be left blank. The common values are as follows: **NORMAL** **OS**: system device **BOOT** **VOLUME_GROUP** **BTRFS** Minimum: **0** Maximum: **255**
mount_point	No	String	The mount point. Minimum: **0** Maximum: **256**
name	No	String	Name. Minimum: **0** Maximum: **1024**
size	No	Long	The size. Minimum: **0** Maximum: **9223372036854775807**
used_size	No	Long	The used space. Minimum: **0** Maximum: **9223372036854775807**
free_size	No	Long	The available space. Minimum: **0** Maximum: **9223372036854775807**

Table 5-167 TargetDisk

Parameter	Mandatory	Type	Description
id	No	Long	The disk ID. Minimum: **0** Maximum: **9223372036854775807**

Parameter	Mandatory	Type	Description
device_use	No	String	The partition function. The partition can be a general, boot, or OS partition. **BOOT**: boot device **OS**: system device **NORMAL**: general device Default: **NORMAL** Enumeration values: ● **NORMAL** ● **OS** ● **BOOT**
disk_id	No	String	The disk ID. Minimum: **0** Maximum: **255**
name	No	String	The disk name. Minimum: **0** Maximum: **255**
physical_volumes	No	Array of **TargetPhysicalVolumes** objects	The logical volume information. Array Length: **0 - 65535**
size	No	Long	The size. Minimum: **0** Maximum: **9223372036854775807**
used_size	No	Long	The used space. Minimum: **0** Maximum: **9223372036854775807**
disk_index	No	String	The disk index. Minimum: **0** Maximum: **255**
os_disk	No	Boolean	Indicates whether the disk is the system disk.

Parameter	Mandatory	Type	Description
partition_style	No	String	The disk partition type. This parameter is mandatory for source server registration. **MBR**: Master Boot Record **GPT**: GUID Partition Table Enumeration values: • **MBR** • **GPT**
relation_name	No	String	The name of the paired disk on the Linux target ECS server. Minimum: **0** Maximum: **255**

Table 5-168 TargetPhysicalVolumes

Parameter	Mandatory	Type	Description
id	No	Long	The logical volume ID. Minimum: **0** Maximum: **9223372036854775807**
device_use	No	String	The partition function. **NORMAL**: general device **OS**: system device **BOOT**: boot device Default: **NORMAL** Enumeration values: • **NORMAL** • **OS** • **BOOT**
file_system	No	String	The file system. Minimum: **0** Maximum: **255**
index	No	Integer	No. Minimum: **0** Maximum: **2147483647**

Parameter	Mandatory	Type	Description
mount_point	No	String	The mount point. Minimum: **0** Maximum: **255**
name	No	String	Name. Minimum: **0** Maximum: **255**
size	No	Long	The size. Minimum: **0** Maximum: **9223372036854775807**
used_size	No	Long	The used space. Minimum: **0** Maximum: **9223372036854775807**
uuid	No	String	uuid Minimum: **0** Maximum: **255**
relation_name	No	String	The name of the paired disk on the Linux target ECS server. Minimum: **0** Maximum: **255**
free_size	No	Long	The idle partition space. Minimum: **0** Maximum: **9223372036854775807**

Table 5-169 CloneServer

Parameter	Mandatory	Type	Description
vm_id	No	String	The ID of the cloned server. Minimum: **0** Maximum: **255**
name	No	String	The name of the cloned server. Minimum: **0** Maximum: **255**

Parameter	Mandatory	Type	Description
clone_error	No	String	The error code returned for a clone failure. Minimum: **0** Maximum: **255**
clone_state	No	String	The clone status. NOT_READY READY PREPARING CREATING ERROR FINISHED Enumeration values: ● **NOT_READY** ● **READY** ● **PREPARING** ● **CREATING** ● **ERROR** ● **FINISHED**
error_msg	No	String	The error message returned for a clone failure. Minimum: **0** Maximum: **1024**

Response Parameters

Status code: 200

Table 5-170 Response body parameters

Parameter	Type	Description
id	String	The task ID. Minimum: **1** Maximum: **255**

Status code: 403

Table 5-171 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **255**
encoded_authorization_message	String	The encrypted authorization information. Minimum: **0** Maximum: **65535**
error_param	Array of strings	Error parameters. Minimum: **0** Maximum: **65535** Array Length: **1 - 20**
details	Array of **details** objects	The error details. Array Length: **1 - 20**

Table 5-172 details

Parameter	Type	Description
error_code	String	The SMS error code. Minimum: **0** Maximum: **65535**
error_msg	String	The SMS error message. Minimum: **0** Maximum: **65535**

Example Requests

This example updates the migration task whose ID is **ef3b9722-07a0-40ae-89b0-889ee96dxxxx**.

```
put https://{endpoint}/v3/tasks/ef3b9722-07a0-40ae-89b0-889ee96dxxxx

{
  "name" : "MigrationTask",
  "type" : "MIGRATE_BLOCK",
  "speed_limit" : 0,
  "migration_ip" : null,
  "vm_template_id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
```



```
 "region_name" : "region name",
 "region_id" : "region id",
 "project_name" : "project name",
 "project_id" : "xxxxxxxxxxxxxxxxxxxxxxxx00000001",
 "target_server" : {
   "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
   "vm_id" : "",
   "name" : "",
   "ip" : null,
   "os_version" : "WINDOWS2016_64BIT",
   "system_dir" : "Y:\\Windows\\System32",
   "disks" : [ {
     "id" : 88008,
     "name" : "Disk 1",
     "relation_name" : null,
     "disk_id" : "0",
     "partition_style" : "MBR",
     "size" : 42949672960,
     "used_size" : 42947575808,
     "device_use" : "OS",
     "os_disk" : true,
     "physical_volumes" : [ {
       "id" : 135055,
       "uuid" : "\\\\?\\Volume{586b7157-0000-0000-0000-100000000000}\\",
       "index" : 1,
       "name" : "Z:",
       "relation_name" : null,
       "device_use" : "BOOT",
       "file_system" : "NTFS",
       "mount_point" : null,
       "size" : 524288000,
       "used_size" : 410275840,
       "free_size" : 114012160
     }, {
       "id" : 135056,
       "uuid" : "\\\\?\\Volume{586b7157-0000-0000-0000-501f00000000}\\",
       "index" : 2,
       "name" : "Y:",
       "relation_name" : null,
       "device_use" : "OS",
       "file_system" : "NTFS",
       "mount_point" : null,
       "size" : 42423287808,
       "used_size" : 23170301952,
       "free_size" : 19252985856
     } ],
     "disk_index" : "0"
   } ],
   "volume_groups" : [ ],
   "image_disk_id" : null,
   "cutovered_snapshot_ids" : null
 },
 "clone_server" : null
}
```

Example Responses

Status code: 200

This API is used to update a migration task with a specified ID.

```
{
 "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
}
```

Status code: 403

Authentication failed.

```
{
  "error_code" : "SMS.9004",
  "error_msg" : "The current account does not have the permission to execute policy. You do not have permission to perform action XXX on resource XXX.",
  "encoded_authorization_message" : "XXXXXX",
  "error_param" : [ "You do not have permission to perform action XXX on resource XXX." ],
  "details" : [ {
    "error_code" : "SMS.9004",
    "error_msg" : "You do not have permission to perform action XXX on resource XXX."
  } ]
}
```


SDK Sample Code

The SDK sample code is as follows.

Java

This example updates the migration task whose ID is **ef3b9722-07a0-40ae-89b0-889ee96dxxxx**.

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;

import java.util.List;
import java.util.ArrayList;

public class UpdateTaskSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        UpdateTaskRequest request = new UpdateTaskRequest();
        request.withTaskId("{task_id}");
        PutTaskReq body = new PutTaskReq();
        List<TargetPhysicalVolumes> listDisksPhysicalVolumes = new ArrayList<>();
        listDisksPhysicalVolumes.add(
            new TargetPhysicalVolumes()
                .withId(135055L)
                .withDeviceUse(TargetPhysicalVolumes.DeviceUseEnum.fromValue("BOOT"))
                .withFileSystem("NTFS")
                .withIndex(1)
                .withName("Z:")
                .withSize(524288000L)
                .withUsedSize(410275840L)
```

```
                .withUuid("\\?\Volume{586b7157-0000-0000-0000-100000000000}\")
                .withFreeSize(114012160L)
        );
        listDisksPhysicalVolumes.add(
            new TargetPhysicalVolumes()
                .withId(135056L)
                .withDeviceUse(TargetPhysicalVolumes.DeviceUseEnum.fromValue("OS"))
                .withFileSystem("NTFS")
                .withIndex(2)
                .withName("Y:")
                .withSize(42423287808L)
                .withUsedSize(23170301952L)
                .withUuid("\\?\Volume{586b7157-0000-0000-0000-501f00000000}\")
                .withFreeSize(19252985856L)
        );
        List<TargetDisk> listTargetServerDisks = new ArrayList<>();
        listTargetServerDisks.add(
            new TargetDisk()
                .withId(88008L)
                .withDeviceUse(TargetDisk.DeviceUseEnum.fromValue("OS"))
                .withDiskId("0")
                .withName("Disk 1")
                .withPhysicalVolumes(listDisksPhysicalVolumes)
                .withSize(42949672960L)
                .withUsedSize(42947575808L)
                .withDiskIndex("0")
                .withOsDisk(true)
                .withPartitionStyle(TargetDisk.PartitionStyleEnum.fromValue("MBR"))
        );
        TargetServer targetServerbody = new TargetServer();
        targetServerbody.withDisks(listTargetServerDisks)
            .withVolumeGroups()
            .withVmId("")
            .withId("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
            .withOsVersion("WINDOWS2016_64BIT")
            .withSystemDir("Y:\Windows\System32")
            .withName("");
        body.withTargetServer(targetServerbody);
        body.withVmTemplateId("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001");
        body.withProjectId("xxxxxxxxxxxxxxxxxxxxxxxx00000001");
        body.withProjectName("project name");
        body.withSpeedLimit(0);
        body.withRegionName("region name");
        body.withRegionId("region id");
        body.withType(PutTaskReq.TypeEnum.fromValue("MIGRATE_BLOCK"));
        body.withName("MigrationTask");
        request.withBody(body);
        try {
            UpdateTaskResponse response = client.updateTask(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

Python

This example updates the migration task whose ID is **ef3b9722-07a0-40ae-89b0-889ee96dxxxx**.

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = UpdateTaskRequest()
        request.task_id = "{task_id}"
        listPhysicalVolumesDisks = [
            TargetPhysicalVolumes(
                id=135055,
                device_use="BOOT",
                file_system="NTFS",
                index=1,
                name="Z:",
                size=524288000,
                used_size=410275840,
                uuid="\\?\Volume{586b7157-0000-0000-0000-100000000000}\",
                free_size=114012160
            ),
            TargetPhysicalVolumes(
                id=135056,
                device_use="OS",
                file_system="NTFS",
                index=2,
                name="Y:",
                size=42423287808,
                used_size=23170301952,
                uuid="\\?\Volume{586b7157-0000-0000-0000-501f00000000}\",
                free_size=19252985856
            )
        ]
        listDisksTargetServer = [
            TargetDisk(
                id=88008,
                device_use="OS",
                disk_id="0",
                name="Disk 1",
                physical_volumes=listPhysicalVolumesDisks,
                size=42949672960,
                used_size=42947575808,
                disk_index="0",
                os_disk=True,
                partition_style="MBR"
            )
        ]
        targetServerbody = TargetServer(
            disks=listDisksTargetServer,
            vm_id="",
            id="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
```

```
            os_version="WINDOWS2016_64BIT",
            system_dir="Y:\Windows\System32",
            name=""
        )
        request.body = PutTaskReq(
            target_server=targetServerbody,
            vm_template_id="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
            project_id="xxxxxxxxxxxxxxxxxxxxxxxx00000001",
            project_name="project name",
            speed_limit=0,
            region_name="region name",
            region_id="region id",
            type="MIGRATE_BLOCK",
            name="MigrationTask"
        )
        response = client.update_task(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

Go

This example updates the migration task whose ID is **ef3b9722-07a0-40ae-89b0-889ee96dxxxx**.

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
         WithRegion(region.ValueOf("<YOUR REGION>")).
         WithCredential(auth).
         SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)
```

```
request := &model.UpdateTaskRequest{}
request.TaskId = "{task_id}"
idPhysicalVolumes:= int64(135055)
deviceUsePhysicalVolumes:= model.GetTargetPhysicalVolumesDeviceUseEnum().BOOT
fileSystemPhysicalVolumes:= "NTFS"
indexPhysicalVolumes:= int32(1)
namePhysicalVolumes:= "Z:"
sizePhysicalVolumes:= int64(524288000)
usedSizePhysicalVolumes:= int64(410275840)
uuidPhysicalVolumes:= "\\?\Volume{586b7157-0000-0000-0000-100000000000}\"
freeSizePhysicalVolumes:= int64(114012160)
idPhysicalVolumes1:= int64(135056)
deviceUsePhysicalVolumes1:= model.GetTargetPhysicalVolumesDeviceUseEnum().OS
fileSystemPhysicalVolumes1:= "NTFS"
indexPhysicalVolumes1:= int32(2)
namePhysicalVolumes1:= "Y:"
sizePhysicalVolumes1:= int64(42423287808)
usedSizePhysicalVolumes1:= int64(23170301952)
uuidPhysicalVolumes1:= "\\?\Volume{586b7157-0000-0000-0000-501f00000000}\"
freeSizePhysicalVolumes1:= int64(19252985856)
var listPhysicalVolumesDisks = []model.TargetPhysicalVolumes{
    {
        Id: &idPhysicalVolumes,
        DeviceUse: &deviceUsePhysicalVolumes,
        FileSystem: &fileSystemPhysicalVolumes,
        Index: &indexPhysicalVolumes,
        Name: &namePhysicalVolumes,
        Size: &sizePhysicalVolumes,
        UsedSize: &usedSizePhysicalVolumes,
        Uuid: &uuidPhysicalVolumes,
        FreeSize: &freeSizePhysicalVolumes,
    },
    {
        Id: &idPhysicalVolumes1,
        DeviceUse: &deviceUsePhysicalVolumes1,
        FileSystem: &fileSystemPhysicalVolumes1,
        Index: &indexPhysicalVolumes1,
        Name: &namePhysicalVolumes1,
        Size: &sizePhysicalVolumes1,
        UsedSize: &usedSizePhysicalVolumes1,
        Uuid: &uuidPhysicalVolumes1,
        FreeSize: &freeSizePhysicalVolumes1,
    },
}
idDisks:= int64(88008)
deviceUseDisks:= model.GetTargetDiskDeviceUseEnum().OS
diskIdDisks:= "0"
nameDisks:= "Disk 1"
sizeDisks:= int64(42949672960)
usedSizeDisks:= int64(42947575808)
diskIndexDisks:= "0"
osDiskDisks:= true
partitionStyleDisks:= model.GetTargetDiskPartitionStyleEnum().MBR
var listDisksTargetServer = []model.TargetDisk{
    {
        Id: &idDisks,
        DeviceUse: &deviceUseDisks,
        DiskId: &diskIdDisks,
        Name: &nameDisks,
        PhysicalVolumes: &listPhysicalVolumesDisks,
        Size: &sizeDisks,
        UsedSize: &usedSizeDisks,
        DiskIndex: &diskIndexDisks,
        OsDisk: &osDiskDisks,
        PartitionStyle: &partitionStyleDisks,
    },
}
vmIdTargetServer:= ""
```

```
    idTargetServer:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
    osVersionTargetServer:= "WINDOWS2016_64BIT"
    systemDirTargetServer:= "Y:\Windows\System32"
    nameTargetServer:= ""
    targetServerbody := &model.TargetServer{
        Disks: &listDisksTargetServer,
        VmId: &vmIdTargetServer,
        Id: &idTargetServer,
        OsVersion: &osVersionTargetServer,
        SystemDir: &systemDirTargetServer,
        Name: &nameTargetServer,
    }
    vmTemplateIdPutTaskReq:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
    projectIdPutTaskReq:= "xxxxxxxxxxxxxxxxxxxxxxxx00000001"
    projectNamePutTaskReq:= "project name"
    speedLimitPutTaskReq:= int32(0)
    regionNamePutTaskReq:= "region name"
    regionIdPutTaskReq:= "region id"
    typePutTaskReq:= model.GetPutTaskReqTypeEnum().MIGRATE_BLOCK
    namePutTaskReq:= "MigrationTask"
    request.Body = &model.PutTaskReq{
        TargetServer: targetServerbody,
        VmTemplateId: &vmTemplateIdPutTaskReq,
        ProjectId: &projectIdPutTaskReq,
        ProjectName: &projectNamePutTaskReq,
        SpeedLimit: &speedLimitPutTaskReq,
        RegionName: &regionNamePutTaskReq,
        RegionId: &regionIdPutTaskReq,
        Type: &typePutTaskReq,
        Name: &namePutTaskReq,
    }
    response, err := client.UpdateTask(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

More

For SDK sample code of more programming languages, see the Sample Code tab in **API Explorer**. SDK sample code can be automatically generated.

Status Codes

Status Code	Description
200	This API is used to update a migration task with a specified ID.
403	Authentication failed.

Error Codes

See **Error Codes**.

5.4.7 Managing Migration Tasks

Function

This API is used to manage migration tasks, including starting, suspending, and synchronizing tasks, uploading logs, and deleting snapshot resources.

Calling Method

For details, see **Calling APIs**.

Authorization Information

Each account has all the permissions required to call all APIs, but IAM users must be assigned the required permissions.

- If you are using role/policy-based authorization, see **Permissions Policies and Supported Actions** for details on the required permissions.
- If you are using identity policy-based authorization, the following identity policy-based permissions are required.

Action	Access Level	Resource Type (*: required)	Condition Key	Alias	Dependencies
sms:server:manageTask	Write	server *	-	sms:server:migrationServer	-
		-	g:EnterpriseProjectId		

URI

POST /v3/tasks/{task_id}/action

Table 5-173 Path Parameters

Parameter	Mandatory	Type	Description
task_id	Yes	String	The migration task ID. Minimum: **1** Maximum: **36**

Request Parameters

Table 5-174 Request header parameters

Parameter	Mandatory	Type	Description
X-Auth-Token	Yes	String	User token. The token can be obtained by calling the IAM API for obtaining a user token. The value of **X-Subject-Token** in the response header is the user token. Minimum: **1** Maximum: **16384**

Table 5-175 Request body parameters

Parameter	Mandatory	Type	Description
operation	Yes	String	Operation on the task. **start** **stop** **test** **cutover**: Start the target server. **clone_test** **restart** **network_check** **skip**: Skip the consistency verification subtask. **clear**: Clear snapshot resources. **migration_test**: Start the migration drill. **error_for_overspeed**: The task is automatically suspended due to overspeed. Enumeration values: • **start** • **stop** • **test** • **clone_test** • **restart** • **network_check** • **clear** • **skip** • **migration_test**
template_id	No	String	The template ID. Minimum: **0** Maximum: **255**
switch_hce	No	Boolean	Whether to switch to HCE. Only Linux migration tasks are supported. Default: **false**
is_need_consistency_check	No	Boolean	Indicates whether consistency verification is enabled. Default: **false**

Response Parameters

Status code: 200

Table 5-176 Response body parameters

Parameter	Type	Description
-	String	The management operation for the migration task succeeded.

Status code: 403

Table 5-177 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **255**
encoded_authorization_message	String	The encrypted authorization information. Minimum: **0** Maximum: **65535**
error_param	Array of strings	Error parameters. Minimum: **0** Maximum: **65535** Array Length: **1 - 20**
details	Array of **details** objects	The error details. Array Length: **1 - 20**

Table 5-178 details

Parameter	Type	Description
error_code	String	The SMS error code. Minimum: **0** Maximum: **65535**

Parameter	Type	Description
error_msg	String	The SMS error message. Minimum: **0** Maximum: **65535**

Example Requests

- Migration tasks can be managed. To clone the target server in a migration task, a template ID is required. This example clones the target server in the task with the ID **7a9a9540-ff28-4869-b9e4-855fbe12xxxx**.

```
POST https://{endpoint}/v3/tasks/7a9a9540-ff28-4869-b9e4-855fbe12xxxx/action

{
  "operation" : "clone_test",
  "template_id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
}
```

- This example launches the target server in a migration task.

```
POST https://{endpoint}/v3/tasks/7a9a9540-ff28-4869-b9e4-855fbe12xxxx/action

{
  "operation" : "cutover"
}
```

- This example starts the task with the ID **7a9a9540-ff28-4869-b9e4-855fbe12xxxx**.

```
POST https://{endpoint}/v3/tasks/7a9a9540-ff28-4869-b9e4-855fbe12xxxx/action

{
  "operation" : "start"
}
```

- This example pauses the task with the ID **7a9a9540-ff28-4869-b9e4-855fbe12xxxx**.

```
POST https://{endpoint}/v3/tasks/7a9a9540-ff28-4869-b9e4-855fbe12xxxx/action

{
  "operation" : "stop"
}
```

- This example starts the task whose ID is **7a9a9540-ff28-4869-b9e4-855fbe12xxxx**.

```
POST https://{endpoint}/v3/tasks/7a9a9540-ff28-4869-b9e4-855fbe12xxxx/action

{
  "operation" : "restart"
}
```

- This example clears snapshot resources for the task whose ID is **7a9a9540-ff28-4869-b9e4-855fbe12xxxx**.

```
POST https://{endpoint}/v3/tasks/7a9a9540-ff28-4869-b9e4-855fbe12xxxx/action

{
  "operation" : "clear"
}
```

- This example skips the consistency verification subtask.

```
POST https://{endpoint}/v3/tasks/7a9a9540-ff28-4869-b9e4-855fbe12xxxx/action

{
```

```
  "operation" : "skip"
}
```

- This example starts the migration drill subtask.

```
POST https://{endpoint}/v3/tasks/7a9a9540-ff28-4869-b9e4-855fbe12xxxx/action

{
  "operation" : "migration_test"
}
```

Example Responses

Status code: 200

The management operation for the migration task succeeded.

```
{ }
```

Status code: 403

Authentication failed.

```
{
  "error_code" : "SMS.9004",
  "error_msg" : "The current account does not have the permission to execute policy. You do not have permission to perform action XXX on resource XXX.",
  "encoded_authorization_message" : "XXXXXX",
  "error_param" : [ "You do not have permission to perform action XXX on resource XXX." ],
  "details" : [ {
    "error_code" : "SMS.9004",
    "error_msg" : "You do not have permission to perform action XXX on resource XXX."
  } ]
}
```

SDK Sample Code

The SDK sample code is as follows.

Java

- Migration tasks can be managed. To clone the target server in a migration task, a template ID is required. This example clones the target server in the task with the ID **7a9a9540-ff28-4869-b9e4-855fbe12xxxx**.

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;


public class UpdateTaskStatusSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
```

```
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        UpdateTaskStatusRequest request = new UpdateTaskStatusRequest();
        request.withTaskId("{task_id}");
        UpdateTaskStatusReq body = new UpdateTaskStatusReq();
        body.withTemplateId("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001");
        body.withOperation(UpdateTaskStatusReq.OperationEnum.fromValue("clone_test"));
        request.withBody(body);
        try {
            UpdateTaskStatusResponse response = client.updateTaskStatus(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

- This example launches the target server in a migration task.

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;


public class UpdateTaskStatusSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great
security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or
environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before
running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local
environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        UpdateTaskStatusRequest request = new UpdateTaskStatusRequest();
        request.withTaskId("{task_id}");
        UpdateTaskStatusReq body = new UpdateTaskStatusReq();
        body.withOperation(UpdateTaskStatusReq.OperationEnum.fromValue("cutover"));
```

```
        request.withBody(body);
        try {
            UpdateTaskStatusResponse response = client.updateTaskStatus(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

- This example starts the task with the ID **7a9a9540-ff28-4869-b9e4-855fbe12xxxx**.

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;


public class UpdateTaskStatusSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great 
security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or 
environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before 
running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local 
environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        UpdateTaskStatusRequest request = new UpdateTaskStatusRequest();
        request.withTaskId("{task_id}");
        UpdateTaskStatusReq body = new UpdateTaskStatusReq();
        body.withOperation(UpdateTaskStatusReq.OperationEnum.fromValue("start"));
        request.withBody(body);
        try {
            UpdateTaskStatusResponse response = client.updateTaskStatus(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
```

```
            System.out.println(e.getErrorMsg());
        }
    }
}
```

- This example pauses the task with the ID **7a9a9540-ff28-4869-b9e4-855fbe12xxxx**.

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;


public class UpdateTaskStatusSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        UpdateTaskStatusRequest request = new UpdateTaskStatusRequest();
        request.withTaskId("{task_id}");
        UpdateTaskStatusReq body = new UpdateTaskStatusReq();
        body.withOperation(UpdateTaskStatusReq.OperationEnum.fromValue("stop"));
        request.withBody(body);
        try {
            UpdateTaskStatusResponse response = client.updateTaskStatus(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

- This example starts the task whose ID is **7a9a9540-ff28-4869-b9e4-855fbe12xxxx**.

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
```

```
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;


public class UpdateTaskStatusSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        UpdateTaskStatusRequest request = new UpdateTaskStatusRequest();
        request.withTaskId("{task_id}");
        UpdateTaskStatusReq body = new UpdateTaskStatusReq();
        body.withOperation(UpdateTaskStatusReq.OperationEnum.fromValue("restart"));
        request.withBody(body);
        try {
            UpdateTaskStatusResponse response = client.updateTaskStatus(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

- This example clears snapshot resources for the task whose ID is **7a9a9540-ff28-4869-b9e4-855fbe12xxxx**.

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;


public class UpdateTaskStatusSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before
```

```
running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local
environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        UpdateTaskStatusRequest request = new UpdateTaskStatusRequest();
        request.withTaskId("{task_id}");
        UpdateTaskStatusReq body = new UpdateTaskStatusReq();
        body.withOperation(UpdateTaskStatusReq.OperationEnum.fromValue("clear"));
        request.withBody(body);
        try {
            UpdateTaskStatusResponse response = client.updateTaskStatus(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

- This example skips the consistency verification subtask.

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;


public class UpdateTaskStatusSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great
security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or
environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before
running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local
environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        UpdateTaskStatusRequest request = new UpdateTaskStatusRequest();
        request.withTaskId("{task_id}");
```

```
        UpdateTaskStatusReq body = new UpdateTaskStatusReq();
        body.withOperation(UpdateTaskStatusReq.OperationEnum.fromValue("skip"));
        request.withBody(body);
        try {
            UpdateTaskStatusResponse response = client.updateTaskStatus(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

- This example starts the migration drill subtask.

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;


public class UpdateTaskStatusSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        UpdateTaskStatusRequest request = new UpdateTaskStatusRequest();
        request.withTaskId("{task_id}");
        UpdateTaskStatusReq body = new UpdateTaskStatusReq();
        body.withOperation(UpdateTaskStatusReq.OperationEnum.fromValue("migration_test"));
        request.withBody(body);
        try {
            UpdateTaskStatusResponse response = client.updateTaskStatus(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
```

```
            System.out.println(e.getErrorMsg());
        }
    }
}
```

Python

- Migration tasks can be managed. To clone the target server in a migration task, a template ID is required. This example clones the target server in the task with the ID **7a9a9540-ff28-4869-b9e4-855fbe12xxxx**.

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = UpdateTaskStatusRequest()
        request.task_id = "{task_id}"
        request.body = UpdateTaskStatusReq(
            template_id="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
            operation="clone_test"
        )
        response = client.update_task_status(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

- This example launches the target server in a migration task.

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]
```

```
    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = UpdateTaskStatusRequest()
        request.task_id = "{task_id}"
        request.body = UpdateTaskStatusReq(
            operation="cutover"
        )
        response = client.update_task_status(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

- This example starts the task with the ID **7a9a9540-ff28-4869-b9e4-855fbe12xxxx**.

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great
security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or
environment variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before
running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local
environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = UpdateTaskStatusRequest()
        request.task_id = "{task_id}"
        request.body = UpdateTaskStatusReq(
            operation="start"
        )
        response = client.update_task_status(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

- This example pauses the task with the ID **7a9a9540-ff28-4869-b9e4-855fbe12xxxx**.

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
```



```
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = UpdateTaskStatusRequest()
        request.task_id = "{task_id}"
        request.body = UpdateTaskStatusReq(
            operation="stop"
        )
        response = client.update_task_status(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

- This example starts the task whose ID is **7a9a9540-ff28-4869-b9e4-855fbe12xxxx**.

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = UpdateTaskStatusRequest()
        request.task_id = "{task_id}"
        request.body = UpdateTaskStatusReq(
            operation="restart"
        )
        response = client.update_task_status(request)
```



```
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

- This example clears snapshot resources for the task whose ID is **7a9a9540-ff28-4869-b9e4-855fbe12xxxx**.

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great
security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or
environment variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before
running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local
environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = UpdateTaskStatusRequest()
        request.task_id = "{task_id}"
        request.body = UpdateTaskStatusReq(
            operation="clear"
        )
        response = client.update_task_status(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

- This example skips the consistency verification subtask.

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great
security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or
environment variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before
running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local
environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)
```

```
    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = UpdateTaskStatusRequest()
        request.task_id = "{task_id}"
        request.body = UpdateTaskStatusReq(
            operation="skip"
        )
        response = client.update_task_status(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

- This example starts the migration drill subtask.

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = UpdateTaskStatusRequest()
        request.task_id = "{task_id}"
        request.body = UpdateTaskStatusReq(
            operation="migration_test"
        )
        response = client.update_task_status(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

Go

- Migration tasks can be managed. To clone the target server in a migration task, a template ID is required. This example clones the target server in the task with the ID **7a9a9540-ff28-4869-b9e4-855fbe12xxxx**.

```
package main

import (
    "fmt"
```

```
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great
security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or
environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before
running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local
environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
         WithRegion(region.ValueOf("<YOUR REGION>")).
         WithCredential(auth).
         SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.UpdateTaskStatusRequest{}
    request.TaskId = "{task_id}"
    templateIdUpdateTaskStatusReq:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
    request.Body = &model.UpdateTaskStatusReq{
        TemplateId: &templateIdUpdateTaskStatusReq,
        Operation: model.GetUpdateTaskStatusReqOperationEnum().CLONE_TEST,
    }
    response, err := client.UpdateTaskStatus(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

- This example launches the target server in a migration task.

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great
security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or
environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before
```

```
running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local
environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
         WithRegion(region.ValueOf("<YOUR REGION>")).
         WithCredential(auth).
         SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.UpdateTaskStatusRequest{}
    request.TaskId = "{task_id}"
    request.Body = &model.UpdateTaskStatusReq{
        Operation: model.GetUpdateTaskStatusReqOperationEnum().CUTOVER,
    }
    response, err := client.UpdateTaskStatus(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

- This example starts the task with the ID **7a9a9540-ff28-4869-b9e4-855fbe12xxxx**.

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great
security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or
environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before
running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local
environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
```

```
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
         WithRegion(region.ValueOf("<YOUR REGION>")).
         WithCredential(auth).
         SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.UpdateTaskStatusRequest{}
    request.TaskId = "{task_id}"
    request.Body = &model.UpdateTaskStatusReq{
        Operation: model.GetUpdateTaskStatusReqOperationEnum().START,
    }
    response, err := client.UpdateTaskStatus(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

- This example pauses the task with the ID **7a9a9540-ff28-4869-b9e4-855fbe12xxxx**.

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great
security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or
environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before
running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local
environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
         WithRegion(region.ValueOf("<YOUR REGION>")).
         WithCredential(auth).
         SafeBuild()


    if err != nil {
```

```
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.UpdateTaskStatusRequest{}
    request.TaskId = "{task_id}"
    request.Body = &model.UpdateTaskStatusReq{
        Operation: model.GetUpdateTaskStatusReqOperationEnum().STOP,
    }
    response, err := client.UpdateTaskStatus(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

- This example starts the task whose ID is **7a9a9540-ff28-4869-b9e4-855fbe12xxxx**.

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
         WithRegion(region.ValueOf("<YOUR REGION>")).
         WithCredential(auth).
         SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.UpdateTaskStatusRequest{}
    request.TaskId = "{task_id}"
    request.Body = &model.UpdateTaskStatusReq{
        Operation: model.GetUpdateTaskStatusReqOperationEnum().RESTART,
    }
```



```
    response, err := client.UpdateTaskStatus(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

- This example clears snapshot resources for the task whose ID is **7a9a9540-ff28-4869-b9e4-855fbe12xxxx**.

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
         WithRegion(region.ValueOf("<YOUR REGION>")).
         WithCredential(auth).
         SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.UpdateTaskStatusRequest{}
    request.TaskId = "{task_id}"
    request.Body = &model.UpdateTaskStatusReq{
        Operation: model.GetUpdateTaskStatusReqOperationEnum().CLEAR,
    }
    response, err := client.UpdateTaskStatus(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

- This example skips the consistency verification subtask.

```
package main
```

```
import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
         WithRegion(region.ValueOf("<YOUR REGION>")).
         WithCredential(auth).
         SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.UpdateTaskStatusRequest{}
    request.TaskId = "{task_id}"
    request.Body = &model.UpdateTaskStatusReq{
        Operation: model.GetUpdateTaskStatusReqOperationEnum().SKIP,
    }
    response, err := client.UpdateTaskStatus(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

- This example starts the migration drill subtask.

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before
```

```
running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local
environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
         WithRegion(region.ValueOf("<YOUR REGION>")).
         WithCredential(auth).
         SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.UpdateTaskStatusRequest{}
    request.TaskId = "{task_id}"
    request.Body = &model.UpdateTaskStatusReq{
        Operation: model.GetUpdateTaskStatusReqOperationEnum().MIGRATION_TEST,
    }
    response, err := client.UpdateTaskStatus(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

More

For SDK sample code of more programming languages, see the Sample Code tab in **API Explorer**. SDK sample code can be automatically generated.

Status Codes

Status Code	Description
200	The management operation for the migration task succeeded.
403	Authentication failed.

Error Codes

See **Error Codes**.

5.4.8 Reporting Migration Progress and Rate

Function

This API is called by the Agent installed on source servers during migration to report the migration progress and speed to SMS.

You do not need to make calls to this API.

Calling Method

For details, see **Calling APIs**.

Authorization Information

Each account has all the permissions required to call all APIs, but IAM users must be assigned the required permissions.

- If you are using role/policy-based authorization, see **Permissions Policies and Supported Actions** for details on the required permissions.
- If you are using identity policy-based authorization, the following identity policy-based permissions are required.

Action	Access Level	Resource Type (*: required)	Condition Key	Alias	Dependencies
sms:server:updateTaskProgress	Write	server *	-	sms:server:migrationServer	-
		-	g:EnterpriseProjectId		

URI

PUT /v3/tasks/{task_id}/progress

Table 5-179 Path Parameters

Parameter	Mandatory	Type	Description
task_id	Yes	String	The migration task ID. Minimum: **1** Maximum: **36**

Request Parameters

Table 5-180 Request header parameters

Parameter	Mandatory	Type	Description
X-Auth-Token	Yes	String	User token. The token can be obtained by calling the IAM API for obtaining a user token. The value of **X-Subject-Token** in the response header is the user token. Minimum: **1** Maximum: **16384**

Table 5-181 Request body parameters

Parameter	Mandatory	Type	Description
subtask_name	Yes	String	The name of the subtask whose progress is being reported. Options: **CREATE_CLOUD_SERVER** **SSL_CONFIG** **ATTACH_AGENT_IMAGE** **DETTACH_AGENT_IMAGE** **FORMAT_DISK_LINUX** **FORMAT_DISK_LINUX_FILE** **FORMAT_DISK_LINUX_BLOCK** **FORMAT_DISK_WINDOWS** **MIGRATE_LINUX_FILE** **MIGRATE_LINUX_BLOCK** **MIGRATE_WINDOWS_BLOCK** **CLONE_VM** **SYNC_LINUX_FILE** **SYNC_LINUX_BLOCK** **SYNC_WINDOWS_BLOCK** **CONFIGURE_LINUX** **CONFIGURE_LINUX_BLOCK** **CONFIGURE_LINUX_FILE** **CONFIGURE_WINDOWS** Minimum: **0** Maximum: **255** Enumeration values: • **CREATE_CLOUD_SERVER** • **SSL_CONFIG** • **ATTACH_AGENT_IMAGE** • **DETTACH_AGENT_IMAGE** • **FORMAT_DISK_LINUX** • **FORMAT_DISK_LINUX_FILE** • **FORMAT_DISK_LINUX_BLOCK** • **FORMAT_DISK_WINDOWS** • **MIGRATE_LINUX_FILE** • **MIGRATE_LINUX_BLOCK**

Parameter	Mandatory	Type	Description
			• **MIGRATE_WINDOWS_BLOCK** • **CLONE_VM** • **SYNC_LINUX_FILE** • **SYNC_LINUX_BLOCK** • **SYNC_WINDOWS_BLOCK** • **CONFIGURE_LINUX** • **CONFIGURE_LINUX_BLOCK** • **CONFIGURE_LINUX_FILE** • **CONFIGURE_WINDOWS**
progress	Yes	Integer	The progress of the subtask, in percentage. Minimum: **0** Maximum: **100**
replicatesize	Yes	Long	The amount of data replicated in the subtask, in bytes. Minimum: **0** Maximum: **9223372036854775807**
totalsize	Yes	Long	The total amount of data to be migrated in the subtask. Minimum: **0** Maximum: **9223372036854775807**
process_trace	Yes	String	The detailed progress of the migration or synchronization. Minimum: **0** Maximum: **2048**
migrate_speed	No	Double	The migration rate, in Mbit/s. Minimum: **0** Maximum: **10000**
compress_rate	No	Double	The file compression rate. Minimum: **0** Maximum: **10000**
remain_time	No	Long	The remaining time. Minimum: **0** Maximum: **2147483647**

Parameter	Mandatory	Type	Description
total_cpu_usage	No	Double	The CPU usage of the server. The value ranges from 0 to 100, in percentage. Minimum: **0** Maximum: **100**
agent_cpu_usage	No	Double	The CPU usage of the Agent. The value ranges from 0 to 100, in percentage. Minimum: **0** Maximum: **100**
total_mem_usage	No	Double	The memory usage of the server, in MB. Minimum: **0** Maximum: **1048576.0**
agent_mem_usage	No	Double	The memory usage of the Agent, in MB. Minimum: **0** Maximum: **1048576.0**
total_disk_io	No	Double	The disk I/O of the server, in Mbit/s. Minimum: **0** Maximum: **10000.0**
agent_disk_io	No	Double	The disk I/O of the Agent, in Mbit/s. Minimum: **0** Maximum: **10000.0**
need_migration_test	No	Boolean	Indicates whether migration drilling is enabled.
agent_time	No	String	The current local time of the source server, which is used for overspeed detection. The speed limits can be configured by time period. Minimum: **0** Maximum: **30**

Response Parameters

Status code: 200

Table 5-182 Response body parameters

Parameter	Type	Description
-	String	The migration progress and speed were reported.

Status code: 403

Table 5-183 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **255**
encoded_authorization_message	String	The encrypted authorization information. Minimum: **0** Maximum: **65535**
error_param	Array of strings	Error parameters. Minimum: **0** Maximum: **65535** Array Length: **1 - 20**
details	Array of **details** objects	The error details. Array Length: **1 - 20**

Table 5-184 details

Parameter	Type	Description
error_code	String	The SMS error code. Minimum: **0** Maximum: **65535**
error_msg	String	The SMS error message. Minimum: **0** Maximum: **65535**

Example Requests

This example reports the migration progress of a task. The current subtask name is **ATTACH_AGENT_IMAGE**, the task progress is **100**, and the total size of data migrated in the current task is **10000**.

```
PUT https://{endpoint}/v3/tasks/7a9a9540-ff28-4869-b9e4-855fbe12xxxx/progress

{
  "subtask_name" : "ATTACH_AGENT_IMAGE",
  "progress" : 100,
  "replicatesize" : 1000,
  "totalsize" : 100000,
  "process_trace" : ""
}
```

Example Responses

Status code: 200

The migration progress and speed were reported.

```
{ }
```

Status code: 403

Authentication failed.

```
{
  "error_code" : "SMS.9004",
  "error_msg" : "The current account does not have the permission to execute policy. You do not have
permission to perform action XXX on resource XXX.",
  "encoded_authorization_message" : "XXXXXX",
  "error_param" : [ "You do not have permission to perform action XXX on resource XXX." ],
  "details" : [ {
    "error_code" : "SMS.9004",
    "error_msg" : "You do not have permission to perform action XXX on resource XXX."
  } ]
}
```

SDK Sample Code

The SDK sample code is as follows.

Java

This example reports the migration progress of a task. The current subtask name is **ATTACH_AGENT_IMAGE**, the task progress is **100**, and the total size of data migrated in the current task is **10000**.

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;


public class UpdateTaskSpeedSolution {
```

```
    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great 
security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or 
environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running 
this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        UpdateTaskSpeedRequest request = new UpdateTaskSpeedRequest();
        request.withTaskId("{task_id}");
        UpdateTaskSpeedReq body = new UpdateTaskSpeedReq();
        body.withProcessTrace("");
        body.withTotalsize(100000L);
        body.withReplicatesize(1000L);
        body.withProgress(100);

body.withSubtaskName(UpdateTaskSpeedReq.SubtaskNameEnum.fromValue("ATTACH_AGENT_IMAGE"));
        request.withBody(body);
        try {
            UpdateTaskSpeedResponse response = client.updateTaskSpeed(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

Python

This example reports the migration progress of a task. The current subtask name is **ATTACH_AGENT_IMAGE**, the task progress is **100**, and the total size of data migrated in the current task is **10000**.

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security 
risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment 
variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this 
example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)
```

```
client = SmsClient.new_builder() \
    .with_credentials(credentials) \
    .with_region(SmsRegion.value_of("<YOUR REGION>")) \
    .build()

try:
    request = UpdateTaskSpeedRequest()
    request.task_id = "{task_id}"
    request.body = UpdateTaskSpeedReq(
        process_trace="",
        totalsize=100000,
        replicatesize=1000,
        progress=100,
        subtask_name="ATTACH_AGENT_IMAGE"
    )
    response = client.update_task_speed(request)
    print(response)
except exceptions.ClientRequestException as e:
    print(e.status_code)
    print(e.request_id)
    print(e.error_code)
    print(e.error_msg)
```

Go

This example reports the migration progress of a task. The current subtask name is **ATTACH_AGENT_IMAGE**, the task progress is **100**, and the total size of data migrated in the current task is **10000**.

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
         WithRegion(region.ValueOf("<YOUR REGION>")).
         WithCredential(auth).
         SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
```

```
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.UpdateTaskSpeedRequest{}
    request.TaskId = "{task_id}"
    request.Body = &model.UpdateTaskSpeedReq{
        ProcessTrace: "",
        Totalsize: int64(100000),
        Replicatesize: int64(1000),
        Progress: int32(100),
        SubtaskName: model.GetUpdateTaskSpeedReqSubtaskNameEnum().ATTACH_AGENT_IMAGE,
    }
    response, err := client.UpdateTaskSpeed(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

More

For SDK sample code of more programming languages, see the Sample Code tab in **API Explorer**. SDK sample code can be automatically generated.

Status Codes

Status Code	Description
200	The migration progress and speed were reported.
403	Authentication failed.

Error Codes

See **Error Codes**.

5.4.9 Querying the Migration Rate Limiting Rules of a Migration Task

Function

This API is used to query the time segment-based migration rate limiting rules of a migration task.

Calling Method

For details, see **Calling APIs**.

Authorization Information

Each account has all the permissions required to call all APIs, but IAM users must be assigned the required permissions.

- If you are using role/policy-based authorization, see **Permissions Policies and Supported Actions** for details on the required permissions.
- If you are using identity policy-based authorization, the following identity policy-based permissions are required.

Action	Access Level	Resource Type (*: required)	Condition Key	Alias	Dependencies
sms:server:getTaskSpeedLimit	Read	server *	-	sms:server:queryServer	-
		-	g:EnterpriseProjectId		

URI

GET /v3/tasks/{task_id}/speed-limit

Table 5-185 Path Parameters

Parameter	Mandatory	Type	Description
task_id	Yes	String	The ID of the task for which you want to query the speed limiting rules Minimum: **1** Maximum: **36**

Request Parameters

Table 5-186 Request header parameters

Parameter	Mandatory	Type	Description
X-Auth-Token	Yes	String	User token. The token can be obtained by calling the IAM API for obtaining a user token. The value of **X-Subject-Token** in the response header is the user token. Minimum: **1** Maximum: **16384**

Response Parameters

Status code: 200

Table 5-187 Response body parameters

Parameter	Type	Description
speed_limit	Array of **SpeedLimitlJson** objects	The information about the time range-based migration rate limiting rules. Array Length: **0 - 5**

Table 5-188 SpeedLimitlJson

Parameter	Type	Description
start	String	The start time of a period. The format is XX:XX. Minimum: **0** Maximum: **255**
end	String	The end time of a period. The format is XX:XX. Minimum: **0** Maximum: **255**
speed	Integer	The migration rate limit for the specified period of time. The value is an integer ranging from 0 to 1,000. The unit is Mbit/s. Minimum: **0** Maximum: **1000** Default: **0**
over_speed_threshold	Double	The overspeed threshold for stopping migration. This is a protection measure. If the migration speed exceeds the threshold, the task is stopped. It is used to control the consumption of resources (especially network bandwidth) during the migration to ensure that the overall system performance is not affected by a single migration task. The unit is percentage. Minimum: **10** Maximum: **100** Default: **10**

Status code: 403

Table 5-189 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **255**
encoded_authorization_message	String	The encrypted authorization information. Minimum: **0** Maximum: **65535**
error_param	Array of strings	Error parameters. Minimum: **0** Maximum: **65535** Array Length: **1 - 20**
details	Array of **details** objects	The error details. Array Length: **1 - 20**

Table 5-190 details

Parameter	Type	Description
error_code	String	The SMS error code. Minimum: **0** Maximum: **65535**
error_msg	String	The SMS error message. Minimum: **0** Maximum: **65535**

Example Requests

This example queries the migration rate limiting rules for a migration task.

```
GET https://{endpoint}/v3/tasks/7a9a9540-ff28-4869-b9e4-855fbe12xxxx/speed-limit
```

Example Responses

Status code: 200

The migration rate limiting rules of a migration task were obtained.


```
{
  "speed_limit" : [ {
    "start" : "00:00",
    "end" : "23:59",
    "speed" : 1000,
    "over_speed_threshold" : 50.0
  } ]
}
```


Status code: 403

Authentication failed.


```
{
  "error_code" : "SMS.9004",
  "error_msg" : "The current account does not have the permission to execute policy. You do not have permission to perform action XXX on resource XXX.",
  "encoded_authorization_message" : "XXXXXX",
  "error_param" : [ "You do not have permission to perform action XXX on resource XXX." ],
  "details" : [ {
    "error_code" : "SMS.9004",
    "error_msg" : "You do not have permission to perform action XXX on resource XXX."
  } ]
}
```


SDK Sample Code

The SDK sample code is as follows.

Java

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;


public class ShowsSpeedLimitsSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        ShowsSpeedLimitsRequest request = new ShowsSpeedLimitsRequest();
        request.withTaskId("{task_id}");
        try {
            ShowsSpeedLimitsResponse response = client.showsSpeedLimits(request);
```

```
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

Python

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = ShowsSpeedLimitsRequest()
        request.task_id = "{task_id}"
        response = client.shows_speed_limits(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

Go

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before running this
```

```
example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
        WithRegion(region.ValueOf("<YOUR REGION>")).
        WithCredential(auth).
        SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.ShowsSpeedLimitsRequest{}
    request.TaskId = "{task_id}"
    response, err := client.ShowsSpeedLimits(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

More

For SDK sample code of more programming languages, see the Sample Code tab in **API Explorer**. SDK sample code can be automatically generated.

Status Codes

Status Code	Description
200	The migration rate limiting rules of a migration task were obtained.
403	Authentication failed.

Error Codes

See **Error Codes**.

5.4.10 Setting Migration Rate Limiting Rules for a Migration Task

Function

This API is used to set migration rate limiting rules for a migration task.

Calling Method

For details, see **Calling APIs**.

Authorization Information

Each account has all the permissions required to call all APIs, but IAM users must be assigned the required permissions.

- If you are using role/policy-based authorization, see **Permissions Policies and Supported Actions** for details on the required permissions.
- If you are using identity policy-based authorization, the following identity policy-based permissions are required.

Action	Access Level	Resource Type (*: required)	Condition Key	Alias	Dependencies
sms:server:updateTaskSpeedLimit	Write	server *	-	sms:server:migrationServer	-
		-	g:EnterpriseProjectId		

URI

POST /v3/tasks/{task_id}/speed-limit

Table 5-191 Path Parameters

Parameter	Mandatory	Type	Description
task_id	Yes	String	The migration task ID. Minimum: **1** Maximum: **36**

Request Parameters

Table 5-192 Request header parameters

Parameter	Mandatory	Type	Description
X-Auth-Token	Yes	String	User token. The token can be obtained by calling the IAM API for obtaining a user token. The value of **X-Subject-Token** in the response header is the user token. Minimum: **1** Maximum: **16384**

Table 5-193 Request body parameters

Parameter	Mandatory	Type	Description
speed_limit	Yes	Array of **SpeedLimitJson** objects	The information about the time range-based migration rate limiting rules. Array Length: **0 - 5**

Table 5-194 SpeedLimitJson

Parameter	Mandatory	Type	Description
start	Yes	String	The start time of a period. The format is XX:XX. Minimum: **0** Maximum: **255**
end	Yes	String	The end time of a period. The format is XX:XX. Minimum: **0** Maximum: **255**
speed	No	Integer	The migration rate limit for the specified period of time. The value is an integer ranging from 0 to 1,000. The unit is Mbit/s. Minimum: **0** Maximum: **1000** Default: **0**

Parameter	Mandatory	Type	Description
over_speed_threshold	No	Double	The overspeed threshold for stopping migration. This is a protection measure. If the migration speed exceeds the threshold, the task is stopped. It is used to control the consumption of resources (especially network bandwidth) during the migration to ensure that the overall system performance is not affected by a single migration task. The unit is percentage. Minimum: **10** Maximum: **100** Default: **10**

Response Parameters

Status code: 200

Table 5-195 Response body parameters

Parameter	Type	Description
-	String	Setting migration rate limiting rules for a migration task succeeded.

Status code: 403

Table 5-196 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **255**

Parameter	Type	Description
encoded_authorization_message	String	The encrypted authorization information. Minimum: **0** Maximum: **65535**
error_param	Array of strings	Error parameters. Minimum: **0** Maximum: **65535** Array Length: **1 - 20**
details	Array of **details** objects	The error details. Array Length: **1 - 20**

Table 5-197 details

Parameter	Type	Description
error_code	String	The SMS error code. Minimum: **0** Maximum: **65535**
error_msg	String	The SMS error message. Minimum: **0** Maximum: **65535**

Example Requests

- This example limits the speed of a migration task to 20 Mbit/s from 0:00 to 8:00, 50 Mbit/s from 8:00 to 15:00, and 25 Mbit/s from 15:00 to 23:59. It also sets an overspeed threshold for the task.

```
POST https://{endpoint}/v3/tasks/7a9a9540-ff28-4869-b9e4-855fbe12xxxx/speed-limit

{
  "speed_limit" : [ {
    "start" : "00:00",
    "end" : "08:00",
    "speed" : 20,
    "over_speed_threshold" : 50
  }, {
    "start" : "08:00",
    "end" : "15:00",
    "speed" : 50,
    "over_speed_threshold" : 50
  }, {
    "start" : "15:00",
    "end" : "23:59",
    "speed" : 25,
    "over_speed_threshold" : 50
  } ]
}
```


- This example updates the migration rate limit rules of the task whose ID is **a45a300b-86b5-4b13-8802-52274fa43016**.

```
POST https://{endpoint}/v3/tasks/a45a300b-86b5-4b13-8802-52274fa43016/speed-limit

{
  "speed_limit" : [ {
    "start" : "00:00",
    "end" : "08:00",
    "speed" : 20
  }, {
    "start" : "08:00",
    "end" : "15:00",
    "speed" : 50
  }, {
    "start" : "15:00",
    "end" : "23:59",
    "speed" : 25
  } ]
}
```

Example Responses

Status code: 200

Setting migration rate limiting rules for a migration task succeeded.

```
{ }
```

Status code: 403

Authentication failed.

```
{
  "error_code" : "SMS.9004",
  "error_msg" : "The current account does not have the permission to execute policy. You do not have
permission to perform action XXX on resource XXX.",
  "encoded_authorization_message" : "XXXXXX",
  "error_param" : [ "You do not have permission to perform action XXX on resource XXX." ],
  "details" : [ {
    "error_code" : "SMS.9004",
    "error_msg" : "You do not have permission to perform action XXX on resource XXX."
  } ]
}
```

SDK Sample Code

The SDK sample code is as follows.

Java

- This example limits the speed of a migration task to 20 Mbit/s from 0:00 to 8:00, 50 Mbit/s from 8:00 to 15:00, and 25 Mbit/s from 15:00 to 23:59. It also sets an overspeed threshold for the task.

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;
```

```
import java.util.List;
import java.util.ArrayList;

public class UpdateSpeedSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        UpdateSpeedRequest request = new UpdateSpeedRequest();
        request.withTaskId("{task_id}");
        SpeedLimit body = new SpeedLimit();
        List<SpeedLimitlJson> listbodySpeedLimit = new ArrayList<>();
        listbodySpeedLimit.add(
            new SpeedLimitlJson()
                .withStart("00:00")
                .withEnd("08:00")
                .withSpeed(20)
                .withOverSpeedThreshold((double)50)
        );
        listbodySpeedLimit.add(
            new SpeedLimitlJson()
                .withStart("08:00")
                .withEnd("15:00")
                .withSpeed(50)
                .withOverSpeedThreshold((double)50)
        );
        listbodySpeedLimit.add(
            new SpeedLimitlJson()
                .withStart("15:00")
                .withEnd("23:59")
                .withSpeed(25)
                .withOverSpeedThreshold((double)50)
        );
        body.withSpeedLimit(listbodySpeedLimit);
        request.withBody(body);
        try {
            UpdateSpeedResponse response = client.updateSpeed(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

- This example updates the migration rate limit rules of the task whose ID is **a45a300b-86b5-4b13-8802-52274fa43016**.

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;

import java.util.List;
import java.util.ArrayList;

public class UpdateSpeedSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        UpdateSpeedRequest request = new UpdateSpeedRequest();
        request.withTaskId("{task_id}");
        SpeedLimit body = new SpeedLimit();
        List<SpeedLimitlJson> listbodySpeedLimit = new ArrayList<>();
        listbodySpeedLimit.add(
            new SpeedLimitlJson()
                .withStart("00:00")
                .withEnd("08:00")
                .withSpeed(20)
        );
        listbodySpeedLimit.add(
            new SpeedLimitlJson()
                .withStart("08:00")
                .withEnd("15:00")
                .withSpeed(50)
        );
        listbodySpeedLimit.add(
            new SpeedLimitlJson()
                .withStart("15:00")
                .withEnd("23:59")
                .withSpeed(25)
        );
        body.withSpeedLimit(listbodySpeedLimit);
        request.withBody(body);
        try {
            UpdateSpeedResponse response = client.updateSpeed(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
```

```
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

Python

- This example limits the speed of a migration task to 20 Mbit/s from 0:00 to 8:00, 50 Mbit/s from 8:00 to 15:00, and 25 Mbit/s from 15:00 to 23:59. It also sets an overspeed threshold for the task.

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great
security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or
environment variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before
running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local
environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = UpdateSpeedRequest()
        request.task_id = "{task_id}"
        listSpeedLimitbody = [
            SpeedLimitJson(
                start="00:00",
                end="08:00",
                speed=20,
                over_speed_threshold=50
            ),
            SpeedLimitJson(
                start="08:00",
                end="15:00",
                speed=50,
                over_speed_threshold=50
            ),
            SpeedLimitJson(
                start="15:00",
                end="23:59",
                speed=25,
                over_speed_threshold=50
            )
        ]
        request.body = SpeedLimit(
            speed_limit=listSpeedLimitbody
        )
        response = client.update_speed(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
```

```
        print(e.error_code)
        print(e.error_msg)
```

- This example updates the migration rate limit rules of the task whose ID is **a45a300b-86b5-4b13-8802-52274fa43016**.

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = UpdateSpeedRequest()
        request.task_id = "{task_id}"
        listSpeedLimitbody = [
            SpeedLimitJson(
                start="00:00",
                end="08:00",
                speed=20
            ),
            SpeedLimitJson(
                start="08:00",
                end="15:00",
                speed=50
            ),
            SpeedLimitJson(
                start="15:00",
                end="23:59",
                speed=25
            )
        ]
        request.body = SpeedLimit(
            speed_limit=listSpeedLimitbody
        )
        response = client.update_speed(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

Go

- This example limits the speed of a migration task to 20 Mbit/s from 0:00 to 8:00, 50 Mbit/s from 8:00 to 15:00, and 25 Mbit/s from 15:00 to 23:59. It also sets an overspeed threshold for the task.

```
package main
```

```
import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great
security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or
environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before
running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local
environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
         WithRegion(region.ValueOf("<YOUR REGION>")).
         WithCredential(auth).
         SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.UpdateSpeedRequest{}
    request.TaskId = "{task_id}"
    speedSpeedLimit:= int32(20)
    overSpeedThresholdSpeedLimit:= float64(50)
    speedSpeedLimit1:= int32(50)
    overSpeedThresholdSpeedLimit1:= float64(50)
    speedSpeedLimit2:= int32(25)
    overSpeedThresholdSpeedLimit2:= float64(50)
    var listSpeedLimitbody = []model.SpeedLimitJson{
        {
            Start: "00:00",
            End: "08:00",
            Speed: &speedSpeedLimit,
            OverSpeedThreshold: &overSpeedThresholdSpeedLimit,
        },
        {
            Start: "08:00",
            End: "15:00",
            Speed: &speedSpeedLimit1,
            OverSpeedThreshold: &overSpeedThresholdSpeedLimit1,
        },
        {
            Start: "15:00",
            End: "23:59",
            Speed: &speedSpeedLimit2,
            OverSpeedThreshold: &overSpeedThresholdSpeedLimit2,
        },
    }
```



```
    request.Body = &model.SpeedLimit{
        SpeedLimit: listSpeedLimitbody,
    }
    response, err := client.UpdateSpeed(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

- This example updates the migration rate limit rules of the task whose ID is **a45a300b-86b5-4b13-8802-52274fa43016**.

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great
security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or
environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before
running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local
environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
        WithRegion(region.ValueOf("<YOUR REGION>")).
        WithCredential(auth).
        SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.UpdateSpeedRequest{}
    request.TaskId = "{task_id}"
    speedSpeedLimit:= int32(20)
    speedSpeedLimit1:= int32(50)
    speedSpeedLimit2:= int32(25)
    var listSpeedLimitbody = []model.SpeedLimitJson{
        {
            Start: "00:00",
            End: "08:00",
            Speed: &speedSpeedLimit,
        },
        {
            Start: "08:00",
```

```
            End: "15:00",
            Speed: &speedSpeedLimit1,
        },
        {
            Start: "15:00",
            End: "23:59",
            Speed: &speedSpeedLimit2,
        },
    }
    request.Body = &model.SpeedLimit{
        SpeedLimit: listSpeedLimitbody,
    }
    response, err := client.UpdateSpeed(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

More

For SDK sample code of more programming languages, see the Sample Code tab in **API Explorer**. SDK sample code can be automatically generated.

Status Codes

Status Code	Description
200	Setting migration rate limiting rules for a migration task succeeded.
403	Authentication failed.

Error Codes

See **Error Codes**.

5.4.11 Querying a Certificate Passphrase

Function

This API is used to query the certificate passphrase of the secure transmission channel for a task.

Calling Method

For details, see **Calling APIs**.

Authorization Information

Each account has all the permissions required to call all APIs, but IAM users must be assigned the required permissions.

- If you are using role/policy-based authorization, see **Permissions Policies and Supported Actions** for details on the required permissions.

- If you are using identity policy-based authorization, the following identity policy-based permissions are required.

Action	Access Level	Resource Type (*: required)	Condition Key	Alias	Dependencies
sms:server:getTaskPassphrase	Read	server *	-	sms:server:queryServer	-
		-	g:EnterpriseProjectId		

URI

GET /v3/tasks/{task_id}/passphrase

Table 5-198 Path Parameters

Parameter	Mandatory	Type	Description
task_id	Yes	String	Task ID. Minimum: **1** Maximum: **36**

Request Parameters

Table 5-199 Request header parameters

Parameter	Mandatory	Type	Description
X-Auth-Token	Yes	String	User token. The token can be obtained by calling the IAM API for obtaining a user token. The value of **X-Subject-Token** in the response header is the user token. Minimum: **1** Maximum: **16384**

Response Parameters

Status code: 200

Table 5-200 Response body parameters

Parameter	Type	Description
task_id	String	The task ID. Minimum: **0** Maximum: **255**
passphrase	String	The certificate passphrase of the secure transmission channel. Minimum: **0** Maximum: **255**

Status code: 403

Table 5-201 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **255**
encoded_authorization_message	String	The encrypted authorization information. Minimum: **0** Maximum: **65535**
error_param	Array of strings	Error parameters. Minimum: **0** Maximum: **65535** Array Length: **1 - 20**
details	Array of **details** objects	The error details. Array Length: **1 - 20**

Table 5-202 details

Parameter	Type	Description
error_code	String	The SMS error code. Minimum: **0** Maximum: **65535**
error_msg	String	The SMS error message. Minimum: **0** Maximum: **65535**

Example Requests

This example queries the certificate passphrase of the secure transmission channel for the task with ID **d7fa81b9-c174-4c0a-a475-51a54c8af8a4**.

```
GET https://{endpoint}/v3/tasks/d7fa81b9-c174-4c0a-a475-51a54c8af8a4/passphrase
X-Auth-Token: XXXX
```

Example Responses

Status code: 200

Request succeeded.

```
{
  "task_id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
  "passphrase" : "********"
}
```

Status code: 403

Authentication failed.

```
{
  "error_code" : "SMS.9004",
  "error_msg" : "The current account does not have the permission to execute policy. You do not have
permission to perform action XXX on resource XXX.",
  "encoded_authorization_message" : "XXXXXX",
  "error_param" : [ "You do not have permission to perform action XXX on resource XXX." ],
  "details" : [ {
    "error_code" : "SMS.9004",
    "error_msg" : "You do not have permission to perform action XXX on resource XXX."
  } ]
}
```

SDK Sample Code

The SDK sample code is as follows.

Java

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
```

```
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;


public class ShowPassphraseSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great
security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or
environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running
this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        ShowPassphraseRequest request = new ShowPassphraseRequest();
        request.withTaskId("{task_id}");
        try {
            ShowPassphraseResponse response = client.showPassphrase(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

Python

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security
risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment
variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this
example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()
```

```
    try:
        request = ShowPassphraseRequest()
        request.task_id = "{task_id}"
        response = client.show_passphrase(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

Go

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
         WithRegion(region.ValueOf("<YOUR REGION>")).
         WithCredential(auth).
         SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.ShowPassphraseRequest{}
    request.TaskId = "{task_id}"
    response, err := client.ShowPassphrase(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

More

For SDK sample code of more programming languages, see the Sample Code tab in **API Explorer**. SDK sample code can be automatically generated.

Status Codes

Status Code	Description
200	Request succeeded.
403	Authentication failed.

Error Codes

See **Error Codes**.

5.4.12 Uploading Migration Task Logs

Function

This API is used to upload the logs of a migration task.

Calling Method

For details, see **Calling APIs**.

Authorization Information

Each account has all the permissions required to call all APIs, but IAM users must be assigned the required permissions.

- If you are using role/policy-based authorization, see **Permissions Policies and Supported Actions** for details on the required permissions.
- If you are using identity policy-based authorization, the following identity policy-based permissions are required.

Action	Access Level	Resource Type (*: required)	Condition Key	Alias	Dependencies
sms:server:collectLog	Write	server *	-	sms:server:migrationServer	-
		-	g:EnterpriseProjectId		

URI

POST /v3/tasks/{task_id}/log

Table 5-203 Path Parameters

Parameter	Mandatory	Type	Description
task_id	Yes	String	The migration task ID. Minimum: **1** Maximum: **36**

Request Parameters

Table 5-204 Request header parameters

Parameter	Mandatory	Type	Description
X-Auth-Token	Yes	String	The user token. Minimum: **1** Maximum: **16384**

Table 5-205 Request body parameters

Parameter	Mandatory	Type	Description
log_bucket	Yes	String	The bucket name. Minimum: **0** Maximum: **255**

Response Parameters

Status code: 200

Table 5-206 Response body parameters

Parameter	Type	Description
-	String	Migration task logs were uploaded successfully.

Example Requests

This example uploads migration task logs to bucket **centos** and sets the URL validity period to **300 seconds**.

```
POST https://{endpoint}/v3/tasks/{task_id}/log

{
```

```
  "log_bucket" : "centos"
}
```

Example Responses

Status code: 200

Migration task logs were uploaded successfully.

```
{ }
```

SDK Sample Code

The SDK sample code is as follows.

Java

This example uploads migration task logs to bucket **centos** and sets the URL validity period to **300 seconds**.

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;


public class CollectLogSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great 
security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or 
environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running 
this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        CollectLogRequest request = new CollectLogRequest();
        request.withTaskId("{task_id}");
        UploadLogRequestBody body = new UploadLogRequestBody();
        body.withLogBucket("centos");
        request.withBody(body);
        try {
            CollectLogResponse response = client.collectLog(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
```

```
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

Python

This example uploads migration task logs to bucket **centos** and sets the URL validity period to **300 seconds**.

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = CollectLogRequest()
        request.task_id = "{task_id}"
        request.body = UploadLogRequestBody(
            log_bucket="centos"
        )
        response = client.collect_log(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

Go

This example uploads migration task logs to bucket **centos** and sets the URL validity period to **300 seconds**.

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security
```

```
risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment
variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before running this
example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
         WithRegion(region.ValueOf("<YOUR REGION>")).
         WithCredential(auth).
         SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.CollectLogRequest{}
    request.TaskId = "{task_id}"
    request.Body = &model.UploadLogRequestBody{
        LogBucket: "centos",
    }
    response, err := client.CollectLog(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

More

For SDK sample code of more programming languages, see the Sample Code tab in **API Explorer**. SDK sample code can be automatically generated.

Status Codes

Status Code	Description
200	Migration task logs were uploaded successfully.

Error Codes

See **Error Codes**.

5.4.13 Obtaining Consistency Verification Results

Function

This API is used to obtain the brief consistency verification results of a task.

Calling Method

For details, see **Calling APIs**.

Authorization Information

Each account has all the permissions required to call all APIs, but IAM users must be assigned the required permissions.

- If you are using role/policy-based authorization, see **Permissions Policies and Supported Actions** for details on the required permissions.
- If you are using identity policy-based authorization, the following identity policy-based permissions are required.

Action	Access Level	Resource Type (*: required)	Condition Key	Alias	Dependencies
sms:server:getConsistencyCheckResult	Read	server *	-	sms:server:queryServer	-

URI

GET /v3/tasks/{task_id}/consistency-result

Table 5-207 Path Parameters

Parameter	Mandatory	Type	Description
task_id	Yes	String	Task ID. Minimum: **1** Maximum: **36**

Request Parameters

Table 5-208 Request header parameters

Parameter	Mandatory	Type	Description
X-Auth-Token	Yes	String	User token. The token can be obtained by calling the IAM API for obtaining a user token. The value of **X-Subject-Token** in the response header is the user token. Minimum: **1** Maximum: **16384**

Response Parameters

Status code: 200

Table 5-209 Response body parameters

Parameter	Type	Description
result_list	Array of strings	The consistency verification results. Minimum: **0** Maximum: **65535** Array Length: **0 - 65535**
task_id	String	The task ID. Minimum: **1** Maximum: **36**

Status code: 400

Table 5-210 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **1024**

Status code: 403

Table 5-211 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **255**
encoded_authorization_message	String	The encrypted authorization information. Minimum: **0** Maximum: **65535**
error_param	Array of strings	Error parameters. Minimum: **0** Maximum: **65535** Array Length: **1 - 20**
details	Array of **details** objects	The error details. Array Length: **1 - 20**

Table 5-212 details

Parameter	Type	Description
error_code	String	The SMS error code. Minimum: **0** Maximum: **65535**
error_msg	String	The SMS error message. Minimum: **0** Maximum: **65535**

Status code: 404

Table 5-213 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **1024**

Status code: 500

Table 5-214 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **1024**

Example Requests

This example obtains the brief consistency verification results of the task whose ID is **137224b7-8d7c-4919-b33e-ed159778xxxx**.

```
GET https://{endpoint}/v3/137224b7-8d7c-4919-b33e-ed159778xxxx/consistency-result
```

Example Responses

Status code: 200

The brief consistency verification results were obtained.

```
{
  "result_list" : [ "{\"finished_time\": 1736854315000,\"check_result\": \"success\",\"consistency_result\":
[{\"dir_check\": \"/root/sync\",\"num_total_files\": 1,\"num_different_files\": 0,\"num_target_miss_files\":
0,\"num_target_more_files\": 0}]}" ],
  "task_id" : "7861c7ab-06c0-4b23-a350-00e5ed361fbb"
}
```

Status code: 403

Authentication failed.

```
{
  "error_code" : "SMS.9004",
  "error_msg" : "The current account does not have the permission to execute policy. You do not have
```

```
permission to perform action XXX on resource XXX.",
  "encoded_authorization_message" : "XXXXXX",
  "error_param" : [ "You do not have permission to perform action XXX on resource XXX." ],
  "details" : [ {
    "error_code" : "SMS.9004",
    "error_msg" : "You do not have permission to perform action XXX on resource XXX."
  } ]
}
```

SDK Sample Code

The SDK sample code is as follows.

Java

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;


public class ShowConsistencyResultSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great
security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or
environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running
this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        ShowConsistencyResultRequest request = new ShowConsistencyResultRequest();
        request.withTaskId("{task_id}");
        try {
            ShowConsistencyResultResponse response = client.showConsistencyResult(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

Python

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = ShowConsistencyResultRequest()
        request.task_id = "{task_id}"
        response = client.show_consistency_result(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

Go

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
```

```
        WithRegion(region.ValueOf("<YOUR REGION>")).
        WithCredential(auth).
        SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.ShowConsistencyResultRequest{}
    request.TaskId = "{task_id}"
    response, err := client.ShowConsistencyResult(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

More

For SDK sample code of more programming languages, see the Sample Code tab in **API Explorer**. SDK sample code can be automatically generated.

Status Codes

Status Code	Description
200	The brief consistency verification results were obtained.
400	The request parameters were missing.
403	Authentication failed.
404	The task was not found.
500	Obtaining the brief consistency verification results failed.

Error Codes

See **Error Codes**.

5.4.14 Uploading Consistency Verification Results

Function

This API is called by the Agent to upload the consistency verification results of a task.

Calling Method

For details, see **Calling APIs**.

Authorization Information

Each account has all the permissions required to call all APIs, but IAM users must be assigned the required permissions.

- If you are using role/policy-based authorization, see **Permissions Policies and Supported Actions** for details on the required permissions.
- If you are using identity policy-based authorization, the following identity policy-based permissions are required.

Action	Access Level	Resource Type (*: required)	Condition Key	Alias	Dependencies
sms:server:updateConsistencyCheckResult	Write	server *	-	sms:server:migrationServer	-

URI

POST /v3/tasks/{task_id}/consistency-result

Table 5-215 Path Parameters

Parameter	Mandatory	Type	Description
task_id	Yes	String	Task ID. Minimum: **1** Maximum: **36**

Request Parameters

Table 5-216 Request header parameters

Parameter	Mandatory	Type	Description
X-Auth-Token	Yes	String	User token. The token can be obtained by calling the IAM API for obtaining a user token. The value of **X-Subject-Token** in the response header is the user token. Minimum: **1** Maximum: **16384**

Table 5-217 Request body parameters

Parameter	Mandatory	Type	Description
consistency_result	No	Array of **ConsistencyResult** objects	The consistency verification results. Array Length: **0 - 30**

Table 5-218 ConsistencyResult

Parameter	Mandatory	Type	Description
dir_check	Yes	String	The directory verified. Minimum: **0** Maximum: **1024**
num_total_files	Yes	Integer	The total number of files verified. Minimum: **0** Maximum: **1000000**
num_different_files	Yes	Integer	The number of inconsistent files. Minimum: **0** Maximum: **1000000**
num_target_miss_files	Yes	Integer	The number of files missing at the target. Minimum: **0** Maximum: **1000000**
num_target_more_files	Yes	Integer	The number of files redundant at the target. Minimum: **0** Maximum: **1000000**

Response Parameters

Status code: 200

Table 5-219 Response body parameters

Parameter	Type	Description
-	String	The consistency verification results were updated.

Status code: 400

Table 5-220 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **1024**

Status code: 403

Table 5-221 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **255**
encoded_authorization_message	String	The encrypted authorization information. Minimum: **0** Maximum: **65535**
error_param	Array of strings	Error parameters. Minimum: **0** Maximum: **65535** Array Length: **1 - 20**
details	Array of **details** objects	The error details. Array Length: **1 - 20**

Table 5-222 details

Parameter	Type	Description
error_code	String	The SMS error code. Minimum: **0** Maximum: **65535**
error_msg	String	The SMS error message. Minimum: **0** Maximum: **65535**

Status code: 404

Table 5-223 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **1024**

Status code: 500

Table 5-224 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **1024**

Example Requests

This example updates the brief consistency verification results of the task whose ID is **137224b7-8d7c-4919-b33e-ed159778xxxx**.

```
POST https://{endpoint}/v3/137224b7-8d7c-4919-b33e-ed159778xxxx/consistency-result
```



```
{
  "consistency_result" : [ {
    "dir_check" : "/root/data",
    "num_total_files" : 1235,
    "num_different_files" : 12,
    "num_target_miss_files" : 12,
    "num_target_more_files" : 12
  }, {
    "dir_check" : "/var",
    "num_total_files" : 1235,
    "num_different_files" : 12,
    "num_target_miss_files" : 12,
    "num_target_more_files" : 12
  } ]
}
```


Example Responses

Status code: 200

The consistency verification results were updated.

```
{ }
```

Status code: 403

Authentication failed.


```
{
  "error_code" : "SMS.9004",
  "error_msg" : "The current account does not have the permission to execute policy. You do not have 
permission to perform action XXX on resource XXX.",
  "encoded_authorization_message" : "XXXXXX",
  "error_param" : [ "You do not have permission to perform action XXX on resource XXX." ],
  "details" : [ {
    "error_code" : "SMS.9004",
    "error_msg" : "You do not have permission to perform action XXX on resource XXX."
  } ]
}
```


SDK Sample Code

The SDK sample code is as follows.

Java

This example updates the brief consistency verification results of the task whose ID is **137224b7-8d7c-4919-b33e-ed159778xxxx**.

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;

import java.util.List;
import java.util.ArrayList;

public class UpdateConsistencyResultSolution {

    public static void main(String[] args) {
```

```
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great
security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or
environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running
this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        UpdateConsistencyResultRequest request = new UpdateConsistencyResultRequest();
        request.withTaskId("{task_id}");
        SetConsistencyResultRequestBody body = new SetConsistencyResultRequestBody();
        List<ConsistencyResult> listbodyConsistencyResult = new ArrayList<>();
        listbodyConsistencyResult.add(
            new ConsistencyResult()
                .withDirCheck("/root/data")
                .withNumTotalFiles(1235)
                .withNumDifferentFiles(12)
                .withNumTargetMissFiles(12)
                .withNumTargetMoreFiles(12)
        );
        listbodyConsistencyResult.add(
            new ConsistencyResult()
                .withDirCheck("/var")
                .withNumTotalFiles(1235)
                .withNumDifferentFiles(12)
                .withNumTargetMissFiles(12)
                .withNumTargetMoreFiles(12)
        );
        body.withConsistencyResult(listbodyConsistencyResult);
        request.withBody(body);
        try {
            UpdateConsistencyResultResponse response = client.updateConsistencyResult(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

Python

This example updates the brief consistency verification results of the task whose ID is **137224b7-8d7c-4919-b33e-ed159778xxxx**.

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
```



```
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = UpdateConsistencyResultRequest()
        request.task_id = "{task_id}"
        listConsistencyResultbody = [
            ConsistencyResult(
                dir_check="/root/data",
                num_total_files=1235,
                num_different_files=12,
                num_target_miss_files=12,
                num_target_more_files=12
            ),
            ConsistencyResult(
                dir_check="/var",
                num_total_files=1235,
                num_different_files=12,
                num_target_miss_files=12,
                num_target_more_files=12
            )
        ]
        request.body = SetConsistencyResultRequestBody(
            consistency_result=listConsistencyResultbody
        )
        response = client.update_consistency_result(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

Go

This example updates the brief consistency verification results of the task whose ID is **137224b7-8d7c-4919-b33e-ed159778xxxx**.

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")
```

```
    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
        WithRegion(region.ValueOf("<YOUR REGION>")).
        WithCredential(auth).
        SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.UpdateConsistencyResultRequest{}
    request.TaskId = "{task_id}"
    var listConsistencyResultbody = []model.ConsistencyResult{
        {
            DirCheck: "/root/data",
            NumTotalFiles: int32(1235),
            NumDifferentFiles: int32(12),
            NumTargetMissFiles: int32(12),
            NumTargetMoreFiles: int32(12),
        },
        {
            DirCheck: "/var",
            NumTotalFiles: int32(1235),
            NumDifferentFiles: int32(12),
            NumTargetMissFiles: int32(12),
            NumTargetMoreFiles: int32(12),
        },
    }
    request.Body = &model.SetConsistencyResultRequestBody{
        ConsistencyResult: &listConsistencyResultbody,
    }
    response, err := client.UpdateConsistencyResult(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

More

For SDK sample code of more programming languages, see the Sample Code tab in **API Explorer**. SDK sample code can be automatically generated.

Status Codes

Status Code	Description
200	The consistency verification results were updated.
400	The request parameters were missing.

Status Code	Description
403	Authentication failed.
404	The task was not found.
500	Uploading the brief consistency verification results failed.

Error Codes

See **Error Codes**.

5.4.15 Obtaining Consistency Verification Results in Batches

Function

This API is used to export consistency verification results in batches.

Calling Method

For details, see **Calling APIs**.

Authorization Information

Each account has all the permissions required to call all APIs, but IAM users must be assigned the required permissions.

- If you are using role/policy-based authorization, see **Permissions Policies and Supported Actions** for details on the required permissions.
- If you are using identity policy-based authorization, the following identity policy-based permissions are required.

Action	Access Level	Resource Type (*: required)	Condition Key	Alias	Dependencies
sms:server:getBatchConsistencyResults	Read	server *	-	sms:server:queryServer	-

URI

POST /v3/tasks/consistency-results/export

Request Parameters

Table 5-225 Request header parameters

Parameter	Mandatory	Type	Description
X-Auth-Token	Yes	String	User token. The token can be obtained by calling the IAM API for obtaining a user token. The value of **X-Subject-Token** in the response header is the user token.
X-Language	No	String	The preferred language. Select English.

Table 5-226 Request body parameters

Parameter	Mandatory	Type	Description
task_info	Yes	Array of **BatchConsistencyReq** objects	All task information. Array Length: **1 - 200**

Table 5-227 BatchConsistencyReq

Parameter	Mandatory	Type	Description
task_id	No	String	The task ID. Minimum: **1** Maximum: **36**
source_id	No	String	The source server ID. Minimum: **0** Maximum: **255**
source_name	No	String	The source server name. Minimum: **0** Maximum: **255**

Response Parameters

Status code: 200

Table 5-228 Response body parameters

Parameter	Type	Description
-	File	OK

Status code: 400

Table 5-229 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **1024**

Status code: 403

Table 5-230 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **255**
encoded_authorization_message	String	The encrypted authorization information. Minimum: **0** Maximum: **65535**
error_param	Array of strings	Error parameters. Minimum: **0** Maximum: **65535** Array Length: **1 - 20**
details	Array of **details** objects	The error details. Array Length: **1 - 20**

Table 5-231 details

Parameter	Type	Description
error_code	String	The SMS error code. Minimum: **0** Maximum: **65535**
error_msg	String	The SMS error message. Minimum: **0** Maximum: **65535**

Status code: 404

Table 5-232 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **1024**

Example Requests

Task information

source_id: specifies the source server ID.

task_id: specifies the task ID.

source_name: specifies the source server name.

```
POST https://{endpoint}/v3/tasks/consistency-results/export

{
  "task_info" : [ {
    "source_id" : "31806fb2-95bd-421f-ae35-b81b750dxxxx",
    "task_id" : "05a41be9-3ffe-4cc6-8d66-07359039xxxx",
    "source_name" : "ecsxxx"
  } ]
}
```

Example Responses

Status code: 200

OK

```
{ }
```

Status code: 403

Authentication failed.


```
{
  "error_code" : "SMS.9004",
  "error_msg" : "The current account does not have the permission to execute policy. You do not have permission to perform action XXX on resource XXX.",
  "encoded_authorization_message" : "XXXXXX",
  "error_param" : [ "You do not have permission to perform action XXX on resource XXX." ],
  "details" : [ {
    "error_code" : "SMS.9004",
    "error_msg" : "You do not have permission to perform action XXX on resource XXX."
  } ]
}
```


SDK Sample Code

The SDK sample code is as follows.

Java

Task information

source_id: specifies the source server ID.

task_id: specifies the task ID.

source_name: specifies the source server name.

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;

import java.util.List;
import java.util.ArrayList;

public class ExportConsistencyResultsSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        ExportConsistencyResultsRequest request = new ExportConsistencyResultsRequest();
        BatchGetConsistencyResultReq body = new BatchGetConsistencyResultReq();
        List<BatchConsistencyReq> listbodyTaskInfo = new ArrayList<>();
        listbodyTaskInfo.add(
```

```
            new BatchConsistencyReq()
                .withTaskId("05a41be9-3ffe-4cc6-8d66-07359039xxxx")
                .withSourceId("31806fb2-95bd-421f-ae35-b81b750dxxxx")
                .withSourceName("ecsxxx")
        );
        body.withTaskInfo(listbodyTaskInfo);
        request.withBody(body);
        try {
            ExportConsistencyResultsResponse response = client.exportConsistencyResults(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

Python

Task information

source_id: specifies the source server ID.

task_id: specifies the task ID.

source_name: specifies the source server name.

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = ExportConsistencyResultsRequest()
        listTaskInfobody = [
            BatchConsistencyReq(
                task_id="05a41be9-3ffe-4cc6-8d66-07359039xxxx",
                source_id="31806fb2-95bd-421f-ae35-b81b750dxxxx",
                source_name="ecsxxx"
            )
        ]
        request.body = BatchGetConsistencyResultReq(
```

```
        task_info=listTaskInfobody
    )
    response = client.export_consistency_results(request)
    print(response)
except exceptions.ClientRequestException as e:
    print(e.status_code)
    print(e.request_id)
    print(e.error_code)
    print(e.error_msg)
```

Go

Task information

source_id: specifies the source server ID.

task_id: specifies the task ID.

source_name: specifies the source server name.

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
         WithRegion(region.ValueOf("<YOUR REGION>")).
         WithCredential(auth).
         SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.ExportConsistencyResultsRequest{}
    taskIdTaskInfo:= "05a41be9-3ffe-4cc6-8d66-07359039xxxx"
    sourceIdTaskInfo:= "31806fb2-95bd-421f-ae35-b81b750dxxxx"
    sourceNameTaskInfo:= "ecsxxx"
    var listTaskInfobody = []model.BatchConsistencyReq{
        {
```

```
            TaskId: &taskIdTaskInfo,
            SourceId: &sourceIdTaskInfo,
            SourceName: &sourceNameTaskInfo,
        },
    }
    request.Body = &model.BatchGetConsistencyResultReq{
        TaskInfo: listTaskInfobody,
    }
    response, err := client.ExportConsistencyResults(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

More

For SDK sample code of more programming languages, see the Sample Code tab in **API Explorer**. SDK sample code can be automatically generated.

Status Codes

Status Code	Description
200	OK
400	Bad Request
403	Authentication failed.
404	Not Found

Error Codes

See **Error Codes**.

5.5 Command Management

5.5.1 Obtaining Commands from SMS

Function

This API is called by the migration Agent on source servers to obtain commands from the SMS server.

Calling Method

For details, see **Calling APIs**.

Authorization Information

Each account has all the permissions required to call all APIs, but IAM users must be assigned the required permissions.

- If you are using role/policy-based authorization, see **Permissions Policies and Supported Actions** for details on the required permissions.
- If you are using identity policy-based authorization, the following identity policy-based permissions are required.

Action	Access Level	Resource Type (*: required)	Condition Key	Alias	Dependencies
sms:server:getCommand	Read	server *	-	sms:server:migrationServer	-
		-	g:EnterpriseProjectId		

URI

GET /v3/sources/{server_id}/command

Table 5-233 Path Parameters

Parameter	Mandatory	Type	Description
server_id	Yes	String	The ID of the source server that the command is sent to. Minimum: **1** Maximum: **36**

Request Parameters

Table 5-234 Request header parameters

Parameter	Mandatory	Type	Description
X-Auth-Token	Yes	String	User token. The token can be obtained by calling the IAM API for obtaining a user token. The value of **X-Subject-Token** in the response header is the user token. Minimum: **1** Maximum: **16384**

Response Parameters

Status code: 200

Table 5-235 Response body parameters

Parameter	Type	Description
command_name	String	The command name. The value can be **START**, **STOP**, **SKIP**, **PREMIGRATE**, **TASK_NOT_STARTED**, **TASK_HAS_STOPPED**, or **TASK_HAS_FINISHED**. Minimum: **0** Maximum: **255**
command_param	**CommandParam** object	Command response parameters.

Table 5-236 CommandParam

Parameter	Type	Description
task_id	String	The task ID. Minimum: **0** Maximum: **255**
bucket	String	The bucket name. Minimum: **0** Maximum: **255**

Status code: 400

Table 5-237 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **1024**

Status code: 401

Table 5-238 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **1024**

Status code: 403

Table 5-239 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **255**
encoded_authorization_message	String	The encrypted authorization information. Minimum: **0** Maximum: **65535**
error_param	Array of strings	Error parameters. Minimum: **0** Maximum: **65535** Array Length: **1 - 20**
details	Array of **details** objects	The error details. Array Length: **1 - 20**

Table 5-240 details

Parameter	Type	Description
error_code	String	The SMS error code. Minimum: **0** Maximum: **65535**

Parameter	Type	Description
error_msg	String	The SMS error message. Minimum: **0** Maximum: **65535**

Status code: 404

Table 5-241 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **1024**

Status code: 500

Table 5-242 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **1024**

Example Requests

This example obtains the command sent to the Agent installed on the source server whose ID is **f32ab4d6-d150-4fb3-aa55-edbb5cf9947f**.

```
GET https://{endpoint}/v3/sources/f32ab4d6-d150-4fb3-aa55-edbb5cf9947f/command
```

Example Responses

Status code: 200

Obtaining commands from SMS succeeded.

```
{
  "command_name" : "START",
  "command_param" : {
    "task_id" : "xxxxxxxxxxxxxxxxxxxxxxxxx00000001",
    "bucket" : "my_bucket"
  }
}
```


Status code: 403

Authentication failed.


```
{
  "error_code" : "SMS.9004",
  "error_msg" : "The current account does not have the permission to execute policy. You do not have permission to perform action XXX on resource XXX.",
  "encoded_authorization_message" : "XXXXXX",
  "error_param" : [ "You do not have permission to perform action XXX on resource XXX." ],
  "details" : [ {
    "error_code" : "SMS.9004",
    "error_msg" : "You do not have permission to perform action XXX on resource XXX."
  } ]
}
```


SDK Sample Code

The SDK sample code is as follows.

Java

```java
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;


public class ShowCommandSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        ShowCommandRequest request = new ShowCommandRequest();
        request.withServerId("{server_id}");
        try {
            ShowCommandResponse response = client.showCommand(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
```

```
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

Python

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = ShowCommandRequest()
        request.server_id = "{server_id}"
        response = client.show_command(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

Go

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")
```

```
    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
         WithRegion(region.ValueOf("<YOUR REGION>")).
         WithCredential(auth).
         SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.ShowCommandRequest{}
    request.ServerId = "{server_id}"
    response, err := client.ShowCommand(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

More

For SDK sample code of more programming languages, see the Sample Code tab in **API Explorer**. SDK sample code can be automatically generated.

Status Codes

Status Code	Description
200	Obtaining commands from SMS succeeded.
400	Bad Request
401	Unauthorized
403	Authentication failed.
404	Not Found
500	Internal Server Error

Error Codes

See **Error Codes**.

5.5.2 Reporting Command Execution Results to SMS

Function

This API is called by the Agent to send the execution result of a specified command to SMS.

Calling Method

For details, see **Calling APIs**.

Authorization Information

Each account has all the permissions required to call all APIs, but IAM users must be assigned the required permissions.

- If you are using role/policy-based authorization, see **Permissions Policies and Supported Actions** for details on the required permissions.
- If you are using identity policy-based authorization, the following identity policy-based permissions are required.

Action	Access Level	Resource Type (*: required)	Condition Key	Alias	Dependencies
sms:server:updateCommandResult	Write	server *	-	sms:server:migrationServer	-
		-	g:EnterpriseProjectId		

URI

POST /v3/sources/{server_id}/command_result

Table 5-243 Path Parameters

Parameter	Mandatory	Type	Description
server_id	Yes	String	The ID of the source server for which the Agent reports the command execution result to SMS. Minimum: **1** Maximum: **36**

Request Parameters

Table 5-244 Request header parameters

Parameter	Mandatory	Type	Description
X-Auth-Token	Yes	String	User token. The token can be obtained by calling the IAM API for obtaining a user token. The value of **X-Subject-Token** in the response header is the user token. Minimum: **1** Maximum: **16384**

Table 5-245 Request body parameters

Parameter	Mandatory	Type	Description
command_name	Yes	String	The command name. The value can be **START**, **STOP**, **DELETE**, **SYNC**, **UPLOAD_LOG**, or **RSET_LOG_ACL**. Minimum: **0** Maximum: **255**
result	Yes	String	The command execution result. **success**: The command is executed successfully. **fail**: The command fails to be executed. Minimum: **0** Maximum: **255**
result_detail	Yes	Object	The command execution result in JSON format. This parameter is used only to save command execution results to the SMS database.

Response Parameters

Status code: 200

Table 5-246 Response body parameters

Parameter	Type	Description
-	String	Reporting command execution results to SMS succeeded.

Status code: 400

Table 5-247 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **1024**

Status code: 401

Table 5-248 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **1024**

Status code: 403

Table 5-249 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**

Parameter	Type	Description
error_msg	String	The error message. Minimum: **0** Maximum: **255**
encoded_authorization_message	String	The encrypted authorization information. Minimum: **0** Maximum: **65535**
error_param	Array of strings	Error parameters. Minimum: **0** Maximum: **65535** Array Length: **1 - 20**
details	Array of **details** objects	The error details. Array Length: **1 - 20**

Table 5-250 details

Parameter	Type	Description
error_code	String	The SMS error code. Minimum: **0** Maximum: **65535**
error_msg	String	The SMS error message. Minimum: **0** Maximum: **65535**

Status code: 404

Table 5-251 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **1024**

Status code: 500

Table 5-252 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **1024**

Example Requests

This example sends the execution result of a specified command to the SMS server. The command is **START**, and the execution result is **success**.

```
POST https://{endpoint}/v3/sources/f32ab4d6-d150-4fb3-aa55-edbb5cf9xxxx/command_result

{
  "command_name" : "START",
  "result" : "success",
  "result_detail" : {
    "msg" : "xxx"
  }
}
```

Example Responses

Status code: 200

Reporting command execution results to SMS succeeded.

```
{ }
```

Status code: 403

Authentication failed.

```
{
  "error_code" : "SMS.9004",
  "error_msg" : "The current account does not have the permission to execute policy. You do not have
permission to perform action XXX on resource XXX.",
  "encoded_authorization_message" : "XXXXXX",
  "error_param" : [ "You do not have permission to perform action XXX on resource XXX." ],
  "details" : [ {
    "error_code" : "SMS.9004",
    "error_msg" : "You do not have permission to perform action XXX on resource XXX."
  } ]
}
```

SDK Sample Code

The SDK sample code is as follows.

Java

This example sends the execution result of a specified command to the SMS server. The command is **START**, and the execution result is **success**.

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;


public class UpdateCommandResultSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great
security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or
environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running
this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        UpdateCommandResultRequest request = new UpdateCommandResultRequest();
        request.withServerId("{server_id}");
        CommandBody body = new CommandBody();
        body.withResultDetail("{\"msg\":\"xxx\"}");
        body.withResult("success");
        body.withCommandName("START");
        request.withBody(body);
        try {
            UpdateCommandResultResponse response = client.updateCommandResult(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

Python

This example sends the execution result of a specified command to the SMS server. The command is **START**, and the execution result is **success**.

```
# coding: utf-8

import os
```

```
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = UpdateCommandResultRequest()
        request.server_id = "{server_id}"
        request.body = CommandBody(
            result_detail="{\"msg\":\"xxx\"}",
            result="success",
            command_name="START"
        )
        response = client.update_command_result(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

Go

This example sends the execution result of a specified command to the SMS server. The command is **START**, and the execution result is **success**.

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
```

```
    }

    hcClient, err := sms.SmsClientBuilder().
        WithRegion(region.ValueOf("<YOUR REGION>")).
        WithCredential(auth).
        SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.UpdateCommandResultRequest{}
    request.ServerId = "{server_id}"
    var resultDetailCommandBody interface{} = "{\"msg\":\"xxx\"}"
    request.Body = &model.CommandBody{
        ResultDetail: &resultDetailCommandBody,
        Result: "success",
        CommandName: "START",
    }
    response, err := client.UpdateCommandResult(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

More

For SDK sample code of more programming languages, see the Sample Code tab in **API Explorer**. SDK sample code can be automatically generated.

Status Codes

Status Code	Description
200	Reporting command execution results to SMS succeeded.
400	Bad Request
401	Unauthorized
403	Authentication failed.
404	Not Found
500	Internal Server Error

Error Codes

See **Error Codes**.

5.6 Template Management

5.6.1 Creating a Template

Function

This API is used to create a template.

Calling Method

For details, see **Calling APIs**.

Authorization Information

Each account has all the permissions required to call all APIs, but IAM users must be assigned the required permissions.

- If you are using role/policy-based authorization, see **Permissions Policies and Supported Actions** for details on the required permissions.
- If you are using identity policy-based authorization, the following identity policy-based permissions are required.

Action	Access Level	Resource Type (*: required)	Condition Key	Alias	Dependencies
sms:template:create	Write	template *	-	sms:server:migrationServer	-

URI

POST /v3/vm/templates

Request Parameters

Table 5-253 Request header parameters

Parameter	Mandatory	Type	Description
X-Auth-Token	Yes	String	User token. The token can be obtained by calling the IAM API for obtaining a user token. The value of **X-Subject-Token** in the response header is the user token. Minimum: **1** Maximum: **16384**

Table 5-254 Request body parameters

Parameter	Mandatory	Type	Description
template	Yes	**TemplateRequest** object	The template information.

Table 5-255 TemplateRequest

Parameter	Mandatory	Type	Description
name	Yes	String	The template name. Only Chinese characters, underscores (_), hyphens (-), digits, and uppercase and lowercase letters are allowed. Minimum: **1** Maximum: **64**
is_template	No	Boolean	Indicates whether the template is general. If the template is associated with a task, the template is not a general template.
region	No	String	The region. Minimum: **0** Maximum: **255**
projectid	No	String	The project ID. Minimum: **0** Maximum: **255**
target_server_name	No	String	The name of the target server. Minimum: **0** Maximum: **255**
availability_zone	No	String	The availability zone. Minimum: **0** Maximum: **255**
volumetype	No	String	Template name. The value can contain only letters, digits, underscores (_), and hyphens (-). Minimum: **0** Maximum: **255**

Parameter	Mandatory	Type	Description
flavor	No	String	The server flavor. Minimum: **0** Maximum: **65535**
vpc	No	**VpcObject** object	The VPC information.
nics	No	Array of **Nics** objects	The NIC information. Multiple NICs are supported. To let the system automatically add a NIC, configure only one NIC entry and set the ID to **autoCreate**. Array Length: **0 - 65535**
security_groups	No	Array of **SgObject** objects	The security group information. Multiple security groups are supported. To let the system automatically create a security group, configure only one security group record and set the ID to **autoCreate**. Array Length: **0 - 65535**
publicip	No	**PublicIp** object	The public IP address.
disk	No	Array of **TemplateDisk** objects	The disk information. Array Length: **0 - 65535**
data_volume_type	No	String	Disk type of the data disk. Its value comes from EVS. Common types include **SAS** (serial attached SCSI), **SSD** (solid-state drive), and **SATA** (serial ATA). For details, see the description of the **volume_type** field in the response parameters in section "Querying Details About a Single EVS Disk" in the EIP API document. Minimum: **0** Maximum: **255**

Parameter	Mandatory	Type	Description
target_password	No	String	The target server password. Minimum: **0** Maximum: **1024**
image_id	No	String	The ID of the image used to create target servers. Minimum: **0** Maximum: **255**

Table 5-256 VpcObject

Parameter	Mandatory	Type	Description
id	No	String	The VPC ID. To let the system automatically create a VPC, set this parameter to **autoCreate**. Minimum: **1** Maximum: **255**
name	No	String	The VPC name. Minimum: **1** Maximum: **255**
cidr	No	String	The VPC CIDR block. The default value is **192.168.0.0/16**. Minimum: **1** Maximum: **255**

Table 5-257 Nics

Parameter	Mandatory	Type	Description
id	Yes	String	The subnet ID. To let the system automatically create a subnet, set this parameter to **autoCreate**. Minimum: **0** Maximum: **255**
name	Yes	String	The subnet name. Minimum: **0** Maximum: **255**

Parameter	Mandatory	Type	Description
cidr	Yes	String	The subnet gateway/mask. Minimum: **0** Maximum: **255**
ip	No	String	The server IP address. If this parameter is not specified, the system will automatically assign an IP address. Minimum: **0** Maximum: **255**

Table 5-258 SgObject

Parameter	Mandatory	Type	Description
id	Yes	String	Security group ID. Minimum: **0** Maximum: **255**
name	Yes	String	Security group name. Minimum: **0** Maximum: **255**

Table 5-259 PublicIp

Parameter	Mandatory	Type	Description
type	Yes	String	The EIP type. The default value is **5_bgp**. For details about the type, see the description of the **type** field in **Response Message** in section "Querying an EIP" of the EIP API reference. Minimum: **0** Maximum: **255**

Parameter	Mandatory	Type	Description
bandwidth_size	Yes	Integer	The bandwidth size, in Mbit/s. The minimum increment for bandwidth adjustment varies depending on the bandwidth range. The minimum increment is 1 Mbit/s if the allowed bandwidth does not exceed 300 Mbit/s. The minimum increment is 50 Mbit/s if the allowed bandwidth ranges from 300 Mbit/s to 1,000 Mbit/s. The minimum increment is 500 Mbit/s if the allowed bandwidth exceeds 1,000 Mbit/s. Minimum: **1** Maximum: **2000**
bandwidth_share_type	No	String	Bandwidth sharing type (long text, non-enumerated data, from EIP) For details about the type, see the description of the **bandwidth_share_type** field in **Response Message** in section "Querying an EIP" of the EIP API reference. Minimum: **0** Maximum: **255**

Table 5-260 TemplateDisk

Parameter	Mandatory	Type	Description
id	No	Long	The disk ID. Minimum: **0** Maximum: **9223372036854775807**
index	Yes	Integer	The disk serial number, starting from 0. Minimum: **0** Maximum: **2147483647**

Parameter	Mandatory	Type	Description
name	Yes	String	The disk name. Minimum: **0** Maximum: **255**
disktype	Yes	String	Disk type, which is the same as the **volumetype** field. Its value comes from EVS. For details, see the description of the **volume_type** field in the response parameters in the section "Querying Details About a Single EVS Disk" in the EIP API document. Minimum: **0** Maximum: **255**
size	Yes	Long	The disk size in GB. Minimum: **0** Maximum: **9223372036854775807**
device_use	No	String	The disk function. Minimum: **0** Maximum: **255**

Response Parameters

Status code: 200

Table 5-261 Response body parameters

Parameter	Type	Description
id	String	The ID of the newly created template returned by SMS. Minimum: **0** Maximum: **255**

Status code: 403

Table 5-262 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **255**
encoded_authorization_message	String	The encrypted authorization information. Minimum: **0** Maximum: **65535**
error_param	Array of strings	Error parameters. Minimum: **0** Maximum: **65535** Array Length: **1 - 20**
details	Array of **details** objects	The error details. Array Length: **1 - 20**

Table 5-263 details

Parameter	Type	Description
error_code	String	The SMS error code. Minimum: **0** Maximum: **65535**
error_msg	String	The SMS error message. Minimum: **0** Maximum: **65535**

Example Requests

- This example creates a migration task template. The template name is **xxxx**, the region information is **region**, and the project ID is **00924d0ad2df4f21ac476dd9f3288xxx**.

```
POST https://{endpoint}/v3/vm/templates

{
  "template" : {
    "name" : "my template",
    "is_template" : false,
    "region" : "region",
```



```
    "target_server_name" : "abcd",
    "availability_zone" : "availability_zone",
    "projectid" : "xxxxxxxxxxxxxxxxxxxxxxxx00000001",
    "volumetype" : "",
    "image_id" : "",
    "vpc" : {
      "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
      "name" : "sms-1566979232(192.168.0.0/16)"
    },
    "security_groups" : [ {
      "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
      "name" : "kubernetes.io-default-sg (Inbound: udp/1-65535; tcp/22,1-65535,3389; Outbound: --)"
    } ],
    "nics" : [ {
      "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
      "name" : "sms-1566979244(192.168.0.0/16)",
      "cidr" : "192.168.0.0/16",
      "ip" : ""
    } ],
    "flavor" : "s2.medium.2",
    "publicip" : {
      "type" : "5_bgp",
      "bandwidth_size" : 5,
      "bandwidth_share_type" : "PER"
    },
    "disk" : [ {
      "index" : 0,
      "name" : "system",
      "disktype" : "",
      "size" : 40
    } ]
  }
}
```

- This example creates a template directly.

```
POST https://{endpoint}/v3/vm/templates

{
  "template" : {
    "name" : "xxxx",
    "is_template" : true,
    "region" : "region",
    "target_server_name" : "ggg-win16-t",
    "availability_zone" : "availability_zone",
    "projectid" : "xxxxxxxxxxxxxxxxxxxxxxxx00000001",
    "target_password" : "********",
    "flavor" : "c3.medium.2",
    "vpc" : {
      "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
      "name" : "Migrate-SSd-1",
      "cidr" : "192.168.0.0/16"
    },
    "nics" : [ {
      "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
      "name" : "Migrate-SSd-35",
      "cidr" : "192.168.0.0/16",
      "ip" : ""
    } ],
    "security_groups" : [ {
      "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
      "name" : "Migrate-dddd"
    } ],
    "disk" : [ {
      "id" : 0,
      "index" : 0,
      "name" : "Disk 0",
      "disktype" : "SATA",
      "size" : 40,
      "device_use" : "BOOT"
    } ],
```

```
    "volumetype" : "SATA",
    "publicip" : {
      "type" : "5_g-vm",
      "bandwidth_size" : 10,
      "bandwidth_share_type" : "PER"
    }
  }
}
```

Example Responses

Status code: 200

The template was created.

```
{
  "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
}
```

Status code: 403

Authentication failed.

```
{
  "error_code" : "SMS.9004",
  "error_msg" : "The current account does not have the permission to execute policy. You do not have permission to perform action XXX on resource XXX.",
  "encoded_authorization_message" : "XXXXXX",
  "error_param" : [ "You do not have permission to perform action XXX on resource XXX." ],
  "details" : [ {
    "error_code" : "SMS.9004",
    "error_msg" : "You do not have permission to perform action XXX on resource XXX."
  } ]
}
```

SDK Sample Code

The SDK sample code is as follows.

Java

- This example creates a migration task template. The template name is **xxxx**, the region information is **region**, and the project ID is **00924d0ad2df4f21ac476dd9f3288xxx**.

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;

import java.util.List;
import java.util.ArrayList;

public class CreateTemplateSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before
```

```
running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local
environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        CreateTemplateRequest request = new CreateTemplateRequest();
        CreateTemplateReq body = new CreateTemplateReq();
        List<TemplateDisk> listTemplateDisk = new ArrayList<>();
        listTemplateDisk.add(
            new TemplateDisk()
                .withIndex(0)
                .withName("system")
                .withDisktype("")
                .withSize(40L)
        );
        PublicIp publicipTemplate = new PublicIp();
        publicipTemplate.withType("5_bgp")
            .withBandwidthSize(5)
            .withBandwidthShareType("PER");
        List<SgObject> listTemplateSecurityGroups = new ArrayList<>();
        listTemplateSecurityGroups.add(
            new SgObject()
                .withId("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
                .withName("kubernetes.io-default-sg (Inbound: udp/1-65535; tcp/22,1-65535,3389;
Outbound: --)")
        );
        List<Nics> listTemplateNics = new ArrayList<>();
        listTemplateNics.add(
            new Nics()
                .withId("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
                .withName("sms-1566979244(192.168.0.0/16)")
                .withCidr("192.168.0.0/16")
                .withIp("")
        );
        VpcObject vpcTemplate = new VpcObject();
        vpcTemplate.withId("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
            .withName("sms-1566979232(192.168.0.0/16)");
        TemplateRequest templatebody = new TemplateRequest();
        templatebody.withName("my template")
            .withIsTemplate(false)
            .withRegion("region")
            .withProjectid("xxxxxxxxxxxxxxxxxxxxxxxx00000001")
            .withTargetServerName("abcd")
            .withAvailabilityZone("availability_zone")
            .withVolumetype("")
            .withFlavor("s2.medium.2")
            .withVpc(vpcTemplate)
            .withNics(listTemplateNics)
            .withSecurityGroups(listTemplateSecurityGroups)
            .withPublicip(publicipTemplate)
            .withDisk(listTemplateDisk)
            .withImageId("");
        body.withTemplate(templatebody);
        request.withBody(body);
        try {
            CreateTemplateResponse response = client.createTemplate(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
```

```
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

- This example creates a template directly.

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;

import java.util.List;
import java.util.ArrayList;

public class CreateTemplateSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        CreateTemplateRequest request = new CreateTemplateRequest();
        CreateTemplateReq body = new CreateTemplateReq();
        List<TemplateDisk> listTemplateDisk = new ArrayList<>();
        listTemplateDisk.add(
            new TemplateDisk()
                .withId(0L)
                .withIndex(0)
                .withName("Disk 0")
                .withDisktype("SATA")
                .withSize(40L)
                .withDeviceUse("BOOT")
        );
        PublicIp publicipTemplate = new PublicIp();
        publicipTemplate.withType("5_g-vm")
            .withBandwidthSize(10)
            .withBandwidthShareType("PER");
        List<SgObject> listTemplateSecurityGroups = new ArrayList<>();
        listTemplateSecurityGroups.add(
            new SgObject()
                .withId("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
                .withName("Migrate-dddd")
        );
        List<Nics> listTemplateNics = new ArrayList<>();
        listTemplateNics.add(
            new Nics()
```

```
                .withId("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
                .withName("Migrate-SSd-35")
                .withCidr("192.168.0.0/16")
                .withIp("")
        );
        VpcObject vpcTemplate = new VpcObject();
        vpcTemplate.withId("xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001")
            .withName("Migrate-SSd-1")
            .withCidr("192.168.0.0/16");
        TemplateRequest templatebody = new TemplateRequest();
        templatebody.withName("xxxx")
            .withIsTemplate(true)
            .withRegion("region")
            .withProjectid("xxxxxxxxxxxxxxxxxxxxxxxx00000001")
            .withTargetServerName("ggg-win16-t")
            .withAvailabilityZone("availability_zone")
            .withVolumetype("SATA")
            .withFlavor("c3.medium.2")
            .withVpc(vpcTemplate)
            .withNics(listTemplateNics)
            .withSecurityGroups(listTemplateSecurityGroups)
            .withPublicip(publicipTemplate)
            .withDisk(listTemplateDisk)
            .withTargetPassword("********");
        body.withTemplate(templatebody);
        request.withBody(body);
        try {
            CreateTemplateResponse response = client.createTemplate(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

Python

- This example creates a migration task template. The template name is **xxxx**, the region information is **region**, and the project ID is **00924d0ad2df4f21ac476dd9f3288xxx**.

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)
```

```
    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = CreateTemplateRequest()
        listDiskTemplate = [
            TemplateDisk(
                index=0,
                name="system",
                disktype="",
                size=40
            )
        ]
        publicipTemplate = PublicIp(
            type="5_bgp",
            bandwidth_size=5,
            bandwidth_share_type="PER"
        )
        listSecurityGroupsTemplate = [
            SgObject(
                id="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
                name="kubernetes.io-default-sg (Inbound: udp/1-65535; tcp/22,1-65535,3389; Outbound:
--)"
            )
        ]
        listNicsTemplate = [
            Nics(
                id="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
                name="sms-1566979244(192.168.0.0/16)",
                cidr="192.168.0.0/16",
                ip=""
            )
        ]
        vpcTemplate = VpcObject(
            id="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
            name="sms-1566979232(192.168.0.0/16)"
        )
        templatebody = TemplateRequest(
            name="my template",
            is_template=False,
            region="region",
            projectid="xxxxxxxxxxxxxxxxxxxxxxxx00000001",
            target_server_name="abcd",
            availability_zone="availability_zone",
            volumetype="",
            flavor="s2.medium.2",
            vpc=vpcTemplate,
            nics=listNicsTemplate,
            security_groups=listSecurityGroupsTemplate,
            publicip=publicipTemplate,
            disk=listDiskTemplate,
            image_id=""
        )
        request.body = CreateTemplateReq(
            template=templatebody
        )
        response = client.create_template(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

- This example creates a template directly.

```
# coding: utf-8

import os
```

```
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = CreateTemplateRequest()
        listDiskTemplate = [
            TemplateDisk(
                id=0,
                index=0,
                name="Disk 0",
                disktype="SATA",
                size=40,
                device_use="BOOT"
            )
        ]
        publicipTemplate = PublicIp(
            type="5_g-vm",
            bandwidth_size=10,
            bandwidth_share_type="PER"
        )
        listSecurityGroupsTemplate = [
            SgObject(
                id="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
                name="Migrate-dddd"
            )
        ]
        listNicsTemplate = [
            Nics(
                id="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
                name="Migrate-SSd-35",
                cidr="192.168.0.0/16",
                ip=""
            )
        ]
        vpcTemplate = VpcObject(
            id="xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
            name="Migrate-SSd-1",
            cidr="192.168.0.0/16"
        )
        templatebody = TemplateRequest(
            name="xxxx",
            is_template=True,
            region="region",
            projectid="xxxxxxxxxxxxxxxxxxxxxxxx00000001",
            target_server_name="ggg-win16-t",
            availability_zone="availability_zone",
            volumetype="SATA",
            flavor="c3.medium.2",
            vpc=vpcTemplate,
            nics=listNicsTemplate,
```

```
            security_groups=listSecurityGroupsTemplate,
            publicip=publicipTemplate,
            disk=listDiskTemplate,
            target_password="********"
        )
        request.body = CreateTemplateReq(
            template=templatebody
        )
        response = client.create_template(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

Go

- This example creates a migration task template. The template name is **xxxx**, the region information is **region**, and the project ID is **00924d0ad2df4f21ac476dd9f3288xxx**.

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
         WithRegion(region.ValueOf("<YOUR REGION>")).
         WithCredential(auth).
         SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.CreateTemplateRequest{}
    var listDiskTemplate = []model.TemplateDisk{
        {
            Index: int32(0),
```

```
            Name: "system",
            Disktype: "",
            Size: int64(40),
        },
    }
    bandwidthShareTypePublicip:= "PER"
    publicipTemplate := &model.PublicIp{
        Type: "5_bgp",
        BandwidthSize: int32(5),
        BandwidthShareType: &bandwidthShareTypePublicip,
    }
    var listSecurityGroupsTemplate = []model.SgObject{
        {
            Id: "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
            Name: "kubernetes.io-default-sg (Inbound: udp/1-65535; tcp/22,1-65535,3389; Outbound:
--)",
        },
    }
    ipNics:= ""
    var listNicsTemplate = []model.Nics{
        {
            Id: "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
            Name: "sms-1566979244(192.168.0.0/16)",
            Cidr: "192.168.0.0/16",
            Ip: &ipNics,
        },
    }
    idVpc:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
    nameVpc:= "sms-1566979232(192.168.0.0/16)"
    vpcTemplate := &model.VpcObject{
        Id: &idVpc,
        Name: &nameVpc,
    }
    isTemplateTemplate:= false
    regionTemplate:= "region"
    projectidTemplate:= "xxxxxxxxxxxxxxxxxxxxxxxx00000001"
    targetServerNameTemplate:= "abcd"
    availabilityZoneTemplate:= "availability_zone"
    volumetypeTemplate:= ""
    flavorTemplate:= "s2.medium.2"
    imageIdTemplate:= ""
    templatebody := &model.TemplateRequest{
        Name: "my template",
        IsTemplate: &isTemplateTemplate,
        Region: &regionTemplate,
        Projectid: &projectidTemplate,
        TargetServerName: &targetServerNameTemplate,
        AvailabilityZone: &availabilityZoneTemplate,
        Volumetype: &volumetypeTemplate,
        Flavor: &flavorTemplate,
        Vpc: vpcTemplate,
        Nics: &listNicsTemplate,
        SecurityGroups: &listSecurityGroupsTemplate,
        Publicip: publicipTemplate,
        Disk: &listDiskTemplate,
        ImageId: &imageIdTemplate,
    }
    request.Body = &model.CreateTemplateReq{
        Template: templatebody,
    }
    response, err := client.CreateTemplate(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

- This example creates a template directly.

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great
security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or
environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before
running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local
environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
         WithRegion(region.ValueOf("<YOUR REGION>")).
         WithCredential(auth).
         SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.CreateTemplateRequest{}
    idDisk:= int64(0)
    deviceUseDisk:= "BOOT"
    var listDiskTemplate = []model.TemplateDisk{
        {
            Id: &idDisk,
            Index: int32(0),
            Name: "Disk 0",
            Disktype: "SATA",
            Size: int64(40),
            DeviceUse: &deviceUseDisk,
        },
    }
    bandwidthShareTypePublicip:= "PER"
    publicipTemplate := &model.PublicIp{
        Type: "5_g-vm",
        BandwidthSize: int32(10),
        BandwidthShareType: &bandwidthShareTypePublicip,
    }
    var listSecurityGroupsTemplate = []model.SgObject{
        {
            Id: "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
            Name: "Migrate-dddd",
        },
    }
    ipNics:= ""
```

```
    var listNicsTemplate = []model.Nics{
        {
            Id: "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
            Name: "Migrate-SSd-35",
            Cidr: "192.168.0.0/16",
            Ip: &ipNics,
        },
    }
    idVpc:= "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
    nameVpc:= "Migrate-SSd-1"
    cidrVpc:= "192.168.0.0/16"
    vpcTemplate := &model.VpcObject{
        Id: &idVpc,
        Name: &nameVpc,
        Cidr: &cidrVpc,
    }
    isTemplateTemplate:= true
    regionTemplate:= "region"
    projectidTemplate:= "xxxxxxxxxxxxxxxxxxxxxxxx00000001"
    targetServerNameTemplate:= "ggg-win16-t"
    availabilityZoneTemplate:= "availability_zone"
    volumetypeTemplate:= "SATA"
    flavorTemplate:= "c3.medium.2"
    targetPasswordTemplate:= "********"
    templatebody := &model.TemplateRequest{
        Name: "xxxx",
        IsTemplate: &isTemplateTemplate,
        Region: &regionTemplate,
        Projectid: &projectidTemplate,
        TargetServerName: &targetServerNameTemplate,
        AvailabilityZone: &availabilityZoneTemplate,
        Volumetype: &volumetypeTemplate,
        Flavor: &flavorTemplate,
        Vpc: vpcTemplate,
        Nics: &listNicsTemplate,
        SecurityGroups: &listSecurityGroupsTemplate,
        Publicip: publicipTemplate,
        Disk: &listDiskTemplate,
        TargetPassword: &targetPasswordTemplate,
    }
    request.Body = &model.CreateTemplateReq{
        Template: templatebody,
    }
    response, err := client.CreateTemplate(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

More

For SDK sample code of more programming languages, see the Sample Code tab in **API Explorer**. SDK sample code can be automatically generated.

Status Codes

Status Code	Description
200	The template was created.
403	Authentication failed.

Error Codes

See **Error Codes**.

5.6.2 Querying the Template List

Function

This API is used to list templates used for setting up target servers. You can use these templates for quick configuration.

Calling Method

For details, see **Calling APIs**.

Authorization Information

Each account has all the permissions required to call all APIs, but IAM users must be assigned the required permissions.

- If you are using role/policy-based authorization, see **Permissions Policies and Supported Actions** for details on the required permissions.
- If you are using identity policy-based authorization, the following identity policy-based permissions are required.

Action	Access Level	Resource Type (*: required)	Condition Key	Alias	Dependencies
sms:template:list	List	template *	-	sms:server:queryServer	-

URI

GET /v3/vm/templates

Table 5-264 Query Parameters

Parameter	Mandatory	Type	Description
name	No	String	The template name. Minimum: **0** Maximum: **255**
availability_zone	No	String	The availability zone. Minimum: **0** Maximum: **255**

Parameter	Mandatory	Type	Description
region	No	String	Region ID Minimum: **0** Maximum: **255**
limit	No	Integer	The page size. If this parameter is not transferred, the default value **50** is used. Minimum: **0** Maximum: **100** Default: **50**
offset	No	Integer	The offset for pagination. If this parameter is not transferred, the default value **0** is used. Minimum: **0** Maximum: **65535** Default: **0**
id	No	String	The template ID. Minimum: **1** Maximum: **36**

Request Parameters

Table 5-265 Request header parameters

Parameter	Mandatory	Type	Description
X-Auth-Token	Yes	String	User token. The token can be obtained by calling the IAM API for obtaining a user token. The value of **X-Subject-Token** in the response header is the user token. Minimum: **1** Maximum: **16384**

Response Parameters

Status code: 200

Table 5-266 Response body parameters

Parameter	Type	Description
count	Integer	The number of templates. Minimum: **0** Maximum: **65535**
templates	Array of **TemplateResponseBody** objects	The template information. Array Length: **0 - 65535**

Table 5-267 TemplateResponseBody

Parameter	Type	Description
id	String	The template ID. Minimum: **0** Maximum: **255**
name	String	The template name. Minimum: **0** Maximum: **255**
is_template	String	Indicates whether the template is general. If the template is associated with a task, the template is not a general template. Minimum: **0** Maximum: **255**
region	String	The region. Minimum: **0** Maximum: **255**
projectid	String	The project ID. Minimum: **0** Maximum: **255**
target_server_name	String	The name of the target server. Minimum: **0** Maximum: **255**
availability_zone	String	The availability zone. Minimum: **0** Maximum: **255**

Parameter	Type	Description
volumetype	String	The data disk type. **SAS**: serial attached SCSI **SSD**: solid-state drive **SATA**: serial advanced technology attachment Enumeration values: • **SAS** • **SSD** • **SATA**
flavor	String	The server flavor. Minimum: **0** Maximum: **255**
vpc	**VpcObject** object	The VPC information.
nics	Array of **Nics** objects	The NIC information. Multiple NICs are supported. To let the system automatically add a NIC, configure only one NIC entry and set the ID to **autoCreate**. Array Length: **0 - 65535**
security_groups	Array of **SgObject** objects	The security group information. Multiple security groups are supported. To let the system automatically create a security group, configure only one security group record and set the ID to **autoCreate**. Array Length: **0 - 65535**
publicip	**PublicIp** object	The public IP address.
disk	Array of **TemplateDisk** objects	The disk information. Array Length: **0 - 65535**
data_volume_type	String	The data disk type. **SAS**: serial attached SCSI **SSD**: solid-state drive **SATA**: serial advanced technology attachment Enumeration values: • **SAS** • **SSD** • **SATA**

Parameter	Type	Description
target_password	String	The target server password. Minimum: **0** Maximum: **1024**
image_id	String	The ID of the selected image. Minimum: **0** Maximum: **255**

Table 5-268 VpcObject

Parameter	Type	Description
id	String	The VPC ID. To let the system automatically create a VPC, set this parameter to **autoCreate**. Minimum: **1** Maximum: **255**
name	String	The VPC name. Minimum: **1** Maximum: **255**
cidr	String	The VPC CIDR block. The default value is **192.168.0.0/16**. Minimum: **1** Maximum: **255**

Table 5-269 Nics

Parameter	Type	Description
id	String	The subnet ID. To let the system automatically create a subnet, set this parameter to **autoCreate**. Minimum: **0** Maximum: **255**
name	String	The subnet name. Minimum: **0** Maximum: **255**
cidr	String	The subnet gateway/mask. Minimum: **0** Maximum: **255**

Parameter	Type	Description
ip	String	The server IP address. If this parameter is not specified, the system will automatically assign an IP address. Minimum: **0** Maximum: **255**

Table 5-270 SgObject

Parameter	Type	Description
id	String	Security group ID. Minimum: **0** Maximum: **255**
name	String	Security group name. Minimum: **0** Maximum: **255**

Table 5-271 PublicIp

Parameter	Type	Description
type	String	The EIP type. The default value is **5_bgp**. For details about the type, see the description of the **type** field in **Response Message** in section "Querying an EIP" of the EIP API reference. Minimum: **0** Maximum: **255**

Parameter	Type	Description
bandwidth_size	Integer	The bandwidth size, in Mbit/s. The minimum increment for bandwidth adjustment varies depending on the bandwidth range. The minimum increment is 1 Mbit/s if the allowed bandwidth does not exceed 300 Mbit/s. The minimum increment is 50 Mbit/s if the allowed bandwidth ranges from 300 Mbit/s to 1,000 Mbit/s. The minimum increment is 500 Mbit/s if the allowed bandwidth exceeds 1,000 Mbit/s. Minimum: **1** Maximum: **2000**
bandwidth_share_type	String	Bandwidth sharing type (long text, non-enumerated data, from EIP) For details about the type, see the description of the **bandwidth_share_type** field in **Response Message** in section "Querying an EIP" of the EIP API reference. Minimum: **0** Maximum: **255**

Table 5-272 TemplateDisk

Parameter	Type	Description
id	Long	The disk ID. Minimum: **0** Maximum: **9223372036854775807**
index	Integer	The disk serial number, starting from 0. Minimum: **0** Maximum: **2147483647**
name	String	The disk name. Minimum: **0** Maximum: **255**

Parameter	Type	Description
disktype	String	Disk type, which is the same as the **volumetype** field. Its value comes from EVS. For details, see the description of the **volume_type** field in the response parameters in the section "Querying Details About a Single EVS Disk" in the EIP API document. Minimum: **0** Maximum: **255**
size	Long	The disk size in GB. Minimum: **0** Maximum: **9223372036854775807**
device_use	String	The disk function. Minimum: **0** Maximum: **255**

Status code: 403

Table 5-273 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **255**
encoded_authorization_message	String	The encrypted authorization information. Minimum: **0** Maximum: **65535**
error_param	Array of strings	Error parameters. Minimum: **0** Maximum: **65535** Array Length: **1 - 20**
details	Array of **details** objects	The error details. Array Length: **1 - 20**

Table 5-274 details

Parameter	Type	Description
error_code	String	The SMS error code. Minimum: **0** Maximum: **65535**
error_msg	String	The SMS error message. Minimum: **0** Maximum: **65535**

Example Requests

This example queries the template list.

```
GET https://{endpoint}/v3/vm/templates
X-Auth-Token: XXXX
```

Example Responses

Status code: 200

Querying the template list succeeded.


```
{
  "count" : 9,
  "templates" : [ {
    "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "name" : "test",
    "region" : "region",
    "availability_zone" : "availability_zone",
    "projectid" : "xxxxxxxxxxxxxxxxxxxxxxxx00000001",
    "flavor" : "s2.large.2",
    "volumetype" : "SATA",
    "image_id" : "",
    "vpc" : {
      "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
      "name" : "vpc-dfdb"
    },
    "nics" : [ {
      "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
      "name" : "subnet-dfdb(192.168.1.0/24)",
      "cidr" : "192.168.1.0/24",
      "ip" : ""
    } ],
    "security_groups" : [ {
      "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
      "name" : "default(Inbound:tcp/8900,8899,3389,22; Outbound:--)"
    } ]
  }, {
    "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "name" : "test1",
    "region" : "region",
    "availability_zone" : "availability_zone",
    "projectid" : "xxxxxxxxxxxxxxxxxxxxxxxx00000001",
    "flavor" : "s6.large.2",
    "volumetype" : "SATA",
    "image_id" : "",
    "vpc" : {
      "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
```

```
    "name" : "vpc-13d6"
  },
  "nics" : [ {
    "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "name" : "subnet-13d6(192.168.1.0/24)",
    "cidr" : "192.168.1.0/24",
    "ip" : ""
  } ],
  "security_groups" : [ {
    "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "name" : "sms-1568190885(Inbound:tcp/8900,8899,3389; Outbound:--)"
  } ]
}, {
  "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
  "name" : "test2",
  "region" : "region",
  "availability_zone" : "availability_zone",
  "projectid" : "xxxxxxxxxxxxxxxxxxxxxxxx00000001",
  "flavor" : "s2.large.2",
  "volumetype" : "SATA",
  "image_id" : "",
  "vpc" : {
    "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "name" : "vpc-testcloud(192.168.0.0/16)"
  },
  "nics" : [ {
    "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "name" : "subnet-testcloud(192.168.0.0/24)",
    "cidr" : "192.168.0.0/24",
    "ip" : ""
  } ],
  "security_groups" : [ ]
}, {
  "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
  "name" : "fdff",
  "region" : "region",
  "availability_zone" : "availability_zone",
  "projectid" : "xxxxxxxxxxxxxxxxxxxxxxxx00000001",
  "flavor" : "s2.large.2",
  "volumetype" : "SATA",
  "image_id" : "",
  "vpc" : {
    "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "name" : "vpc-migration(192.168.0.0/16)"
  },
  "nics" : [ {
    "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "name" : "subnet-cf42(192.168.5.0/24)",
    "cidr" : "192.168.5.0/24",
    "ip" : ""
  } ],
  "security_groups" : [ {
    "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "name" : "sg-smt-test(Inbound:tcp/3389,8899,22,8900; Outbound:--)"
  } ]
}, {
  "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
  "name" : "test3",
  "region" : "region",
  "availability_zone" : "availability_zone",
  "projectid" : "xxxxxxxxxxxxxxxxxxxxxxxx00000001",
  "flavor" : "s2.medium.2",
  "volumetype" : "SATA",
  "image_id" : "",
  "vpc" : { },
  "nics" : [ ],
  "security_groups" : [ ]
}, {
  "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
```

```
  "name" : "test_linux_childproj",
  "region" : "region",
  "availability_zone" : "availability_zone",
  "projectid" : "xxxxxxxxxxxxxxxxxxxxxxxx00000001",
  "flavor" : "s2.small.1",
  "volumetype" : "SATA",
  "image_id" : "",
  "vpc" : {
    "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "name" : "sms-1567992634(192.168.0.0/16)"
  },
  "nics" : [ {
    "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "name" : "sms-1567992646(192.168.0.0/16)",
    "cidr" : "192.168.0.0/16",
    "ip" : ""
  } ],
  "security_groups" : [ {
    "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "name" : "sg-7e50(Inbound:tcp/8900,8899,3389,22; Outbound:--)"
  } ]
}, {
  "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
  "name" : "12212",
  "region" : "region",
  "availability_zone" : "availability_zone",
  "projectid" : "xxxxxxxxxxxxxxxxxxxxxxxx00000001",
  "flavor" : "s2.large.2",
  "volumetype" : "SATA",
  "image_id" : "",
  "vpc" : {
    "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "name" : "vpc-migration(192.168.0.0/16)"
  },
  "nics" : [ {
    "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "name" : "subnet-migration(192.168.1.0/24)",
    "cidr" : "192.168.1.0/24",
    "ip" : ""
  } ],
  "security_groups" : [ {
    "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "name" : "SMT-Windows(Inbound:tcp/8443,8899,8900,22,3389;icmp; Outbound:--)"
  } ]
}, {
  "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
  "name" : "test4",
  "region" : "region",
  "availability_zone" : "availability_zone",
  "projectid" : "xxxxxxxxxxxxxxxxxxxxxxxx00000001",
  "flavor" : "s2.medium.2",
  "volumetype" : "SATA",
  "image_id" : "",
  "vpc" : { },
  "nics" : [ ],
  "security_groups" : [ ]
}, {
  "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
  "name" : "dddd",
  "region" : "region",
  "availability_zone" : "availability_zone",
  "projectid" : "xxxxxxxxxxxxxxxxxxxxxxxx00000001",
  "flavor" : "s2.large.2",
  "volumetype" : "SATA",
  "image_id" : "",
  "vpc" : {
    "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "name" : "sms-1566979232(192.168.0.0/16)"
  },
```

```
    "nics" : [ {
      "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
      "name" : "sms-1566979244(192.168.0.0/16)",
      "cidr" : "192.168.0.0/16",
      "ip" : ""
    } ],
    "security_groups" : [ ]
  } ]
}
```


Status code: 403

Authentication failed.


```
{
  "error_code" : "SMS.9004",
  "error_msg" : "The current account does not have the permission to execute policy. You do not have permission to perform action XXX on resource XXX.",
  "encoded_authorization_message" : "XXXXXX",
  "error_param" : [ "You do not have permission to perform action XXX on resource XXX." ],
  "details" : [ {
    "error_code" : "SMS.9004",
    "error_msg" : "You do not have permission to perform action XXX on resource XXX."
  } ]
}
```


SDK Sample Code

The SDK sample code is as follows.

Java

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;


public class ListTemplatesSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        ListTemplatesRequest request = new ListTemplatesRequest();
        try {
            ListTemplatesResponse response = client.listTemplates(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
```

```
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

Python

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = ListTemplatesRequest()
        response = client.list_templates(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

Go

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")
```



```
    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
        WithRegion(region.ValueOf("<YOUR REGION>")).
        WithCredential(auth).
        SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.ListTemplatesRequest{}
    response, err := client.ListTemplates(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

More

For SDK sample code of more programming languages, see the Sample Code tab in **API Explorer**. SDK sample code can be automatically generated.

Status Codes

Status Code	Description
200	Querying the template list succeeded.
403	Authentication failed.

Error Codes

See **Error Codes**.

5.6.3 Batch Deleting Templates

Function

This API is used to delete templates with specified IDs in batches.

Calling Method

For details, see **Calling APIs**.

Authorization Information

Each account has all the permissions required to call all APIs, but IAM users must be assigned the required permissions.

- If you are using role/policy-based authorization, see **Permissions Policies and Supported Actions** for details on the required permissions.
- If you are using identity policy-based authorization, no identity policy-based permission required for calling this API.

URI

POST /v3/vm/templates/delete

Request Parameters

Table 5-275 Request header parameters

Parameter	Mandatory	Type	Description
X-Auth-Token	Yes	String	User token. The token can be obtained by calling the IAM API for obtaining a user token. The value of **X-Subject-Token** in the response header is the user token. Minimum: **1** Maximum: **16384**

Table 5-276 Request body parameters

Parameter	Mandatory	Type	Description
ids	Yes	Array of strings	The IDs of the templates to be deleted. Minimum: **1** Maximum: **36** Array Length: **1 - 50**

Response Parameters

Status code: 200

Table 5-277 Response body parameters

Parameter	Type	Description
-	String	Templates with specified IDs were deleted in batches.

Status code: 403

Table 5-278 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **255**
encoded_authorization_message	String	The encrypted authorization information. Minimum: **0** Maximum: **65535**
error_param	Array of strings	Error parameters. Minimum: **0** Maximum: **65535** Array Length: **1 - 20**
details	Array of **details** objects	The error details. Array Length: **1 - 20**

Table 5-279 details

Parameter	Type	Description
error_code	String	The SMS error code. Minimum: **0** Maximum: **65535**
error_msg	String	The SMS error message. Minimum: **0** Maximum: **65535**

Example Requests

This example deletes templates with IDs **3db302e8-95de-478c-a892-8a083f2dxxxx** and **708847ae-f013-4b1a-8ea8-6cfa1e94xxxx** in a batch.

```
POST https://{endpoint}/v3/vm/templates/delete

{
  "ids" : [ "3db302e8-95de-478c-a892-8a083f2dxxxx", "708847ae-f013-4b1a-8ea8-6cfa1e94xxxx" ]
}
```

Example Responses

Status code: 200

Templates with specified IDs were deleted in batches.

```
{ }
```

Status code: 403

Authentication failed.

```
{
  "error_code" : "SMS.9004",
  "error_msg" : "The current account does not have the permission to execute policy. You do not have permission to perform action XXX on resource XXX.",
  "encoded_authorization_message" : "XXXXXX",
  "error_param" : [ "You do not have permission to perform action XXX on resource XXX." ],
  "details" : [ {
    "error_code" : "SMS.9004",
    "error_msg" : "You do not have permission to perform action XXX on resource XXX."
  } ]
}
```

SDK Sample Code

The SDK sample code is as follows.

Java

This example deletes templates with IDs **3db302e8-95de-478c-a892-8a083f2dxxxx** and **708847ae-f013-4b1a-8ea8-6cfa1e94xxxx** in a batch.

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;

import java.util.List;
import java.util.ArrayList;

public class DeleteTemplatesSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running
```

```
this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        DeleteTemplatesRequest request = new DeleteTemplatesRequest();
        DeletetemplatesReq body = new DeletetemplatesReq();
        List<String> listbodyIds = new ArrayList<>();
        listbodyIds.add("3db302e8-95de-478c-a892-8a083f2dxxxx");
        listbodyIds.add("708847ae-f013-4b1a-8ea8-6cfa1e94xxxx");
        body.withIds(listbodyIds);
        request.withBody(body);
        try {
            DeleteTemplatesResponse response = client.deleteTemplates(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

Python

This example deletes templates with IDs **3db302e8-95de-478c-a892-8a083f2dxxxx** and **708847ae-f013-4b1a-8ea8-6cfa1e94xxxx** in a batch.

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security
risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment
variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this
example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = DeleteTemplatesRequest()
        listIdsbody = [
            "3db302e8-95de-478c-a892-8a083f2dxxxx",
```

```
            "708847ae-f013-4b1a-8ea8-6cfa1e94xxxx"
        ]
        request.body = DeletetemplatesReq(
            ids=listIdsbody
        )
        response = client.delete_templates(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

Go

This example deletes templates with IDs **3db302e8-95de-478c-a892-8a083f2dxxxx** and **708847ae-f013-4b1a-8ea8-6cfa1e94xxxx** in a batch.

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
         WithRegion(region.ValueOf("<YOUR REGION>")).
         WithCredential(auth).
         SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.DeleteTemplatesRequest{}
    var listIdsbody = []string{
        "3db302e8-95de-478c-a892-8a083f2dxxxx",
        "708847ae-f013-4b1a-8ea8-6cfa1e94xxxx",
    }
    request.Body = &model.DeletetemplatesReq{
        Ids: listIdsbody,
    }
    response, err := client.DeleteTemplates(request)
```

```
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

More

For SDK sample code of more programming languages, see the Sample Code tab in **API Explorer**. SDK sample code can be automatically generated.

Status Codes

Status Code	Description
200	Templates with specified IDs were deleted in batches.
403	Authentication failed.

Error Codes

See **Error Codes**.

5.6.4 Deleting a Template

Function

This API is used to delete a template with a specified ID.

Calling Method

For details, see **Calling APIs**.

Authorization Information

Each account has all the permissions required to call all APIs, but IAM users must be assigned the required permissions.

- If you are using role/policy-based authorization, see **Permissions Policies and Supported Actions** for details on the required permissions.
- If you are using identity policy-based authorization, the following identity policy-based permissions are required.

Action	Access Level	Resource Type (*: required)	Condition Key	Alias	Dependencies
sms:template:delete	Write	template *	-	sms:server:migrationServer	-

URI

DELETE /v3/vm/templates/{id}

Table 5-280 Path Parameters

Parameter	Mandatory	Type	Description
id	Yes	String	The ID of the template to be deleted. Minimum: **1** Maximum: **36**

Request Parameters

Table 5-281 Request header parameters

Parameter	Mandatory	Type	Description
X-Auth-Token	Yes	String	User token. The token can be obtained by calling the IAM API for obtaining a user token. The value of **X-Subject-Token** in the response header is the user token. Minimum: **1** Maximum: **16384**

Response Parameters

Status code: 200

Table 5-282 Response body parameters

Parameter	Type	Description
-	String	The template with the specified ID was deleted.

Status code: 403

Table 5-283 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **255**
encoded_authorization_message	String	The encrypted authorization information. Minimum: **0** Maximum: **65535**
error_param	Array of strings	Error parameters. Minimum: **0** Maximum: **65535** Array Length: **1 - 20**
details	Array of **details** objects	The error details. Array Length: **1 - 20**

Table 5-284 details

Parameter	Type	Description
error_code	String	The SMS error code. Minimum: **0** Maximum: **65535**
error_msg	String	The SMS error message. Minimum: **0** Maximum: **65535**

Example Requests

This example deletes the template whose ID is **2bf4344f-8f1f-414e-bb1b-8c2f59ada67f**.

```
DELETE https://{endpoint}/v3/vm/templates/2bf4344f-8f1f-414e-bb1b-8c2f59ada67f
X-Auth-Token: XXXX
```

Example Responses

Status code: 403

Authentication failed.

```
{
  "error_code" : "SMS.9004",
  "error_msg" : "The current account does not have the permission to execute policy. You do not have permission to perform action XXX on resource XXX.",
  "encoded_authorization_message" : "XXXXXX",
  "error_param" : [ "You do not have permission to perform action XXX on resource XXX." ],
  "details" : [ {
    "error_code" : "SMS.9004",
    "error_msg" : "You do not have permission to perform action XXX on resource XXX."
  } ]
}
```

SDK Sample Code

The SDK sample code is as follows.

Java

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;


public class DeleteTemplateSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        DeleteTemplateRequest request = new DeleteTemplateRequest();
        request.withId("{id}");
        try {
```



```
            DeleteTemplateResponse response = client.deleteTemplate(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

Python

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = DeleteTemplateRequest()
        request.id = "{id}"
        response = client.delete_template(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

Go

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
```

```
    // In this example, AK and SK are stored in environment variables for authentication. Before running this
example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
         WithRegion(region.ValueOf("<YOUR REGION>")).
         WithCredential(auth).
         SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.DeleteTemplateRequest{}
    request.Id = "{id}"
    response, err := client.DeleteTemplate(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

More

For SDK sample code of more programming languages, see the Sample Code tab in **API Explorer**. SDK sample code can be automatically generated.

Status Codes

Status Code	Description
200	The template with the specified ID was deleted.
403	Authentication failed.

Error Codes

See **Error Codes**.

5.6.5 Querying a Template

Function

This API is used to query information about an ECS template with a specified ID.

Calling Method

For details, see **Calling APIs**.

Authorization Information

Each account has all the permissions required to call all APIs, but IAM users must be assigned the required permissions.

- If you are using role/policy-based authorization, see **Permissions Policies and Supported Actions** for details on the required permissions.
- If you are using identity policy-based authorization, the following identity policy-based permissions are required.

Action	Access Level	Resource Type (*: required)	Condition Key	Alias	Dependencies
sms:template:get	Read	template *	-	sms:server:queryServer	-

URI

GET /v3/vm/templates/{id}

Table 5-285 Path Parameters

Parameter	Mandatory	Type	Description
id	Yes	String	The ID of the template to be queried. Minimum: **1** Maximum: **36**

Request Parameters

Table 5-286 Request header parameters

Parameter	Mandatory	Type	Description
X-Auth-Token	Yes	String	User token. The token can be obtained by calling the IAM API for obtaining a user token. The value of **X-Subject-Token** in the response header is the user token. Minimum: **1** Maximum: **16384**

Response Parameters

Status code: 200

Table 5-287 Response body parameters

Parameter	Type	Description
template	**TemplateResponseBody** object	The template information.

Table 5-288 TemplateResponseBody

Parameter	Type	Description
id	String	The template ID. Minimum: **0** Maximum: **255**
name	String	The template name. Minimum: **0** Maximum: **255**
is_template	String	Indicates whether the template is general. If the template is associated with a task, the template is not a general template. Minimum: **0** Maximum: **255**

Parameter	Type	Description
region	String	The region. Minimum: **0** Maximum: **255**
projectid	String	The project ID. Minimum: **0** Maximum: **255**
target_server_name	String	The name of the target server. Minimum: **0** Maximum: **255**
availability_zone	String	The availability zone. Minimum: **0** Maximum: **255**
volumetype	String	The data disk type. **SAS**: serial attached SCSI **SSD**: solid-state drive **SATA**: serial advanced technology attachment Enumeration values: • **SAS** • **SSD** • **SATA**
flavor	String	The server flavor. Minimum: **0** Maximum: **255**
vpc	**VpcObject** object	The VPC information.
nics	Array of **Nics** objects	The NIC information. Multiple NICs are supported. To let the system automatically add a NIC, configure only one NIC entry and set the ID to **autoCreate**. Array Length: **0 - 65535**
security_groups	Array of **SgObject** objects	The security group information. Multiple security groups are supported. To let the system automatically create a security group, configure only one security group record and set the ID to **autoCreate**. Array Length: **0 - 65535**

Parameter	Type	Description
publicip	**PublicIp** object	The public IP address.
disk	Array of **TemplateDisk** objects	The disk information. Array Length: **0 - 65535**
data_volume_type	String	The data disk type. **SAS**: serial attached SCSI **SSD**: solid-state drive **SATA**: serial advanced technology attachment Enumeration values: ● **SAS** ● **SSD** ● **SATA**
target_password	String	The target server password. Minimum: **0** Maximum: **1024**
image_id	String	The ID of the selected image. Minimum: **0** Maximum: **255**

Table 5-289 VpcObject

Parameter	Type	Description
id	String	The VPC ID. To let the system automatically create a VPC, set this parameter to **autoCreate**. Minimum: **1** Maximum: **255**
name	String	The VPC name. Minimum: **1** Maximum: **255**
cidr	String	The VPC CIDR block. The default value is **192.168.0.0/16**. Minimum: **1** Maximum: **255**

Table 5-290 Nics

Parameter	Type	Description
id	String	The subnet ID. To let the system automatically create a subnet, set this parameter to **autoCreate**. Minimum: **0** Maximum: **255**
name	String	The subnet name. Minimum: **0** Maximum: **255**
cidr	String	The subnet gateway/mask. Minimum: **0** Maximum: **255**
ip	String	The server IP address. If this parameter is not specified, the system will automatically assign an IP address. Minimum: **0** Maximum: **255**

Table 5-291 SgObject

Parameter	Type	Description
id	String	Security group ID. Minimum: **0** Maximum: **255**
name	String	Security group name. Minimum: **0** Maximum: **255**

Table 5-292 PublicIp

Parameter	Type	Description
type	String	The EIP type. The default value is **5_bgp**. For details about the type, see the description of the **type** field in **Response Message** in section "Querying an EIP" of the EIP API reference. Minimum: **0** Maximum: **255**
bandwidth_size	Integer	The bandwidth size, in Mbit/s. The minimum increment for bandwidth adjustment varies depending on the bandwidth range. The minimum increment is 1 Mbit/s if the allowed bandwidth does not exceed 300 Mbit/s. The minimum increment is 50 Mbit/s if the allowed bandwidth ranges from 300 Mbit/s to 1,000 Mbit/s. The minimum increment is 500 Mbit/s if the allowed bandwidth exceeds 1,000 Mbit/s. Minimum: **1** Maximum: **2000**
bandwidth_share_type	String	Bandwidth sharing type (long text, non-enumerated data, from EIP) For details about the type, see the description of the **bandwidth_share_type** field in **Response Message** in section "Querying an EIP" of the EIP API reference. Minimum: **0** Maximum: **255**

Table 5-293 TemplateDisk

Parameter	Type	Description
id	Long	The disk ID. Minimum: **0** Maximum: **9223372036854775807**

Parameter	Type	Description
index	Integer	The disk serial number, starting from 0. Minimum: **0** Maximum: **2147483647**
name	String	The disk name. Minimum: **0** Maximum: **255**
disktype	String	Disk type, which is the same as the **volumetype** field. Its value comes from EVS. For details, see the description of the **volume_type** field in the response parameters in the section "Querying Details About a Single EVS Disk" in the EIP API document. Minimum: **0** Maximum: **255**
size	Long	The disk size in GB. Minimum: **0** Maximum: **9223372036854775807**
device_use	String	The disk function. Minimum: **0** Maximum: **255**

Status code: 403

Table 5-294 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **255**

Parameter	Type	Description
encoded_authorization_message	String	The encrypted authorization information. Minimum: **0** Maximum: **65535**
error_param	Array of strings	Error parameters. Minimum: **0** Maximum: **65535** Array Length: **1 - 20**
details	Array of **details** objects	The error details. Array Length: **1 - 20**

Table 5-295 details

Parameter	Type	Description
error_code	String	The SMS error code. Minimum: **0** Maximum: **65535**
error_msg	String	The SMS error message. Minimum: **0** Maximum: **65535**

Example Requests

This example queries the details about the template whose ID is **6874cb49-48bb-4875-975d-4bca464d8472**.

```
GET https://{endpoint}/v3/vm/templates/6874cb49-48bb-4875-975d-4bca464d8472
```

Example Responses

Status code: 200

Querying details about a template succeeded.

```
{
  "template" : {
    "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "name" : "test1025",
    "region" : "region",
    "target_server_name" : "",
    "availability_zone" : "availability_zone",
    "projectid" : "xxxxxxxxxxxxxxxxxxxxxxxx00000001",
    "flavor" : "s2.large.2",
    "volumetype" : "SATA",
    "image_id" : "",
    "vpc" : {
```



```
      "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
      "name" : "vpc-testcloud(192.168.0.0/16)"
    },
    "nics" : [ {
      "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
      "name" : "subnet-testcloud(192.168.0.0/24)",
      "cidr" : "192.168.0.0/24",
      "ip" : ""
    } ],
    "security_groups" : [ ],
    "publicip" : {
      "type" : "5_bgp",
      "bandwidth_size" : 5,
      "bandwidth_share_type" : "PER"
    },
    "disk" : [ {
      "index" : 0,
      "name" : "system",
      "disktype" : "",
      "size" : 40
    } ]
  }
}
```


Status code: 403

Authentication failed.


```
{
  "error_code" : "SMS.9004",
  "error_msg" : "The current account does not have the permission to execute policy. You do not have permission to perform action XXX on resource XXX.",
  "encoded_authorization_message" : "XXXXXX",
  "error_param" : [ "You do not have permission to perform action XXX on resource XXX." ],
  "details" : [ {
    "error_code" : "SMS.9004",
    "error_msg" : "You do not have permission to perform action XXX on resource XXX."
  } ]
}
```


SDK Sample Code

The SDK sample code is as follows.

Java

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;


public class ShowTemplateSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");
```

```
        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        ShowTemplateRequest request = new ShowTemplateRequest();
        request.withId("{id}");
        try {
            ShowTemplateResponse response = client.showTemplate(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

Python

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = ShowTemplateRequest()
        request.id = "{id}"
        response = client.show_template(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

Go

```
package main
```

```
import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
         WithRegion(region.ValueOf("<YOUR REGION>")).
         WithCredential(auth).
         SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.ShowTemplateRequest{}
    request.Id = "{id}"
    response, err := client.ShowTemplate(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

More

For SDK sample code of more programming languages, see the Sample Code tab in **API Explorer**. SDK sample code can be automatically generated.

Status Codes

Status Code	Description
200	Querying details about a template succeeded.
403	Authentication failed.

Error Codes

See **Error Codes**.

5.6.6 Modifying a Template

Function

This API is used to modify a template.

Calling Method

For details, see **Calling APIs**.

Authorization Information

Each account has all the permissions required to call all APIs, but IAM users must be assigned the required permissions.

- If you are using role/policy-based authorization, see **Permissions Policies and Supported Actions** for details on the required permissions.
- If you are using identity policy-based authorization, the following identity policy-based permissions are required.

Action	Access Level	Resource Type (*: required)	Condition Key	Alias	Dependencies
sms:template:update	Write	template *	-	sms:server:migrationServer	-

URI

PUT /v3/vm/templates/{id}

Table 5-296 Path Parameters

Parameter	Mandatory	Type	Description
id	Yes	String	The ID of the template to be modified. Minimum: **1** Maximum: **36**

Request Parameters

Table 5-297 Request header parameters

Parameter	Mandatory	Type	Description
X-Auth-Token	Yes	String	User token. The token can be obtained by calling the IAM API for obtaining a user token. The value of **X-Subject-Token** in the response header is the user token. Minimum: **1** Maximum: **16384**

Table 5-298 Request body parameters

Parameter	Mandatory	Type	Description
template	No	**TemplateRequest** object	The template information.

Table 5-299 TemplateRequest

Parameter	Mandatory	Type	Description
name	Yes	String	The template name. Only Chinese characters, underscores (_), hyphens (-), digits, and uppercase and lowercase letters are allowed. Minimum: **1** Maximum: **64**
is_template	No	Boolean	Indicates whether the template is general. If the template is associated with a task, the template is not a general template.
region	No	String	The region. Minimum: **0** Maximum: **255**

Parameter	Mandatory	Type	Description
projectid	No	String	The project ID. Minimum: **0** Maximum: **255**
target_server_name	No	String	The name of the target server. Minimum: **0** Maximum: **255**
availability_zone	No	String	The availability zone. Minimum: **0** Maximum: **255**
volumetype	No	String	Template name. The value can contain only letters, digits, underscores (_), and hyphens (-). Minimum: **0** Maximum: **255**
flavor	No	String	The server flavor. Minimum: **0** Maximum: **65535**
vpc	No	**VpcObject** object	The VPC information.
nics	No	Array of **Nics** objects	The NIC information. Multiple NICs are supported. To let the system automatically add a NIC, configure only one NIC entry and set the ID to **autoCreate**. Array Length: **0 - 65535**
security_groups	No	Array of **SgObject** objects	The security group information. Multiple security groups are supported. To let the system automatically create a security group, configure only one security group record and set the ID to **autoCreate**. Array Length: **0 - 65535**
publicip	No	**PublicIp** object	The public IP address.

Parameter	Mandatory	Type	Description
disk	No	Array of **TemplateDisk** objects	The disk information. Array Length: **0 - 65535**
data_volume_type	No	String	Disk type of the data disk. Its value comes from EVS. Common types include **SAS** (serial attached SCSI), **SSD** (solid-state drive), and **SATA** (serial ATA). For details, see the description of the **volume_type** field in the response parameters in section "Querying Details About a Single EVS Disk" in the EIP API document. Minimum: **0** Maximum: **255**
target_password	No	String	The target server password. Minimum: **0** Maximum: **1024**
image_id	No	String	The ID of the image used to create target servers. Minimum: **0** Maximum: **255**

Table 5-300 VpcObject

Parameter	Mandatory	Type	Description
id	No	String	The VPC ID. To let the system automatically create a VPC, set this parameter to **autoCreate**. Minimum: **1** Maximum: **255**
name	No	String	The VPC name. Minimum: **1** Maximum: **255**

Parameter	Mandatory	Type	Description
cidr	No	String	The VPC CIDR block. The default value is **192.168.0.0/16**. Minimum: **1** Maximum: **255**

Table 5-301 Nics

Parameter	Mandatory	Type	Description
id	Yes	String	The subnet ID. To let the system automatically create a subnet, set this parameter to **autoCreate**. Minimum: **0** Maximum: **255**
name	Yes	String	The subnet name. Minimum: **0** Maximum: **255**
cidr	Yes	String	The subnet gateway/mask. Minimum: **0** Maximum: **255**
ip	No	String	The server IP address. If this parameter is not specified, the system will automatically assign an IP address. Minimum: **0** Maximum: **255**

Table 5-302 SgObject

Parameter	Mandatory	Type	Description
id	Yes	String	Security group ID. Minimum: **0** Maximum: **255**
name	Yes	String	Security group name. Minimum: **0** Maximum: **255**

Table 5-303 PublicIp

Parameter	Mandatory	Type	Description
type	Yes	String	The EIP type. The default value is **5_bgp**. For details about the type, see the description of the **type** field in **Response Message** in section "Querying an EIP" of the EIP API reference. Minimum: **0** Maximum: **255**
bandwidth_size	Yes	Integer	The bandwidth size, in Mbit/s. The minimum increment for bandwidth adjustment varies depending on the bandwidth range. The minimum increment is 1 Mbit/s if the allowed bandwidth does not exceed 300 Mbit/s. The minimum increment is 50 Mbit/s if the allowed bandwidth ranges from 300 Mbit/s to 1,000 Mbit/s. The minimum increment is 500 Mbit/s if the allowed bandwidth exceeds 1,000 Mbit/s. Minimum: **1** Maximum: **2000**
bandwidth_share_type	No	String	Bandwidth sharing type (long text, non-enumerated data, from EIP) For details about the type, see the description of the **bandwidth_share_type** field in **Response Message** in section "Querying an EIP" of the EIP API reference. Minimum: **0** Maximum: **255**

Table 5-304 TemplateDisk

Parameter	Mandatory	Type	Description
id	No	Long	The disk ID. Minimum: **0** Maximum: **9223372036854775807**
index	Yes	Integer	The disk serial number, starting from 0. Minimum: **0** Maximum: **2147483647**
name	Yes	String	The disk name. Minimum: **0** Maximum: **255**
disktype	Yes	String	Disk type, which is the same as the **volumetype** field. Its value comes from EVS. For details, see the description of the **volume_type** field in the response parameters in the section "Querying Details About a Single EVS Disk" in the EIP API document. Minimum: **0** Maximum: **255**
size	Yes	Long	The disk size in GB. Minimum: **0** Maximum: **9223372036854775807**
device_use	No	String	The disk function. Minimum: **0** Maximum: **255**

Response Parameters

Status code: 200

Table 5-305 Response body parameters

Parameter	Type	Description
-	String	The template was modified.

Status code: 403

Table 5-306 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **255**
encoded_authorization_message	String	The encrypted authorization information. Minimum: **0** Maximum: **65535**
error_param	Array of strings	Error parameters. Minimum: **0** Maximum: **65535** Array Length: **1 - 20**
details	Array of **details** objects	The error details. Array Length: **1 - 20**

Table 5-307 details

Parameter	Type	Description
error_code	String	The SMS error code. Minimum: **0** Maximum: **65535**
error_msg	String	The SMS error message. Minimum: **0** Maximum: **65535**

Example Requests

This example changes the name of the template whose ID is **dfbdd142-985f-4a4f-93e1-c535b46bxxxx** to **test1025** and changes its project ID to **00924d0ad2df4f21ac476dd9f3288xxxx**.

```
PUT https://{endpoint}/v3/vm/templates/dfbdd142-985f-4a4f-93e1-c535b46bxxxx

{
```

```
  "template" : {
    "name" : "test1025",
    "is_template" : false,
    "region" : "region",
    "projectid" : "00924d0ad2df4f21ac476dd9f3288xxxx",
    "vpc" : {
      "id" : "autoCreate",
      "name" : "autoCreate"
    },
    "nics" : [ {
      "id" : "autoCreate",
      "name" : "autoCreate",
      "cidr" : "192.168.0.0/24"
    } ],
    "security_groups" : [ {
      "id" : "autoCreate",
      "name" : "autoCreate"
    } ],
    "publicip" : {
      "type" : "5_bgp",
      "bandwidth_size" : 1
    },
    "disk" : [ {
      "index" : 0,
      "name" : "index1",
      "disktype" : "type",
      "size" : 111
    } ]
  }
}
```


Example Responses

Status code: 200

The template was modified.


```
{ }
```


Status code: 403

Authentication failed.


```
{
  "error_code" : "SMS.9004",
  "error_msg" : "The current account does not have the permission to execute policy. You do not have permission to perform action XXX on resource XXX.",
  "encoded_authorization_message" : "XXXXXX",
  "error_param" : [ "You do not have permission to perform action XXX on resource XXX." ],
  "details" : [ {
    "error_code" : "SMS.9004",
    "error_msg" : "You do not have permission to perform action XXX on resource XXX."
  } ]
}
```


SDK Sample Code

The SDK sample code is as follows.

Java

This example changes the name of the template whose ID is **dfbdd142-985f-4a4f-93e1-c535b46bxxxx** to **test1025** and changes its project ID to **00924d0ad2df4f21ac476dd9f3288xxxx**.


```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;

import java.util.List;
import java.util.ArrayList;

public class UpdateTemplateSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great
security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or
environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running
this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        UpdateTemplateRequest request = new UpdateTemplateRequest();
        request.withId("{id}");
        UpdateTemplateReq body = new UpdateTemplateReq();
        List<TemplateDisk> listTemplateDisk = new ArrayList<>();
        listTemplateDisk.add(
            new TemplateDisk()
                .withIndex(0)
                .withName("index1")
                .withDisktype("type")
                .withSize(111L)
        );
        PublicIp publicipTemplate = new PublicIp();
        publicipTemplate.withType("5_bgp")
            .withBandwidthSize(1);
        List<SgObject> listTemplateSecurityGroups = new ArrayList<>();
        listTemplateSecurityGroups.add(
            new SgObject()
                .withId("autoCreate")
                .withName("autoCreate")
        );
        List<Nics> listTemplateNics = new ArrayList<>();
        listTemplateNics.add(
            new Nics()
                .withId("autoCreate")
                .withName("autoCreate")
                .withCidr("192.168.0.0/24")
        );
        VpcObject vpcTemplate = new VpcObject();
        vpcTemplate.withId("autoCreate")
            .withName("autoCreate");
        TemplateRequest templatebody = new TemplateRequest();
        templatebody.withName("test1025")
            .withIsTemplate(false)
            .withRegion("region")
            .withProjectid("00924d0ad2df4f21ac476dd9f3288xxxx")
            .withVpc(vpcTemplate)
```

```
            .withNics(listTemplateNics)
            .withSecurityGroups(listTemplateSecurityGroups)
            .withPublicip(publicipTemplate)
            .withDisk(listTemplateDisk);
        body.withTemplate(templatebody);
        request.withBody(body);
        try {
            UpdateTemplateResponse response = client.updateTemplate(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

Python

This example changes the name of the template whose ID is **dfbdd142-985f-4a4f-93e1-c535b46bxxxx** to **test1025** and changes its project ID to **00924d0ad2df4f21ac476dd9f3288xxxx**.

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = UpdateTemplateRequest()
        request.id = "{id}"
        listDiskTemplate = [
            TemplateDisk(
                index=0,
                name="index1",
                disktype="type",
                size=111
            )
        ]
        publicipTemplate = PublicIp(
            type="5_bgp",
            bandwidth_size=1
        )
```

```
        listSecurityGroupsTemplate = [
            SgObject(
                id="autoCreate",
                name="autoCreate"
            )
        ]
        listNicsTemplate = [
            Nics(
                id="autoCreate",
                name="autoCreate",
                cidr="192.168.0.0/24"
            )
        ]
        vpcTemplate = VpcObject(
            id="autoCreate",
            name="autoCreate"
        )
        templatebody = TemplateRequest(
            name="test1025",
            is_template=False,
            region="region",
            projectid="00924d0ad2df4f21ac476dd9f3288xxxx",
            vpc=vpcTemplate,
            nics=listNicsTemplate,
            security_groups=listSecurityGroupsTemplate,
            publicip=publicipTemplate,
            disk=listDiskTemplate
        )
        request.body = UpdateTemplateReq(
            template=templatebody
        )
        response = client.update_template(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

Go

This example changes the name of the template whose ID is **dfbdd142-985f-4a4f-93e1-c535b46bxxxx** to **test1025** and changes its project ID to **00924d0ad2df4f21ac476dd9f3288xxxx**.

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()
```

```
if err != nil {
    fmt.Println(err)
    return
}

hcClient, err := sms.SmsClientBuilder().
     WithRegion(region.ValueOf("<YOUR REGION>")).
     WithCredential(auth).
     SafeBuild()


if err != nil {
    fmt.Println(err)
    return
}

client := sms.NewSmsClient(hcClient)

request := &model.UpdateTemplateRequest{}
request.Id = "{id}"
var listDiskTemplate = []model.TemplateDisk{
    {
        Index: int32(0),
        Name: "index1",
        Disktype: "type",
        Size: int64(111),
    },
}
publicipTemplate := &model.PublicIp{
    Type: "5_bgp",
    BandwidthSize: int32(1),
}
var listSecurityGroupsTemplate = []model.SgObject{
    {
        Id: "autoCreate",
        Name: "autoCreate",
    },
}
var listNicsTemplate = []model.Nics{
    {
        Id: "autoCreate",
        Name: "autoCreate",
        Cidr: "192.168.0.0/24",
    },
}
idVpc:= "autoCreate"
nameVpc:= "autoCreate"
vpcTemplate := &model.VpcObject{
    Id: &idVpc,
    Name: &nameVpc,
}
isTemplateTemplate:= false
regionTemplate:= "region"
projectidTemplate:= "00924d0ad2df4f21ac476dd9f3288xxxx"
templatebody := &model.TemplateRequest{
    Name: "test1025",
    IsTemplate: &isTemplateTemplate,
    Region: &regionTemplate,
    Projectid: &projectidTemplate,
    Vpc: vpcTemplate,
    Nics: &listNicsTemplate,
    SecurityGroups: &listSecurityGroupsTemplate,
    Publicip: publicipTemplate,
    Disk: &listDiskTemplate,
}
request.Body = &model.UpdateTemplateReq{
    Template: templatebody,
}
```

```
    response, err := client.UpdateTemplate(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

More

For SDK sample code of more programming languages, see the Sample Code tab in **API Explorer**. SDK sample code can be automatically generated.

Status Codes

Status Code	Description
200	The template was modified.
403	Authentication failed.

Error Codes

See **Error Codes**.

5.6.7 Querying the Target Server Password in a Template

Function

This API is used to query the target server password configured in a template.

Calling Method

For details, see **Calling APIs**.

Authorization Information

Each account has all the permissions required to call all APIs, but IAM users must be assigned the required permissions.

- If you are using role/policy-based authorization, see **Permissions Policies and Supported Actions** for details on the required permissions.
- If you are using identity policy-based authorization, the following identity policy-based permissions are required.

Action	Access Level	Resource Type (*: required)	Condition Key	Alias	Dependencies
sms:template:getTargetPassword	Read	template *	-	sms:server:queryServer	-

URI

GET /v3/vm/templates/{id}/target-password

Table 5-308 Path Parameters

Parameter	Mandatory	Type	Description
id	Yes	String	The template ID. Minimum: **1** Maximum: **36**

Request Parameters

Table 5-309 Request header parameters

Parameter	Mandatory	Type	Description
X-Auth-Token	Yes	String	User token. The token can be obtained by calling the IAM API for obtaining a user token. The value of **X-Subject-Token** in the response header is the user token. Minimum: **1** Maximum: **16384**

Response Parameters

Status code: 200

Table 5-310 Response body parameters

Parameter	Type	Description
template_id	String	The template ID. Minimum: **0** Maximum: **255**
target_password	String	The target server password. Minimum: **0** Maximum: **255**

Status code: 403

Table 5-311 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **255**
encoded_authorization_message	String	The encrypted authorization information. Minimum: **0** Maximum: **65535**
error_param	Array of strings	Error parameters. Minimum: **0** Maximum: **65535** Array Length: **1 - 20**
details	Array of **details** objects	The error details. Array Length: **1 - 20**

Table 5-312 details

Parameter	Type	Description
error_code	String	The SMS error code. Minimum: **0** Maximum: **65535**

Parameter	Type	Description
error_msg	String	The SMS error message. Minimum: **0** Maximum: **65535**

Example Requests

This example queries the target server password configured in the specified template.

```
GET https://{endpoint}/v3/vm/templates/ef3b9722-07a0-40ae-89b0-889ee96dfc56/target-password
```

Example Responses

Status code: 200

The target server password was obtained.


```
{
  "template_id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
  "target_password" : "********"
}
```


Status code: 403

Authentication failed.


```
{
  "error_code" : "SMS.9004",
  "error_msg" : "The current account does not have the permission to execute policy. You do not have permission to perform action XXX on resource XXX.",
  "encoded_authorization_message" : "XXXXXX",
  "error_param" : [ "You do not have permission to perform action XXX on resource XXX." ],
  "details" : [ {
    "error_code" : "SMS.9004",
    "error_msg" : "You do not have permission to perform action XXX on resource XXX."
  } ]
}
```


SDK Sample Code

The SDK sample code is as follows.

Java

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;


public class ShowTargetPasswordSolution {
```

```
    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great
security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or
environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running
this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        ShowTargetPasswordRequest request = new ShowTargetPasswordRequest();
        request.withId("{id}");
        try {
            ShowTargetPasswordResponse response = client.showTargetPassword(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

Python

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security
risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment
variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this
example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = ShowTargetPasswordRequest()
        request.id = "{id}"
        response = client.show_target_password(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
```

```
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

Go

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
         WithRegion(region.ValueOf("<YOUR REGION>")).
         WithCredential(auth).
         SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.ShowTargetPasswordRequest{}
    request.Id = "{id}"
    response, err := client.ShowTargetPassword(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

More

For SDK sample code of more programming languages, see the Sample Code tab in **API Explorer**. SDK sample code can be automatically generated.

Status Codes

Status Code	Description
200	The target server password was obtained.
403	Authentication failed.

Error Codes

See **Error Codes**.

5.7 Key Management

5.7.1 Obtaining an SSL Certificate and Private Key

Function

If the block-level migration method is used, the Agent installed on the source server communicates with the target server through an SSL socket. This API is used to download the certificate and private key (in PEM format) required for data migration.

Calling Method

For details, see **Calling APIs**.

Authorization Information

Each account has all the permissions required to call all APIs, but IAM users must be assigned the required permissions.

- If you are using role/policy-based authorization, see **Permissions Policies and Supported Actions** for details on the required permissions.
- If you are using identity policy-based authorization, the following identity policy-based permissions are required.

<table>
<tr><th>Action</th><th>Access Level</th><th>Resource Type (*: required)</th><th>Condition Key</th><th>Alias</th><th>Dependencies</th></tr>
<tr><td rowspan="2">sms:server:getCert</td><td rowspan="2">Read</td><td>server *</td><td>-</td><td rowspan="2">sms:server:queryServer</td><td rowspan="2">-</td></tr>
<tr><td>-</td><td>g:EnterpriseProjectId</td></tr>
</table>

URI

GET /v3/tasks/{task_id}/certkey

Table 5-313 Path Parameters

Parameter	Mandatory	Type	Description
task_id	Yes	String	The migration task ID. Minimum: **1** Maximum: **36**

Table 5-314 Query Parameters

Parameter	Mandatory	Type	Description
enable_ca_cert	No	Boolean	This API is used to indicate whether to generate a CA certificate. Default: **false**

Request Parameters

Table 5-315 Request header parameters

Parameter	Mandatory	Type	Description
X-Auth-Token	Yes	String	User token. The token can be obtained by calling the IAM API for obtaining a user token. The value of **X-Subject-Token** in the response header is the user token. Minimum: **1** Maximum: **16384**

Response Parameters

Status code: 200

Table 5-316 Response body parameters

Parameter	Type	Description
cert	String	The source certificate. Minimum: **1** Maximum: **1048576**
private_key	String	The source private key. Minimum: **1** Maximum: **1048576**
ca	String	The CA certificate. Minimum: **1** Maximum: **1048576**
target_mgmt_cert	String	The certificate of the target server for migration task management. Minimum: **1** Maximum: **1048576**
target_mgmt_private_key	String	The private key of the target server for migration task management. Minimum: **1** Maximum: **1048576**
target_data_cert	String	The certificate of the target server for data migration. Minimum: **1** Maximum: **1048576**
target_data_private_key	String	The private key of the target server for data migration. Minimum: **1** Maximum: **1048576**

Status code: 400

Table 5-317 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**

Parameter	Type	Description
error_msg	String	The error message. Minimum: **0** Maximum: **1024**

Status code: 401

Table 5-318 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **1024**

Status code: 403

Table 5-319 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **255**
encoded_authorization_message	String	The encrypted authorization information. Minimum: **0** Maximum: **65535**
error_param	Array of strings	Error parameters. Minimum: **0** Maximum: **65535** Array Length: **1 - 20**

Parameter	Type	Description
details	Array of **details** objects	The error details. Array Length: **1 - 20**

Table 5-320 details

Parameter	Type	Description
error_code	String	The SMS error code. Minimum: **0** Maximum: **65535**
error_msg	String	The SMS error message. Minimum: **0** Maximum: **65535**

Status code: 404

Table 5-321 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **1024**

Status code: 500

Table 5-322 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **1024**

Example Requests

This example obtains the certificate and private key required for a migration task.

```
GET https://{endpoint}/v3/tasks/{task_id}/certkey
```

Example Responses

Status code: 200

Obtaining an SSL certificate and private key succeeded.

```
{
  "ca" : "-----BEGIN CERTIFICATE-----\n********************************\n-----END CERTIFICATE-----",
  "cert" : "-----BEGIN CERTIFICATE-----\n********************************\n-----END CERTIFICATE-----",
  "private_key" : "-----BEGIN RSA PRIVATE KEY-----\n********************************\n-----END RSA PRIVATE
KEY-----",
  "target_mgmt_cert" : "-----BEGIN CERTIFICATE-----\n********************************\n-----END
CERTIFICATE-----",
  "target_mgmt_private_key" : "-----BEGIN RSA PRIVATE KEY-----\n********************************\n-----END RSA
PRIVATE KEY-----",
  "target_data_cert" : "-----BEGIN CERTIFICATE-----\n********************************\n-----END CERTIFICATE-----",
  "target_data_private_key" : "-----BEGIN RSA PRIVATE KEY-----\n********************************\n-----END RSA
PRIVATE KEY-----"
}
```

Status code: 403

Authentication failed.

```
{
  "error_code" : "SMS.9004",
  "error_msg" : "The current account does not have the permission to execute policy. You do not have
permission to perform action XXX on resource XXX.",
  "encoded_authorization_message" : "XXXXXX",
  "error_param" : [ "You do not have permission to perform action XXX on resource XXX." ],
  "details" : [ {
    "error_code" : "SMS.9004",
    "error_msg" : "You do not have permission to perform action XXX on resource XXX."
  } ]
}
```

SDK Sample Code

The SDK sample code is as follows.

Java

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;


public class ShowCertKeySolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great
security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or
environment variables and decrypted during use to ensure security.
```

```
        // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        ShowCertKeyRequest request = new ShowCertKeyRequest();
        request.withTaskId("{task_id}");
        try {
            ShowCertKeyResponse response = client.showCertKey(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

Python

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = ShowCertKeyRequest()
        request.task_id = "{task_id}"
        response = client.show_cert_key(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

Go

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
         WithRegion(region.ValueOf("<YOUR REGION>")).
         WithCredential(auth).
         SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.ShowCertKeyRequest{}
    request.TaskId = "{task_id}"
    response, err := client.ShowCertKey(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

More

For SDK sample code of more programming languages, see the Sample Code tab in **API Explorer**. SDK sample code can be automatically generated.

Status Codes

Status Code	Description
200	Obtaining an SSL certificate and private key succeeded.

Status Code	Description
400	Bad Request
401	Unauthorized
403	Authentication failed.
404	Not Found
500	Internal Server Error

Error Codes

See **Error Codes**.

5.8 Migration Project Management

5.8.1 Creating a Migration Project

Function

This API is used to create a migration project.

Constraints

The migration project name must be unique.

Calling Method

For details, see **Calling APIs**.

Authorization Information

Each account has all the permissions required to call all APIs, but IAM users must be assigned the required permissions.

- If you are using role/policy-based authorization, see **Permissions Policies and Supported Actions** for details on the required permissions.
- If you are using identity policy-based authorization, the following identity policy-based permissions are required.

Action	Access Level	Resource Type (*: required)	Condition Key	Alias	Dependencies
sms:migproject:create	Write	migproject *	-	sms:server:migrationServer	-

URI

POST /v3/migprojects

Request Parameters

Table 5-323 Request header parameters

Parameter	Mandatory	Type	Description
X-Auth-Token	Yes	String	User token. The token can be obtained by calling the IAM API for obtaining a user token. The value of **X-Subject-Token** in the response header is the user token. Minimum: **1** Maximum: **16384**

Table 5-324 Request body parameters

Parameter	Mandatory	Type	Description
name	Yes	String	The migration project name. Minimum: **1** Maximum: **100**
description	No	String	The migration project description. Minimum: **0** Maximum: **255**
isdefault	No	Boolean	Indicates whether the migration project is the default project. Default: **false**

Parameter	Mandatory	Type	Description
region	Yes	String	The region name. Minimum: **0** Maximum: **100**
start_target_server	No	Boolean	Indicates whether the target server is started after the migration is complete. Default: **true**
speed_limit	No	Integer	The migration rate limit, in Mbit/s. **0** indicates no limit. Minimum: **0** Maximum: **1000** Default: **0**
use_public_ip	No	Boolean	Indicates whether a public IP address is used for migration. Default: **true**
exist_server	No	Boolean	Specifies whether an existing server is as the target server. Default: **true**
type	Yes	String	The migration project type. **MIGRATE_BLOCK** **MIGRATE_FILE** Minimum: **0** Maximum: **255** Enumeration values: • **MIGRATE_BLOCK** • **MIGRATE_FILE**
enterprise_project	No	String	The enterprise project name. Default: **default** Minimum: **0** Maximum: **255**
syncing	No	Boolean	Indicates whether continuous synchronization is performed after the first replication or synchronization is complete. Default: **false**
start_network_check	No	Boolean	Indicates whether network performance measurement is enabled.

Response Parameters

Status code: 200

Table 5-325 Response body parameters

Parameter	Type	Description
id	String	The ID of the newly added migration project. Minimum: **0** Maximum: **255**

Status code: 403

Table 5-326 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **255**
encoded_authorization_message	String	The encrypted authorization information. Minimum: **0** Maximum: **65535**
error_param	Array of strings	Error parameters. Minimum: **0** Maximum: **65535** Array Length: **1 - 20**
details	Array of **details** objects	The error details. Array Length: **1 - 20**

Table 5-327 details

Parameter	Type	Description
error_code	String	The SMS error code. Minimum: **0** Maximum: **65535**
error_msg	String	The SMS error message. Minimum: **0** Maximum: **65535**

Example Requests

This API is used to create a migration project. It sets the project name to **N121**, the region to **region**, the public IP address to **true**, and the migration project type to **MIGRATE_BLOCK**.

```
POST https://{endpoint}/v3/migprojects

{
  "name" : "N121",
  "description" : "",
  "region" : "region",
  "start_target_server" : true,
  "speed_limit" : 0,
  "use_public_ip" : true,
  "exist_server" : true,
  "isdefault" : true,
  "type" : "MIGRATE_BLOCK",
  "syncing" : false,
  "enterprise_project" : "default"
}
```

Example Responses

Status code: 200

Creating a migration project succeeded.

```
{
  "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
}
```

Status code: 403

Authentication failed.

```
{
  "error_code" : "SMS.9004",
  "error_msg" : "The current account does not have the permission to execute policy. You do not have 
permission to perform action XXX on resource XXX.",
  "encoded_authorization_message" : "XXXXXX",
  "error_param" : [ "You do not have permission to perform action XXX on resource XXX." ],
  "details" : [ {
    "error_code" : "SMS.9004",
    "error_msg" : "You do not have permission to perform action XXX on resource XXX."
  } ]
}
```

SDK Sample Code

The SDK sample code is as follows.

Java

This API is used to create a migration project. It sets the project name to **N121**, the region to **region**, the public IP address to **true**, and the migration project type to **MIGRATE_BLOCK**.

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;


public class CreateMigprojectSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great
security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or
environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running
this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        CreateMigprojectRequest request = new CreateMigprojectRequest();
        PostMigProjectBody body = new PostMigProjectBody();
        body.withSyncing(false);
        body.withEnterpriseProject("default");
        body.withType(PostMigProjectBody.TypeEnum.fromValue("MIGRATE_BLOCK"));
        body.withExistServer(true);
        body.withUsePublicIp(true);
        body.withSpeedLimit(0);
        body.withStartTargetServer(true);
        body.withRegion("region");
        body.withIsdefault(true);
        body.withDescription("");
        body.withName("N121");
        request.withBody(body);
        try {
            CreateMigprojectResponse response = client.createMigproject(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
```

```
            System.out.println(e.getErrorMsg());
        }
    }
}
```

Python

This API is used to create a migration project. It sets the project name to **N121**, the region to **region**, the public IP address to **true**, and the migration project type to **MIGRATE_BLOCK**.

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = CreateMigprojectRequest()
        request.body = PostMigProjectBody(
            syncing=False,
            enterprise_project="default",
            type="MIGRATE_BLOCK",
            exist_server=True,
            use_public_ip=True,
            speed_limit=0,
            start_target_server=True,
            region="region",
            isdefault=True,
            description="",
            name="N121"
        )
        response = client.create_migproject(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

Go

This API is used to create a migration project. It sets the project name to **N121**, the region to **region**, the public IP address to **true**, and the migration project type to **MIGRATE_BLOCK**.

```
package main
```

```
import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
         WithRegion(region.ValueOf("<YOUR REGION>")).
         WithCredential(auth).
         SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.CreateMigprojectRequest{}
    syncingPostMigProjectBody:= false
    enterpriseProjectPostMigProjectBody:= "default"
    existServerPostMigProjectBody:= true
    usePublicIpPostMigProjectBody:= true
    speedLimitPostMigProjectBody:= int32(0)
    startTargetServerPostMigProjectBody:= true
    isdefaultPostMigProjectBody:= true
    descriptionPostMigProjectBody:= ""
    request.Body = &model.PostMigProjectBody{
        Syncing: &syncingPostMigProjectBody,
        EnterpriseProject: &enterpriseProjectPostMigProjectBody,
        Type: model.GetPostMigProjectBodyTypeEnum().MIGRATE_BLOCK,
        ExistServer: &existServerPostMigProjectBody,
        UsePublicIp: &usePublicIpPostMigProjectBody,
        SpeedLimit: &speedLimitPostMigProjectBody,
        StartTargetServer: &startTargetServerPostMigProjectBody,
        Region: "region",
        Isdefault: &isdefaultPostMigProjectBody,
        Description: &descriptionPostMigProjectBody,
        Name: "N121",
    }
    response, err := client.CreateMigproject(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

More

For SDK sample code of more programming languages, see the Sample Code tab in **API Explorer**. SDK sample code can be automatically generated.

Status Codes

Status Code	Description
200	Creating a migration project succeeded.
403	Authentication failed.

Error Codes

See **Error Codes**.

5.8.2 Listing Migration Projects

Function

SMS enables you to use migration projects to manage source servers. This API is used to obtain the list of all migration projects under the current account.

Calling Method

For details, see **Calling APIs**.

Authorization Information

Each account has all the permissions required to call all APIs, but IAM users must be assigned the required permissions.

- If you are using role/policy-based authorization, see **Permissions Policies and Supported Actions** for details on the required permissions.
- If you are using identity policy-based authorization, the following identity policy-based permissions are required.

Action	Access Level	Resource Type (*: required)	Condition Key	Alias	Dependencies
sms:migproject:list	List	migproject *	-	sms:server:queryServer	-

URI

GET /v3/migprojects

Table 5-328 Query Parameters

Parameter	Mandatory	Type	Description
limit	No	Integer	The number of migration projects recorded on each page Minimum: **0** Maximum: **100** Default: **50**
offset	No	Integer	The offset. Minimum: **0** Maximum: **65535** Default: **0**

Request Parameters

Table 5-329 Request header parameters

Parameter	Mandatory	Type	Description
X-Auth-Token	Yes	String	User token. The token can be obtained by calling the IAM API for obtaining a user token. The value of **X-Subject-Token** in the response header is the user token. Minimum: **1** Maximum: **16384**

Response Parameters

Status code: 200

Table 5-330 Response body parameters

Parameter	Type	Description
count	Integer	The number of queried migration projects. Minimum: **0** Maximum: **2147483647**

Parameter	Type	Description
migprojects	Array of **MigprojectsResponseBody** objects	The details of queried migration projects. Array Length: **0 - 65535**

Table 5-331 MigprojectsResponseBody

Parameter	Type	Description
id	String	The migration project ID. Minimum: **1** Maximum: **255**
name	String	The migration project name. Minimum: **1** Maximum: **20**
use_public_ip	Boolean	Indicates whether a public IP address is used for migration.
isdefault	Boolean	Indicates whether the migration project is the default project.
start_target_server	Boolean	Indicates whether the target server is started after the migration is complete.
region	String	The region name. Minimum: **0** Maximum: **255**
speed_limit	Integer	The migration rate limit configured in the project. The unit is Mbit/s. Minimum: **0** Maximum: **10000**
exist_server	Boolean	Indicates whether there are servers in the migration project.
description	String	The migration project description. Minimum: **0** Maximum: **255**

Parameter	Type	Description
type	String	The migration project type. **MIGRATE_BLOCK** **MIGRATE_FILE** Enumeration values: ● **MIGRATE_BLOCK** ● **MIGRATE_FILE**
enterprise_project	String	The name of the enterprise project to which the migration project belongs. Minimum: **0** Maximum: **255**
syncing	Boolean	Indicate whether continuous synchronization is performed after the full replication is complete.
start_network_check	Boolean	Indicates whether network performance measurement is enabled.

Status code: 403

Table 5-332 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **255**
encoded_authorization_message	String	The encrypted authorization information. Minimum: **0** Maximum: **65535**
error_param	Array of strings	Error parameters. Minimum: **0** Maximum: **65535** Array Length: **1 - 20**
details	Array of **details** objects	The error details. Array Length: **1 - 20**

Table 5-333 details

Parameter	Type	Description
error_code	String	The SMS error code. Minimum: **0** Maximum: **65535**
error_msg	String	The SMS error message. Minimum: **0** Maximum: **65535**

Example Requests

This example obtains the list of migration projects.

```
GET https://{endpoint}/v3/migprojects
```

Example Responses

Status code: 200

The list of migration projects was obtained.


```
{
  "count" : 6,
  "migprojects" : [ {
    "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "name" : "sms_test",
    "use_public_ip" : true,
    "isdefault" : true,
    "start_target_server" : true,
    "region" : "06334e957c80d2642f39c0030856abdb",
    "speed_limit" : 0,
    "exist_server" : true,
    "description" : "",
    "type" : "MIGRATE_BLOCK",
    "enterprise_project" : "default"
  }, {
    "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "name" : "SystemProject",
    "use_public_ip" : true,
    "isdefault" : false,
    "start_target_server" : true,
    "region" : "region",
    "speed_limit" : 0,
    "exist_server" : true,
    "description" : "",
    "type" : "MIGRATE_BLOCK",
    "enterprise_project" : "default"
  }, {
    "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "name" : "sms_test",
    "use_public_ip" : true,
    "isdefault" : false,
    "start_target_server" : true,
    "region" : "region",
    "speed_limit" : 0,
    "exist_server" : true,
    "description" : "",
    "type" : "MIGRATE_BLOCK",
```

```
    "enterprise_project" : "default"
  }, {
    "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "name" : "sms_test_Test",
    "use_public_ip" : true,
    "isdefault" : false,
    "start_target_server" : true,
    "region" : "region",
    "speed_limit" : 0,
    "exist_server" : true,
    "description" : "",
    "type" : "MIGRATE_BLOCK",
    "enterprise_project" : "default"
  }, {
    "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "name" : "sms_test002",
    "use_public_ip" : true,
    "isdefault" : false,
    "start_target_server" : true,
    "region" : "region",
    "speed_limit" : 0,
    "exist_server" : true,
    "description" : "",
    "type" : "MIGRATE_BLOCK",
    "enterprise_project" : "default"
  }, {
    "id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001",
    "name" : "sms_test003",
    "use_public_ip" : true,
    "isdefault" : false,
    "start_target_server" : true,
    "region" : "region",
    "speed_limit" : 0,
    "exist_server" : true,
    "description" : "",
    "type" : "MIGRATE_BLOCK",
    "enterprise_project" : "default"
  } ]
}
```


Status code: 403

Authentication failed.


```
{
  "error_code" : "SMS.9004",
  "error_msg" : "The current account does not have the permission to execute policy. You do not have permission to perform action XXX on resource XXX.",
  "encoded_authorization_message" : "XXXXXX",
  "error_param" : [ "You do not have permission to perform action XXX on resource XXX." ],
  "details" : [ {
    "error_code" : "SMS.9004",
    "error_msg" : "You do not have permission to perform action XXX on resource XXX."
  } ]
}
```


SDK Sample Code

The SDK sample code is as follows.

Java

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
```

```
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;


public class ListMigprojectsSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        ListMigprojectsRequest request = new ListMigprojectsRequest();
        try {
            ListMigprojectsResponse response = client.listMigprojects(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

Python

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()
```



```
    try:
        request = ListMigprojectsRequest()
        response = client.list_migprojects(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

Go

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
         WithRegion(region.ValueOf("<YOUR REGION>")).
         WithCredential(auth).
         SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.ListMigprojectsRequest{}
    response, err := client.ListMigprojects(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

More

For SDK sample code of more programming languages, see the Sample Code tab in **API Explorer**. SDK sample code can be automatically generated.

Status Codes

Status Code	Description
200	The list of migration projects was obtained.
403	Authentication failed.

Error Codes

See **Error Codes**.

5.8.3 Querying Details About a Migration Project with a Specified ID

Function

This API is used to query details about a migration project with a specified ID.

Calling Method

For details, see **Calling APIs**.

Authorization Information

Each account has all the permissions required to call all APIs, but IAM users must be assigned the required permissions.

- If you are using role/policy-based authorization, see **Permissions Policies and Supported Actions** for details on the required permissions.
- If you are using identity policy-based authorization, the following identity policy-based permissions are required.

Action	Access Level	Resource Type (*: required)	Condition Key	Alias	Dependencies
sms:migproject:get	Read	migproject *	-	sms:server:queryServer	-

URI

GET /v3/migprojects/{mig_project_id}

Table 5-334 Path Parameters

Parameter	Mandatory	Type	Description
mig_project_id	Yes	String	The migration project ID. Minimum: **1** Maximum: **36**

Request Parameters

Table 5-335 Request header parameters

Parameter	Mandatory	Type	Description
X-Auth-Token	Yes	String	User token. The token can be obtained by calling the IAM API for obtaining a user token. The value of **X-Subject-Token** in the response header is the user token. Minimum: **1** Maximum: **16384**

Response Parameters

Status code: 200

Table 5-336 Response body parameters

Parameter	Type	Description
name	String	Migration project name. The value can contain only letters, digits, underscores (_), and hyphens (-). Minimum: **1** Maximum: **100**
description	String	The migration project description. Minimum: **0** Maximum: **255**
isdefault	Boolean	Indicates whether the migration project is the default project. Default: **false**

Parameter	Type	Description
region	String	The region name. Minimum: **0** Maximum: **255**
start_target_server	Boolean	Indicates whether the target server is started after the migration is complete. Default: **true**
speed_limit	Integer	The migration rate limit, in Mbit/s. Minimum: **0** Maximum: **1000**
use_public_ip	Boolean	Indicates whether a public IP address is used for migration. Default: **true**
exist_server	Boolean	Specifies whether an existing server is as the target server. Default: **true**
type	String	The migration project type. **MIGRATE_BLOCK** **MIGRATE_FILE** Enumeration values: ● **MIGRATE_BLOCK** ● **MIGRATE_FILE**
enterprise_project	String	The enterprise project name. Default: **default** Minimum: **0** Maximum: **255**
syncing	Boolean	Indicates whether continuous synchronization is performed after the first replication or synchronization is complete. Default: **false**
start_network_check	Boolean	Indicates whether network performance measurement is enabled.

Status code: 403

Table 5-337 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **255**
encoded_authorization_message	String	The encrypted authorization information. Minimum: **0** Maximum: **65535**
error_param	Array of strings	Error parameters. Minimum: **0** Maximum: **65535** Array Length: **1 - 20**
details	Array of **details** objects	The error details. Array Length: **1 - 20**

Table 5-338 details

Parameter	Type	Description
error_code	String	The SMS error code. Minimum: **0** Maximum: **65535**
error_msg	String	The SMS error message. Minimum: **0** Maximum: **65535**

Example Requests

This example queries the migration project whose ID is **137224b7-8d7c-4919-b33e-ed159778d7a7**.

```
GET  https://{endpoint}/v3/migprojects/137224b7-8d7c-4919-b33e-ed159778d7a7
```

Example Responses

Status code: 200

The details about the migration project with a specified ID were queried.


```
{
  "name" : "456",
  "isdefault" : true,
  "region" : "cn-north-7",
  "start_target_server" : false,
  "speed_limit" : 0,
  "type" : "MIGRATE_BLOCK",
  "use_public_ip" : true
}
```


Status code: 403

Authentication failed.


```
{
  "error_code" : "SMS.9004",
  "error_msg" : "The current account does not have the permission to execute policy. You do not have permission to perform action XXX on resource XXX.",
  "encoded_authorization_message" : "XXXXXX",
  "error_param" : [ "You do not have permission to perform action XXX on resource XXX." ],
  "details" : [ {
    "error_code" : "SMS.9004",
    "error_msg" : "You do not have permission to perform action XXX on resource XXX."
  } ]
}
```


SDK Sample Code

The SDK sample code is as follows.

Java

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;


public class ShowMigprojectSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        ShowMigprojectRequest request = new ShowMigprojectRequest();
        request.withMigProjectId("{mig_project_id}");
        try {
            ShowMigprojectResponse response = client.showMigproject(request);
            System.out.println(response.toString());
```

```
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

Python

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = ShowMigprojectRequest()
        request.mig_project_id = "{mig_project_id}"
        response = client.show_migproject(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

Go

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
```

```
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
         WithRegion(region.ValueOf("<YOUR REGION>")).
         WithCredential(auth).
         SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.ShowMigprojectRequest{}
    request.MigProjectId = "{mig_project_id}"
    response, err := client.ShowMigproject(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

More

For SDK sample code of more programming languages, see the Sample Code tab in **API Explorer**. SDK sample code can be automatically generated.

Status Codes

Status Code	Description
200	The details about the migration project with a specified ID were queried.
403	Authentication failed.

Error Codes

See **Error Codes**.

5.8.4 Deleting a Migration Project

Function

This API is used to delete a migration project with a specified ID.

Constraints

Only projects that have no servers can be deleted.

Calling Method

For details, see **Calling APIs**.

Authorization Information

Each account has all the permissions required to call all APIs, but IAM users must be assigned the required permissions.

- If you are using role/policy-based authorization, see **Permissions Policies and Supported Actions** for details on the required permissions.
- If you are using identity policy-based authorization, the following identity policy-based permissions are required.

Action	Access Level	Resource Type (*: required)	Condition Key	Alias	Dependencies
sms:migproject:delete	Write	migproject *	-	sms:server:migrationServer	-

URI

DELETE /v3/migprojects/{mig_project_id}

Table 5-339 Path Parameters

Parameter	Mandatory	Type	Description
mig_project_id	Yes	String	The ID of the migration project to be deleted. Minimum: **1** Maximum: **36**

Request Parameters

Table 5-340 Request header parameters

Parameter	Mandatory	Type	Description
X-Auth-Token	Yes	String	User token. The token can be obtained by calling the IAM API for obtaining a user token. The value of **X-Subject-Token** in the response header is the user token. Minimum: **1** Maximum: **16384**

Response Parameters

Status code: 200

Table 5-341 Response body parameters

Parameter	Type	Description
-	String	Deleting a migration project succeeded.

Status code: 403

Table 5-342 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **255**
encoded_authorization_message	String	The encrypted authorization information. Minimum: **0** Maximum: **65535**

Parameter	Type	Description
error_param	Array of strings	Error parameters. Minimum: **0** Maximum: **65535** Array Length: **1 - 20**
details	Array of **details** objects	The error details. Array Length: **1 - 20**

Table 5-343 details

Parameter	Type	Description
error_code	String	The SMS error code. Minimum: **0** Maximum: **65535**
error_msg	String	The SMS error message. Minimum: **0** Maximum: **65535**

Example Requests

This example deletes the project whose ID is **137224b7-8d7c-4919-b33e-ed159778d7a7**.

```
DELETE https://{endpoint}/v3/migprojects/137224b7-8d7c-4919-b33e-ed159778d7a7
X-Auth-Token: XXXX
```

Example Responses

Status code: 403

Authentication failed.

```
{
  "error_code" : "SMS.9004",
  "error_msg" : "The current account does not have the permission to execute policy. You do not have
permission to perform action XXX on resource XXX.",
  "encoded_authorization_message" : "XXXXXX",
  "error_param" : [ "You do not have permission to perform action XXX on resource XXX." ],
  "details" : [ {
    "error_code" : "SMS.9004",
    "error_msg" : "You do not have permission to perform action XXX on resource XXX."
  } ]
}
```

SDK Sample Code

The SDK sample code is as follows.

Java

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;


public class DeleteMigprojectSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great
security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or
environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running
this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        DeleteMigprojectRequest request = new DeleteMigprojectRequest();
        request.withMigProjectId("{mig_project_id}");
        try {
            DeleteMigprojectResponse response = client.deleteMigproject(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

Python

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security
risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment
variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this
example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
```

```
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = DeleteMigprojectRequest()
        request.mig_project_id = "{mig_project_id}"
        response = client.delete_migproject(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

Go

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
         WithRegion(region.ValueOf("<YOUR REGION>")).
         WithCredential(auth).
         SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.DeleteMigprojectRequest{}
    request.MigProjectId = "{mig_project_id}"
    response, err := client.DeleteMigproject(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
```

```
    } else {
        fmt.Println(err)
    }
}
```

More

For SDK sample code of more programming languages, see the Sample Code tab in **API Explorer**. SDK sample code can be automatically generated.

Status Codes

Status Code	Description
200	Deleting a migration project succeeded.
403	Authentication failed.

Error Codes

See **Error Codes**.

5.8.5 Modifying a Migration Project

Function

This API is used to modify a migration project.

Calling Method

For details, see **Calling APIs**.

Authorization Information

Each account has all the permissions required to call all APIs, but IAM users must be assigned the required permissions.

- If you are using role/policy-based authorization, see **Permissions Policies and Supported Actions** for details on the required permissions.
- If you are using identity policy-based authorization, the following identity policy-based permissions are required.

Action	Access Level	Resource Type (*: required)	Condition Key	Alias	Dependencies
sms:migproject:update	Write	migproject *	-	sms:server:migrationServer	-

URI

PUT /v3/migprojects/{mig_project_id}

Table 5-344 Path Parameters

Parameter	Mandatory	Type	Description
mig_project_id	Yes	String	The migration project ID. Minimum: **1** Maximum: **36**

Request Parameters

Table 5-345 Request header parameters

Parameter	Mandatory	Type	Description
X-Auth-Token	Yes	String	User token. The token can be obtained by calling the IAM API for obtaining a user token. The value of **X-Subject-Token** in the response header is the user token. Minimum: **1** Maximum: **16384**

Table 5-346 Request body parameters

Parameter	Mandatory	Type	Description
name	No	String	Migration project name. The value can contain only letters, digits, underscores (_), and hyphens (-). Minimum: **1** Maximum: **100**
description	No	String	The migration project description. Minimum: **0** Maximum: **255**

Parameter	Mandatory	Type	Description
isdefault	No	Boolean	Indicates whether the migration project is the default project. Default: **false**
region	No	String	The region name. Minimum: **0** Maximum: **255**
start_target_server	No	Boolean	Indicates whether the target server is started after the migration is complete. Default: **true**
speed_limit	No	Integer	The migration rate limit, in Mbit/s. Minimum: **0** Maximum: **1000**
use_public_ip	No	Boolean	Indicates whether a public IP address is used for migration. Default: **true**
exist_server	No	Boolean	Specifies whether an existing server is as the target server. Default: **true**
type	No	String	The migration project type. **MIGRATE_BLOCK** **MIGRATE_FILE** Enumeration values: • **MIGRATE_BLOCK** • **MIGRATE_FILE**
enterprise_project	No	String	The enterprise project name. Default: **default** Minimum: **0** Maximum: **255**
syncing	No	Boolean	Indicates whether continuous synchronization is performed after the first replication or synchronization is complete. Default: **false**
start_network_check	No	Boolean	Indicates whether network performance measurement is enabled.

Response Parameters

Status code: 200

Table 5-347 Response body parameters

Parameter	Type	Description
-	String	The default migration project was modified.

Status code: 403

Table 5-348 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **255**
encoded_authorization_message	String	The encrypted authorization information. Minimum: **0** Maximum: **65535**
error_param	Array of strings	Error parameters. Minimum: **0** Maximum: **65535** Array Length: **1 - 20**
details	Array of **details** objects	The error details. Array Length: **1 - 20**

Table 5-349 details

Parameter	Type	Description
error_code	String	The SMS error code. Minimum: **0** Maximum: **65535**
error_msg	String	The SMS error message. Minimum: **0** Maximum: **65535**

Example Requests

This example modifies the information about the migration project whose ID is **9879f7aa-3347-47fb-8f89-6070f9e0xxxx**. The new migration project name is **225**, the region information is **region**, the speed limit is **100 Mbit/s**, and the migration type is **MIGRATE_FILE**.

```
PUT  https://{endpoint}/v3/migprojects/9879f7aa-3347-47fb-8f89-6070f9e0xxxx

{
  "name" : 225,
  "region" : "region",
  "description" : "hello",
  "start_target_server" : true,
  "speed_limit" : 100,
  "use_public_ip" : true,
  "exist_server" : true,
  "type" : "MIGRATE_FILE",
  "syncing" : false
}
```

Example Responses

Status code: 200

The default migration project was modified.

```
{ }
```

Status code: 403

Authentication failed.

```
{
  "error_code" : "SMS.9004",
  "error_msg" : "The current account does not have the permission to execute policy. You do not have permission to perform action XXX on resource XXX.",
  "encoded_authorization_message" : "XXXXXX",
  "error_param" : [ "You do not have permission to perform action XXX on resource XXX." ],
  "details" : [ {
    "error_code" : "SMS.9004",
    "error_msg" : "You do not have permission to perform action XXX on resource XXX."
  } ]
}
```

SDK Sample Code

The SDK sample code is as follows.

Java

This example modifies the information about the migration project whose ID is **9879f7aa-3347-47fb-8f89-6070f9e0xxxx**. The new migration project name is **225**, the region information is **region**, the speed limit is **100 Mbit/s**, and the migration type is **MIGRATE_FILE**.

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;


public class UpdateMigprojectSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great 
security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or 
environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running 
this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        UpdateMigprojectRequest request = new UpdateMigprojectRequest();
        request.withMigProjectId("{mig_project_id}");
        MigProject body = new MigProject();
        body.withSyncing(false);
        body.withType(MigProject.TypeEnum.fromValue("MIGRATE_FILE"));
        body.withExistServer(true);
        body.withUsePublicIp(true);
        body.withSpeedLimit(100);
        body.withStartTargetServer(true);
        body.withRegion("region");
        body.withDescription("hello");
        body.withName("225");
        request.withBody(body);
        try {
            UpdateMigprojectResponse response = client.updateMigproject(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
```

```
            System.out.println(e.getErrorMsg());
        }
    }
}
```

Python

This example modifies the information about the migration project whose ID is **9879f7aa-3347-47fb-8f89-6070f9e0xxxx**. The new migration project name is **225**, the region information is **region**, the speed limit is **100 Mbit/s**, and the migration type is **MIGRATE_FILE**.

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = UpdateMigprojectRequest()
        request.mig_project_id = "{mig_project_id}"
        request.body = MigProject(
            syncing=False,
            type="MIGRATE_FILE",
            exist_server=True,
            use_public_ip=True,
            speed_limit=100,
            start_target_server=True,
            region="region",
            description="hello",
            name="225"
        )
        response = client.update_migproject(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

Go

This example modifies the information about the migration project whose ID is **9879f7aa-3347-47fb-8f89-6070f9e0xxxx**. The new migration project name is **225**, the region information is **region**, the speed limit is **100 Mbit/s**, and the migration type is **MIGRATE_FILE**.

```
package main
```

```
import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
         WithRegion(region.ValueOf("<YOUR REGION>")).
         WithCredential(auth).
         SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.UpdateMigprojectRequest{}
    request.MigProjectId = "{mig_project_id}"
    syncingMigProject:= false
    typeMigProject:= model.GetMigProjectTypeEnum().MIGRATE_FILE
    existServerMigProject:= true
    usePublicIpMigProject:= true
    speedLimitMigProject:= int32(100)
    startTargetServerMigProject:= true
    regionMigProject:= "region"
    descriptionMigProject:= "hello"
    nameMigProject:= "225"
    request.Body = &model.MigProject{
        Syncing: &syncingMigProject,
        Type: &typeMigProject,
        ExistServer: &existServerMigProject,
        UsePublicIp: &usePublicIpMigProject,
        SpeedLimit: &speedLimitMigProject,
        StartTargetServer: &startTargetServerMigProject,
        Region: &regionMigProject,
        Description: &descriptionMigProject,
        Name: &nameMigProject,
    }
    response, err := client.UpdateMigproject(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

More

For SDK sample code of more programming languages, see the Sample Code tab in **API Explorer**. SDK sample code can be automatically generated.

Status Codes

Status Code	Description
200	The default migration project was modified.
403	Authentication failed.

Error Codes

See **Error Codes**.

5.8.6 Changing the Default Migration Project

Function

This API is used to change the default migration project. If you change the default migration project, the source server will be registered under the changed project.

Calling Method

For details, see **Calling APIs**.

Authorization Information

Each account has all the permissions required to call all APIs, but IAM users must be assigned the required permissions.

- If you are using role/policy-based authorization, see **Permissions Policies and Supported Actions** for details on the required permissions.
- If you are using identity policy-based authorization, the following identity policy-based permissions are required.

Action	Access Level	Resource Type (*: required)	Condition Key	Alias	Dependencies
sms:migproject:update	Write	migproject *	-	sms:server:migrationServer	-

URI

PUT /v3/migprojects/{mig_project_id}/default

Table 5-350 Path Parameters

Parameter	Mandatory	Type	Description
mig_project_id	Yes	String	The migration project ID. Minimum: **1** Maximum: **36**

Request Parameters

Table 5-351 Request header parameters

Parameter	Mandatory	Type	Description
X-Auth-Token	Yes	String	User token. The token can be obtained by calling the IAM API for obtaining a user token. The value of **X-Subject-Token** in the response header is the user token. Minimum: **1** Maximum: **16384**

Response Parameters

Status code: 200

Table 5-352 Response body parameters

Parameter	Type	Description
-	String	The default migration project was changed.

Status code: 403

Table 5-353 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**

Parameter	Type	Description
error_msg	String	The error message. Minimum: **0** Maximum: **255**
encoded_authorization_message	String	The encrypted authorization information. Minimum: **0** Maximum: **65535**
error_param	Array of strings	Error parameters. Minimum: **0** Maximum: **65535** Array Length: **1 - 20**
details	Array of **details** objects	The error details. Array Length: **1 - 20**

Table 5-354 details

Parameter	Type	Description
error_code	String	The SMS error code. Minimum: **0** Maximum: **65535**
error_msg	String	The SMS error message. Minimum: **0** Maximum: **65535**

Example Requests

This example changes the default migration project to **137224b7-8d7c-4919-b33e-ed159778xxx**.

```
PUT https://{endpoint}/v3/migprojects/137224b7-8d7c-4919-b33e-ed159778xxxx/default
```

Example Responses

Status code: 200

The default migration project was changed.

```
{ }
```

Status code: 403

Authentication failed.


```
{
  "error_code" : "SMS.9004",
  "error_msg" : "The current account does not have the permission to execute policy. You do not have
permission to perform action XXX on resource XXX.",
  "encoded_authorization_message" : "XXXXXX",
  "error_param" : [ "You do not have permission to perform action XXX on resource XXX." ],
  "details" : [ {
    "error_code" : "SMS.9004",
    "error_msg" : "You do not have permission to perform action XXX on resource XXX."
  } ]
}
```


SDK Sample Code

The SDK sample code is as follows.

Java

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;


public class UpdateDefaultMigprojectSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great
security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or
environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running
this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        UpdateDefaultMigprojectRequest request = new UpdateDefaultMigprojectRequest();
        request.withMigProjectId("{mig_project_id}");
        try {
            UpdateDefaultMigprojectResponse response = client.updateDefaultMigproject(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

Python

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = UpdateDefaultMigprojectRequest()
        request.mig_project_id = "{mig_project_id}"
        response = client.update_default_migproject(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

Go

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
```

```
        WithRegion(region.ValueOf("<YOUR REGION>")).
        WithCredential(auth).
        SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.UpdateDefaultMigprojectRequest{}
    request.MigProjectId = "{mig_project_id}"
    response, err := client.UpdateDefaultMigproject(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

More

For SDK sample code of more programming languages, see the Sample Code tab in **API Explorer**. SDK sample code can be automatically generated.

Status Codes

Status Code	Description
200	The default migration project was changed.
403	Authentication failed.

Error Codes

See **Error Codes**.

5.9 Network Measurement Management

5.9.1 Updating Network Measurement Information

Function

The API is called by the Agent to report network measurement information.

Calling Method

For details, see **Calling APIs**.

Authorization Information

Each account has all the permissions required to call all APIs, but IAM users must be assigned the required permissions.

- If you are using role/policy-based authorization, see **Permissions Policies and Supported Actions** for details on the required permissions.
- If you are using identity policy-based authorization, the following identity policy-based permissions are required.

Action	Access Level	Resource Type (*: required)	Condition Key	Alias	Dependencies
sms:server:updateNetworkCheckInfo	Write	task *	-	sms:server:migrationServer	-
		-	g:EnterpriseProjectId		

URI

POST /v3/{task_id}/update-network-check-info

Table 5-355 Path Parameters

Parameter	Mandatory	Type	Description
task_id	Yes	String	Task ID. Minimum: **1** Maximum: **36**

Request Parameters

Table 5-356 Request header parameters

Parameter	Mandatory	Type	Description
X-Auth-Token	Yes	String	User token. The token can be obtained by calling the IAM API for obtaining a user token. The value of **X-Subject-Token** in the response header is the user token. Minimum: **1** Maximum: **16384**

Table 5-357 Request body parameters

Parameter	Mandatory	Type	Description
domain_connectivity	No	Boolean	The connectivity to domain names.
destination_connectivity	No	Boolean	The connectivity to the target server.
network_delay	Yes	Double	Network latency (ms). Minimum: **0** Maximum: **10000.0**
network_jitter	Yes	Double	Network jitter (ms). Minimum: **0** Maximum: **10000**
migration_speed	Yes	Double	Bandwidth (Mbit/s). Minimum: **0** Maximum: **10000**
loss_percentage	Yes	Double	Packet loss rate (%). Minimum: **0** Maximum: **100**
cpu_usage	Yes	Double	CPU usage (%). Minimum: **0** Maximum: **100**
mem_usage	Yes	Double	Memory usage (%). Minimum: **0** Maximum: **100**
evaluation_result	Yes	String	The network evaluation result. Minimum: **5** Maximum: **9**

Response Parameters

Status code: 200

Table 5-358 Response body parameters

Parameter	Type	Description
task_id	String	The task ID. Minimum: **1** Maximum: **255**

Status code: 400

Table 5-359 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **1024**

Status code: 403

Table 5-360 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **255**
encoded_authorization_message	String	The encrypted authorization information. Minimum: **0** Maximum: **65535**
error_param	Array of strings	Error parameters. Minimum: **0** Maximum: **65535** Array Length: **1 - 20**
details	Array of **details** objects	The error details. Array Length: **1 - 20**

Table 5-361 details

Parameter	Type	Description
error_code	String	The SMS error code. Minimum: **0** Maximum: **65535**
error_msg	String	The SMS error message. Minimum: **0** Maximum: **65535**

Status code: 404

Table 5-362 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **1024**

Status code: 500

Table 5-363 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **1024**

Example Requests

This example updates the network performance measurement results for the task **137224b7-8d7c-4919-b33e-ed159778xxxx**.

```
POST https://{endpoint}/v3/137224b7-8d7c-4919-b33e-ed159778xxxx/update-network-check-info
```

```
{
  "network_delay" : 20,
  "network_jitter" : 2,
  "migration_speed" : 100,
  "loss_percentage" : 0,
  "cpu_usage" : 20,
  "mem_usage" : 20,
  "evaluation_result" : "SUCCESS"
}
```

Example Responses

Status code: 200

The network measurement results were updated.

```
{
  "task_id" : "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxx0001"
}
```

Status code: 403

Authentication failed.

```
{
  "error_code" : "SMS.9004",
  "error_msg" : "The current account does not have the permission to execute policy. You do not have
permission to perform action XXX on resource XXX.",
  "encoded_authorization_message" : "XXXXXX",
  "error_param" : [ "You do not have permission to perform action XXX on resource XXX." ],
  "details" : [ {
    "error_code" : "SMS.9004",
    "error_msg" : "You do not have permission to perform action XXX on resource XXX."
  } ]
}
```

SDK Sample Code

The SDK sample code is as follows.

Java

This example updates the network performance measurement results for the task **137224b7-8d7c-4919-b33e-ed159778xxxx**.

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;


public class UpdateNetworkCheckInfoSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great
security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or
environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running
this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
```

```
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        UpdateNetworkCheckInfoRequest request = new UpdateNetworkCheckInfoRequest();
        request.withTaskId("{task_id}");
        NetworkCheckInfoRequestBody body = new NetworkCheckInfoRequestBody();
        body.withEvaluationResult("SUCCESS");
        body.withMemUsage((double)20);
        body.withCpuUsage((double)20);
        body.withLossPercentage((double)0);
        body.withMigrationSpeed((double)100);
        body.withNetworkJitter((double)2);
        body.withNetworkDelay((double)20);
        request.withBody(body);
        try {
            UpdateNetworkCheckInfoResponse response = client.updateNetworkCheckInfo(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

Python

This example updates the network performance measurement results for the task **137224b7-8d7c-4919-b33e-ed159778xxxx**.

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = UpdateNetworkCheckInfoRequest()
```

```
        request.task_id = "{task_id}"
        request.body = NetworkCheckInfoRequestBody(
            evaluation_result="SUCCESS",
            mem_usage=20,
            cpu_usage=20,
            loss_percentage=0,
            migration_speed=100,
            network_jitter=2,
            network_delay=20
        )
        response = client.update_network_check_info(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

Go

This example updates the network performance measurement results for the task **137224b7-8d7c-4919-b33e-ed159778xxxx**.

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
         WithRegion(region.ValueOf("<YOUR REGION>")).
         WithCredential(auth).
         SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.UpdateNetworkCheckInfoRequest{}
    request.TaskId = "{task_id}"
    request.Body = &model.NetworkCheckInfoRequestBody{
        EvaluationResult: "SUCCESS",
```

```
            MemUsage: float64(20),
            CpuUsage: float64(20),
            LossPercentage: float64(0),
            MigrationSpeed: float64(100),
            NetworkJitter: float64(2),
            NetworkDelay: float64(20),
        }
        response, err := client.UpdateNetworkCheckInfo(request)
        if err == nil {
            fmt.Printf("%+v\n", response)
        } else {
            fmt.Println(err)
        }
}
```

More

For SDK sample code of more programming languages, see the Sample Code tab in **API Explorer**. SDK sample code can be automatically generated.

Status Codes

Status Code	Description
200	The network measurement results were updated.
400	The request parameters were missing.
403	Authentication failed.
404	The task was not found.
500	Updating the network measurement results failed.

Error Codes

See **Error Codes**.

5.10 Advanced Migration Options Management

5.10.1 Configuring Advanced Migration Options

Function

This API is used to configure advanced migration options for a task, for example, specifying the files or paths to be synchronized.

Calling Method

For details, see **Calling APIs**.

Authorization Information

Each account has all the permissions required to call all APIs, but IAM users must be assigned the required permissions.

- If you are using role/policy-based authorization, see **Permissions Policies and Supported Actions** for details on the required permissions.
- If you are using identity policy-based authorization, the following identity policy-based permissions are required.

Action	Access Level	Resource Type (*: required)	Condition Key	Alias	Dependencies
sms:server:updateTaskConfig	Write	task *	-	sms:server:migrationServer	-
		-	g:EnterpriseProjectId		

URI

POST /v3/tasks/{task_id}/configuration-setting

Table 5-364 Path Parameters

Parameter	Mandatory	Type	Description
task_id	Yes	String	Task ID. Minimum: **1** Maximum: **36**

Request Parameters

Table 5-365 Request header parameters

Parameter	Mandatory	Type	Description
X-Auth-Token	Yes	String	User token. The token can be obtained by calling the IAM API for obtaining a user token. The value of **X-Subject-Token** in the response header is the user token. Minimum: **1** Maximum: **16384**

Table 5-366 Request body parameters

Parameter	Mandatory	Type	Description
configurations	Yes	Array of **ConfigBody** objects	The list of advanced migration options. Array Length: **0 - 50**

Table 5-367 ConfigBody

Parameter	Mandatory	Type	Description
config_key	Yes	String	The advanced migration options to be configured. **MIGRATE_EXCLUDE_DIR**: indicates the directories to be excluded from the migration. **SYNC_EXCLUDE_DIR**: indicates the directories to be excluded from the synchronization. **ONLY_SYNC_DIR**: indicates the directories to be included in the synchronization. **CONSISTENCY_DIR**: indicates the directories for consistency verification. **CONSISTENCY_DIR_ILLEGAL**: indicates illegal directories found after consistency verification. **LINUX_BLOCK_COMPRESS_THREAD_NUM**: indicates the number of compression threads for Linux block-level migration. **MIGRATE_DST_IP**: indicates the target IP address used for the migration. **LINUX_BLOCK_CACHE_SIZE**: indicates the cache size for Linux block-level migration. **LINUX_CPU_LIMIT**: indicates the CPU limit for Linux migration. **LINUX_MEM_LIMIT**: indicates the memory limit for Linux migration. **LINUX_IO_LIMIT**: indicates the I/O limit for Linux migration. **NUM_PROCESS_MIGRATE**: indicates the number of migration processes. **NUM_PROCESS_SYNC**: indicates the number of synchronization processes.

Parameter	Mandatory	Type	Description
			CONSISTENCY_RECHECK: indicates the re-verification of inconsistencies found. **CONSISTENCY_MODE**: indicates the consistency verification mode. **DYNAMIC_PORT**: indicates a dynamic port. **INSTALL_PWD_AGENT**: indicates whether to install the password reset plug-in. The value **true** indicates that the plug-in will be installed. The default value is **true**. Enumeration values: ● **MIGRATE_EXCLUDE_DIR** ● **SYNC_EXCLUDE_DIR** ● **ONLY_SYNC_DIR** ● **CONSISTENCY_DIR** ● **CONSISTENCY_DIR_ILLEGAL** ● **LINUX_BLOCK_COMPRESS_THREAD_NUM** ● **MIGRATE_DST_IP** ● **LINUX_BLOCK_CACHE_SIZE** ● **LINUX_CPU_LIMIT** ● **LINUX_MEM_LIMIT** ● **LINUX_IO_LIMIT** ● **NUM_PROCESS_MIGRATE** ● **NUM_PROCESS_SYNC** ● **CONSISTENCY_RECHECK** ● **CONSISTENCY_MODE** ● **DYNAMIC_PORT**
config_value	Yes	String	The value specified for the advanced migration option. It is stored in the database and parsed on the Agent. Minimum: **0** Maximum: **1024**

Parameter	Mandatory	Type	Description
config_status	No	String	The reserved field that describes the configuration status. Minimum: **0** Maximum: **255**

Response Parameters

Status code: 200

Table 5-368 Response body parameters

Parameter	Type	Description
-	String	Advanced migration options were configured successfully.

Status code: 403

Table 5-369 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **255**
encoded_authorization_message	String	The encrypted authorization information. Minimum: **0** Maximum: **65535**
error_param	Array of strings	Error parameters. Minimum: **0** Maximum: **65535** Array Length: **1 - 20**
details	Array of **details** objects	The error details. Array Length: **1 - 20**

Table 5-370 details

Parameter	Type	Description
error_code	String	The SMS error code. Minimum: **0** Maximum: **65535**
error_msg	String	The SMS error message. Minimum: **0** Maximum: **65535**

Status code: 404

Table 5-371 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **1024**

Status code: 500

Table 5-372 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **1024**

Example Requests

This example sets advanced option **LINUX_CPU_LIMIT** to **50** for a migration task.

```
POST https://{endpoint}/v3/tasks/{task_id}/configuration-setting
```



```
{
  "configurations" : [ {
    "config_key" : "LINUX_CPU_LIMIT",
    "config_value" : "50"
  } ]
}
```

Example Responses

Status code: 200

Advanced migration options were configured successfully.

```
{ }
```

Status code: 403

Authentication failed.

```
{
  "error_code" : "SMS.9004",
  "error_msg" : "The current account does not have the permission to execute policy. You do not have permission to perform action XXX on resource XXX.",
  "encoded_authorization_message" : "XXXXXX",
  "error_param" : [ "You do not have permission to perform action XXX on resource XXX." ],
  "details" : [ {
    "error_code" : "SMS.9004",
    "error_msg" : "You do not have permission to perform action XXX on resource XXX."
  } ]
}
```

SDK Sample Code

The SDK sample code is as follows.

Java

This example sets advanced option **LINUX_CPU_LIMIT** to **50** for a migration task.

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;

import java.util.List;
import java.util.ArrayList;

public class UploadSpecialConfigurationSettingSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
```

```
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        UploadSpecialConfigurationSettingRequest request = new
UploadSpecialConfigurationSettingRequest();
        request.withTaskId("{task_id}");
        ConfigurationRequestBody body = new ConfigurationRequestBody();
        List<ConfigBody> listbodyConfigurations = new ArrayList<>();
        listbodyConfigurations.add(
            new ConfigBody()
                .withConfigKey(ConfigBody.ConfigKeyEnum.fromValue("LINUX_CPU_LIMIT"))
                .withConfigValue("50")
        );
        body.withConfigurations(listbodyConfigurations);
        request.withBody(body);
        try {
            UploadSpecialConfigurationSettingResponse response =
client.uploadSpecialConfigurationSetting(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

Python

This example sets advanced option **LINUX_CPU_LIMIT** to **50** for a migration task.

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security
risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment
variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this
example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = UploadSpecialConfigurationSettingRequest()
        request.task_id = "{task_id}"
        listConfigurationsbody = [
            ConfigBody(
```

```
                config_key="LINUX_CPU_LIMIT",
                config_value="50"
            )
        ]
        request.body = ConfigurationRequestBody(
            configurations=listConfigurationsbody
        )
        response = client.upload_special_configuration_setting(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

Go

This example sets advanced option **LINUX_CPU_LIMIT** to **50** for a migration task.

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
         WithRegion(region.ValueOf("<YOUR REGION>")).
         WithCredential(auth).
         SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.UploadSpecialConfigurationSettingRequest{}
    request.TaskId = "{task_id}"
    var listConfigurationsbody = []model.ConfigBody{
        {
            ConfigKey: model.GetConfigBodyConfigKeyEnum().LINUX_CPU_LIMIT,
            ConfigValue: "50",
        },
    }
```

```
    request.Body = &model.ConfigurationRequestBody{
        Configurations: listConfigurationsbody,
    }
    response, err := client.UploadSpecialConfigurationSetting(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

More

For SDK sample code of more programming languages, see the Sample Code tab in **API Explorer**. SDK sample code can be automatically generated.

Status Codes

Status Code	Description
200	Advanced migration options were configured successfully.
403	Authentication failed.
404	The task was not found.
500	Advanced migration options failed to be configured.

Error Codes

See **Error Codes**.

5.10.2 Querying the Settings of Advanced Migration Options of a Task

Function

This API is used to query the settings of advanced migration options of a task.

Calling Method

For details, see **Calling APIs**.

Authorization Information

Each account has all the permissions required to call all APIs, but IAM users must be assigned the required permissions.

- If you are using role/policy-based authorization, see **Permissions Policies and Supported Actions** for details on the required permissions.
- If you are using identity policy-based authorization, the following identity policy-based permissions are required.

Action	Access Level	Resource Type (*: required)	Condition Key	Alias	Dependencies
sms:server:getTaskConfig	Read	task *	-	sms:server:queryServer	-
		-	g:EnterpriseProjectId		

URI

GET /v3/tasks/{task_id}/configuration-setting

Table 5-373 Path Parameters

Parameter	Mandatory	Type	Description
task_id	Yes	String	Task ID. Minimum: **1** Maximum: **36**

Table 5-374 Query Parameters

Parameter	Mandatory	Type	Description
config_key	No	String	The advanced migration options to be queried.

Request Parameters

Table 5-375 Request header parameters

Parameter	Mandatory	Type	Description
X-Auth-Token	Yes	String	User token. The token can be obtained by calling the IAM API for obtaining a user token. The value of **X-Subject-Token** in the response header is the user token. Minimum: **1** Maximum: **16384**

Response Parameters

Status code: 200

Table 5-376 Response body parameters

Parameter	Type	Description
task_id	String	The task ID. Minimum: **0** Maximum: **100**
migrate_type	String	The migration method. WINDOWS_MIGRATE_BLOCK LINUX_MIGRATE_FILE LINUX_MIGRATE_BLOCK Enumeration values: ● **WINDOWS_MIGRATE_BLOCK** ● **LINUX_MIGRATE_FILE** ● **LINUX_MIGRATE_BLOCK**
configurations	Array of **ConfigBody** objects	The settings of advanced migration options. Array Length: **0 - 1000**

Table 5-377 ConfigBody

Parameter	Type	Description
config_key	String	The advanced migration options to be configured. **MIGRATE_EXCLUDE_DIR**: indicates the directories to be excluded from the migration. **SYNC_EXCLUDE_DIR**: indicates the directories to be excluded from the synchronization. **ONLY_SYNC_DIR**: indicates the directories to be included in the synchronization. **CONSISTENCY_DIR**: indicates the directories for consistency verification. **CONSISTENCY_DIR_ILLEGAL**: indicates illegal directories found after consistency verification. **LINUX_BLOCK_COMPRESS_THREAD_NUM**: indicates the number of compression threads for Linux block-level migration. **MIGRATE_DST_IP**: indicates the target IP address used for the migration. **LINUX_BLOCK_CACHE_SIZE**: indicates the cache size for Linux block-level migration. **LINUX_CPU_LIMIT**: indicates the CPU limit for Linux migration. **LINUX_MEM_LIMIT**: indicates the memory limit for Linux migration. **LINUX_IO_LIMIT**: indicates the I/O limit for Linux migration. **NUM_PROCESS_MIGRATE**: indicates the number of migration processes. **NUM_PROCESS_SYNC**: indicates the number of synchronization processes. **CONSISTENCY_RECHECK**: indicates the re-verification of inconsistencies found. **CONSISTENCY_MODE**: indicates the consistency verification mode. **DYNAMIC_PORT**: indicates a dynamic port. **INSTALL_PWD_AGENT**: indicates whether to install the password reset

Parameter	Type	Description
		plug-in. The value **true** indicates that the plug-in will be installed. The default value is **true**. Enumeration values: • **MIGRATE_EXCLUDE_DIR** • **SYNC_EXCLUDE_DIR** • **ONLY_SYNC_DIR** • **CONSISTENCY_DIR** • **CONSISTENCY_DIR_ILLEGAL** • **LINUX_BLOCK_COMPRESS_THREAD_NUM** • **MIGRATE_DST_IP** • **LINUX_BLOCK_CACHE_SIZE** • **LINUX_CPU_LIMIT** • **LINUX_MEM_LIMIT** • **LINUX_IO_LIMIT** • **NUM_PROCESS_MIGRATE** • **NUM_PROCESS_SYNC** • **CONSISTENCY_RECHECK** • **CONSISTENCY_MODE** • **DYNAMIC_PORT**
config_value	String	The value specified for the advanced migration option. It is stored in the database and parsed on the Agent. Minimum: **0** Maximum: **1024**
config_status	String	The reserved field that describes the configuration status. Minimum: **0** Maximum: **255**

Status code: 400

Table 5-378 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**

Parameter	Type	Description
error_msg	String	The error message. Minimum: **0** Maximum: **1024**

Status code: 403

Table 5-379 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **255**
encoded_authorization_message	String	The encrypted authorization information. Minimum: **0** Maximum: **65535**
error_param	Array of strings	Error parameters. Minimum: **0** Maximum: **65535** Array Length: **1 - 20**
details	Array of **details** objects	The error details. Array Length: **1 - 20**

Table 5-380 details

Parameter	Type	Description
error_code	String	The SMS error code. Minimum: **0** Maximum: **65535**
error_msg	String	The SMS error message. Minimum: **0** Maximum: **65535**

Status code: 404

Table 5-381 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **1024**

Status code: 500

Table 5-382 Response body parameters

Parameter	Type	Description
error_code	String	The error code. Minimum: **0** Maximum: **255**
error_msg	String	The error message. Minimum: **0** Maximum: **1024**

Example Requests

This example queries the setting of **MIGRATE_EXCLUDE_PATH** of the task **0867ef5f3xxxxxxxxxxxxxxx**.

```
GET https://{endpoint}/v3/tasks/0867ef5f3xxxxxxxxxxxxxxx/configuration-setting?
config_key=MIGRATE_EXCLUDE_PATH
```

Example Responses

Status code: 200

The settings of advanced migration options of a task were queried.

```
{
  "task_id" : "0867ef5f3xxxxxxxxxxxxxxx",
  "migrate_type" : "LINUX_FILE_MIGRATE",
  "configurations" : [ {
    "config_key" : "MIGRATE_EXCLUDE_PATH",
    "config_value" : "/test",
    "config_status" : ""
  } ]
}
```


Status code: 403

Authentication failed.


```
{
  "error_code" : "SMS.9004",
  "error_msg" : "The current account does not have the permission to execute policy. You do not have permission to perform action XXX on resource XXX.",
  "encoded_authorization_message" : "XXXXXX",
  "error_param" : [ "You do not have permission to perform action XXX on resource XXX." ],
  "details" : [ {
    "error_code" : "SMS.9004",
    "error_msg" : "You do not have permission to perform action XXX on resource XXX."
  } ]
}
```


SDK Sample Code

The SDK sample code is as follows.

Java

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;


public class ShowConfigSettingSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        ShowConfigSettingRequest request = new ShowConfigSettingRequest();
        request.withTaskId("{task_id}");
        try {
            ShowConfigSettingResponse response = client.showConfigSetting(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
```

```
        }
    }
}
```

Python

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = ShowConfigSettingRequest()
        request.task_id = "{task_id}"
        response = client.show_config_setting(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

Go

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
```

```
        return
    }

    hcClient, err := sms.SmsClientBuilder().
        WithRegion(region.ValueOf("<YOUR REGION>")).
        WithCredential(auth).
        SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.ShowConfigSettingRequest{}
    request.TaskId = "{task_id}"
    response, err := client.ShowConfigSetting(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

More

For SDK sample code of more programming languages, see the Sample Code tab in **API Explorer**. SDK sample code can be automatically generated.

Status Codes

Status Code	Description
200	The settings of advanced migration options of a task were queried.
400	Bad Request
403	Authentication failed.
404	The task was not found.
500	Querying the settings of advanced migration options failed.

Error Codes

See **Error Codes**.

5.11 Privacy Agreement Management

5.11.1 Signing the Privacy Agreement

Function

This API is used to agree to the privacy agreement.

Calling Method

For details, see **Calling APIs**.

Authorization Information

Each account has all the permissions required to call all APIs, but IAM users must be assigned the required permissions.

- If you are using role/policy-based authorization, see **Permissions Policies and Supported Actions** for details on the required permissions.
- If you are using identity policy-based authorization, no identity policy-based permission required for calling this API.

URI

POST /v3/privacy-agreements

Request Parameters

Table 5-383 Request header parameters

Parameter	Mandatory	Type	Description
X-Auth-Token	Yes	String	User token. The token can be obtained by calling the IAM API for obtaining a user token. The value of **X-Subject-Token** in the response header is the user token.

Response Parameters

Status code: 200

Table 5-384 Response body parameters

Parameter	Type	Description
-	String	Request succeeded.

Example Requests

This example signs the privacy agreement.

```
POST https://{endpoint}/v3/privacy-agreements
X-Auth-Token: XXXX
```

Example Responses

Status code: 200

Request succeeded.

```
""
```

SDK Sample Code

The SDK sample code is as follows.

Java

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;


public class CreatePrivacyAgreementsSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great 
security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or 
environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running 
this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        CreatePrivacyAgreementsRequest request = new CreatePrivacyAgreementsRequest();
        try {
            CreatePrivacyAgreementsResponse response = client.createPrivacyAgreements(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

Python

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
```



```
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()

    try:
        request = CreatePrivacyAgreementsRequest()
        response = client.create_privacy_agreements(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

Go

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
         WithRegion(region.ValueOf("<YOUR REGION>")).
         WithCredential(auth).
         SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }
```

```
    client := sms.NewSmsClient(hcClient)

    request := &model.CreatePrivacyAgreementsRequest{}
    response, err := client.CreatePrivacyAgreements(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

More

For SDK sample code of more programming languages, see the Sample Code tab in **API Explorer**. SDK sample code can be automatically generated.

Status Codes

Status Code	Description
200	Request succeeded.

Error Codes

See **Error Codes**.

5.11.2 Checking Whether a User Has Agreed to the Privacy Agreement

Function

This API is used to query whether a user has signed the privacy agreement.

Calling Method

For details, see **Calling APIs**.

Authorization Information

Each account has all the permissions required to call all APIs, but IAM users must be assigned the required permissions.

- If you are using role/policy-based authorization, see **Permissions Policies and Supported Actions** for details on the required permissions.
- If you are using identity policy-based authorization, no identity policy-based permission required for calling this API.

URI

GET /v3/privacy-agreements

Request Parameters

Table 5-385 Request header parameters

Parameter	Mandatory	Type	Description
X-Auth-Token	Yes	String	User token. The token can be obtained by calling the IAM API for obtaining a user token. The value of **X-Subject-Token** in the response header is the user token.

Response Parameters

Status code: 200

Table 5-386 Response body parameters

Parameter	Type	Description
flag	Boolean	The result of the request to query whether the user has signed the privacy agreement.

Example Requests

This example queries whether a user has signed the privacy agreement.

```
GET https://{endpoint}/v3/privacy-agreements
```

Example Responses

Status code: 200

Querying whether a user has agreed to the privacy agreement succeeded.

```
{
  "flag" : true
}
```

SDK Sample Code

The SDK sample code is as follows.

Java

```
package com.huaweicloud.sdk.test;

import com.huaweicloud.sdk.core.auth.ICredential;
import com.huaweicloud.sdk.core.auth.GlobalCredentials;
import com.huaweicloud.sdk.core.exception.ConnectionException;
import com.huaweicloud.sdk.core.exception.RequestTimeoutException;
```

```
import com.huaweicloud.sdk.core.exception.ServiceResponseException;
import com.huaweicloud.sdk.sms.v3.region.SmsRegion;
import com.huaweicloud.sdk.sms.v3.*;
import com.huaweicloud.sdk.sms.v3.model.*;


public class ShowPrivacyAgreementsSolution {

    public static void main(String[] args) {
        // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great
security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or
environment variables and decrypted during use to ensure security.
        // In this example, AK and SK are stored in environment variables for authentication. Before running
this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
        String ak = System.getenv("CLOUD_SDK_AK");
        String sk = System.getenv("CLOUD_SDK_SK");

        ICredential auth = new GlobalCredentials()
                .withAk(ak)
                .withSk(sk);

        SmsClient client = SmsClient.newBuilder()
                .withCredential(auth)
                .withRegion(SmsRegion.valueOf("<YOUR REGION>"))
                .build();
        ShowPrivacyAgreementsRequest request = new ShowPrivacyAgreementsRequest();
        try {
            ShowPrivacyAgreementsResponse response = client.showPrivacyAgreements(request);
            System.out.println(response.toString());
        } catch (ConnectionException e) {
            e.printStackTrace();
        } catch (RequestTimeoutException e) {
            e.printStackTrace();
        } catch (ServiceResponseException e) {
            e.printStackTrace();
            System.out.println(e.getHttpStatusCode());
            System.out.println(e.getRequestId());
            System.out.println(e.getErrorCode());
            System.out.println(e.getErrorMsg());
        }
    }
}
```

Python

```
# coding: utf-8

import os
from huaweicloudsdkcore.auth.credentials import GlobalCredentials
from huaweicloudsdksms.v3.region.sms_region import SmsRegion
from huaweicloudsdkcore.exceptions import exceptions
from huaweicloudsdksms.v3 import *

if __name__ == "__main__":
    # The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security
risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment
variables and decrypted during use to ensure security.
    # In this example, AK and SK are stored in environment variables for authentication. Before running this
example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak = os.environ["CLOUD_SDK_AK"]
    sk = os.environ["CLOUD_SDK_SK"]

    credentials = GlobalCredentials(ak, sk)

    client = SmsClient.new_builder() \
        .with_credentials(credentials) \
        .with_region(SmsRegion.value_of("<YOUR REGION>")) \
        .build()
```

```
    try:
        request = ShowPrivacyAgreementsRequest()
        response = client.show_privacy_agreements(request)
        print(response)
    except exceptions.ClientRequestException as e:
        print(e.status_code)
        print(e.request_id)
        print(e.error_code)
        print(e.error_msg)
```

Go

```
package main

import (
    "fmt"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/core/auth/global"
    sms "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3"
    "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/model"
    region "github.com/huaweicloud/huaweicloud-sdk-go-v3/services/sms/v3/region"
)

func main() {
    // The AK and SK used for authentication are hard-coded or stored in plaintext, which has great security risks. It is recommended that the AK and SK be stored in ciphertext in configuration files or environment variables and decrypted during use to ensure security.
    // In this example, AK and SK are stored in environment variables for authentication. Before running this example, set environment variables CLOUD_SDK_AK and CLOUD_SDK_SK in the local environment
    ak := os.Getenv("CLOUD_SDK_AK")
    sk := os.Getenv("CLOUD_SDK_SK")

    auth, err := global.NewCredentialsBuilder().
        WithAk(ak).
        WithSk(sk).
        SafeBuild()

    if err != nil {
        fmt.Println(err)
        return
    }

    hcClient, err := sms.SmsClientBuilder().
        WithRegion(region.ValueOf("<YOUR REGION>")).
        WithCredential(auth).
        SafeBuild()


    if err != nil {
        fmt.Println(err)
        return
    }

    client := sms.NewSmsClient(hcClient)

    request := &model.ShowPrivacyAgreementsRequest{}
    response, err := client.ShowPrivacyAgreements(request)
    if err == nil {
        fmt.Printf("%+v\n", response)
    } else {
        fmt.Println(err)
    }
}
```

More

For SDK sample code of more programming languages, see the Sample Code tab in **API Explorer**. SDK sample code can be automatically generated.

Status Codes

Status Code	Description
200	Querying whether a user has agreed to the privacy agreement succeeded.

Error Codes

See **Error Codes**.

6 Permissions and Supported Actions

6.1 Introduction

This topic describes fine-grained permissions management for your SMS resources. If your account does not need individual IAM users, you can skip this section.

With IAM, you can control access to specific Huawei Cloud resources from principals (IAM users, user groups, agencies, or trust agencies). IAM supports role/policy-based authorization and identity policy-based authorization.

The following table describes the differences between the two authorization models.

Table 6-1 Differences between role/policy-based and identity policy-based authorization

Authorization Model	Core Relationship	Permissions	Authorization Method	Scenario
Role/Policy	User-permission-authorization scope	• System-defined roles • System-defined policies • Custom policies	Assigning roles or policies to principals	To authorize a user, you need to add it to a user group first and then specify the scope of authorization. It is hard to provide fine-grained permissions control using authorization by user groups and a limited number of condition keys. This method is suitable for small- and medium-sized enterprises.

Authorization Model	Core Relationship	Permissions	Authorization Method	Scenario
Identity policy	User-policy	• System-defined identity policies • Custom identity policies	• Assigning identity policies to principals • Attaching identity policies to principals	You can authorize a user by attaching an identity policy to it. User-specific authorization and a variety of key conditions allow for more fine-grained permissions control. However, this model can be hard to set up. It requires a certain amount of expertise and is suitable for medium- and large-sized enterprises.

Assume that you want to grant IAM users the permissions to create ECSs in CN North-Beijing4 and OBS buckets in CN South-Guangzhou. With role/policy-based authorization, the administrator needs to create two custom policies and attach both to the IAM users. With identity policy-based authorization, the administrator only needs to create one custom policy, configure the condition key **g:RequestedRegion** for the policy, and then attach the policy to the principals or grant the principals the access permissions to the specified regions. Identity policy-based authorization is more flexible than role/policy-based authorization.

Policies and actions in the two authorization models are not interoperable. You are advised to use the identity policy-based authorization model.

If you use IAM users in your account to call an API, the IAM users must be granted the required permissions. The required permissions are determined by the actions supported by the API. Only users with the permissions allowing for those actions can call the API successfully.

Assume that an IAM user wants to call an API to query SMS resources. With policy-based authorization, the IAM user must be granted the permissions allowing for action **sms:server:queryServer**. With identity policy-based authorization, the IAM user must be granted the permissions allowing for action **sms:server:list**.

6.2 Actions Supported by Policy-based Authorization

This section describes the actions supported by SMS in policy-based authorization.

Supported Actions

SMS provides system-defined policies that can be directly used in IAM. You can also create custom policies to supplement system-defined policies for more refined access control. Operations supported by policies are specific to APIs. The following are common concepts related to policies:

- Permissions: statements in a policy that allow or deny certain operations.
- APIs: REST APIs that can be called by a user who has been granted specific permissions.
- Actions: specific operations that are allowed or denied.
- Related actions: actions on which a specific action depends to take effect. When assigning permissions for the action to a user, you also need to assign permissions for the related actions.
- IAM projects/Enterprise projects: the authorization scope of a custom policy. A custom policy can be applied to IAM projects or enterprise projects or both. Policies that contain actions for both IAM and enterprise projects can be used and take effect for both IAM and Enterprise Management. Policies that only contain actions for IAM projects can be used and only take effect for IAM. Administrators can check whether an action supports IAM projects or enterprise projects in the action list. "√" indicates that the action supports the project and "×" indicates that the action does not support the project. For details about the differences between IAM and enterprise management, see **Differences Between IAM and Enterprise Management**

SMS supports the following actions in custom policies:

- **Template management** actions
- **Source server management** actions
- **Task management** actions
- **Command management** actions
- **Key management** actions
- **Project management** actions
- **API version management** actions
- **Agent running** actions
- **Network measurement management** actions
- **Advanced migration options management** actions
- **Privacy agreement management** actions

Template Management

Table 6-2 Template management

Permission	API	Action	IAM Project	Enterprise Project
Listing templates	GET /v3/vm/templates	sms:server:queryServer	√	×

Permission	API	Action	IAM Project	Enterprise Project
Creating a template	POST /v3/vm/templates	sms:server:migrationServer	√	×
Batch deleting templates	POST /v3/vm/templates/delete	sms:server:migrationServer	√	×
Querying details about a template	GET /v3/vm/templates/{id}	sms:server:queryServer	√	×
Modifying a template	PUT /v3/vm/templates/{id}	sms:server:migrationServer	√	×
Querying the target server password in a template	GET /v3/vm/templates/{id}/target-password	sms:server:queryServer	√	×
Deleting a template	DELETE /v3/vm/templates/{id}	sms:server:migrationServer	√	×

Source Server Management

Table 6-3 Source server management

Permission	API	Action	IAM Project	Enterprise Project
Listing failed source servers	GET /v3/errors	sms:server:queryServer	√	×
Listing source servers	GET /v3/sources	sms:server:queryServer	√	√
Registering a source server with SMS	POST /v3/sources	sms:server:registerServer	√	√
Batch deleting source server records	POST /v3/sources/delete	sms:server:migrationServer	√	√

Permission	API	Action	IAM Project	Enterprise Project
Querying details about a source server	GET /v3/sources/{source_id}	sms:server:queryServer	√	√
Modifying a source server name	PUT /v3/sources/{source_id}	sms:server:migrationServer	√	√
Deleting a source server record	DELETE /v3/sources/{source_id}	sms:server:migrationServer	√	√
Updating disk information	PUT /v3/sources/{source_id}/diskinfo	sms:server:registerServer	√	√
Obtaining a source server summary	GET /v3/sources/overview	sms:server:queryServer	√	×
Updating the migration task status of a source server	PUT /v3/sources/{source_id}/changestate	sms:server:migrationServer	√	√

Task Management

Table 6-4 Task management

Permission	API	Action	IAM Project	Enterprise Project
Listing migration tasks	GET /v3/tasks	sms:server:queryServer	√	√
Creating a migration task	POST /v3/tasks	sms:server:migrationServer	√	√
Batch deleting migration tasks	POST /v3/tasks/delete	sms:server:migrationServer	√	√
Querying details about a migration task	GET /v3/tasks/{task_id}	sms:server:queryServer	√	√

Permission	API	Action	IAM Project	Enterprise Project
Updating a migration task	PUT /v3/tasks/{task_id}	sms:server:migrationServer	√	√
Deleting a migration task	DELETE /v3/tasks/{task_id}	sms:server:migrationServer	√	√
Managing migration tasks	POST /v3/tasks/{task_id}/action	sms:server:migrationServer	√	√
Reporting migration progress and rate	PUT /v3/tasks/{task_id}/progress	sms:server:migrationServer	√	√
Unlocking a target server	POST /v3/tasks/{task_id}/unlock	sms:server:migrationServer	√	√
Uploading migration task logs	POST /v3/tasks/{task_id}/log	sms:server:migrationServer	√	√
Querying a certificate passphrase	GET /v3/tasks/{task_id}/passphrase	sms:server:queryServer	√	√
Checking NICs and security groups	GET /v3/tasks/{t_project_id}/networkacl/{t_network_id}/check	sms:server:queryServer	√	×
Querying the migration rate limiting rules of a migration task	GET /v3/tasks/{task_id}/speed-limit	sms:server:queryServer	√	√
Setting migration rate limit rules for a migration task	POST /v3/tasks/{task_id}/speed-limit	sms:server:migrationServer	√	√

Command Management

Table 6-5 Command management

Permission	API	Action	IAM Project	Enterprise Project
Obtaining commands from SMS	GET /v3/sources/{server_id}/command	sms:server:queryServer	√	√
Reporting command execution results to SMS	POST /v3/sources/{server_id}/command_result	sms:server:migrationServer	√	√

Key Management

Table 6-6 Key management

Permission	API	Action	IAM Project	Enterprise Project
Obtaining an SSL certificate and private key	GET /v3/tasks/{task_id}/certkey	sms:server:queryServer	√	√
Calculating an SHA256 hash	GET /v3/sha256/{key}	sms:server:queryServer	√	×

Migration Project Management

Table 6-7 Migration project management

Permission	API	Action	IAM Project	Enterprise Project
Listing migration projects	GET /v3/migprojects	sms:server:queryServer	√	×

Permission	API	Action	IAM Project	Enterprise Project
Creating a migration project	POST /v3/migprojects	sms:server:migrationServer	√	×
Querying details about a migration project	GET /v3/migprojects/{mig_project_id}	sms:server:queryServer	√	×
Modifying a migration project	PUT /v3/migprojects/{mig_project_id}	sms:server:migrationServer	√	×
Deleting a migration project	DELETE /v3/migprojects/{mig_project_id}	sms:server:migrationServer	√	×
Changing the default migration project	PUT /v3/migprojects/{mig_project_id}/default	sms:server:migrationServer	√	×

API Version Management

Table 6-8 API version management

Permission	API	Action	IAM Project	Enterprise Project
Querying an API version	GET /{version}	sms:server:queryServer	√	×
Listing API versions	GET /	sms:server:queryServer	√	×

Agent Running

Table 6-9 Agent running

Permission	API	Action	IAM Project	Enterprise Project
Obtaining the Agent configuration information	GET /v3/config	sms:server:queryServer	√	×

Network Measurement Management

Table 6-10 Network measurement management

Permission	API	Action	IAM Project	Enterprise Project
Updating network measurement information	POST /v3/{task_id}/update-network-check-info	sms:server:migrationServer	√	√

Advanced Migration Options Management

Table 6-11 Advanced migration options management

Permission	API	Action	IAM Project	Enterprise Project
Querying the settings of advanced migration options of a task	GET /v3/tasks/{task_id}/configuration-setting	sms:server:queryServer	√	√

Permission	API	Action	IAM Project	Enterprise Project
Configuring advanced migration options	POST /v3/tasks/{task_id}/configuration-setting	sms:server:migrationServer	√	√

Privacy Agreement Management

Table 6-12 Privacy agreement management

Permission	API	Action	IAM Project	Enterprise Project
Querying whether a user has signed the privacy agreement.	GET /v3/privacy-agreements	sms:server:queryServer	√	×
Signing the privacy agreement	POST /v3/privacy-agreements	sms:server:registerServer	√	√

6.3 Actions Supported by Identity Policy-based Authorization

IAM provides system-defined identity policies to define common actions supported by cloud services. You can also create custom identity policies using the actions supported by cloud services for more refined access control.

In addition to IAM, the **Organizations** service also provides **service control policies (SCPs)** to set access control policies.

SCPs do not actually grant any permissions to a principal. They only set the permissions boundary for the principal. When SCPs are attached to an organizational unit (OU) or a member account, the SCPs do not directly grant permissions to that OU or member account. Instead, the SCPs just determine what permissions are available for that member account or those member accounts under that OU. The granted permissions can be applied only if they are allowed by the SCPs.

To learn more about how IAM is different from Organizations for access control, see **What Are the Differences in Access Control Between IAM and Organizations?**

This section describes the elements used by IAM custom identity policies and Organizations SCPs. The elements include actions, resources, and conditions.

- For details about how to use these elements to create an IAM custom identity policy, see **Creating a Custom Identity Policy**.
- For details about how to use these elements to create a custom SCP, see **Creating an SCP**.

Actions

Actions are specific operations that are allowed or denied in an identity policy.

- The **Access Level** column describes how the action is classified (**List**, **Read**, or **Write**). This classification helps you understand the level of access that an action grants when you use it in an identity policy.
- The **Resource Type** column indicates whether the action supports resource-level permissions.
  - You can use a wildcard (*) to indicate all resource types. If this column is empty (-), the action does not support resource-level permissions and you must specify all resources ("*") in your identity policy statements.
  - If this column includes a resource type, you must specify a URN for the Resource element in your identity policy statements.
  - Required resources are marked with asterisks (*) in the table. If you specify a resource in a statement using this action, then it must be of this type.

  For details about the resource types defined by SMS, see **Resources**.
- The **Condition Key** column contains keys that you can specify in the Condition element of an identity policy statement.
  - If the **Resource Type** column has values for an action, the condition key takes effect only for the listed resource types.
  - If the **Resource Type** column is empty (-) for an action, the condition key takes effect for all resources that action supports.
  - If the **Condition Key** column is empty (-) for an action, the action does not support any condition keys.

  For details about the condition keys defined by SMS, see **Conditions**.
- The **Alias** column lists the policy actions that are configured in identity policies. With these actions, you can use APIs for policy-based authorization. For details, see **Policies and Identity Policies**.

The following table lists the actions that you can define in identity policy statements for SMS.

Table 6-13 Actions supported by SMS

Action	Description	Access Level	Resource Type (*: required)	Condition Key	Alias
sms:template:list	Grants the permission to list templates.	list	template	-	sms:server:queryServer
sms:template:create	Grants the permission to create a server template.	write	template	-	sms:server:migrationServer
sms:template:batchDelete	Grants the permission to delete templates in batches.	write	template	-	-
sms:template:get	Grants the permission to query templates.	read	template	-	sms:server:queryServer
sms:template:update	Grants the permission to update templates.	write	template	-	sms:server:migrationServer
sms:template:getTargetPassword	Grants the permission to query the target server password configured in a template.	read	template	-	sms:server:queryServer
sms:template:delete	Grants the permission to delete templates.	write	template	-	sms:server:migrationServer
sms:server:listErrors	Grants the permission to query server errors.	list	server	-	server:queryServer
sms:server:list	Grants the permission to list servers.	list	server	**g:EnterpriseProjectId**	sms:server:queryServer

Action	Description	Access Level	Resource Type (*: required)	Condition Key	Alias
sms:server:register	Grants the permission to report the source server list.	write	server	g:EnterpriseProjectId	sms:server:registerServer
sms:server:batchDelete	Grants the permission to delete servers in batches.	write	server	g:EnterpriseProjectId	-
sms:server:get	Grants the permission to query servers.	read	server	g:EnterpriseProjectId	sms:server:queryServer
sms:server:update	Grants the permission to update servers.	write	server	g:EnterpriseProjectId	sms:server:migrationServer
sms:server:delete	Grants the permission to delete servers.	write	server	g:EnterpriseProjectId	sms:server:migrationServer
sms:server:updateDiskInfo	Grants the permission to update server disk information.	write	server	g:EnterpriseProjectId	sms:server:registerServer
sms:server:overview	Grants the permission to query general information of servers.	read	server	-	sms:server:queryServer
sms:server:updateState	Grants the permission to update the server status.	write	server	g:EnterpriseProjectId	sms:server:migrationServer
sms:server:listTask	Grants the permission to query the task list.	list	server	g:EnterpriseProjectId	sms:server:queryServer
sms:server:createTask	Grants the permission to create tasks.	write	server	g:EnterpriseProjectId	sms:server:migrationServer

Action	Description	Access Level	Resource Type (*: required)	Condition Key	Alias
sms:server:batchDeleteTask	Grants the permission to delete tasks in batches.	write	server	**g:EnterpriseProjectId**	-
sms:server:getTask	Grants the permission to query tasks.	read	server	**g:EnterpriseProjectId**	sms:server:queryServer
sms:server:updateTask	Grants the permission to update a task.	write	server	**g:EnterpriseProjectId**	sms:server:migrationServer
sms:server:deleteTask	Grants the permission to delete tasks.	write	server	**g:EnterpriseProjectId**	sms:server:deleteTask
sms:server:manageTask	Grants the permission to manage tasks.	write	server	**g:EnterpriseProjectId**	sms:server:migrationServer
sms:server:updateTaskProgress	Grants the permission to report the migration progress.	write	server	**g:EnterpriseProjectId**	sms:server:migrationServer
sms:server:unlock	Grants the permission to unlock the target endpoint.	write	server	**g:EnterpriseProjectId**	-
sms:server:collectLog	Grants the permission to upload logs.	write	server	**g:EnterpriseProjectId**	sms:server:migrationServer
sms:server:getTaskPassphrase	Grants the permission to query certificates' passphrases.	read	server	**g:EnterpriseProjectId**	sms:server:queryServer
sms::checkNetwork	Grants the network check permission.	read	server	-	-
sms:server:getTaskSpeedLimit	Grants the permission to query the task rate limit.	read	server	**g:EnterpriseProjectId**	sms:server:getTaskSpeedLimit

Action	Description	Access Level	Resource Type (*: required)	Condition Key	Alias
sms:server:updateTaskSpeedLimit	Grants the permission to set the rate limit of migration tasks.	write	server	g:EnterpriseProjectId	sms:server:migrationServer
sms:server:getCommand	Grants the permission to query server commands.	read	server	g:EnterpriseProjectId	sms:server:migrationServer
sms:server:updateCommandResult	Grants the permission to update the server command execution result.	write	server	g:EnterpriseProjectId	sms:server:migrationServer
sms:server:getCert	Grants the permission to obtain keys.	read	server	g:EnterpriseProjectId	sms:server:queryServer
sms:migproject:list	Grants the permission to list migration projects.	list	migproject	-	sms:server:queryServer
sms:migproject:create	Grants the permission to create migration projects.	write	migproject	-	sms:server:migrationServer
sms:migproject:get	Grants the permission to query migration projects.	read	migproject	-	sms:server:queryServer
sms:migproject:update	Grants the permission to update migration projects.	write	migproject	-	sms:server:migrationServer
sms:migproject:delete	Grants the permission to delete migration projects.	write	migproject	-	sms:server:migrationServer
sms::getConfig	Grants the permission to obtain Agent configurations.	read	-	-	sms:server:queryServer

Action	Description	Access Level	Resource Type (*: required)	Condition Key	Alias
sms:server:updateNetworkCheckInfo	Grants the permission to update network check information.	write	task	**g:EnterpriseProjectId**	sms:server:migrationServer
sms:server:getTaskConfig	Grants the permission to query task configurations.	read	task	**g:EnterpriseProjectId**	sms:server:queryServer
sms:server:updateTaskConfig	Grants the permission to update task configurations.	write	task	**g:EnterpriseProjectId**	sms:server:migrationServer
sms:server:getConsistencyCheckResult	Grants the permission to query consistency check results.	read	server	**g:EnterpriseProjectId**	sms:server:queryServer
sms:server:updateConsistencyCheckResult	Grants the permission to update consistency check results.	write	server	**g:EnterpriseProjectId**	sms:server:migrationServer

Each API of SMS usually supports one or more actions. **Table 6-14** lists the supported actions and dependencies.

Table 6-14 Actions and dependencies supported by SMS APIs

API	Action	Dependencies
GET /v3/vm/templates	sms:template:list	-
POST /v3/vm/templates	sms:template:create	-
POST /v3/vm/templates/delete	sms:template:batchDelete	-
GET /v3/vm/templates/{id}	sms:template:get	-
PUT /v3/vm/templates/{id}	sms:template:update	-

API	Action	Dependencies
GET /v3/vm/templates/{id}/target-password	sms:template:getTargetPassword	-
DELETE /v3/vm/templates/{id}	sms:template:delete	-
GET /v3/errors	sms:server:listErrors	-
GET /v3/sources	sms:server:list	-
POST /v3/sources	sms:server:register	-
POST /v3/sources/delete	sms:server:batchDelete	• ecs:cloudServers:showServer • ecs:cloudServers:attach • evs:volumes:use • ecs:cloudServers:stop • ecs:cloudServers:start • ecs:cloudServers:detachVolume • evs:volumes:delete • evs:snapshots:delete • evs:volumes:get
GET /v3/sources/{source_id}	sms:server:get	-
PUT /v3/sources/{source_id}	sms:server:update	-
DELETE /v3/sources/{source_id}	sms:server:delete	• ecs:cloudServers:showServer • ecs:cloudServers:attach • evs:volumes:use • ecs:cloudServers:stop • ecs:cloudServers:start • ecs:cloudServers:detachVolume • evs:volumes:delete • evs:snapshots:delete • evs:volumes:get
PUT /v3/sources/{source_id}/diskinfo	sms:server:updateDiskInfo	-
GET /v3/sources/overview	sms:server:overview	-
PUT /v3/sources/{source_id}/changestate	sms:server:updateState	-

API	Action	Dependencies
GET /v3/tasks	sms:server:listTask	-
POST /v3/tasks	sms:server:createTask	-
POST /v3/tasks/delete	sms:server:batchDeleteTask	• ecs:cloudServers:showServer • ecs:cloudServers:attach • evs:volumes:use • ecs:cloudServers:stop • ecs:cloudServers:start • ecs:cloudServers:detachVolume • evs:volumes:delete • evs:snapshots:delete • evs:volumes:get
GET /v3/tasks/{task_id}	sms:server:getTask	-
PUT /v3/tasks/{task_id}	sms:server:updateTask	-
DELETE /v3/tasks/{task_id}	sms:server:deleteTask	• ecs:cloudServers:showServer • ecs:cloudServers:attach • evs:volumes:use • ecs:cloudServers:stop • ecs:cloudServers:start • ecs:cloudServers:detachVolume • evs:volumes:delete • evs:snapshots:delete • evs:volumes:get
POST /v3/tasks/{task_id}/action	sms:server:manageTask	-
PUT /v3/tasks/{task_id}/progress	sms:server:updateTaskProgress	-
POST /v3/tasks/{task_id}/unlock	sms:server:unlock	-
POST /v3/tasks/{task_id}/log	sms:server:collectLog	-

API	Action	Dependencies
GET /v3/tasks/{task_id}/passphrase	sms:server:getTaskPassphrase	-
GET /v3/tasks/{t_project_id}/networkacl/{t_network_id}/check	sms::checkNetwork	-
GET /v3/tasks/{task_id}/speed-limit	sms:server:getTaskSpeedLimit	-
POST /v3/tasks/{task_id}/speed-limit	sms:server:updateTaskSpeedLimit	-
GET /v3/sources/{server_id}/command	sms:server:getCommand	-
POST /v3/sources/{server_id}/command_result	sms:server:updateCommandResult	-
GET /v3/tasks/{task_id}/certkey	sms:server:getCert	-
GET /v3/migprojects	sms:migproject:list	-
POST /v3/migprojects	sms:migproject:create	-
GET /v3/migprojects/{mig_project_id}	sms:migproject:get	-
PUT /v3/migprojects/{mig_project_id}	sms:migproject:update	-
DELETE /v3/migprojects/{mig_project_id}	sms:migproject:delete	-
PUT /v3/migprojects/{mig_project_id}/default	sms:migproject:update	-
GET /v3/config	sms::getConfig	-
POST /v3/{task_id}/update-network-check-info	sms:server:updateNetworkCheckInfo	-
POST /v3/tasks/{task_id}/configuration-setting	sms:server:updateTaskConfig	-
GET /v3/tasks/{task_id}/consistency-result	sms:server:getConsistencyCheckResult	-
POST /v3/tasks/{task_id}/consistency-result	sms:server:updateConsistencyCheckResult	-

Resources

A resource type (Resource) defines the resources on which an identity policy takes effect. In **Table 6-15**, if a specific operation has an associated resource type, then in the identity policy statement containing that operation, the policy will only take effect on the specified resource when the corresponding resource URN is provided. If no resource URN is specified, the policy will apply to all resources under the designated resource type. If no resource type is defined, the default value of **Resource** is *****, meaning the policy applies to all resources. In addition, you can set conditions in the identity policy to further refine and restrict the resource type.

Table 6-15 Resource types supported by SMS

Resource Type	URN
server	sms::<account-id>:server:*
	sms::<account-id>:server:<server-id>
Task	sms::<account-id>:task:*
	sms::<account-id>:task:<task-id>
template	sms::<account-id>:template:*
	sms::<account-id>:template:<template-id>
migproject	sms::<account-id>:migproject:*
	sms::<account-id>:migproject:<migproject-id>

Conditions

SMS does not support service-specific condition keys in identity policies.

It can only use global condition keys applicable to all services. For details, see **Global Condition Keys**.

7 Appendix

7.1 Error Codes

If an error code starting with APIGW is returned after you call an API, rectify the fault by referring to the instructions provided in **API Gateway Error Codes**.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.0007	Agent error.	The Agent program is abnormal.	Contact technical support.
400	SMS.0201	Network busy. Source server failed to connect to API Gateway.	The source server's network is busy, and the API gateway cannot be connected.	Check the network between the source server and the API gateway.
400	SMS.0202	AK/SK authentication failed. Ensure that the AK and SK are correct.	AK/SK authentication failed. Ensure that the AK/SK pair is correct.	Check whether you are using the correct AK/SK pair for the target platform account and whether the account is frozen.
400	SMS.0203	Connection from source server to API Gateway timed out.	The connection from source server to the API gateway timed out.	Check the network between the source server and the API gateway.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.0204	Insufficient permissions. Cause: %s. Obtain the required permissions.	Insufficient permissions. Cause: %s. Obtain the required permissions.	Obtain the required permissions by referring to "SMS.0204 Insufficient Permissions. Obtain the Required Fine-grained Permissions."
400	SMS.0205	AK/SK authentication failed. The time or time zone of the source server is incorrect.	AK/SK authentication failed. The time or time zone of the source server is incorrect.	Change the system time or time zone on the source server.
400	SMS.0206	Only x86 servers can be migrated.	Only x86 servers can be migrated.	Use another migration method.
400	SMS.0207	KVM driver not available on source server.	KVM drivers are not available on the source server.	Install the KVM drivers.
400	SMS.0208	Failed to send your service statement confirmation to SMS	Your service statement confirmation failed to be sent to SMS.	Check the network between the source server and the API gateway.
400	SMS.0210	Failed to create file %s on target server	File %s could not be created on the target server.	Check the network and restart the migration task.
400	SMS.0211	Failed to send certificate decryption key to target server	The certificate decryption key failed to be sent to the target server.	Check the network, restart the target server, and restart the migration task.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.0212	Agent restarted. Delete the current target configuration and configure the target server again.	The Agent was restarted. Delete the current target server configuration and reconfigure the migration task.	Delete the target configuration and reconfigure the migration task.
400	SMS.0301	Failed to send the network request. Cause: %s	Failed to send the network request. Cause: %s	Modify the proxy information.
400	SMS.0302	Failed to resolve domain name %s.	Failed to resolve domain name %s.	Check whether the domain name is reachable.
400	SMS.0303	Unable to access domain name %s. Cause: %s	Unable to access domain name %s. Cause: %s	Check whether the domain name is reachable. If it is not, rectify the fault by referring to "SMS.0303 Unable to access domain name".
400	SMS.0304	Network request TLS/SSL authentication failed. Cause: %s	TLS/SSL authentication failed during the network request. Cause: %s	Rectify the fault by referring to "SMS.0304 SSL/TLS authentication failed."
400	SMS.0401	No project ID found. Cause: %s	Failed to obtain the project ID. Cause: %s	Contact technical support.
400	SMS.0406	No image found. Cause: %s	Failed to obtain the image. Cause: %s.	Contact technical support.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.0407	No target server information found. Cause: %s	Failed to obtain target server information. Cause: %s.	Contact technical support.
400	SMS.0408	Failed to obtain details about target server %s. Cause: %s	Failed to obtain details about target server %s. Cause: %s	Locate the root cause based on the returned error details and rectify the fault by referring to "Error Codes" in the ECS documentation.
400	SMS.0409	Failed to obtain volume information of target server %s. Cause %s	Failed to obtain volume information of target server %s. Cause %s	Locate the root cause based on the returned error details and rectify the fault by referring to "Error Codes" in the ECS documentation.
400	SMS.0410	Failed to obtain %s information of source server. Cause: %s	Failed to obtain %s information of source server. Cause %s	Contact technical support.
400	SMS.0411	Disk %s does not exist.	Disk %s does not exist.	The disk has been deleted. Perform the migration again.
400	SMS.0412	Target server %s does not exist.	Target server %s does not exist.	The target server has been deleted. Perform the migration again.
400	SMS.0413	Failed to query the region where the bucket locates. Cause: %s	Failed to query the region where the bucket locates. Cause: %s	Locate the root cause based on the returned error details and rectify the fault by referring to "Error Codes" in the OBS documentation.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.0414	Failed to copy logs from the customer's OBS bucket to the destination bucket. Cause: %s	Failed to copy logs from the customer's OBS bucket to the destination bucket. Cause: %s	Contact technical support.
400	SMS.0415	Failed to obtain the target server specifications. Cause: %s	Failed to obtain the target server specifications. Cause: %s	Locate the root cause based on the returned error details and rectify the fault by referring to "Error Codes" in the ECS documentation.
400	SMS.0416	No VPC found. Cause: %s	No VPC found. Cause: %s	Locate the root cause based on the returned error details and rectify the fault by referring to "Error Codes" in the VPC documentation.
400	SMS.0417	No security group found. Cause: %s	No security group found. Cause: %s	Locate the root cause based on the returned error details and rectify the fault by referring to "Error Codes" in the VPC documentation.
400	SMS.0418	No subnet found. Cause: %s	No subnet found. Cause: %s	Locate the root cause based on the returned error details and rectify the fault by referring to "Error Codes" in the VPC documentation.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.0419	Failed to obtain the volume details of target server %s by calling the combined API. Cause: %s	Failed to obtain the volume details of target server %s by calling the combined API. Cause: %s	Locate the root cause based on the returned error details and rectify the fault by referring to "Error Codes" in the ECS documentation.
400	SMS.0420	Failed to obtain the disk details of the target server. Cause: %s	Failed to obtain the disk details of the target server. Cause: %s	Locate the root cause based on the returned error details and rectify the fault by referring to "Error Codes" in the EVS documentation.
400	SMS.0421	Failed to obtain the domain ID. Cause: %s	Failed to obtain the domain ID. Cause: %s	Locate the root cause based on the returned error details and rectify the fault by referring to "Error Codes" in the IAM documentation.
400	SMS.0422	Failed to obtain the details about the VPC with the specified ID.	Failed to obtain the details about the VPC with the specified ID.	Locate the root cause based on the returned error details and rectify the fault by referring to "Error Codes" in the VPC documentation.
400	SMS.0423	The VPC with the specified ID is abnormal.	The VPC with the specified ID is abnormal.	Locate the root cause based on the returned error details and rectify the fault by referring to "Error Codes" in the VPC documentation.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.0424	The obtained list of disks on the target server has no content.	The obtained list of disks on the target server has no content.	Locate the root cause based on the returned error details and rectify the fault by referring to "Error Codes" in the EVS documentation.
400	SMS.0425	Failed to obtain the JSON configuration file.	Failed to obtain the JSON configuration file.	Contact technical support.
400	SMS.0426	Response body too long. Maximum length: %s	Response body too long. Maximum length: %s	Modify the maximum length limit in the configuration file.
400	SMS.0501	Failed to report the task progress to SMS.	Failed to report the task progress to SMS.	Contact technical support.
400	SMS.0502	Source server registration failed. Cause: %s	Source server registration failed. Cause: %s	Obtain the required permissions based on the returned information or contact technical support.
400	SMS.0503	Log upload failed. Cause: %s	Log upload failed. Cause: %s	Locate the root cause based on the returned error details and rectify the fault by referring to "Error Codes" in the OBS documentation.
400	SMS.0504	Failed to obtain task details. Cause: %s	Failed to obtain task details. Cause: %s	Start the task again or contact technical support.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.0505	Task %s not found. Check whether the task has been deleted.	Task %s not found. Check whether the task has been deleted.	Create a migration task again.
400	SMS.0506	Failed to obtain template details. Template ID: %s	Failed to obtain template details. Template ID: %s	Contact technical support.
400	SMS.0507	Source server details deleted. Register it with SMS again.	Source server details deleted. Register it with SMS again.	Register the source server with SMS again.
400	SMS.0508	Failed to obtain commands from the source server. Cause %s	Failed to obtain commands from source server %s. Cause: %s	Restart the task.
400	SMS.0509	The application failed to start because its side-by-side configuration is incorrect.	The application failed to start because its side-by-side configuration is incorrect.	Contact technical support.
400	SMS.0510	Failed to update the replication status to %s.	Failed to update the replication status to %s.	Restart the task.
400	SMS.0511	Failed to obtain the private key. Cause: %s	Failed to obtain the private key. Cause: %s	Contact technical support.
400	SMS.0512	Failed to update task details.	Failed to update task details.	Restart the task.
400	SMS.0513	Failed to add the subtask.	Failed to add the subtask.	Contact technical support.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.0514	Updating source server information failed. Cause: %s	Failed to update source server information. Cause: %s	Restart the task.
400	SMS.0515	Migration failed. Source disk information has changed. Delete target server configuration and restart the Agent.	Migration failed. Source disk information has changed. Delete target server configuration and restart the Agent.	Delete the target server configuration and restart the Agent.
400	SMS.0516	Failed to execute the synchronization task, because the I/O monitoring module failed to run.	Failed to execute the synchronization task because the I/O monitoring module failed to run.	Contact technical support.
400	SMS.0519	Not enough space. Delete target configuration, restart Agent, and expand target partition.	Not enough space. Delete target configuration, restart Agent, and expand target partition.	Delete the target configuration, restart the Agent, and expand the target partition.
400	SMS.0520	Failed to obtain the target server password. Cause: %s	Failed to obtain the target server password. Cause: %s	Contact technical support.
400	SMS.0521	Failed to obtain the certificate password. Cause: %s	Failed to obtain the certificate password. Cause: %s	Check the network connection, restart the target server, and restart the migration task.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.0522	The parameter contains invalid characters: %s	The parameter contains invalid characters: %s	Check whether the JSON body of the API call is correct.
400	SMS.0601	Target server creation failed. Cause: %s	Target server creation failed. Cause: %s	Locate the root cause based on the returned error details and rectify the fault by referring to "Error Codes" in the ECS documentation.
400	SMS.0602	VPC creation failed. Cause: %s	VPC creation failed. Cause: %s	Locate the root cause based on the returned error details and rectify the fault by referring to "Error Codes" in the VPC documentation.
400	SMS.0603	Security group creation failed. Cause: %s	Security group creation failed. Cause: %s	Contact VPC technical support.
400	SMS.0604	Failed to add the security group rule. Cause: %s	Failed to add the security group rule. Cause: %s	Contact VPC technical support.
400	SMS.0605	Subnet creation failed. Cause: %s	Subnet creation failed. Cause: %s	Locate the root cause based on the returned error details and rectify the fault by referring to "Error Codes" in the VPC documentation.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.0606	No general-computing server flavors are available in this AZ. Select another flavor.	No general-computing server flavors are available in this AZ. Select another flavor.	Select another flavor.
400	SMS.0607	cloud-region.json does not contain details about the current region.	cloud-region.json does not contain details about the current region.	Check the destination region information.
400	SMS.0608	Volume creation failed. Cause: %s	Volume creation failed. Cause: %s	Locate the root cause based on the returned error details and rectify the fault by referring to "Error Codes" in the EVS documentation.
400	SMS.0804	File upload failed.	File upload failed.	Check whether the network is functional.
400	SMS.0805	Failed to migrate partition %s to target server %s.	Failed to migrate partition %s to target server %s.	Handle the problem as instructed in the corresponding case.
400	SMS.0806	Failed to synchronize partition %s to target server %s.	Failed to synchronize partition %s to target server %s.	Handle the problem as instructed in the corresponding case.
400	SMS.0807	Network error between source and target servers.	Network error between source and target servers.	Fix the issue by referring to "SMS.0807 Network Error Between Source and Target Servers."

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.0808	Failed to initialize sync task. Target disks or partitions not found.	Failed to initialize sync task. Target disks not found.	Contact technical support.
400	SMS.1101	Failed to start target server %s.	Failed to start target server %s.	Try again later. Check whether your account has the permission to stop the ECS.
400	SMS.1102	Failed to stop target server %s.	Failed to stop target server %s.	Try again later. Check whether your account has the permission to stop the ECS.
400	SMS.1103	Failed to attach disk %s. Cause: %s	Failed to attach disk %s. Cause: %s	Ensure that the network is connected and try again. If the fault persists, rectify the fault by referring to "Why Can't I Attach My Disk to a Server?" in the EVS documentation.
400	SMS.1104	Failed to detach disk %s. Cause: %s	Failed to detach disk %s. Cause: %s	Rectify the fault by referring to "SMS.1104 Failed to Detach Disk xxx" in the documentation.
400	SMS.1105	Disk creation failed. Cause: %s	Disk creation failed. Cause: %s	Rectify the fault by referring to "SMS.1105 Disk Creation Failed" in the documentation.
400	SMS.1106	Failed to delete disk %s. Cause: %s	Failed to delete disk %s. Cause: %s	Rectify the fault by referring to "SMS.1106 Failed to Delete Disk ***" in the documentation.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.1107	Failed to upload the private key and certificate to target server %s. Cause: %s	Failed to upload the private key and certificate to target server %s. Cause: %s	Ensure that the source server can access the target server through SSH and check whether the agent image version is correct.
400	SMS.1108	Failed to detach disk %s.	Failed to detach disk %s.	Refer to "SMS.1104 Failed to Detach disk xx" in the SMS documentation.
400	SMS.1109	An exception occurred when the private key and certificate are uploaded to target server %s	An exception occurred when the private key and certificate were uploaded to target server %s.	Contact technical support.
400	SMS.1110	Failed to attach disk %s.	Failed to attach disk %s.	Contact technical support. Ensure stable network connectivity and retry the task. If the fault persists, rectify the fault by referring to "Why Can't I Attach My Disk to a Server?" in the EVS documentation.
400	SMS.1111	Failed to start target server %s. Cause %s	Failed to start target server %s. Cause %s	Contact technical support.
400	SMS.1112	Failed to stop target server %s. Cause: %s	Failed to stop target server %s. Cause: %s	Contact technical support.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.1113	Failed to reconfigure partition details on the target server.	Failed to reconfigure partition details on the target server.	Rectify the fault by referring to "SMS.1113 Failed to Reconfigure Partition Details on the Target Server" in the documentation.
400	SMS.1114	Failed to send command to the target server.	Failed to send command to the target server.	Contact technical support.
400	SMS.1116	Failed to set type for disk %s. Cause: %s	Failed to set type for disk %s. Cause: %s	Wait and try again. If the issue persists, check whether the agent image version is correct.
400	SMS.1117	Failed to generate the RSA key pair on the target server.	Failed to generate the RSA key pair on the target server.	Wait and try again. If the issue persists, check whether the agent image version is correct.
400	SMS.1118	Failed to query the content of file %s.	Failed to query the content of file %s.	Contact technical support.
400	SMS.1119	Failed to restart SSHD on the target server.	Failed to restart SSHD on the target server.	Contact technical support.
400	SMS.1120	Command execution on the Windows server failed. Cause: missing command or command parameter.	Failed to run the command on Windows. The cause is that the command or command parameter is empty.	Wait and try again. If the issue persists, check whether the agent image version is correct.
400	SMS.1201	%s not installed.	The specified service %s is not installed.	Contact technical support.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.1202	Agent image ID is empty. Create agent image and reconfigure the task.	The agent image ID is empty. Create an agent image and perform the migration again.	Create an agent image according to the documentation and perform the migration again.
400	SMS.1203	Insufficient memory on the source server.	The source server memory is insufficient.	Close unneeded programs on the source server to release memory.
400	SMS.1204	Failed to create a file on the source server. Cause: %s	Failed to create a file on the source server. Cause: %s	Rectify the fault by referring to "SMS.1204 Failed to Create File on Source Server" in the documentation.
400	SMS.1205	Failed to load WMI. Go to the official website to view the solution.	Failed to load WMI. Go to the official website to view the solution.	Rectify the fault by referring to "SMS.1205 Failed to Load WMI" in the documentation.
400	SMS.1301	Failed to write partition information to the configuration file.	Failed to write partition information to the configuration file.	Ensure that the network connection is normal and try again.
400	SMS.1311	Insufficient disks on the target server.	Insufficient disks on the target server.	Add disks to the target server.
400	SMS.1312	Partition failed. Disk: %s. Cause: %s	Partition failed. Disk: %s. Cause: %s	Wait and try again. If the issue persists, check whether the agent image version is correct.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.1313	Failed to create physical volume %s. Cause: %s	Failed to create physical volume %s. Cause: %s	Wait and try again. If the issue persists, check whether the agent image version is correct.
400	SMS.1314	Formatting failed. Partition: %s. Cause: %s	Formatting failed. Partition: %s. Cause: %s	Wait and try again. If the issue persists, check whether the agent image version is correct.
400	SMS.1315	Insufficient free space on the target partition %s.	Insufficient free space on the target partition %s.	Fix the issue by referring to "SMS.1315 Insufficient Free Space on the Target Partition xxxx."
400	SMS.1351	A directory used as a mount point on the source server is full.	A directory used as a mount point on the source server is full.	Fix the issue by referring to "SMS.1351: Mount Point /xxx Detected on the Source Server, Which Has No Free Space. Ensure that There Is at Least 5 MB of Space."
400	SMS.1401	Failed to copy the I/O monitoring module.	Failed to copy the I/O monitoring module.	Contact technical support.
400	SMS.1402	SSH client not installed. Install the openssh-clients package and check the installation with ssh -V.	The SSH client is not installed. Install the openssh-clients package and check the installation with ssh -V.	Rectify the fault by referring to "SMS.1402 SSH client not installed" in the documentation.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.1403	No grub file found. Find it under /boot.	The grub file could not be found. Search for the file in the /boot directory.	Contact technical support.
400	SMS.1404	Disk %s is abnormal. Expected status: %s	The disk %s status is abnormal. The current status is %s, and the expected status is %s.	Check the target disk status. If the status is expected, try the task again.
400	SMS.1405	Failed to obtain information about disk %s. Cause: %s	Failed to obtain information about disk %s. Cause: %s	The network fluctuates or the ECS API responds slowly. Ensure stable network connectivity, wait for a while, and try the task again. Check whether the target disk exists and whether the target server information in the task is correct. If the disk does not exist or the information is incorrect, correctly configure the target server and perform the migration again.
400	SMS.1406	Failed to obtain the EIP. Cause: %s	Failed to obtain the EIP. Cause: %s	Check whether the EIP status is normal and whether the EIP has been modified or deleted.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.1407	Failed to obtain the target server specifications. Cause: %s	Failed to obtain the target server specifications. Cause: %s	Contact technical support.
400	SMS.1408	Snapshot creation failed. Disk: %s. Cause: %s	Failed to create a snapshot for disk %s. Cause %s.	Rectify the issue by referring to "SMS.1105 Disk creation failed" in the documentation.
400	SMS.1409	Failed to delete snapshot %s. Cause: %s	Failed to delete snapshot %s. Cause: %s	Ensure stable network connectivity and retry the task later. If the error persists, check whether the disk snapshot can be deleted based on the failure cause and whether the disk snapshot exists or has been deleted.
400	SMS.1410	Snapshot %s does not exist.	Snapshot %s does not exist.	Ensure stable network connectivity and retry the task later. If the error persists, check whether the snapshot has been deleted.
400	SMS.1411	Snapshot %s is in the %s state.	Snapshot %s is in the %s state.	Ensure stable network connectivity and retry the task later. If the error persists, check whether the snapshot status is normal.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.1412	Snapshot %s rollback failed. Cause: %s	Snapshot %s rollback failed. Cause: %s	Ensure stable network connectivity and retry the task later. If the error persists, rectify the fault based on the returned error information or contact technical support.
400	SMS.1413	Failed to obtain the status of snapshot %s. Cause: %s	Failed to obtain the status of snapshot %s. Cause: %s	Ensure stable network connectivity and retry the task later. If the error persists, rectify the fault based on the returned error information or contact technical support.
400	SMS.1414	The migration module stopped abnormally and cannot synchronize data.	The migration module stopped abnormally and cannot synchronize data.	Rectify the fault by referring to "SMS.1414 The migration module stops abnormally and cannot synchronize data" in the documentation.
400	SMS.1415	Failed to load the IO monitoring module.	Failed to load the I/O monitoring module.	Rectify the issues by referring to "SMS.1902 Failed to Start the I/O Monitoring Module" in the documentation.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.1416	Failed to create tag for server %s. Cause: %s	Failed to tag server %s. Cause: %s	Ensure stable network connectivity and retry the task later. If the error persists, rectify the fault based on the returned error information or contact technical support.
400	SMS.1417	Operation %s failed on server. Cause: %s	Failed to perform operation %s on the server. Cause: %s	Ensure stable network connectivity and retry the task later. If the error persists, rectify the fault based on the returned error information or contact technical support.
400	SMS.1423	Failed to obtain details about DNS server. Cause: %s	Failed to obtain the DNS server information. Cause: %s	Check the DNS configuration on the source server.
400	SMS.1501	OS version not supported.	The OS is not supported.	Check whether the source OS is supported based on the compatibility list.
400	SMS.1502	Target server %s is abnormal. Expected status: %s	The VM %s status is abnormal. The expected status is %s.	Check the status of the target server. If the status is expected, wait and try again.
400	SMS.1503	Disk %s is not the system boot disk.	Disk %s is not the system boot disk.	SMS considers the first disk as the system disk by default. Adjust the disk attachment order.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.1504	The sizes of disks on the target server do not match those on the source server.	The sizes of disks on the target server do not match those on the source server.	Ensure that the disks on the source server match those on the target server.
400	SMS.1505	No boot partition found.	The boot partition was not found.	Check whether the system disk on the source server is correctly attached.
400	SMS.1506	No boot disk found.	The boot disk was not found.	Check whether the disks on the source server are correctly attached.
400	SMS.1507	No system partition found.	The system partition was not found.	Check whether the system partition on the source server is correctly mounted.
400	SMS.1508	Mount point, partition, or disk configured in disk.cfg not found.	The mount point, partition, or disk information written in the custom disk configuration file was not found.	Check the disk configuration file.
400	SMS.1509	Invalid disk.cfg file.	The custom disk configuration file is not feasible.	Check the disk configuration file.
400	SMS.1510	Configure advanced option %s failed. Cause: %s.	Failed to configure advanced option %s. Cause: %s	Check and modify the settings.
400	SMS.1511	failed to check the consistency,Cause: %s	Failed to verify data consistency. Cause: %s	Reconfigure and execute the task, or contact technical support.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.1701	The installed Agent cannot migrate the server to this region. Install the latest Agent.	The installed Agent cannot migrate the server to this region. Install the latest Agent.	The installed Agent cannot migrate the server to this region. Install the latest Agent.
400	SMS.1702	An error occurred when converting the string into the JSON format.	An error occurred during the conversion of the string to JSON.	Reconfigure and execute the task, or contact technical support.
400	SMS.1801	Migration or synchronization task failed.	The migration or synchronization failed.	Contact technical support.
400	SMS.1802	Migration or synchronization task paused.	The migration or synchronization task is suspended.	Contact technical support.
400	SMS.1805	Migration or synchronization task failed because memory allocation on the target server failed.	The migration or synchronization task failed because memory allocation on the target server failed.	Contact technical support.
400	SMS.1807	Failed to connect to the target server. Check whether its IP address %s is reachable and confirm that port 8900 is enabled.	Failed to connect to the target server. Check whether its IP address %s is reachable and confirm that port 8900 is enabled.	Contact technical support.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.1901	Agent could not read disk information.	The Agent could not read disk information.	Rectify the issue by referring to "SMS.1901 Agent could not read disk information" in the documentation.
400	SMS.1902	Failed to start the I/O monitoring module.	Failed to start the I/O monitoring module.	Rectify the issue by referring to "SMS.1902 Failed to start the I/O monitoring module" in the documentation.
400	SMS.1903	No valid block data found.	Failed to query valid block data.	Try the task again or contact technical support.
400	SMS.1904	Failed to create a snapshot for the Windows server.	Failed to create a snapshot for the Windows server.	Rectify the issue by referring to "SMS.1904 Failed to create a snapshot for the Windows server" in the documentation.
400	SMS.1905	Failed to read the disk information.	Failed to read the volume information.	Try the task again or contact technical support.
400	SMS.1907	Failed to send the migration command.	The Agent failed to send the command.	Try the task again or contact technical support.
400	SMS.1908	Data transmission failed.	The Agent failed to transmit data.	Try the task again or contact technical support.
400	SMS.1909	Data compression failed.	Data compression failed.	Try the task again or contact technical support.
400	SMS.1910	Failed to obtain the I/O monitoring data.	Failed to obtain the I/O monitoring data.	Try the task again or contact technical support.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.1911	Boot sector read error.	Failed to read the boot sector.	Try the task again or contact technical support.
400	SMS.1913	Failed to check IO monitoring disks.	Failed to check I/O monitoring disks.	Rectify the issues by referring to "SMS.1913 Failed to check I/O monitoring disks" in the documentation.
400	SMS.2003	Volume reconfiguratio n failed.	The volume reconfiguratio n failed.	Try the task again or contact technical support.
400	SMS.2004	Failed to back up volume sector details on the target server.	Failed to back up volume sector details on the target server.	Try the task again or contact technical support.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.2007	Failed to write data to disks on the target server.	Failed to write data to disks on the target server.	1. Ensure that the disks on the target server are at least as large as those on the source server. If you expand the source server's disks or add new ones after the Agent is registered, register the Agent again and create a new migration task. 2. Out-of-bounds write may occur on the target server. Ensure that each disk on the target server is 1 GB larger than the paired disk on the source server.
400	SMS.2009	Data decompression failed.	Data decompression failed.	Try the task again or contact technical support.
400	SMS.2011	Failed to receive data.	Failed to receive data.	Try the task again or contact technical support.
400	SMS.2012	Failed to read or parse the volume information from the configuration file.	Failed to read or parse the volume information from the configuration file.	Try the task again or contact technical support.
400	SMS.2013	Failed to open the disk.	Failed to open the disk.	Try the task again or contact technical support.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.2014	Failed to hide the partition information.	Failed to hide the partition information.	Try the task again or contact technical support.
400	SMS.2015	Failed to restore the partition information.	Failed to restore the partition information.	Try the task again or contact technical support.
400	SMS.2016	Drive letter restoration failed.	Drive letter restoration failed.	Try the task again or contact technical support.
400	SMS.2017	Failed to traverse all volumes.	Failed to traverse all volumes.	Try the task again or contact technical support.
400	SMS.2018	Error in reading the volume boot sector.	Failed to read the boot sector of the volume.	Try the task again or contact technical support.
400	SMS.2101	Registry modification failed.	Registry modification failed on Windows.	Try the task again or contact technical support.
400	SMS.2102	Failed to modify the device information.	Failed to modify the device information on Windows.	Try the task again or contact technical support.
400	SMS.2103	Failed to set active partitions.	Failed to set active partitions on Windows.	Try the task again or contact technical support.
400	SMS.2104	Failed to modify BCD configuration.	Failed to modify BCD configuration on Windows.	Try the task again or contact technical support.
400	SMS.2105	Failed to modify boot configuration file.	Failed to modify the boot configuration file on Windows.	Try the task again or contact technical support.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.2106	Check disk failed when modifying Windows server configurations.	Check disk failed when modifying Windows server configurations.	Contact technical support.
400	SMS.2201	Failed to clear disk information.	Failed to clear disk information during the partitioning and formatting on Windows.	Log in to the target server, run the diskpart clean command, and analyze the error cause based on the command output.
400	SMS.2202	Failed to convert the format of disk %s.	Failed to convert the format of disk %s during the partitioning and formatting on Windows.	1. Go to the config directory in the Agent installation directory on drive C. 2. Change the value of target.disk.pci.mapping in the g-property.cfg file in the directory to False. 3. Delete the disk_mapping.record file from the directory. 4. Delete the target configuration and reconfigure the migration task.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.2203	Failed to create partition %s.	Failed to create partition %s during the partitioning and formatting on Windows.	You can try the following solutions: 1. Restart the target server and retry the migration. 2. Re-attach the faulty disk, log in to the target server, manually delete the faulty partition on the Disk Management page, and retry the migration.
400	SMS.2204	Failed to format partition %s.	Failed to format partition %s on Windows.	Try the task again or contact technical support.
400	SMS.2205	Failed to update the partition information.	Failed to update the partition information on Windows.	Try the task again or contact technical support.
400	SMS.2301	Failed to start ntcldst.	Failed to start ntcldst.	Try the task again or contact technical support.
400	SMS.2802	Failed to connect to the target server. Check whether its IP address %s is reachable and port 8899 is enabled.	Failed to connect to the target server. Check whether its IP address %s is reachable and port 8899 is enabled.	Rectify the issue by referring to "SMS.2802 Failed to Connect to the Target Server. Check Whether Its IP Address Is Reachable and Port 8899 Is Enabled" in the documentation.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.3101	Failed to modify the configuration file of the target server. Cause: %s	Failed to modify the configuration file on the target server. Cause: %s	Create an ECS that uses the same or similar OS as the source server, overwrite the /etc/default/grub file on the target server with the content of the file on the ECS, and try the task again.
400	SMS.3102	Failed to modify the initrd file on the target server. Cause: %s	Failed to modify the initrd file on the target server. Cause: %s	Try the task again or contact technical support.
400	SMS.3103	Failed to execute bootloader on the target server. Cause: %s	Failed to rebuild the bootloader on the target server. Cause: %s	The /usr/lib/grub/i386-pc directory does not exist on the source server. Contact technical support.
400	SMS.3104	Failed to read /etc/fstab on the source server. Cause: %s	Failed to read /etc/fstab on the source server. Cause: %s	Failed to read /etc/fstab on the source server. Restore the /etc/fstab file on the source server and try again.
400	SMS.3105	fstab record "%s" is invalid	The fstab record %s is invalid.	Try the task again or contact technical support.
400	SMS.3202	Failed to create volume group %s. Cause: %s	Failed to create volume group %s. Cause: %s	Try the task again or contact technical support.
400	SMS.3203	Failed to create logical volume %s. Cause: %s	Failed to create logical volume %s. Cause: %s	Try the task again or contact technical support.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.3204	Failed to create swap partition %s. Cause: %s	Failed to create swap partition %s. Cause: %s	Try the task again or contact technical support.
400	SMS.3205	Failed to mount partition %s to directory %s. Cause: %s	Failed to mount partition %s to directory %s. Cause: %s	Rectify the issue by referring to "SMS.3205 Failed to Mount Partition XXX to Directory XXX" in the documentation.
400	SMS.3206	Failed to obtain the UUID of the partition on the target server. Cause: %s	Failed to query the UUID of the partition on the target server. Cause: %s	1. Check whether the disk is correctly mounted. 2. Check whether the file system is supported for migration. 3. Retry the task or contact technical support.
400	SMS.3207	Failed to create file %s on the target server. Cause: %s	Failed to create file %s on the target server. Cause: %s	Check whether the disk space is insufficient or contact technical support.
400	SMS.3208	Failed to write file %s on the target server. Cause: %s	Failed to write file %s on the target server. Cause: %s	Check whether the disk space is insufficient or contact technical support.
400	SMS.3209	Failed to assign execution permission to the script on the target server. Cause: %s	Failed to assign execution permission to the script on the target server. Cause: %s	Check whether the security group or firewall of the remote server has restrictions.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.3210	Failed to format partition %s to %s. Cause: %s	Failed to format partition %s to %s. Cause: %s	Check whether the security group or firewall of the remote server has restrictions.
400	SMS.3300	Failed to read the configuration file for Linux block-level migration.	Failed to read the configuration file for Linux block-level migration.	Download the latest Agent.
400	SMS.3301	Failed to load the SSL certificate.	Failed to load the SSL certificate.	Create a migration task again.
400	SMS.3401	Download of RSA public and private keys failed. HTTP Status Code: %s	Download of RSA public and private keys failed. HTTP Status Code: %s	Ensure stable network connectivity and retry the task, or contact technical support.
400	SMS.3402	Download of RSA public and private keys failed.	Download of RSA public and private keys failed.	Ensure stable network connectivity and retry the task, or contact technical support.
400	SMS.3802	Failed to establish an SSH connection with the target server.	Failed to establish an SSH connection with the target server.	Rectify the issue by referring to "SMS.3802 Failed to establish an SSH connection with the target server" in the documentation.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.3803	The connection to the target server failed, because an error occurred during the public key verification on the target server.	The connection to the target server failed because an error occurred during the public key verification on the target server.	Rectify the issue by referring to "SMS.3803 The connection to the port 22 of target server failed because an error occurred during the public key verification on the target server" in the documentation.
400	SMS.3804	The connection to the target server failed due to invalid connection credential.	The connection to the target server failed due to invalid connection credentials.	Rectify the issue by referring to "SMS.3804 The connection to the port 22 of target server failed due to invalid connection credentials" in the documentation.
400	SMS.3805	The connection to the target server timed out.	The connection to the target server timed out.	Rectify the issue by referring to "SMS.3805 The connection to the port 22 of target server timed out" in the documentation.
400	SMS.3806	The connection to the target server was rejected.	The connection to the target server was rejected.	Rectify the issue by referring to "SMS.3806 The connection to the port 22 of target server was rejected" in the documentation.
400	SMS.3807	Failed to decompress %s.	Failed to decompress file %s.	Check whether the disk space is insufficient or contact technical support.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.3808	Failed to ping the target server. Please check whether the ICMP port is enabled.	Failed to ping the target server. Check whether the ICMP port is enabled.	Ensure stable network connectivity and retry the task, or contact technical support.
400	SMS.3809	Failed to replace the RSA certificate on the target server.	Failed to replace the RSA certificate on the target server.	Ensure stable network connectivity and retry the task, or contact technical support.
400	SMS.5101	Agent installation failed.	Agent installation failed.	1. The Agent for Windows could not be launched by double-clicking. 2. The Agent installation failed because files failed to be written. Right-click SMS-Agent-py*.exe and view the properties. Select Unlock. Ensure that there is at least 500 MB of free space on drive C for installing the Agent. This requirement also applies to the Agent for Linux. Ensure that sufficient space is available for decompressing the Agent package to avoid errors during extraction.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.5102	Agent startup failed because the noexec permission is unavailable on /tmp in Linux.	Agent startup failed because the noexec permission is unavailable on /tmp in Linux.	Rectify the issue by referring to "SMS.5102 Agent startup failed because the noexec permission is unavailable on /tmp in Linux" in the documentation.
400	SMS.5103	Agent startup failed due to insufficient space on /tmp in Linux.	Agent startup failed due to insufficient space on /tmp in Linux.	Check whether the /tmp folder exists. If it does not, create the folder. If it does, increase the size of the /tmp folder.
400	SMS.5104	Failed to paste AK or SK.	Failed to paste AK or SK.	Restart the Agent.
400	SMS.5105	Agent startup failed. Insufficient permissions to add files to or delete files from ~.	Agent startup failed. Insufficient permissions to add files to or delete files from the root directory.	Rectify the issue by referring to "Agent Startup Failed. Insufficient Permissions to Add Files to or Delete Files from root" in the documentation.
400	SMS.5106	Internal error. Failed to create a temporary directory.	Internal error. Failed to create a temporary directory.	Run the Agent as an administrator.
400	SMS.5107	Failed to open the file for writing.	The file to be written could not be opened.	Try the task again or contact technical support.
400	SMS.5108	Failed to execute df -TH.	Failed to execute df -TH.	Rectify the issue by referring to "SMS.5108 Failed to execute df -TH" in the documentation.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.5109	The application cannot be started due to incorrect parallel configuration.	The application failed to start because its side-by-side configuration is incorrect.	Rectify the issue by referring to "SMS.5109 The application cannot be started due to incorrect parallel configuration" in the documentation.
400	SMS.5110	The Windows Agent unable to launch.	The Agent for Windows could not be launched by double-clicking.	Unlock the file and try again.
400	SMS.5111	Agent startup failed. Multiple volume groups found with the same name.	Agent startup failed. Multiple volume groups were found with the same name.	Try the task again or contact technical support.
400	SMS.5112	Agent main program: linuxmain could not run.	The main program linuxmain of the Agent could not run.	For details, see "SMS.5112 Agent main program linuxmain could not run" in the documentation.
400	SMS.5113	Check %s on Linux timed out.	Checking %s on Linux timed out.	For details, see "SMS.5113 Checking %s on Linux timed out" in the documentation.
400	SMS.5301	Booting with GRUB failed.	Booting with GRUB failed.	Reinstall the GRUB bootloader.
400	SMS.5302	Failed to install the KVM driver.	Failed to install the KVM drivers.	Uninstall the Xen drivers and install the KVM drivers.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.5401	Disks of the Windows target server are offline.	Disks of the Windows target server are offline.	This issue may be caused by the disk online policy. Contact technical support.
400	SMS.5402	Network error on the target server running Windows 2003.	Network error on the target server running Windows Server 2003.	Use another migration method.
400	SMS.5403	Progress is slow or suspended.	The migration progress is stuck or too slow.	For details, see "Why the Migration Progress Is Suspended or Slow?" in the documentation.
400	SMS.5404	The target server running CentOS 6.x cannot access the Internet.	After the CentOS 6.x migration is complete, the target server cannot connect to the Internet.	Check whether the dhclient command is missing in the system. If it is missing, the IP address cannot be dynamically obtained. In this case, configure a static IP address for the target server to enable Internet connectivity.
400	SMS.5601	The disk usage changes greatly after migration.	The disk usage changes greatly after migration.	Compare the source and target servers or contact technical support.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.5602	Time error occurred in the target server running Linux.	The time on the Linux server is incorrect after migration.	Refer to "What Do I Do If the Linux Source Server's Clocks Are Not Synchronized to the Target Server?" in the documentation.
400	SMS.5603	The target server running Windows cannot access the Internet.	The Windows server cannot access the Internet after the migration.	Refer to "Why Can't a Windows Target Server Access the Internet After the Migration Is Complete?" in the documentation.
400	SMS.5604	The MySQL service on the target server cannot be started.	The MySQL service on the target server cannot be started.	Refer to "How Do I Troubleshoot a MySQL Startup Failure on the Target Server After the Migration?" in the documentation.
400	SMS.5605	After the migration is complete, the target server starts and the System Recovery Options window is displayed.	After the migration is complete, the target server starts and the System Recovery Options window is displayed.	Refer to "Why Is the System Recovery Options Window Displayed When the Target Server Is Started?" in the documentation.
400	SMS.5606	Windows fails to start because the Xen driver is abnormal.	The Windows system cannot be started because the Xen drivers are abnormal.	Delete the Xen drivers and synchronize data again.
400	SMS.6000	Service error.	Service error.	Contact technical support.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.6001	The source server name is a required field.	The source name is empty.	The source server name must be specified.
400	SMS.6002	Source server name must consist of 1 to 64 characters.	A source server name can contain 1 to 64 characters.	A source server name can contain 1 to 64 characters.
400	SMS.6003	Only letters, digits, underscores (_), dot (.), and hyphens (-) are allowed.	Only letters, digits, underscores (_), hyphens (-), and periods (.) are allowed.	Only letters, digits, underscores (_), hyphens (-), and periods (.) are allowed.
400	SMS.6004	Invalid subtask name.	The subtask name is invalid.	Enter a valid subtask name.
400	SMS.6005	The status of the sourceServer does not meet the requirements.	The source server status is not expected.	Check the task status, restart SMS Agent, and restart the task.
400	SMS.6006	Source platform information format error.	The format of the source platform information is incorrect.	Restart the SMS Agent.
400	SMS.6010	The request body is not in the JSON format.	The request body is not in JSON format.	Set the request body as required.
400	SMS.6011	The JSON request body does not meet the specifications.	The JSON in the request body is invalid.	Set the request body as required.
400	SMS.6012	Empty request body.	The request body is empty.	Set the request body as required.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.6013	Incomplete request parameters.	Request parameters are incomplete.	Add all required request parameters.
400	SMS.6014	The parameter part in the request body is too long.	The request body is too long.	Modify the request parameters.
400	SMS.6015	The request header is missing.	The request header is missing.	Add a request header.
400	SMS.6016	The header type must be application/ json.	The header type must be application/ json.	Ensure that the request header type is application/json.
400	SMS.6020	The command name is missing.	The command name is missing.	Ensure stable network connectivity and retry the task, or contact technical support.
400	SMS.6021	The command from the Agent is inconsistent with that from SMS.	The command provided by the client is inconsistent with that on the server.	Enter the command from the server correctly.
400	SMS.6022	Incorrect result format.	The format of "result" is incorrect.	Set "result" in the correct format.
400	SMS.6023	Incorrect result_detail format.	The format of "result_detail" is incorrect.	Set "result_detail" in the correct format.
400	SMS.6030	Invalid source server ID.	The source server ID is invalid.	Enter a correct source server ID.
400	SMS.6031	The target server information is missing.	The target server information is missing.	Enter the required target server information.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.6032	The target server ID is missing.	The parameter "vm_id" is missing from the target server parameters.	Add the target server ID.
400	SMS.6033	The disk information is missing.	The disk information is missing from the target server parameters.	Add disk information of the target server.
400	SMS.6034	Invalid region name.	The region name is invalid.	Enter a valid region name.
400	SMS.6035	Domain ID is missing.	The request body does not contain a domain ID.	Add a domain ID.
400	SMS.6101	Invalid migration progress.	The migration progress is out of range.	Enter an integer from 0 to 100.
400	SMS.6102	Parameters do not meet the JSON format requirements.	Parameters do not meet the JSON format requirements.	Check whether the JSON format requirements are met.
400	SMS.6103	Disk ID is missing.	The disk ID is missing.	Enter a disk ID.
400	SMS.6104	Physical volume name is missing.	The volume name is missing.	Enter a volume name.
400	SMS.6105	Physical volume size is missing or invalid.	The volume size information is missing or invalid.	Enter a valid volume size.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.6106	Failed to obtain migration progress.	Failed to obtain migration progress.	Ensure stable network connectivity and retry the task, or contact technical support.
400	SMS.6301	Database read or write error.	Database read or write error.	An exception occurred during network request parameter verification. Keep the network connection normal and retry the task, or contact technical support.
400	SMS.6302	No migration speed limit found.	No rate limiting information related the task was found.	Reset the rate limit.
400	SMS.6303	The installed Agent is of an earlier version. Download the latest version.	The Agent version is too early. Download the latest Agent.	Download the latest Agent.
400	SMS.6304	Fail to get config information.	Obtaining the configuration information failed.	Ensure stable network connectivity and retry the task, or contact technical support.
400	SMS.6401	Unsupported encoding mode.	The encoding mode is not supported.	Use a correct encoding mode.
400	SMS.6402	The algorithm does not exist.	The algorithm does not exist.	Ensure that the parameters for accessing the API are correct.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.6403	Failed to split into strings of the specified size.	The string cannot be evenly divided into the specified length.	Internal server error. Contact technical support.
400	SMS.6404	An error occurred during integer conversion.	An error occurred during integer conversion.	An exception occurred during network request parameter verification. Keep the network connection normal and retry the task, or contact technical support.
400	SMS.6405	Invalid time format.	Invalid time format.	Check the time format.
400	SMS.6407	Current quota is not enough, please use MgC.	The quota is insufficient. Please use MgC.	Use MgC for migration.
400	SMS.6501	You can add up to 1,000 source servers.	You can add up to 1,000 source servers.	Delete the source servers that have been migrated.
400	SMS.6502	Environment check cannot be performed on the source server in the CHECKING state.	Environment check cannot be performed on the source server in the CHECKING state.	Try the task again or contact technical support.
400	SMS.6503	Failed to obtain the OS version of the source server.	The source server OS version cannot be obtained.	Provide a correct source OS type.
400	SMS.6504	Unsupported source server OS version or firmware type.	The source server OS version or firmware type is not supported.	Check the compatibility list.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.6505	Unknown source firmware type. It must be BIOS or UEFI.	The source server firmware failed the check. It must be BIOS or UEFI.	Use another migration method.
400	SMS.6506	Failed to obtain the number of source server CPUs.	Failed to obtain the number of source server CPUs.	Failed to obtain the number of CPUs. Restart the SMS Agent or contact technical support.
400	SMS.6507	Failed to open the system directory of the source server.	Failed to open the system directory.	Try the task again or contact technical support.
400	SMS.6508	Incompatible disk format on the source server. Only GPT and MBR are compatible.	The disk format on the source server is not supported. Only GPT and MBR are supported.	Use another migration method.
400	SMS.6509	Incompatible file system of the source server.	File systems on the source server are not supported.	Use another migration method.
400	SMS.6510	The source server OS is an OEM system.	The source server OS is an OEM system.	You need to activate the system after migration.
400	SMS.6511	The source server lacks driver files.	The source end lacks necessary driver files.	Refer to "SMS.6511 The source server lacks driver files" in the documentation.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.6512	The system services required for the migration on the source server are not running normally.	The system services required for the migration are not running properly.	Try the task again or contact technical support.
400	SMS.6513	No administrator permissions on the source server.	Administrator permissions are not granted.	Check whether the current account on the source server has the administrator permissions.
400	SMS.6514	Failed to obtain the memory size of the source server.	The memory size could not be obtained.	Restart the SMS Agent or contact technical support.
400	SMS.6515	The source server is paravirtualize d.	Paravirtualize d source servers are not supported.	Use another migration method.
400	SMS.6516	The Linux bootloader must be GRUB.	Only Linux servers using GRUB are supported.	Check whether the Linux boot mode is GRUB.
400	SMS.6517	rsync not installed on the source server.	rsync is not installed on the source server.	Install rsync on the source server.
400	SMS.6518	The source server has a raw device.	Raw devices are not supported.	Use another migration method.
400	SMS.6519	Failed to obtain disk information.	No disk information is available.	Restart the SMS Agent or contact technical support.
400	SMS.6520	The source server is not available.	The source server is not available.	Fix the issue by referring to "SMS.6520 The Source Server Is Not Available."

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.6521	Token-based authentication failed.	Token authentication failed.	Use the correct token.
400	SMS.6522	The X-Auth-Token header field is missing.	X-Auth-Token was not found in the request header.	Add X-Auth-Token to the request header.
400	SMS.6523	Apply for the permission required to operate SMS.	You do not have the permission to use SMS.	Ensure that you have the sms_administrator permissions.
400	SMS.6524	This operation is not allowed because you are not an OBT user.	This operation is restricted to OBT users.	Apply for the OBT.
400	SMS.6525	Account is frozen or restricted. Check account balance and status.	The account or resource is frozen or restricted. Check the account balance and status.	Ensure that the account status is normal and the account balance is sufficient.
400	SMS.6526	Insufficient quota to clone the target server.	The available quota is insufficient to complete the target server cloning operation.	Increase the cloud server quota.
400	SMS.6527	The resource in the task does not belong to you.	The current resource does not belong to the current user.	Check the resources of the current user.
400	SMS.6528	Complete real-name authentication to invoke the SMS APIs.	Users who have not completed real-name authentication cannot call SMS APIs.	Complete real-name authentication first.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.6529	Only accounts with sufficient balance and do not have security risks can invoke the SMS APIs.	Accounts in arrears, with an insufficient balance, or with security risks cannot invoke the SMS APIs.	Ensure that the account status is normal and the account balance is sufficient.
400	SMS.6530	Invalid source server IP address.	The source server's IP address is invalid.	Enter a correct source server IP address.
400	SMS.6531	The boot partition does not exist.	The boot partition does not exist.	Check whether the boot partition of the source server exists.
400	SMS.6532	The system disk must be on the first disk.	The system disk must be on the first disk.	Check whether the system disk is on the first disk.
400	SMS.6533	VSS not installed on the source server.	VSS was not found on the source server.	Check whether VSS has been installed and started.
400	SMS.6534	No system directory information read.	The system directory information is empty.	Restart the SMS Agent or contact technical support.
400	SMS.6535	Servers booted with UEFI do not support auto-creation of target servers for migration.	Automatic creation of target servers is not supported for source servers booted with UEFI.	Create a target server booted with UEFI manually and use it as the target server.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.6536	The source server runs Windows 2003. Network may be unavailable after migration.	The source server runs Windows Server 2003. It may experience network exceptions after migration.	Upgrade the system and perform the migration.
400	SMS.6537	SMS cannot migrate system disks larger than 1 TB.	The system disk is larger than 1 TB.	Downsize the source system disk.
400	SMS.6538	The new partition or logical volume size is less than the used space size.	The new partition or logical volume size is less than the used space size.	Ensure that the new partition or logical volume size is at least 1 GB larger than the used space size.
400	SMS.6539	The disk space is less than the total size of all partitions.	The disk space is less than the total size of all partitions.	The disk space is less than the total size of all partitions.
400	SMS.6540	Partitions on target servers running Windows cannot be reduced.	Partitions on Windows target servers cannot be downsized.	Resize the disks and partitions on the source server and restart the Agent
400	SMS.6541	The VG size is smaller than the combined size of all LVs.	The size of the volume group is smaller than the total size of all logical volumes.	Downsize logical volumes or contact technical support.
400	SMS.6542	Your account has been frozen.	Your account has been frozen.	Unfreeze your account.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.6543	There is more than 1 TB of data to be migrated. It may slow down the migration or synchronization.	The total amount of data to be migrated exceeds 1 TB. The migration and synchronization may be slow.	Continue the migration after confirming risks.
400	SMS.6544	Network condition for migration is poor.	The migration network is in poor conditions.	Continue the migration after confirming risks.
400	SMS.6547	Excessive read/write latency has been detected for the source disks. It may slow down the migration or synchronization.	Excessive read/write latency has been detected for the source disks. It may slow down the migration or synchronization.	Optimize the disk performance.
400	SMS.6548	The migration network performs poorly. It may slow down the migration or synchronization.	The migration network performs poorly. It may slow down the migration or synchronization.	Optimize the network performance.
400	SMS.6560	Failed to obtain the project-level token.	Failed to obtain the project-level token.	The token is called unexpectedly and may pose security risks. Contact technical support.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.6561	The target server is not booted with UEFI. Create a target server using the image whose boot mode is UEFI.	The target server is not booted with UEFI. Create a target server using the image whose boot mode is UEFI.	Create a target server using an UEFI image.
400	SMS.6562	Failed to obtain target server IP address.	Failed to obtain target server IP address.	Refer to "SMS.6564: Component i386-pc not found on source server. For solution, see SMS API Reference." in the documentation.
400	SMS.6601	Invalid OS type.	Invalid OS type.	Set "os_type" by referring to the SMS API Reference.
400	SMS.6602	Invalid target server EIP.	Invalid target server EIP.	Enter an IP address in the correct format.
400	SMS.6603	The connection to SMS was lost.	The connection to SMS was lost.	Re-establish the connection between the source server and SMS. For details, see "Migration Network" in SMS FAQs.
400	SMS.6604	Failed to obtain ACL or firewall details for the target server.	Failed to obtain ACL or firewall details for the target server.	Network request error. Retry the task later or contact technical support.
400	SMS.6605	ACL or security group configuration error.	The ACL or security group settings are incorrect.	Modify the ACL or security group settings.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.6610	The OS in the task conflicts with the source server OS.	The task type does not match the source server OS.	Check whether the task type matches the source server OS.
400	SMS.6611	Insufficient partition space.	Insufficient partition space.	Increase the partition space.
400	SMS.6612	Invalid partition size. The partition size must be an integer multiple of MB.	Invalid partition size. The partition size must be an integer multiple of 1 MB.	Enter a valid partition size.
400	SMS.6613	33 MB of space must be reserved for GPT disks.	33 MB space must be reserved for disks in GPT format.	Reserve 33 MB of space for GPT disks.
400	SMS.6614	The target disk space must be greater than the total space of all partitions.	The target disk space must be greater than the total space of all partitions.	Increase the target disk space.
400	SMS.6615	The target server has been added to a migration task.	The target server has been used in a migration task.	Change another server.
400	SMS.6616	The current OS does not support block-level migration.	Block-level migration is not supported for the current OS release.	Use another migration method.
400	SMS.6617	The current kernel does not support block-level migration.	Block-level migration is not supported for the current kernel release.	Use another migration method.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.6618	Failed to obtain kernel details.	Failed to obtain kernel details.	Try the task again or contact technical support.
400	SMS.6619	Disk encryption key not available.	The disk encryption key is unavailable.	Check whether the key status is normal and ensure that the AES_256 encryption algorithm is used.
400	SMS.6620	Key %d is not enabled.	Key %d is not enabled.	Check the key status.
400	SMS.7101	Template does not exist.	The template does not exist.	Check whether the template exists.
400	SMS.7111	All tasks associated with this template must be deleted first.	All tasks associated with this template must be deleted first.	Delete the associated tasks first.
400	SMS.7112	Duplicate template name.	Duplicate template name.	Change the template name.
400	SMS.7113	Maximum templates reached.	The maximum number of templates is reached.	Go to the SMS console and delete unneeded migration templates.
400	SMS.7301	The specified migration template does not exist.	The specified migration template does not exist.	Check whether the migration template exists.
400	SMS.7303	The default migration template cannot be deleted.	The default project cannot be deleted.	The specified migration template does not exist. Check whether the migration template exists.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.7304	The default migration template does not exist.	The default migration template does not exist.	Create a migration template and set it as the default one.
400	SMS.7311	All source servers associated with the migration project must be deleted first.	All source servers associated with the migration project must be deleted first.	Delete all associated source servers first.
400	SMS.7312	Duplicate migration project name.	Duplicate migration project name.	Change the migration project name.
400	SMS.7313	The default migration project already exists.	The default migration project already exists.	The specified migration template already exists. Use another template.
400	SMS.7321	The region name in the template is inconsistent with that in the migration project.	The region name in the template is inconsistent with that in the migration project.	Try the task again or contact technical support.
400	SMS.7501	No application ID generated.	Failed to generate the application ID.	Try the task again or contact technical support.
400	SMS.7601	Failed to delete the source server because a migration task is running on it.	There is a task on the source server, and the server cannot be deleted.	Try the task again or contact technical support.
400	SMS.7602	The specified source server does not exist.	The specified source server does not exist.	Check whether the source server exists.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.7604	Failed to generate the source server ID.	Failed to generate the source server ID.	Restart the SMS Agent or contact technical support.
400	SMS.7605	The server has been set as the target server of another migration task. Select another server or delete the task associated with the server.	The server has been set as the target server in another migration task. Select another server or delete the migration task associated with the server.	Select another server or delete the migration task associated with the server.
400	SMS.7606	No migration task running on the source server.	No migration task runs on the source server.	The migration task was deleted unexpectedly. Restart the Agent on the source server and reconfigure the migration task.
400	SMS.7703	Task does not exist.	The task does not exist.	Check whether the task exists.
400	SMS.7705	An error occurred when processing JSON files during task creation.	An error occurred during JSON file processing for task creation.	The network request parameters are incorrect. Keep the network connection normal and retry the task, or contact technical support.
400	SMS.7706	Task update failed.	Task update failed.	Network request error. Retry the task or contact technical support.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.7707	The current Agent version does not support log upload.	The Agent version does not support log upload.	Ensure that the Agent version is 25.2.0 or later.
400	SMS.7711	Invalid task name.	Invalid task name.	Enter a valid task name.
400	SMS.7712	Invalid task type.	Invalid task type.	Enter a valid task type.
400	SMS.7713	Only tasks in the Ready for next, Paused, or Error state can be started.	Only tasks in the "Ready for next", "Paused", or "Error" state can be started.	Change the task status.
400	SMS.7714	Only running tasks can be paused.	Only running tasks can be paused.	Change the task status.
400	SMS.7715	Only tasks in the Finished state or in the Continuous synchronization stage can be synchronized.	Only tasks in the "Finished" state or in the "Continuous synchronization" stage can be synchronized.	Change the task status.
400	SMS.7716	Logs can be collected only after the current log collection is completed.	Logs can be collected only after the current log collection is completed.	Wait until the current log collection is complete.
400	SMS.7717	Logs can be collected after the migration task is started.	Logs can be collected after the migration task is started.	Start the task first.
400	SMS.7718	Only tasks in the Synchronization failed state can be rolled back.	Only tasks in the "Synchronization failed" state can be rolled back.	Change the task status.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.7719	Tasks in current state cannot be deleted.	Tasks in current state cannot be deleted.	Change the task status.
400	SMS.7721	Only target servers in tasks in the Continuous synchronization state can be cloned or started.	Only target servers in tasks in the "Continuous synchronization" state can be cloned or launched.	Change the task status.
400	SMS.7722	Only tasks where the target servers have successfully started can be restarted.	Only tasks where the target servers have successfully launched can be restarted.	Check whether the target server for the task has been launched successfully.
400	SMS.7723	Failed to delete the task. Ensure that the Agent is connected.	Failed to delete the task. Ensure that the Agent is connected.	Ensure that the Agent is connected.
400	SMS.7724	No project ID found.	No project ID found.	An exception occurred during the verification of task parameters. Restart the Agent on the source server and reconfigure the migration task, or contact technical support.
400	SMS.7725	No region information found.	No region information found.	Try the task again or contact technical support.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.7726	Failed to update the migration status or stage of the source server. Field copystate or migrationcycle is missing in the request body.	Failed to update the migration status or stage for the source server. The request body lacks the "copystate" or "migrationcycle" field.	Add the "copystate" or "migrationcycle" field to the request body.
400	SMS.7727	Failed to update the migration status or stage of the source server. Field copystate or migrationcycle in the request body is empty.	Failed to update the migration status or stage for the source server. The "copystate" or "migrationcycle" field in the request body is empty.	Specify a value for "copystate" or "migrationcycle" in the request body.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.7728	Change server replication state or migration cycle failed. Values of copystate and migrationcycle do not match.	Failed to update the migration status or stage for the source server. Fields copystate and migrationcycle do not match.	Set copystate and migrationcycle according to the following rules: - If the value of migrationcycle is checking, the value of copystate can be unavailable or waiting. - If the value of migrationcycle is setting, the value of copystate can be waiting, premigready, premiging, premiged or premigfailed - If the value of migrationcycle is replicating, the value of copystate can be init, replicate, deleting, stopping, error or stopped - If the value of migrationcycle is syncing, the value of copystate can be syncing, cloning, deleting, stopping, error or stopped. - If the value of migrationcycle is cutovering, the value of copystate can be cutovering, deleting, stopping, error or stopped. - If the value of migrationcycle is cutovered, the value of copystate can be finished,

Status Code	Error Codes	Error Message	Description	Solution
				deleting, stopping, error or stopped.
400	SMS.8017	Clone server not found.	No clone server associated with the migration task was found.	Contact technical support.
400	SMS.8101	Failed to stop the target server.	Failed to stop the target server.	Try the task again or contact technical support.
400	SMS.8102	The target server is missing.	The target server is missing from the SMS task.	Set target server parameters.
400	SMS.8103	The target server does not exist.	The target server does not exist.	Check whether the server has been deleted.
400	SMS.8104	Failed to obtain details about the target server.	Failed to obtain details about the target server.	Check whether the target server was deleted. If it was not, contact technical support.
400	SMS.8105	Failed to obtain the details about disks on the target server.	Failed to obtain the details about disks on the target server.	Check whether disks are attached to the target server. If yes, contact technical support.
400	SMS.8106	Invalid information about disks on the target server.	Invalid information about disks on the target server.	Check whether the target disk information is correct.
400	SMS.8107	No agent image found on the target server.	No agent image found on the target server.	Check whether the image has been deleted.
400	SMS.8108	Disk attachment failed.	Disk attachment failed.	Network request error. Retry the task or contact technical support.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.8109	No source server found.	Source server not found.	Check whether the server record has been deleted.
400	SMS.8110	Disk detachment failed.	Disk detachment failed.	Network request error. Retry the task or contact technical support.
400	SMS.8111	No disk information found.	No disk information found.	Check whether the target disk information is correct.
400	SMS.8112	No snapshot information found.	No snapshot information found.	Check whether the snapshot exists.
400	SMS.8113	The snapshot does not exist.	The snapshot does not exist.	Check whether the snapshot has been deleted.
400	SMS.8114	Failed to unlock the target server.	Failed to unlock the target server.	Check whether you have the permissions to perform this operation.
400	SMS.8115	Maximum templates: 50	A maximum of 50 templates can be created.	Go to the SMS console, delete unneeded migration templates, and try again.
400	SMS.9001	Invalid parameters.	Invalid parameters.	Check whether the parameters are correct.
400	SMS.9002	User tags are not supported.	User tags are not supported.	User tags are not supported.
400	SMS.9003	Insufficient permissions.	Insufficient permissions.	Obtain the required permissions.

Status Code	Error Codes	Error Message	Description	Solution
400	SMS.9004	The current account does not have the permission to execute policy %s.	Insufficient permissions to perform %s.	Obtain the required permissions.
400	SMS.9005	Failed to bind the enterprise project.	Failed to bind the enterprise project.	Check whether the enterprise project exists and retry the task, or contact technical support.
400	SMS.9905	Password query is not allowed on tasks in the current state.	Password query is not supported for tasks in the current state.	Query passwords only when the task is in a replicating or syncing state.

7.2 Status Codes

See **Table 7-1**.

Table 7-1 Status codes

Status Code	Error Message	Description
100	Continue	The client continues sending the request. This interim response is used to inform the client that the initial part of the request has been received and has not yet been rejected by the server.
101	Switching Protocols	Switching protocols. The target protocol must be more advanced than the source protocol. For example, the current HTTP protocol is switched to a later version.
200	OK	The results of GET and PUT operations are returned as expected.
201	Created	The request for creating a resource has been fulfilled.
202	Accepted	The request has been accepted, but the processing has not been completed.

Status Code	Error Message	Description
203	Non-Authoritative Information	The server successfully processed the request, but is returning information that may be from another source.
204	NoContent	The server has successfully processed the request, but has not returned any content. The status code is returned in response to an HTTP OPTIONS request.
205	Reset Content	The server has fulfilled the request, but the requester is required to reset the content.
206	Partial Content	The server has processed certain GET requests.
300	Multiple Choices	There are multiple options for the location of the requested resource. The response contains a list of resource characteristics and addresses from which the user or user agent (such as a browser) can choose the most appropriate one.
301	Moved Permanently	The requested resource has been assigned a new permanent URI, and the new URI is contained in the response.
302	Found	The requested resource resides temporarily under a different URI.
303	See Other	Retrieve a location. The response to the request can be found under a different URI and should be retrieved using a GET or POST method.
304	Not Modified	The requested resource has not been modified. In such a case, there is no need to retransmit the resource since the client still has a previously-downloaded copy.
305	Use Proxy	The requested resource must be accessed through a proxy.
306	Unused	The HTTP status code is no longer used.
400	BadRequest	Invalid request. The client should not repeat the request without modifications.
401	Unauthorized	The status code is returned after the client provides the authentication information, indicating that the authentication information is incorrect or invalid.
402	Payment Required	This status code is reserved for future use.

Status Code	Error Message	Description
403	Forbidden	The server understood the request, but is refusing to fulfill it. The client should not repeat the request without modifications.
404	NotFound	The requested resource cannot be found. The client should not repeat the request without modifications.
405	MethodNotAllowed	The method specified in the request is not supported for the requested resource. The client should not repeat the request without modifications.
406	Not Acceptable	The server cannot fulfill the request according to the content characteristics of the request.
407	Proxy Authentication Required	This status code is similar to 401, but indicates that the client must first authenticate itself with the proxy.
408	Request Time-out	The request timed out. The client may repeat the request without modifications at any later time.
409	Conflict	The request cannot be processed due to a conflict. This status code indicates that the resource that the client attempts to create already exists, or the requested update failed due to a conflict.
410	Gone	The requested resource is no longer available. The status code indicates that the requested resource has been deleted permanently.
411	Length Required	The server refuses to process the request without a defined Content-Length.
412	Precondition Failed	The server does not meet one of the preconditions that the requester puts on the request.
413	Request Entity Too Large	The request is larger than that a server is able to process. The server may close the connection to prevent the client from continuing the request. If the server cannot process the request temporarily, the response will contain a Retry-After header field.
414	Request-URI Too Large	The URI provided was too long for the server to process.
415	Unsupported Media Type	The server is unable to process the media format in the request.

Status Code	Error Message	Description
416	Requested range not satisfiable	The requested range is invalid.
417	Expectation Failed	The server fails to meet the requirements of the Expect request-header field.
422	UnprocessableEntity	The request is well-formed but is unable to be processed due to semantic errors.
429	TooManyRequests	The client has sent more requests than its rate limit is allowed within a given amount of time, or the server has received more requests than it is able to process within a given amount of time. In this case, it is advisable for the client to re-initiate requests after the time specified in the Retry-After header of the response expires.
500	InternalServerError	The server is able to receive the request but it cannot understand the request.
501	Not Implemented	The server does not support the requested function.
502	Bad Gateway	The server is acting as a gateway or proxy and receives an invalid request from a remote server.
503	ServiceUnavailable	The requested service is invalid. The client should not repeat the request without modifications.
504	ServerTimeout	The request cannot be fulfilled within a given time. This status code is returned to the client only when the **Timeout** parameter is specified in the request.
505	HTTP Version not supported	The server does not support the HTTP protocol version used in the request.

In the message body, the error information is described in JSON format as follows:

{"error_code":"S3M.XXXX","error_msg":"Error description"}

7.3 Obtaining a Project ID

Scenarios

A project ID is required for some URLs. You need to obtain a project ID before you call an API. Two methods are available:

- **Obtaining a Project ID by Calling an API**
- **Obtain a Project ID from the Console**

Obtaining a Project ID by Calling an API

You can obtain the project ID by calling the API used to **query projects based on specified criteria**.

The API used to obtain a project ID is **GET https://*{Endpoint}*/v3/projects**. *{Endpoint}* is the IAM endpoint and can be obtained from **Regions and Endpoints**. For details about API authentication, see **Authentication**.

The following is an example response. The value of **id** is the project ID.

```
{
    "projects": [
        {
            "domain_id": "65382450e8f64ac0870cd180d14e6xxx",
            "is_domain": false,
            "parent_id": "65382450e8f64ac0870cd180d14e6xxx",
            "name": "project_name",
            "description": "",
            "links": {
                "next": null,
                "previous": null,
                "self": "https://www.example.com/v3/projects/a4a5d4098fb4474fa22cd05f897d6xxx"
            },
            "id": "a4a5d4098fb4474fa22cd05f897d6xx",
            "enabled": true
        }
    ],
    "links": {
        "next": null,
        "previous": null,
        "self": "https://www.example.com/v3/projects"
    }
}
```

Obtain a Project ID from the Console

To obtain a project ID from the console, perform the following operations:

1. Sign in to **Huawei Cloud console**.
2. Hover the mouse over the username in the upper right corner and select **My Credentials** from the drop-down list.

   On the **API Credentials** page, view the project ID in the project list.

Figure 7-1 Viewing the project ID